B. Tech.

ELECTRICAL ENGINEERING

Syllabus

AY: 2023-24

CHARUSAT

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Authored By: Charusat

ACADEMIC REGULATIONS & SYLLABUS

2023-24

(Choice Based Credit System)

Faculty of Technology & Engineering

M & V Patel Department of Electrical Engineering

Bachelor of Technology Programme

(Electrical Engineering)

Chandubhai S Patel Institute of Technology

Vision

To become a leading institute for creation and dissemination of knowledge in the frontiers of technology

Mission

To prepare world-class technocrats and researchers and facilitate enhanced deployment of technology for betterment of lives

M & V Patel Department of Electrical Engineering

Vision

To become an internationally recognized, research intensive education entity, serving as a workforce developer for the nation by providing the best education in close collaboration with society.

Mission

To create world class technocrats with strong theoretical foundations and practical engineering skills who can contribute through the technological applications and research for the betterment of lives.

THE PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO 1	**Preparation:** To prepare the students with sound academically background, intensive development of knowledge and skills, and with an uncompromising attitude to achieve excellence in the electrical engineering.
PEO 2	**Core Competence:** To build a broad engineering core competence and to enable the students to solve the problems systematically by providing them exposure to different dimensions of specializations.
PEO 3	**Learning Environment:** To provide conducive learning environment instilling sound foundation in mathematics, computing techniques, science and humanities along with professional courses for futuristic thinking in an increasingly technology dependent society.
PEO 4	**Professionalism:** To provide a platform to the students to plan and shape their career in the chosen field and to imbibe ethical values with thorough professionalism and encouragement towards entrepreneurial thinking and traits.

PROGRAM ARTICULATION MATRIX

(-Not Matching, 1-Low, 2-Medium, 3-High)

	PEO 1	PEO 2	PEO 3	PEO 4
To create world class technocrats with strong theoretical foundations and practical engineering skills who can contribute through the technological applications and research for the betterment of lives.	3	3	3	3

PROGRAM OUTCOMES (POs)

At the end of the program, the student will be able to:

PO 1	**Engineering knowledge:** Apply knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO 2	**Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO 3	**Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health and safety, and the cultural, societal, and environmental considerations.
PO 4	**Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO 5	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO 6	**The engineer and society:** Apply reasoning obtained by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO 7	**Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO 8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO 9	**Individual and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO 10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO 11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO 12	**Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

At the end of the program, the student will be able to:

PSO 1	Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.
PSO 2	Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

ACADEMIC REGULATIONS

Bachelor of Technology (Electrical Engineering) Programme

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
Phone: 02697-265011, Email: info@charusat.ac.in
www.charusat.ac.in

Year – 2023-2024

CHARUSAT

FACULTY OF TECHNOLOGY AND ENGINEERING
ACADEMIC REGULATIONS
Bachelor of Technology Programmes

Choice Based Credit System

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. *System of Education*

Choice based Credit System with Semester pattern of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a course works in the chosen subject of specialization and also complete a project/dissertation if any. Apart from the Programme Core courses, provision for choosing University level electives and Programme/Institutional level electives are available under the Choice based credit system.

2. *Duration of Programme*

Undergraduate programme (B. Tech.)

Minimum	8 semesters (4 academic years)
Maximum	12 semesters (6 academic years)

3. *Eligibility for admissions:*

As enacted by Govt. of Gujarat from time to time.

4. *Mode of admissions*

As enacted by Govt. of Gujarat from time to time.

5. *Programme structure and Credits*

As per annexure – 1 attached

6. *Attendance*

6.1 All activities prescribed under these regulations and enlisted by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student regarding attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Principal.

6.2 Student's attendance in a course should be 80%.

7 *Course Evaluation*

7.1 The performance of every student in each course will be evaluated as follows:

7.1.1. Internal evaluation by the course faculty member(s) based on continuous assessment, the continuous assessment will be conducted by the respective department /institute.

7.1.2 Final end-semester examination by the University through written paper or practical test or oral test or presentation by the student or a combination of these.

7.1.3 The weightages of continuous assessment and End-semester university examination in overall assessment shall depend on individual course as approved by Academic Council through Board of Studies.

7.1.4 The performance of candidate in continuous assessment and in end-semester examination together (if applicable) shall be considered for deciding the final grade in a course.

7.1.5 In order to earn the credit in a course a student has to obtain grade other than FF.

7.2 Performance in continuous assessment and end-semester University Examination

7.2.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations (internal/external) are as follows

Minimum marks in University Exam per course	Minimum marks Overall per course
40%	45%

7.2.2 If a candidate obtains minimum required marks in each course but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

8 Grade Point System

8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade **on** a ten-point scale as per the following scheme:

Table: Grading Scheme (UG)

Range of Marks (%)	≥80	**<80 ≥73**	**<73 ≥66**	**<66 ≥60**	**<60 ≥55**	**<55 ≥50**	**<50 ≥45**	**<45**
Corresponding Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his/her performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

(i) SGPA = $\Sigma C_i G_i / \Sigma C_i$ where C_i is the number of credits of course i
G_i is the Grade Point for the course i
and i = 1 to n, n = number of courses in the semester

(ii) CGPA = $\Sigma C_i G_i / \Sigma C_i$ where C_i is the number of credits of course i
G_i is the Grade Point for the course i
and i = 1 to n, n = number of courses of all semesters up to which CGPA is computed.

9 Award of Class

☞ The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Award of Class	CGPA Range
Distinction	CGPA ≥ 7.5 & ≤ 10.0
First Class	CGPA ≥ 6.0 & < 7.5
Second Class	CGPA ≥ 5.0 & < 6.0
Pass Class	CGPA < 5.0

Grade sheets of only the final semester shall indicate the class. In case of all the other semesters, it will simply indicate as Pass / Fail.

9.1 Maximum duration allowed for Completion of a programme

- Maximum duration to allow for completion of a particular programme shall not be more than twice the normal duration of the respective programme. For example, a 6-Semester programme should be completed within not more than 12 semesters.

10 Detention Criteria

- No student will be allowed to move further in next semester if CGPA is less than 3 at the end of an academic year.
- A student will not be allowed to move to third year if he/she has not cleared all the courses of second year.
- A student will not be allowed to move to fourth year if he/she has not cleared all the courses of third year.

11 Transcript:

- The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

Annexure – 1

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY												
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN ELECTRICAL ENGINEERING												
Sem	Course Code	Course Title	Teaching Scheme					Examination Scheme				
			Contact Hours				Credit	Theory		Practical		Total
			Theory	Practical	Tutorial	Total		Internal	External	Internal	External	
SY Sem-III	XXXXX	University Elective I	0	2	0	2	2			30	70	100
	MA252	Advanced Engineering Mathematics	4	0	0	4	4	30	70	0	0	100
	EE251	Analog and Digital Electronics	3	2	0	5	4	30	70	25	25	150
	EE252.01	Circuit Theory	3	2	0	5	4	30	70	25	25	150
	EE243.01	Electrical Measurement and Industrial Instrumentation	4	2	0	6	5	30	70	25	25	150
	HS121.02A	Creativity, Problem Solving and Innovation		2		2	2	0	0	30	70	100
			14	**10**	**0**	**24**	**21**	**120**	**280**	**135**	**215**	**750**
SY Sem-IV	XXXXX	University Elective II	0	2	0	2	2			30	70	100
	EE254	Electrical Power Generation and Its Economics	3	0	0	3	3	30	70	0	0	100
	EE246	Microprocessor and Microcontrollers	4	2	0	6	5	30	70	25	25	150
	EE247	Transformers and Induction Machines	4	4	0	8	6	30	70	50	50	200
	EE255.01	Control Systems	3	2	0	5	4	30	70	25	25	150
	HS111.02A	Human Values and Professional Ethics		2		2	2	0	0	30	70	100
			14	**12**	**0**	**26**	**22**	**120**	**280**	**160**	**240**	**800**

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

List of University Elective Courses

for 3rd Semester Undergraduate Programme

(Effective from Academic year 2023-24)

Sr. No.	Course Code & Course Name	Department / Faculty offering the Course
1	BM231: Banking and Insurance	I2IM / FMS
2	CA224: Introduction to Web Designing	CMPICA / FCA
3	CE281.01: Art of Programming	CE / FTE
4	CL281.01: Environmental Sustainability and Climate Change	CL / FTE
5	CL283: SDG Handprint Laboratory	CL / FTE
6	EC281.01: Introduction to MATLAB Programming	EC / FTE
7	EE284: Python Programming	EE/FTE
8	IT283: Web Designing & UI/UX	IT/FTE
9	ME281.01: Engineering Drawing	ME/FTE
10	NR251.01: First Aid & Life Support	NURSING / FMD
11	PD260.01: Basic Laboratory Techniques	PDPIAS/FAS
12	PH233.01: Fundamentals of Packaging	RPCP/FPH
13	PT191.01: Health Promotion and Fitness	ARIP / FMD

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

List of University Elective Courses for 4th Semester Undergraduate Programme (Effective from Academic year 2023-24)

Sr. No.	Course Code & Course Name	Department / Faculty offering the Course
1	BM241: Health Care Management	I2IM / FMS
2	CA225: Programming the Internet	CMPICA / FCA
3	CE282.01: Web Designing	CE / FTE
4	CL282.01: Basics of Environmental Impact Assessment	CL / FTE
5	EC282.01: Prototyping Electronics with Arduino	EC / FTE
6	EE287: MATLAB Programming	EE/FTE
7	EE288: Maintenance of Household Apparatus	EE/FTE
8	IT284: DATA Visualization	IT/FTE
9	ME282.01: Material Science	ME/FTE
10	NR261.01: Life Style Diseases & Management	NURSING / FMD
11	PH238.01: Cosmetics in daily life	RPCP/FPH
12	PT192.01: Occupational Health & Ergonomics	ARIP / FMD

B. Tech. (Electrical Engineering) Programme

SYLLABI
(Semester – III)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Charotar University of Science and Technology, Gujarat

Re-designed Course Structure of UG in Electrical and Electronics Engineering

Course Code: **EE284**
Course Title: **Python Programming**

UG Engineering Programme	Semester
Electrical Engineering	Third

I RATIONALE

Python programming is an essential subject to learn in today's technological landscape. Python is a high-level, general-purpose programming language that is easy to learn and widely used in various fields, including data science, machine learning, web development, and automation. The language's simplicity and readability make it an ideal choice for beginners, while its extensive libraries and frameworks make it a powerful tool for experts. By learning Python programming, students can acquire valuable skills that can lead to numerous career opportunities, as Python is in high demand in industries such as finance, healthcare, and technology.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Proficiently write, debug and deploy Python code for various applications and industries.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an <u>integrated performance</u> of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) ***<u>Write</u>*** *program for real world problems*
2) ***<u>Describe</u>*** *the Math functions, Strings, List, Tuples and Dictionaries in Python.*
3) ***<u>Implement</u>*** *solution for mathematical problems, processing data, and handling strings.*
4) ***<u>Debug</u>*** *and* ***<u>Troubleshoot</u>*** *python code.*

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme (Anna 2021)				
				Theory Marks		Practical Marks		Total Marks
L	**T**	**P**	**C**	**ESE**	PA	**ESE**	**PA**	100
0	0	2	2	0	0*	70	30	

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

***(*):** Under the Theory PA, out of 30 marks, 10 marks are for micro-project assessment to facilitate integration of COs and the remaining 20 marks is based on lab activities by assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	2,4	Hands-on practice to use print, input command; variables; operators.	2
2	2,4	Hands-on practice and use cases of if, if-else, elif, nested if-else, for, while statement/loop.	2
3	1,3	Develop Python program to find odd/even year.	1
4	1,3	Develop Python program to compute the GCD of two numbers.	1
5	1,3	Develop Python program to find the square root of a number.	2
6	1,3	Develop Python program to find the exponentiation (Power of a number).	2
7	1,3	Develop Python program to find the first n prime numbers.	2
8	1,3	Develop Python program to multiply two matrices.	2
9	2,4	Hands-on practice to create list and various list methods.	2
10	1,3	Develop Python program to find the maximum of a list of numbers.	2
11	1,3	Develop Python program to search an element in an array.	2
12	1,3	Develop Python program to sort the elements.	2
13	1,3	Hands-on practice to use string and string methods, and file handling.	2
14	1,3	Develop Python program to take command line arguments(word count)	2
15	1,3	Develop Python program to find the most frequent words in a text read from a file.	2
16	2,4	Hands-on practice and use cases for Tuples and Dictionaries.	2
		Total	**30**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Software tool: Python v3.0 or higher	1,2,3,4,5, 6,7,8,9,10, 11,12,13, 14,15,16

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Function as a team member.
3) Function as a team leader.
4) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

IX SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Develop a program that simulates rolling dice. The user should be able to specify the number of dice to roll and the number of sides on each die.
b) Develop a program to construct calculator.
c) Create a weather app based on python programming.
d) Create a chatbot using python programming.

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Think Python	Allen Downey	O'Reilly Media 978-1449330729
2	Python Programming: A Complete Guide for Beginners to Master, Python Programming Language	Brian Draper	Createspace Independent Pub 978-1539434375
3	Python: The Complete Reference	Martin C. Brown	Tata McGraw Hill 978-9387572942

XV SOFTWARE/LEARNING WEBSITES

1. https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-0001-introduction-to-computer-science-and-programming-in-python-fall-2016
2. https://www.python.org/about/gettingstarted
3. https://onlinecourses.nptel.ac.in/noc18_cs21/preview

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE284	**Semester IV**	**Python Programming** **Mark 'H'=3** for high, '**M**'=2 for medium or **'L'=1'** for the relevant correlation of each PO or PSO, Competency or CO													
284	*Proficiently write, debug and deploy Python code for various applications and industries.*	***2***	***2***	***3***	***1***	***3***								***2***	***1***
284.1	Write program for real world problems	1	1	2		3								3	1
284.2	Describe the Math functions, Strings, List, Tuples and Dictionaries in Python.	3	2												
284.3	Implement solution for mathematical problems, processing data, and handling strings.	1	1	3	1									2	1
284.	Debug and		3	1	2	1								1	

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
4	Troubleshoot python code.														

XVII NAMES OF RESOURCE PERSONS

a) Jignesh S Patel

b) ………………………………………………….

Course Code: **MA252**
Course Title: **Advance Engineering Mathematics**

UG Engineering Programme	Semester
Electrical Engineering, Electronics & Communication Engineering	Third

I RATIONALE

Mathematical concepts play a significant role in engineering for comprehending core technical subjects. This course structure is developed to familiarize engineering students with some advanced concepts in mathematics which includes the calculus of vector valued functions, basic transformation such as Laplace Transform, series expansion of periodic functions and curve fitting techniques.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Explain mathematical techniques for the solution of various problems of engineering.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) ***Solve*** the complex problems through the concepts of Laplace transforms.
2) ***Compute*** the problems related to the Fourier series.
3) ***Explain*** the vector space, linear span, linear dependence, and basis**.**
4) ***Explain*** the concepts of vectors, directional derivative, gradient, vector fields, divergence and curl.
5) **Solve** the problems through the concepts of Z- transforms.
6) ***Find*** the Fourier Integral and Fourier Transform of various functions.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			**Total Credits (L+T+P)**	**Examination Scheme**				
				Theory Marks		**Practical Marks**		**Total Marks**
L	**T**	**P**	**C**	**ESE**	**PA**	**ESE**	**PA**	
4	0	0	4	70	30*	-	-	100

Legends: ***L***-*Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

(*): *Under the* ***Theory PA****, out of 30 marks,* ***15 marks*** *are for* ***micro-project assessment*** *to facilitate integration of COs and the remaining 15 marks is the average of 2 tests to be taken*

during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Laplace Transforms and Its Applications	**CO 1. *Solve the complex problems through the concepts of Laplace transforms.*** 1a. Compute the Laplace transform of the given functions 1b. Explain various properties of the Laplace transform 1c. Find the inverse Laplace transform of some functions 1d. Explain various properties of the inverse Laplace transform 1e. Evaluate the inverse Laplace transform of the given functions using convolution theorem 1f. Use the Method of Laplace transforms to solve initial-value problems for linear differential equations	1.1 Laplace transforms as an improper integral and its existence 1.2 Laplace transforms of elementary functions and its properties 1.3 Inverse Laplace transforms and its properties 1.4 First and second shifting theorems. Laplace transforms of derivatives and integrals 1.5 Convolution theorem and its application to obtain inverse Laplace transform. 1.6 Laplace transform of periodic functions, Unit step function, Unit impulse function (Dirac delta function). 1.7 Solving differential equations using Laplace transform 1.8 Applications of Laplace transform to ODE: Electric circuits (RLC and RL)

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit– II **Fourier Series**	**CO 2. *Compute the problems related to the Fourier Series.*** 2a. Explain the existence condition for Fourier series 2b. Compute the Fourier series of the given periodic functions 2c. Find the Fourier series of discontinuous functions 2d. Determine the Half range Fourier sine and cosine series of the given periodic functions	2.1 Periodic functions, Dirichlet's conditions, Trigonometric series 2.2 Euler formulae, Fourier series of periodic functions of period p 2.3 Fourier series of Discontinuous functions 2.4 Fourier series of Even and odd functions; half range Fourier series
Unit– III **Linear Algebra**	**CO 3. *Explain the vector space, linear span, linear dependence, and basis.*** 3a. Verify either a structure forms a vector space or not 3b. Define Linear dependence of set of vectors 3c. Explain basis of vector space	3.1 Introduction of Vector Space and its examples 3.2 Linear dependence and independence and its examples 3.3 Basis and its examples
Unit-IV **Vector Differential Calculus**	**CO 4. *Explain the concepts of vectors, directional derivative, gradient, vector fields, divergence and curl.*** 4a. Interpret the concepts of vectors 4b. Evaluate the gradient and directional derivatives of problems 4c. Find the divergence and curl of vector fields 4d. Justify the terms irrotational and solenoidal 4e. Evaluate the scalar potential function of vectors	4.1 Revision of concepts of Vector algebra, Scalar and Vector fields 4.2 Gradient of a scalar functions, Directional derivatives 4.3 Divergence and Curl of a vector field and their properties 4.4 Physical interpretations of gradient, divergence and curl:Irrotational and solenoidal vector fields 4.5 Scalar potential function
Unit – V **Z-transforms**	**CO 5. *Solve the problems through the concepts of Z- transforms.*** 5a. Find the Z-transform of the given sequences 5b. Explain various properties of the Z-transform 5c. Compute the inverse Z-transform of some functions	5.1 Introduction to Z transform. Representation of sequences 5.2 Z-transforms of basic sequences 5.3 Unit impulse and Unit step sequence. Basic operations on sequence 5.4 Properties of Z-transforms: Damping rule

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
		and shifting property 5.5 Introduction to inverse Ztransforms. Inverse Z-transform using partial fraction and geometric series
Unit-VI Fourier Transforms	**CO 6.** ***Find the Fourier Integral and Fourier Transform of various functions.*** 6a. Compute the Fourier integral of the given functions 6b. Find the Fourier transform of the given functions 6c. Explain various properties of the Fourier transform 6d. Determine the Fourier sine and cosine transform of some functions	6.1 Introduction to Fourier integral: sine and cosine integrals 6.2 Introduction to Fourier Transform and inverse Fourier Transform 6.3 Properties and examples of Fourier transforms 6.4 Fourier sine and cosine transforms

VI TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Laplace Transforms and Its Applications	15	2	3	4	4	5	0	18
II	Fourier Series	10	1	2	2	3	4	0	12
III	Linear Algebra	07	1	1	2	4	0	0	08
IV	Vector Differential Calculus	08	1	1	2	5	0	0	09
V	Z-transforms	10	1	2	2	3	4	0	12
VI	Fourier Transforms	10	1	1	2	3	4	0	11
Total		**60**	**07**	**10**	**14**	**22**	**17**	**0**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

VII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Laplace Transforms and Its Applications	1	1a. Compute the Laplace transform of the given functions	Conventional, ICT based	Green board, PPTs
	2	1b. Explain various properties of the Laplace transform 1c. Find the inverse Laplace transform of some functions	Conventional, ICT based	Green board, PPTs
	3	1d. Explain various properties of the inverse Laplace transform 1e. Evaluate the inverse Laplace transform of the given functions using convolution theorem	Conventional, ICT based	Green board, PPTs
	4	1f. Use the Method of Laplace transforms to solve initial-value problems for linear differential equations	Conventional, ICT based	Green board, PPTs
Unit– II Fourier Series	**5**	2a. Choose the relevant electromagnetic relay for the given situation. 2b. Explain the existence condition for Fourier series	Conventional, ICT based	Green board, PPTs
	6	2c. Compute the Fourier series of the given periodic functions 2d. Find the Fourier series of discontinuous functions	Conventional, ICT based	Green board, PPTs
	7	2e. Determine the Half range Fourier sine and cosine series of the given periodic functions	Conventional, ICT based	Green board, PPTs
Unit– III Linear Algebra	**8**	3a. Verify either a structure forms a vector space or not	Conventional, ICT based	Green board, PPTs
	9	3b. Define Linear dependence of set of vectors 3c. Explain basis of vector space	Conventional, ICT based	Green board, PPTs
Unit-IV Vector Differential Calculus	**10**	4a. Interpret the concepts of vectors 4b. Evaluate the gradient and directional derivatives of problems	Conventional, ICT based	Green board, PPTs
	11	4c. Find the divergence and curl of vector fields 4d. Justify the terms irrotational and solenoidal 4e. Evaluate the scalar potential function of vectors	Conventional, ICT based	Green board, PPTs
Unit – V Z-transforms	**12**	5a. Find the Z-transform of the given sequences 5b. Explain various properties of the Z-transform	Conventional, ICT based	Green board, PPTs
	13	5c. Compute the inverse Z-transform of some functions	Conventional, ICT based	Green board, PPTs
Unit-VI Fourier	**14**	6a. Compute the Fourier integral of the given functions	Conventional, ICT	Green board,

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Transforms		6b. Find the Fourier transform of the given functions	based	PPTs
	15	6c. Explain various properties of the Fourier transform 6d. Determine the Fourier sine and cosine transform of some functions	Conventional, ICT based	Green board, PPTs

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Advanced Engineering Mathematics	Erwin Kreyszig	8th Ed., Jhon Wiley & Sons, India, 1999
2	Integral transforms and their applications	Debnath, Lokenath, and Dambaru Bhatta	CRC press, 2014
3	Advanced Engineering Mathematics	K. A. Stroud and D. J. Booth	Palgrave Macmillan, 2006
4	Elementary linear algebra	Howard Anton and Chris Rorres	John Wiley & Sons, 2010
5	Advanced engineering mathematics	H. K. Dass	S. Chand, 2008
6	Advanced Engineering Mathematics	M. D. Greenberg	2nd ed., Pearson, 1998
7	Higher engineering mathematics	B. S. Grewal	43^{rd} Ed., Khanna Publishers, New Delhi, 2014
8	Thomas' Calculus	M. D. Weir et. al	11^{th} Ed., Pearson Education, 2008
9	Advanced Engineering Mathematics	C. Ray Wylie and L. C. Barrett	McGraw Hill pub., 1982

XV SOFTWARE/LEARNING WEBSITES

1. http://nptel.ac.in/courses/111107098/
2. http://www1.maths.leeds.ac.uk/~frank/math2420/notes.pdf

XVI PO-COMPETENCY-CO MAPPING

		POs and PSOs

Course Code	Competency and COs	**PO 1** Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** EthicsEnvironment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1**	**PSO2**	**PSO3**
MA 252	**Semester III**	**Protection and Switchgear** Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1' for the relevant correlation of each PO or PSO, Competency or CO														
252	**Competency** *Apply mathematical techniques for the solution of various problems of engineering*	*3*	*2*	*2*	*1*	*1*	-	*1*	-	-	*1*	-	*1*	*2*	*2*	-
252.1	Solve the complex problems through the concepts of Laplace transforms	3	2	1	1	1	-	1	-	-	1	-	1	2	2	-
252.2	Compute the problems related to the Fourier Series	3	2	1	1	1	-	1	-	-	1	-	1	2	2	-
252.3	Explain the vector space, linear span, linear dependence, basis	3	2	-	1	1	-	-	-	-	1	-	1	2	2	-
252.4	Explain the concepts of vectors, directional derivative, gradient, vector fields, divergence and curl	3	2	-	1	1	-	-	-	-	1	-	1	2	2	-
252.5	Solve the problems through the concepts of Z-transforms	3	2	2	1	1	-	1	-	-	1	-	1	2	2	-
252.6	Find the Fourier Integral and Fourier Transform of various functions	2	2	1	-	-	-	-	-	-	1	-	1	1	1	-

XVII NAMES OF RESOURCE PERSONS

a) Pragnesh Makwana
b) Rajendra Chauhan

Course Code: **EE251**
Course Title: **Analog and Digital Electronics**

UG Engineering Programme	Semester
Electrical Engineering	Third

I RATIONALE

The course has been designed to introduce fundamental principles of analog and digital electronics. The students completing this course will understand basic analog and digital electronics, including semiconductor properties, operational amplifiers, combinational and sequential logic and analog-to-digital digital-to-analog conversion techniques. Finally, students will gain experience in with the design of analog amplifiers, power supplies and logic devices.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course:

- **Evaluate the performance of various analog and digital circuits used in Industrial applications**.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

5) Design & Enhance their skills in basics of analog and digital circuits.
6) Evaluate the performance and classification of various components and ICs.
7) State linear and non-linear applications of operational amplifiers.
8) Design and troubleshoot hardware circuits like filters, comparators, PID controllers, oscillators, digital logic gates (combinational circuits), timer and counters (sequential circuits).
9) Analyze characteristics of popular MOS and bipolar logic families, with emphasis on CMOS and TTL technologies.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			**Total Credits (L+T+P)**	**Examination Scheme**				
				Theory Marks		**Practical Marks**		**Total Marks**
L	**T**	**P**	**C**	**ESE**	PA	**ESE**	**PA**	150
3	0	2	4	70	30	25	25	

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

(*): *Under the* ***Theory PA****, out of 30 marks,* ***15 marks*** *are for* ***micro-project assessment*** *to facilitate integration of COs and the remaining 15 marks is the average of 2 tests to be taken*

during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	3	(A) Apply two input voltages and find output difference voltage and Gain of an op-amp. (B) Get Summing, Scaling and Averaging Amplifier of given signal using Inverting configuration.	04
2	3	Apply sine wave, square wave and check output waveforms of Integrator and Differentiator circuit.	02
3	3	(A) Compare sine wave input and DC input using op-amp. (B) Identify Zero Crossing of sine wave using op-amp.	02
4	4	(A) Get the frequency response curve of first order low pass filter. (B) plot the frequency response curve of all-Pass filter.	04
5	3,4	(A) find the frequency for building the Square Wave using op-amp. (B) find the quasi stable states for astable Multivibrator using 555 timer	04
6	2	(A) Observe the waveform using single transistor. (B) Observe the waveform using multi transistor.	04
7	4	Get the Truth table for logic gates.	02
8	4	Build the half-adder and half subs tractor circuit.	02
9	4	Build the Full Adder circuit diagram using two- half adders circuit diagram.	02
10	4	Verify truth tables for J-K Flip-Flop and D Flip-Flop.	02
11	4	Apply input switching conditions for universal Bi-directional Shift Register.	
		Total	**30**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Analog Trainer kit, Digital Multimeter, Digital Oscilloscope	1-6
2	Digital trainer kit	7-11

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Practice building of analog & digital circuits.
2) Design and analyze combinational & sequential circuit.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I BJT and FET	**CO 2. *Evaluate the performance and classification of various components and ICs.*** PrO 6. (A)Plot the waveform using single transistor. (B) Observe the waveform using multi transistor 1a. construction and working principle of transistor, VI characteristics of transistor in cutoff, saturation and active region. 1b. Transistor as a switch, commonly used transistor connection, Transistor as an amplifier in CE arrangement, Analysis of collector current. 1c. Transistor DC load line and operating point, Examples of DC load line and operating point. 1d. Semiconductor devices & numbering system, Power rating of transistor, Transistor testing, Transistor as an differential amplifier, faithful amplification. 1e. Transistor biasing methods:Base resistor method, feedback resistor and voltage divider. 1f. Transistor Stabilization, Examples of Transistor biasing methods. 1g. Design of biasing circuits, Transistor stability and stability factor, Testing and transistor data sheet. 1h. Types of FET, construction and working principle of JFET, Difference between	1.1 Transistor DC load line and Operating point, V-I Characteristics 1.2 Transistor as a switch and an amplifier, Faithful Amplification 1.3 Transistor biasing and its Methods, Transistor Stability and Stability Factor, Testing and Transistor data-sheet 1.4 Types of FET, Construction and working principle of JFET, Difference between BJT and JFET, V-I characteristics, Pinch-off Volt age, Shorted-gate Drain Current, Parameters of JFET, Biasing of JFET 1.5 Working of MOSFET and V-I Characteristics, Difference between JFET and MOSFET.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	BJT and JFET, Voltampere characteristics. 1i. Pinch off voltage, Shorted gate drain current, gate source cut off voltage, Parameters of JFET, Examples. 1j. Biasing of JFET, Working of MOSFET and VI characteristics, Difference between JFET and MOSFET.	
Unit– II Operational Amplifiers	**CO 2.** ***Evaluate the performance and classification of various components and ICs.*** 2a. Integrated Circuits, IC classifications, packages and symbols. 2b. Pin diagram of 741 Opamp, Internal block diagram of a Typical opamp, Ideal differential amplifier, Examples of finding voltage gain & CMRR. 2c. Ideal opamp and its equivalent circuit, Specification and Datasheet parameters of opamp. 2d. Differential Input Resistance(Ri), Output Resistance(Ro), Common Mode Rejection, Ratio (CMRR), Slew Rate, Power Supply Rejection Ratio(PSRR), Input offset, Voltage, input offset Current, input bias Current, Output offset Voltage, Bandwidth. 2e. Opamp in openloop and closed loop, DC and AC Amplifier, Voltage follower.	2.1 Introduction to Integrated Circuits, IC classifications, packages and symbols 2.2 Pin diagram of 741 Op-amp, Internal block diagram of a Typical op-amp, Ideal op-amp and its equivalent circuit 2.3 Specification and Datasheet parameters of op-amp (Differential Input Resistance (Ri), Output Resistance(Ro), Common Mode Rejection Ratio (CMRR), Slew Rate, Power Supply Rejection Ratio (PSRR), Input offset Volt age, Input offset Current, Input bias Current, Output offset Voltage) 2.4 Feedback Configurations, An op-amp with and without Feedback, Closed-loop Voltage Gain.
Unit– III Applications of op-amp	***CO 3. State linear and non-linear applications of operational amplifiers.*** PrO 1._(A) Apply two input voltages and find output difference voltage and Gain of an op-amp. (B) Get Summing, Scaling and Averaging Amplifier of given signal using Inverting configuration. PrO 2. Apply sine wave, square wave and	3.1. **Linear applications:** DC and AC Amplifier, Volt age follower, Peaking, Summing, Scaling, Averaging Amplifier, Differential

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	check output waveforms of Integrator and Differentiator circuit. PrO 3. (A) Compare sine wave input and DC input using op-amp. (B) Identify Zero Crossing of sine wave using op-amp. PrO 5. (A) find the frequency for building the Square Wave using op-amp. 3a. Differential Amplifiers and Instrumentation Amplifier, Peaking, Summing, Scaling, Averaging Amplifier. 3b. Integrator, Differentiator Using op-amp. 3c. V to I converter, I to V converter. 3d. Basic Comparator, Zero crossing Detector, Schmitt trigger. 3e. Basic Positive and Negative Clippers and Clampers.	Amplifiers and Instrumentation Amplifier, V to I converter, I to V converter, Integrator, Differentiator **3.2. Active filters & Oscillators:** First order Low-Pass and High-Pass Filters, second order Low-Pass and High-Pass Filters, Band-Pass and Band-Reject filters, All-Pass filters, Phase Shift, Wien Bridge Oscillators; Square, Triangular, Saw-tooth Wave Generators. **3.3. Comparators & Converters:** Basic Comparator, zero crossing Detector, Schmitt Trigger, Basic Positive and Negative Clippers and Clampers, Absolute-Value Output Circuit, Peak Detector, ADC and DAC using op-amp.
Unit-IV Specialized ICs	***CO 4. Design and troubleshoot hardware circuits like filters, comparators, PID controllers, oscillators, digital logic gates, timer and counters.*** PrO 4. (Get the frequency response curve of first order low pass filter. (B) plot the frequency response curve of all-Pass filter. PrO 5. (B) find the quasi stable states for	4.1 555 Timer IC pin diagram, Block diagram 4.2 555 timer IC as Monostable, bistable and astable Multivibrator. 4.3 Three terminal regulator ICs (Fixed Voltage Regulator ICs, Adjustable

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	astable Multivibrator using 555 timer 4a. 555 Timer IC pin diagram & block diagram. 4b. 555 timer IC as Monostable, Bistable Multivibrator,555 timer IC as an Astable Multivibrator. 4c. Voltage regulator, Positive & Negative Fixed voltage regulators using three terminals, adjustable Positive & Negative voltage regulators.	output voltage regulator ICs, their use and basic design considerations for designing regulated power supplies) 4.4 Analog Multiplier MPY634.
Unit – V Introduction to Digital Electronics and Numbering Systems	***CO 4. Design and troubleshoot hardware circuits like filters, comparators, PID controllers, oscillators, digital logic gates (combinational circuits), timer and counters (sequential circuits).*** 5a. Digital circuits, Digital signals, advantages, applications. 5b. Introduction, Binary number system, conversion to other systems, Decimal, Hexa decimal number system, Octal number system, Conversion to other number systems 5c. Binary arithmetic: addition, subtraction, division, multiplication, Signed & unsigned binary number system, advantages & Disadvantages. 5d. Complements:1's and 2's complement, Binary subtraction using 1's and 2's complement. 5e. Advantages, applications, and classifications, BCD code conversion, Gray code conversion, applications of gray code, Alphanumeric codes, ASCII codes, BCD to seven segment display.	5.1 Digital circuits, Digital signals. 5.2 Binary number system, octal, hexadecimal and decimal Number systems and their inter conversion, Binary arithmetic 5.3 Signed & unsigned binary number system 5.4 1's complement and 2's complement 5.5 BCD numbers, ASCII and EBCDIC codes, Grey codes.
Unit – VI Digital Logic families and Design	***CO 4. Design and troubleshoot hardware circuits like filters, comparators, PID controllers, oscillators, digital logic gates (combinational circuits), timer and counters (sequential circuits).*** **PrO 7.** Get the Truth table for logic gates. 6a. Boolean laws, Boolean expression simplification, logic gates building circuit. 6b. Universal gates 6c. Karnaugh Map simplification, Kmap structure, SOP form, POS form, simplification	6.1 Basic Logic gates and their truth tables, Universal Gates **6.2** Laws of Boolean algebra, De-Morgan's theorem 6.3 Min term, Max term, POS, SOP, Karnaugh Maps, Minimization of logical function, Don't care condition

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	of Boolean expressions using Kmap, Way of grouping, examples 6d. Examples of Minimization of SOP expressions. 6e. Don't care condition, Examples of Minimization of POS expressions. 6f. classification based on circuit complexity: SSI, MSI, LSI, VLSI, classification based on devices used, voltage & current parameters, Fanin and fanout, Noise margin, Propagation delay, Current sourcing and sinking. 6g. Resistor transistor logic, Diode transistor logic, Integrated injection logic circuit. 6h. Two input TTL NAND gate logic, TTL NOR gate logic circuit, CMOS inverter, PMOS inverter, CMOS NAND gate logic, CMOS NOR gate logic circuit.	6.4 Introduction, Resistor transistor logic, Direct coupled transistor logic, Integrated injection logic, DTL logic , TTL logic, CMOS Logic
Unit – VII Combinational Logic	***CO 4. Design and troubleshoot hardware circuits like filters, comparators, PID controllers, oscillators, digital logic gates (combinational circuits), timer and counters (sequential circuits).*** PrO 8. Build the half-adder and half subs tractor circuit. PrO 9. Build the Full Adder circuit diagram using two- half adders circuit diagram. 7a. Analysis of a combinational circuit, Types of binary adders, Half adder logic circuit, Full adder logic circuit, Binary subtraction: Half & full subtractor 7b. Multiplexer (data selector), Necessity of Multiplexers, advantages of Multiplexers, Types of Multiplexer, Examples of implementation of combinational circuit using Multiplexer, demultiplexer. 7c. Priority encoder, Decoders: BCD to seven segment deoders.	7.1 Half adder, Full adder, Binary subtraction, Arithmetic logic unit 7.2 Digital comparators, Priority encoder, decoders, Multiplexer & demultiplexer.
Unit – VIII Flip-flops, Timers and Counters	***CO 4. Design and troubleshoot hardware circuits like filters, comparators, PID controllers, oscillators, digital logic gates (combinational circuits), timer and counters (sequential circuits).*** PrO 10. Verify truth tables for J-K Flip-Flop and D Flip-Flop. PrO 11. Apply input switching conditions for universal Bi-directional Shift	8.1 S-R flip-flop, D- flip-flop, T-flip-flop, J-K flip-flop, Clocked flip-flop design 8.2 Registers 8.3Asynchronouscounter, Synchronous counter

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	Register. 8a. Priority encoder, Decoders: BCD to seven segment deoders. 8b. NAND latch (SR latch using NAND gates), triggering methods, concept of level triggering, types of level triggered flipflops, concept of Edge triggering. 8c. Level triggered SR flipflop, Level triggered D flipflop, 8d. Level triggered JK flipflop, Race around condition in JK latch, difference between latch & FlipFlop, MasterSlave JK FlipFlop. 8e. Registers, Data formats, Classification of registers, Shift register. 8f. Types of counters, Classification of counters, comparison of synchronous and asynchronous counters, Applications of counters, comparison of counters & Registers.	

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	BJT and FET	04	2	3	0	0	2	0	07
II	Operational Amplifiers	04	2	2	0	0	0	0	7
III	Applications of op-amp	09	0	4	0	3	7	0	14
IV	Specialized ICs	05	0	2	4	4	6	0	7
V	Introduction to Digital Electronics and Numbering Systems	04	2	1	0	0	0	3	6
VI	Digital Logic families and Design	07	0	2	2	4	3	0	11
VII	Combinational Logic	06	1	2	1	2	3	0	9
VIII	Flip-flops, Timers and Counters	06	0	2	2	2	3	0	9
Total		**45**	**0**	**10**	**14**	**20**	**16**	**0**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom’s taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **Transistor:** choose the circuit using BJT, FET, MOSFET FOR Electronics applications,
2) **555 Timer:** build the circuit using 555 timers and operate in different modes and electronics applications.
3) **Op-amp:** make the circuit and perform the various applications.
4) **Different types of sensor:** use any electronics component along with sensor and apply for electronics applications.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

e) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
f) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
g) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
h) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical

evidences for their (student's) portfolio which will be useful for their placement interviews:

e) Library survey regarding different analog and digital devices and circuits are used in industries.
f) Prepare 15 min. power point presentation on any analog and digital devices application.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I BJT and FET	1	*PrO 6.* (A) Observe the waveform using single transistor. (B) Observe the waveform using multi transistor. 1a. construction and working principle of transistor, V-I characteristics of transistor in cut-off, saturation and active region. 1b. Transistor as a switch, commonly used transistor connection, Transistor as an amplifier in CE arrangement, Analysis of collector current. 1c. Transistor DC load line and operating point, Examples of DC load line and operating point.	Lecture with media	PPTs, black board
	2	PrO 1. (A) Apply two input voltages and find output difference voltage and Gain of an op-amp. 1d. Semiconductor devices & numbering system, Power rating of transistor, Transistor testing, Transistor as a differential amplifier, faithful amplification. 1e. Transistor biasing Methods-Base resistor method, feedback resistor and voltage divider. 1f. Transistor Stabilization, Examples of Transistor biasing methods	Lecture With media	PPTs, black board
	3	PrO 1. (B) Get Summing, Scaling and Averaging Amplifier of given signal using Inverting configuration. 1g. Design of biasing circuits, Transistor stability and stability factor, Testing and transistor data sheet. 1h. Types of FET, construction and working principle of JFET, Difference between BJT and JFET, Volt-ampere characteristics. 1i. Pinch off voltage, Shorted gate drain current, gate-source cut off voltage, Parameters of JFET, Examples.	Lecture with media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– II Operational Amplifiers	**4**	PrO 2 Apply sine wave, square wave and check output waveforms of Integrator and Differentiator circuit. 2a. Biasing of JFET, Working of MOSFET and V-I characteristics, Difference between JFET and MOSFET. 2b. Integrated Circuits, IC classifications, packages and symbols. 2c. Pin diagram of 741 Op-amp, Internal block diagram of a Typical op-amp, Ideal differential amplifier, Examples of finding voltage gain & CMRR.	Lecture with media	PPTs, black board
	5	PrO 3. (A) Compare sine wave input and DC input using op-amp. (B) Identify Zero Crossing of sine wave using op-amp. 2d. Ideal op-amp and its equivalent circuit, Specification and Datasheet parameters of op-amp. 2e. Differential Input Resistance(Ri), Output Resistance(Ro), Common Mode Rejection, Ratio (CMRR), Slew Rate, Power Supply Rejection Ratio(PSRR), Input offset, Voltage, input offset Current, input bias Current, Output offset Voltage, Bandwidth. 2f. Op-amp in open-loop and closed loop, DC and AC Amplifier, Voltage follower.	Lecture	black board
Unit– III Applications of op-amp	**6**	PrO 4. (A) Get the frequency response curve of first order low pass filter. (B) plot the frequency response curve of all-Pass filter. 3a. Differential Amplifiers and Instrumentation Amplifier, Peaking, Summing, Scaling, Averaging Amplifier. 3b. Integrator, Differentiator Using op-amp. 3c. V to I converter, I to V converter with floating load & grounded load.	Lecture with media	PPTs, black board
	7	3d. Basic Comparator, Zero crossing Detector, Schmitt trigger 3e. Basic Positive and Negative Clippers and Clampers.	Lecture with media	PPTs, black board
Unit– IV Specialized ICs	**8**	PrO 5. (A) find the frequency for building the Square Wave using op-amp. (B) find the quasi stable states for astable Multivibrator using 555 timer 4a. 555 Timer IC pin diagram & block diagram. 4b. 555 timer IC as Monostable, Bistable Multivibrator. 4c. 555 timer IC as an Astable Multivibrator. 4d. Voltage regulator, Block diagram, Positive &	Lecture with media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		Negative Fixed voltage regulators using three terminal, adjustable Positive & Negative voltage regulators		
Unit– V Introduction to Digital Electronics and Numbering Systems	**9**	PrO 7. Get the Truth table for logic gates. 5a. Digital circuits, Digital signals, advantages, applications. 5b. Binary number system, conversion to other systems. 5c. Decimal, Hexa decimal number system, Octal number system, Conversion to other number systems	Lecture	PPTs, black board
	10	PrO 8. Build the half-adder and half subs tractor circuit. 5d. Binary arithmetic: addition, subtraction, division, multiplication. 5e. Signed & unsigned binary number system, advantages & Disadvantages. 5f. Complements:1's and 2's complement, Binary subtraction using 1's and 2's complement 5g. Binary codes: Advantages, applications, and classifications, BCD code conversion, Gray code conversion, applications of gray code, Alphanumeric codes, ASCII codes, BCD to seven segment display.	Lecture	PPTs, black board
Unit– VI Digital Logic families and Design	**11**	PrO 9. Build the Full Adder circuit diagram using two- half adders circuit diagram. **6a.** classification based on circuit complexity: SSI, MSI, LSI, VLSI, classification based on devices used, voltage & current parameters, Fan-in and fan-out, Noise margin, Propagation delay, Current sourcing and sinking. **6b.** Resistor transistor logic, Diode transistor logic, Integrated injection logic circuit. **6c.** Two input TTL NAND gate logic, TTL NOR gate logic circuit. **6d.** CMOS inverter, PMOS inverter, CMOS NAND gate logic, CMOS NOR gate logic circuit.	Lecture with media	PPTs, black board
Unit– VII Combinational Logic	**12**	PrO 10. Verify truth tables for J-K Flip-Flop and D Flip-Flop. **7a.** Boolean laws, Boolean expression simplification, logic gates building circuit. **7b.** Karnaugh Map simplification, K-map structure, SOP form, POS form, simplification of Boolean expressions	Lecture	black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		using K-map, Way of grouping, **7c.** Sop form examples		
	13	7d. Examples of Minimization of SOP expressions, Don't care condition, 7e. Examples of Minimization of POS expressions. 7f. Analysis of a combinational circuit, Types of binary adders, Half adder logic circuit, Full adder logic circuit, Binary subtraction: Half & full subtractor.	Lecture	black board
	14	7g. Binary subtraction: Half & full subtractor. 7h. Multiplexer (data selector), Necessity of Multiplexers, advantages of Multiplexers, Types of Multiplexer, Examples of implementation of combinational circuit using Multiplexer, demultiplexer.	Lecture	black board
Unit–VIII Flip-flops, Timers and Counters	**15**	Pro 10. Apply input combinations and switching conditions for universal Bi-directional Shift Register. 8a. Difference between combination circuit & sequential circuit, clock signal, clock skew, 1-Bit memory cell, Latch. 8b. NAND latch (S-R latch using NAND gates), triggering methods, concept of level triggering, types of level triggered flip-flops, concept of Edge triggering. Level triggered S-R flip-flop, Level triggered D-flip-flop 8c. Level triggered J-K flip-flop, Race around condition in JK latch, difference between latch & Flip-Flop, Master-Slave JK Flip-Flop. 8e. Registers, Data formats, Classification of registers, Shift register, Counters: Types of counters, Classification of counters, comparison of synchronous and asynchronous counters, Applications of counters, comparison of counters & Registers.	Lecture with media	PPTs, black board

XVIII SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Principles of Electronics	V. K. Mehta	S. Chand & Company Ltd. ISBN:81-219-2450-2

S. No.	Title of Book	Author	Publication including ISBN
2	Op-Amp and Linear Integrated Circuit technology	Ramakant A Gayakwad	PHI Publication ISBN: 0132808684, 9780132808682
3	Analog and Digital Electronics	J. S. Katre	J. S. Katre ISBN: 978-81-8492-140-3
4	Fundamental of digital circuits	A. ANANDKUMAR	PHI Publication ISBN: 8120352688, 9788120352681
5	Operational Amplifier and Linear Integrated Circuits	James M. fiore	Jaico Publishing House ISBN:8172247788, 9788172247782
6	Electronics Devices and Circuits an Introduction	A. Mottershed	PHI Publication ISBN:8120301242, 9788120301245
7	Digital Fundamentals	Morris and Mano	PHI Publication ISBN-13: 978-0-13-277420-8 ISBN-10: 0-13-277420-8

XIX SOFTWARE/LEARNING WEBSITES

1. http://nptel.ac.in/courses/108106068/
   (Videos and Notes by Prof. K. Radhakrishna Rao, IIT Madras)
2. http://nptel.ac.in/courses/108106069/
   (Videos and Notes by Dr. Amitava Dasgupta, IIT Madras)

XX PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs					POs and PSOs										
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Project management and finance	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.	**PSO 3**
EE251	**Semester III**					Analog and Digital Electronics **Mark 'H'=3** for high, **'M'**=2 for medium or **'L'=1'** for the relevant correlation of each PO or PSO, Competency or CO										
251	**Competency** *Evaluate the performance of*	*3*	*3*	*3*	*2*	*2*	*0*	*0*	*0*	*0*	*0*	-	-	*3*	*1*	-

Course Code	Competency and COs					POs and PSOs										
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Project management and finance	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.	**PSO 3**
	various analog and digital circuits used in Industrial applications.															
251.1	Design & Enhance their skills in basics of analog and digital circuits.	3	2	-	-	-	-	-	-	-	-	-	-	3	-	-
251.2	Understand the performance and classification of various components and ICs.	2	3	-	-	-	-	-	-	-	-	-	-	2	-	-
251.3	State linear and non-linear applications of operational amplifiers.	1	3	-	-	-	-	-	-	-	-	-	-	2	-	-
251.4	Design and troubleshoot hardware circuits like filters, comparators, PID controllers, oscillators, digital logic gates (combinational circuits), timer and counters (sequential circuits).	1	2	2	3	3	-	-	-	-	-	-	-	3	1	-
251.5	Analyze characteristics of popular MOS and bipolar logic families, with emphasis on CMOS and TTL technologies.	1	1	3	3	-	-	-	-	-	-	-	-	2	1	-

XXI NAMES OF RESOURCE PERSONS

a) **Ankur H Patel**

Course Code: **EE252.01**

Course Title: **Circuit Theory**

UG Engineering Programme	**Semester**
Circuit Theory	Third

I RATIONALE

Electrical engineers have to work on different electrical circuits consisting of electrical elements. This course is developed such that students should be able to analyze, evaluate, design, develop, implement various circuits efficiently for circuit analysis and synthesis. Student should be able to develop, design the circuit parameter according to the system need.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course:

- **Analyze the performance of various circuit elements used in electrical circuits for electrical systems.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

Use the different methods to find the solutions of circuit parameters.
Evaluate the circuit performance with the help of network topology.
Evaluate *the performance of the circuits for Two Port parameters.*
Evaluate the performance of circuits with Time and Laplace domain for circuits having initial conditions.
Analyze *the 1st,2nd order RL, RC and RLC series and parallel circuits.*

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			**Total Credits (L+T+P)**	**Examination Scheme**				
				Theory Marks		**Practical Marks**		**Total Marks**
L	**T**	**P**	**C**	**ESE**	**PA**	**ESE**	**PA**	
3	2	0	4	70	30	25	25	150

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

(*): *Under the* ***Theory PA****, out of 40 marks,* ***20 marks*** *are for* ***micro-project assessment*** *to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Determine the Nodal voltage for DC resistive circuits and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink.	02
2	1	Determine the Mesh current for DC resistive circuits and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink.	02
3	1	Determine the current and voltage for DC resistive circuits with the help of superposition theorem and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink.	02
4	1	Determine the load current and voltage for DC resistive circuits by designing the Thevenin's Equivalent circuit and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink.	02
5	1	Determine the load current and voltage for DC resistive circuits by designing the Norton's Equivalent circuit and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink.	02
6	1	Determine the load resistance value for DC resistive circuits by using Maximum Power Transfer Theorem and check the effect of change in the load resistor for the power variation.	02
7	1	Determine the load current and voltage for DC resistive circuits by using Reciprocity theorem and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink.	02
8	3	Examine the performance of the circuit by using Z parameters concept for Two Port network with the help of graphical representation using MATLAB software.	02
9	3	Examine the performance of the circuit by using Y parameters concept for Two Port network with the help of graphical representation using MATLAB software.	02
10	3	Examine the performance of the circuit by using ABCD and A'B'C'D' parameters concept for Two Port network with the help of graphical representation using MATLAB software.	02
11	4, 5	Analyze the transient and steady state response of R-L, R-C, R-L-C series and parallel circuits using MATLAB software.	04
		Total	**24**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Bread board, DC power supply,	1 to 10
2	MATLAB Simulation software (version 2021a)	1 to 11

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I DC and AC Circuit Simplification Techniques.	**CO 1.** ***Use*** the different methods to find the solutions of circuit parameters. *PrO 1.* Determine the Nodal voltage for DC resistive circuits and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink . *PrO 2.* Determine the Mesh current for DC resistive circuits and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink. PrO 3. Determine the current and voltage for	1.1 Circuit Elements And Their Characteristics, Energy Sources, Source Transformation 1.2 Kirchhoff's laws, Current division, Voltage division in series circuits, Nodal and mesh analysis of electric circuits in DC and AC excitations. 1.3 Superposition theorem, Thevenin theorem, Norton Theorem, Reciprocity and

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	DC resistive circuits with the help of superposition theorem and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink.. PrO 4. Determine the load current and voltage for DC resistive circuits by designing the Thevenin's Equivalent circuit and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink. PrO 5. Determine the load current and voltage for DC resistive circuits by designing the Norton's Equivalent circuit and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink. PrO 6. Determine the load resistance value for DC resistive circuits by using Maximum Power Transfer Theorem and check the effect of change in the load resistor for the power variation. PrO 7. Determine the load current and voltage for DC resistive circuits by using Reciprocity theorem and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink. 1a. Describe the Circuit Elements And Their Characteristics, Energy Sources. 1b. Apply Source Transformation technique for circuit simplification. 1c. Illustrate KCL, KVL, current division rule and voltage division rule in electrical circuits. 1d. Determine nodal voltage and mesh current with nodal and mesh analysis technique for DC and AC circuits. 1e. Examine the circuit performance with the help of theorems. 1f. Design the simple circuits with the help of theorems from the complex circuits. 1g. Choose the optimized circuit parameters for the maximum power transfer with the	Maximum power transfer theorem in Electrical Circuits with DC and AC excitations.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	help of MPT theorem.	
Unit– II Network Topology	**CO 2. *Evaluate*** the circuit performance with the help of network topology. 2a. Identify the terminology used in network graph. 2b. Prepare Incidence, Reduced incidence matrix, Tie-Set, Fundamental Tie-Set matrix, Fundamental Cut-Set, Cut-Set matrix . 2c. Solve the circuits with Dot Convention method for coupled circuit. 2d. Apply Principle Of Duality for creating dual networks.	2.1 Concept Of Network Graph, Terminology Used In Network Graph 2.2 Incidence, Reduced incidence matrix ,Tie-Set Matrix, Fundamental Tie-Set Matrix, Fundamental Cut-Set, Cut-Set Matrix, 2.3 The Dot Convention For Coupled Circuit, Principle Of Duality
Unit– III Two Port Parameters	**CO 3. Evaluate** *the performance of the circuits for Two Port parameters.* *PrO 8.* Examine the performance of the circuit by using Z parameters concept for Two Port network with the help of graphical representation using MATLAB software. *PrO 9.* Examine the performance of the circuit by using Y parameters concept for Two Port network with the help of graphical representation using MATLAB software. *PrO 10.* Examine the performance of the circuit by using ABCD and A'B'C'D' parameters concept for Two Port network with the help of graphical representation using MATLAB software. 3a. Identify the Network Elements and its configuration for two poet networks. 3b. Evaluate Open circuit (Z) parameter, Short circuit (Y) parameter for two port network and review the conditions for reciprocity and symmetricity. 3c. Evaluate Transmission (ABCD) parameter, Inverse Transmission (A'B'C'D') parameter for two port network and review the conditions for reciprocity and symmetricity. 3d. Express Z, Y ,T and T' parameters by each parameters.	3.1 Network Elements, Classification of Network, Network Configuration, Recurrent Network. 3.2 Z-parameters, Y-parameters, ABCD Parameters, Inverse ABCD Parameters. 3.3 Interrelationship between parameters. 3.4 Concept of Reciprocity & Symmetricity for network parameters

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit-IV Laplace transformation and Initial Conditions	**CO 4.*<u>Evaluate</u>*** the performance of circuits with Time and Laplace domain for circuits having initial conditions. *PrO 11.* <u>Analyze</u> the transient and steady state response of R-L, R-C, R-L-C series and parallel circuits using MATLAB software. 4a. <u>List</u> basic types of special ignals and Laplace transformation formulas used in circuit analysis. 4b. <u>Apply</u> Laplace transformation technique for representing the transform impedance and transform admittance circuits. 4c. <u>Illustrate the</u> factors of initial conditions in the circuits and its procedure to evaluate the derivatives. 4d. <u>Analyze</u> the circuit performance under initial condition.	4.1 Basic Types Of Special Signals, Laplace Transformation Of Special Signal Waveforms. 4.2 Transform impedance and transform admittance circuits. 4.3 Initial conditions in elements, Geometrical Interpretation of Derivatives. A Procedure of Evaluating Initial conditions. 4.4 R-L , R-C & R-L-C Circuits analysis using Initial Conditions.
Unit – V Steady state and Transient Response Analysis of circuit	**CO 5.*<u>Analyze</u>*** *the 1st,2nd order RL, RC and RLC series and parallel circuits.* *PrO 11.* <u>Analyze</u> the transient and steady state response of R-L, R-C, R-L-C series and parallel circuits using MATLAB software. 5a. <u>Identify</u> the 1st and 2nd order circuits and its behaviors for step input. 5b. <u>Examine</u> the performance of the series and parallel circuits under step response. 5c. <u>Analyze</u> the R-L, R-C and R-L-C circuit performance with DC excitation. 5d. <u>Analyze</u> the R-L, R-C and R-L-C circuit performance with AC excitation.	5.1 Step Response 1st order and 2nd order circuits. 5.2 Step response of Series & Parallel Circuit 5.3 Transient Response of Series R-L , R-C, R-L-C With DC Excitation. 5.4 Transient Response of Series R-L , R-C, R-L-C With AC Excitation.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	DC and AC Circuit Simplification Techniques	12	02	04	04	04	04	02	20

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
II	Network Topology	05	02	02	02	00	00	00	06
III	Two Port Parameters	12	04	04	04	02	04	00	18
IV	Laplace transformation and Initial Conditions	10	02	02	02	02	06	02	16
V	Steady state and Transient Response Analysis of circuits	06	00	02	02	04	02	02	10
Total		**45**	**10**	**12**	**12**	**10**	**20**	**06**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **Resistors, Inductors & Capacitors:** Collect the specifications of different types of electromagnetic relays used in different protection situations in different types of industries.
2) **Voltage and Current Sources:** Collect the specifications of different types of static relays used in different protection situations in different types of industries.
3) **Initial Condition based performance of RLC circuits:** Prepare case studies of protections failures in different types of industries.
4) **Two port network in power systems:** Prepare case studies of circuit breakers used in the protection schemes in different types of industries.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different circuit elements used in electrical circuits.
b) Prepare 15 min. power point presentation on the various parameter configurations used modern power system analysis.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I DC and AC Circuit Simplification Techniques.	1	**CO 1. *Use*** the different methods to find the solutions of circuit parameters. *PrO 1.* Determine the Nodal voltage for DC resistive circuits and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink. 1a. Describe the Circuit Elements And Their Characteristics, Energy Sources. 1b. Apply Source Transformation technique for circuit simplification. 1c. Illustrate KCL, KVL, current division rule and voltage division rule in electrical circuits.	Lecture with media	PPTs, black board
	2	**CO 1. *Use*** the different methods to find the solutions of circuit parameters.	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		PrO 2. Determine the Mesh current for DC resistive circuits and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink. PrO 3. Determine the current and voltage for DC resistive circuits with the help of superposition theorem and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink. 1d. Determine nodal voltage and mesh current with nodal and mesh analysis technique for DC and AC circuits. 1e. Examine the circuit performance with the help of theorems.		
	3	PrO 4. Determine the load current and voltage for DC resistive circuits by designing the Thevenin's Equivalent circuit and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink. PrO 5. Determine the load current and voltage for DC resistive circuits by designing the Norton's Equivalent circuit and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink. 1e. Examine the circuit performance with the help of theorems. 1f. Design the simple circuits with the help of theorems from the complex circuits	Lecture with media Tutorial and Field visit	PPTs, black board
	4	PrO 6. Determine the load resistance value for DC resistive circuits by using Maximum Power Transfer Theorem and check the effect of change in the load resistor for the power variation. PrO 7. Determine the load current and voltage for DC resistive circuits by using Reciprocity theorem and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink 1f. Design the simple circuits with the help of theorems from the complex circuits. 1g. Choose the optimized circuit parameters for the		

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		maximum power transfer with the help of MPT theorem.		
Unit– II Network Topology	**5**	**CO 2. *Evaluate*** the circuit performance with the help of network topology. 2a. Identify the terminology used in network graph. 2b. Prepare Incidence, Reduced incidence matrix, Tie-Set, Fundamental Tie-Set matrix, Fundamental Cut-Set, Cut-Set matrix.	Lecture with media, and Demonstration	PPTs, Video
	6	**CO 2. *Evaluate*** the circuit performance with the help of network topology. 2c. Solve the circuits with Dot Convention method for coupled circuit. 2d. Apply Principle Of Duality for creating dual networks.	Lecture with media, Tutorial	PPTs
Unit– III Two Port Parameters	**7**	**CO 3. *Evaluate*** the *performance of the circuits for Two Port parameters.* *PrO 8.* Examine the performance of the circuit by using Z parameters concept for Two Port network with the help of graphical representation using MATLAB software. 3a. Identify the Network Elements and its configuration for two poet networks. 3b. Evaluate Open circuit (Z) parameter, Short circuit (Y) parameter for two port network and review the conditions for reciprocity and symmetricity.	Lecture with media, Field visit	PPTs, Video
	8	*PrO 9.* Examine the performance of the circuit by using Y parameters concept for Two Port network with the help of graphical representation using MATLAB software. 3b. Evaluate Open circuit (Z) parameter, Short circuit (Y) parameter for two port network and review the conditions for reciprocity and symmetricity.	Lecture with media, and Tutorial	PPTs
	9	*PrO 8.* Examine the performance of the circuit by using ABCD and A’B’C’D’ parameters concept for Two Port network with the help of graphical representation using MATLAB software.	Lecture with media, and Tutorial	PPTs

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		3c. Evaluate Transmission (ABCD) parameter, Inverse Transmission (A'B'C'D') parameter for two port network and review the conditions for reciprocity and symmetricity.		
	10	*PrO 8.* Examine the performance of the circuit by using ABCD and A'B'C'D' parameters concept for Two Port network with the help of graphical representation using MATLAB software. 3d. Evaluate Transmission (ABCD) parameter, Inverse Transmission (A'B'C'D') parameter for two port network and review the conditions for reciprocity and symmetricity. 3e. Express Z, Y ,T and T' parameters by each parameters.	Lecture with media, and Tutorial	PPTs
Unit– IV Laplace transformation and Initial Conditions	**11**	**CO 4.*Evaluate*** the performance of circuits with Time and Laplace domain for circuits having initial conditions. *PrO 11.* Analyze the transient and steady state response of R-L, R-C, R-L-C series and parallel circuits using MATLAB software. 4a. List basic types of special Signals and Laplace transformation formulas used in circuit analysis. 4b. Apply Laplace transformation technique for representing the transform impedance and transform admittance circuits.	Lecture with media, and Tutorial	PPTs
	12	**CO 4.*Evaluate*** the performance of circuits with Time and Laplace domain for circuits having initial conditions. *PrO 11.* Analyze the transient and steady state response of R-L, R-C, R-L-C series and parallel circuits using MATLAB software. 4c. Illustrate the factors of initial conditions in the circuits and its procedure to evaluate the derivatives.	Lecture with media, and Tutorial	PPTs

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
	13	**CO 4.*<u>Evaluate</u>*** the performance of circuits with Time and Laplace domain for circuits having initial conditions. *PrO 11.* <u>Analyze</u> the transient and steady state response of R-L, R-C, R-L-C series and parallel circuits using MATLAB software. 4d. <u>Analyze</u> the circuit performance under initial condition.	Lecture with media, and Tutorial	PPTs
Unit– V Steady state and Transient Response Analysis of circuits	13	**CO 4. *<u>Analyze</u>*** *the 1st,2nd order RL, RC and RLC series and parallel circuits.* *PrO 12.* <u>Analyze</u> the transient and steady state response of R-L, R-C, R-L-C series and parallel circuits using MATLAB software. 5a. <u>Identify</u> the 1st and 2nd order circuits and its behaviors for step input.	Lecture with media, and Tutorial	PPTs
	14	**CO 5. *<u>Analyze</u>*** *the 1st,2nd order RL, RC and RLC series and parallel circuits.* *PrO 13.* <u>Analyze</u> the transient and steady state response of R-L, R-C, R-L-C series and parallel circuits using MATLAB software. 5b. <u>Examine</u> the performance of the series and parallel circuits under step response. 5c. <u>Analyze</u> the R-L, R-C and R-L-C circuit performance with DC excitation.	Lecture with media, and Tutorial	PPTs
	15	**CO 5. *<u>Analyze</u>*** *the 1st,2nd order RL, RC and RLC series and parallel circuits.* *PrO 14.* <u>Analyze</u> the transient and steady state response of R-L, R-C, R-L-C series and parallel circuits using MATLAB software. 5c. <u>Analyze</u> the R-L, R-C and R-L-C circuit performance with DC excitation. 5d. <u>Analyze</u> the R-L, R-C and R-L-C circuit performance with AC excitation.	Lecture with media, and Tutorial	PPTs

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Circuit Theory Analysis & Synthesis	Abhijit Chakrabarti	Dhanpat Rai Publishing Company (P) Limited, 2008, 8177000004, 9788177000009
2	Network analysis	Mac Elwyn Van Valkenburg	Pearson Publications. 9789332550131, 9332550131
	Electrical Circuit Theory and Technology	John Bird	Taylor & Francis 9781136347115, 1136347119
3	Network Analysis and Synthesis	by Ravish R Singh,	McGraw Hill Education Pvt. Ltd 9781259062957, 1259062953
4	Network analysis	G. K. Mithal,	Khanna publications 9788174090706, 8174090703
5	Electric Circuits and Networks	K. S. Suresh Kumar	Pearson Education 9788131713907, 8131713903
6	U. A. Patel, Network analysis and Synthesis	Late Ajay V. Bakshi, Uday A. Bakshi	Mahajan Publisher 9788189050443

XIV SOFTWARE/LEARNING WEBSITES

1. http://nptel.ac.in/courses/108102042/9
2. http://freevideolectures.com/Course/2350/Networks-Signals-and-Systems

XV PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs		POs and PSOs											
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE252.01	**Semester III**			**Protection and Switchgear** **Mark 'H'=3** for high, '**M**'=2 for medium or **'L'=1'** for the relevant correlation of each PO or PSO, Competency or CO										
252.01	**Competency** *Analyze the performance of*	***3***	***3***	***3***	***2***	***2***	-	-	-	-	-	-	***2***	***3***	***1***

Course Code	Competency and COs			POs and PSOs											
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	various circuit elements used in electrical circuits for electrical systems.														
CO1	Use the different methods to find the solutions of circuit parameters.	3	3	2	-	1	-	-	-	-	-	-	2	3	-
CO2	Evaluate the circuit performance with the help of network topology.	3	2	2	-	-	-	-	-	-	-	-	-	1	1
CO3	Evaluate the performance of the circuits for Two Port parameters.	3	2	2	2	-	-	-	-	-	-	-	2	3	1
CO4	Evaluate the performance of circuits with Time and Laplace domain for circuits having initial conditions.	3	2	3	2	1	-	-	-	-	-	-	-	2	1
CO5	Analyze the 1st,2nd order RL, RC and RLC series and parallel circuits.	3	3	3	2	3	-	-	-	-	-	-	-	2	1

XVI NAMES OF RESOURCE PERSONS

a) Mahammadsoaib Saiyad

b) ……………………………………………………

Course Code: **EE243.01**

Course Title: **Electrical Measurement and Industrial Instrumentation**

UG Engineering Programme	Semester
Electrical Engineering	Third

I RATIONALE

The process of measuring and controlling various quantities in industries by utilizing various industrial instruments is called as industrial instrumentation. Sensors and instrumentation devices are often used in industrial applications for condition monitoring, flow monitoring, stability monitoring, pressure and temperature data collection for process control. In view of this, study of principles of electrical measurement and industrial instruments become mandatory for all electrical engineers. This course deals with the principles of measurement, sensors and transducers.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- Memorize different types of electrical instrumentation systems and transducers.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) ***Memorize*** *the different sensors, transducers and various measuring instruments.*
2) ***Memorize*** *basic principle, working, characteristics and applications of the various measuring instruments and transducers.*
3) ***Interpret*** *the measured results and cause of any possible error.*

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme (Anna 2021)				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	
4	0	2	5	70	30*	25	25	150

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

(*): *Under the* ***Theory PA****, out of 30 marks,* ***15 marks*** *are for* ***micro-project assessment*** *to facilitate integration of COs and the remaining 15 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	**CO No.**	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	**Approx. Hrs. required**
1	1	Memorize different sensors, transducers and various measuring instruments used in industries.	02
2	2	Perform the stress and strain analysis of industrial equipment.	02
3	2	Select a temperature sensor based on its operating characteristic.	06
4	2	Explain the operating principle of flow restriction sensors.	02
5	3	Interpret the condition of rotating equipment based on torque analysis.	02
6	2	Explain the operating principle of level sensors.	02
7	2	Explain the operating principle of humidity sensors.	02
8	1	Memorize single phase energy meter.	02
9	3	Use DC ammeter and voltmeter for different ranges.	04
10	3	Calibrate ammeter, voltmeter and wattmeter as per IS.	02
11	3	Determine the different ratios and errors for potential transformer.	02
12	3	Determine the different ratios and errors for current transformer.	02
		Total	**30**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	**Equipment Name with Broad Specifications**	**PrO No.**
1	Strain Gauge trainer kit	2
2	Temperature trainer kit	3
3	Orifice meter with flow arrangement	4
4	Air Purge System	5
5	Resistive Hygrometer trainer kit	6
6	Strain Gauge Torque transducer with single phase induction motor arrangement	7
7	DMMs	9
8	Potential Transformer	10
9	Current Transformer	11
10	Calibration Tester	12

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Function as a team member.
3) Function as a team leader.
4) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organizing Level' in 3rd year
d) 'Characterizing Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Introduction of Instrumentations and Measurement	**CO 1.** ***Memorize*** *the different sensors, transducers and various measuring instruments.* *PrO 1. Memorize different sensors, transducers and various measuring instruments used in industries.* 1a. Justify the requirement of sensors/transducers for the given electrical equipment. 1b. Memorize the different sensors, transducers and various measuring instruments. 1c. Select the sensor/transducer based on its performance characteristics for the given application.	1.1 History of Instrumentation and Measurements 1.2 Application of Instrument systems 1.3 Classification of transducers 1.4 Performance characteristics of instruments
Unit– II Temperature and Strain Measurement	**CO 2.** ***Explain*** *basic principle, working, characteristics and applications of the various measuring instruments and transducers.* *PrO 2. Perform the stress and strain analysis of industrial equipment.* *PrO 3. Select a temperature sensor based on its operating characteristic.* 2a. Explain the basic principle, working, characteristics and applications of strain gauge transducers. 2b. Explain the basic principle, working, characteristics and application of temperature sensors. 2c. Select a temperature sensor based on its operating characteristic.	2.1 Temperature Scales 2.2 Non-electrical methods for temperature measurement 2.3 Electrical methods for temperature measurement 2.4 Radiation Pyrometers 2.5 Strain Measurement using Piezoresistive strain gauge 2.6 Applications of Strain Gauges 2.7 Load Cells
Unit– III Flow and Pressure	**CO 2.** ***Explain*** *basic principle, working, characteristics and applications of the various measuring instruments and transducers.*	3.1 Flow Restriction Sensors – Orifice meter, Venturi

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Measurement	*PrO 2. Perform the stress and strain analysis of industrial equipment.* *PrO 4. Explain the operating principle of flow restriction sensors.* 3a. Explain the basic principle, working, characteristics and application of flow sensors. 3b. Explain the basic principle, working, characteristics and application of pressure sensors.	meter, Flow nozzle, Rotameter, Pitot Tube 3.2 Electrical Flow Sensors – Turbine Flow meter, Electromagnetic Flow meter, Hot wire anemometer, Ultrasonic flow meter 3.3 Elastic Pressure Sensors – Diaphragm, Bourdon Tube, Bellows 3.4 Electrical Pressure Sensors – Variable inductance and capacitance transducers, Piezoelectric Sensor
Unit –IV Measurement of Liquid Level & Humidity	**CO 2.** ***Explain*** *basic principle, working, characteristics and applications of the various measuring instruments and transducers.* *PrO 5. Explain the operating principle of level sensors.* *PrO 6. Explain the operating principle of humidity sensors.* 4a Explain the basic principle, working, characteristics and application of level sensors. 4b Explain the basic principle, working, characteristics and application of humidity sensors.	4.1 Level Measurement – Direct Methods (Dip stick, sight glass, and float), Air purge method, inductive and capacitive transducers, ultrasonic level gauge, and radiation method. 4.2 Humidity measurement – Dew point meter, Hair hygrometer, Psychrometer, Resistive and capacitive hygrometer
Unit – V Measurement of Electrical Parameters	**CO 2.** ***Explain*** *basic principle, working, characteristics and applications of the various measuring instruments and transducers.* **CO 3.** ***Interpret*** *the measured results and cause of any possible error.* *PrO 7. Memorize single phase energy meter.* *PrO 8. Use DC ammeter and voltmeter for different ranges.* *PrO 9. Calibrate ammeter, voltmeter and wattmeter as per IS.* 5a Explain the basic principle, working, characteristics and application of analog instruments. 5b Explain the basic principle, working, characteristics and application of different bridge circuits. 5c Calibrate ammeter, voltmeter and	5.1 Instruments for measuring voltage, current, power, energy, frequency, power factor 5.2 Extension of Instrument Ranges and Multipliers, problems 5.3 Calibration of ammeter, voltmeter, wattmeter and energy meter. 5.4 Measurement of low, medium and high resistances - Wheatstone Bridge, Making balanced Wheatstone Bridge measurement, Kelvin Double Bridge, problems

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	wattmeter.	5.5 Measurement of inductance & capacitance with the help of AC Bridges (Hays Bridge, Schering Bridge, Maxwell bridge, Anderson Bridge), problems
Unit – VI Instrument Transformer	**CO 2.** ***Explain*** *basic principle, working, characteristics and applications of the various measuring instruments and transducers.* **CO 3.** ***Interpret*** *the measured results and cause of any possible error.* *PrO 10. Determine the different ratios and errors for potential transformer.* *PrO 11. Determine the different ratios and errors for current transformer.* 6a Explain the basic principle, working, characteristics and application of potential transformers. 6b Explain the causes of errors and its solution in potential transformers. 6c Explain the basic principle, working, characteristics and application of current transformers. 6d Explain the causes of errors and its solution in current transformers.	6.1 Current Transformer – Types, Characteristics, Errors, Vector diagram, problems 6.2 Potential Transformer - Types, Characteristics, Errors, Vector diagram, problems

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction of Instrumentations and Measurement	04	2	2	2	0	0	0	06
II	Temperature and Strain Measurement	12	2	2	4	0	8	0	16
III	Flow and Pressure Measurement	08	2	2	4	0	0	0	08
IV	Measurement of Liquid Level & Humidity	04	2	2	0	0	0	0	04
V	Measurement of Electrical Parameters	16	2	2	4	4	6	0	18
VI	Instrument Transformer	16	2	2	4	2	8	0	18

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
Total		**60**	**12**	**12**	**16**	**08**	**20**	**02**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XI SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different sensors used in industries.
b) Prepare 15 min. power point presentation on how to choose sensor for the particular application.

XII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Introduction of Instrumentations and Measurement	**1**	**PrO 1.** Memorize different sensors, transducers and various measuring instruments used in industries. 1a. Justify the requirement of sensors/transducers for the given electrical equipment. 1b. Memorize the different sensors, transducers and various measuring instruments.	Lecture with media	PPTs, black board
Unit– II Temperature and Strain Measurement	**2**	***PrO 2.*** *Perform the stress and strain analysis of industrial equipment.* 2a. Explain the basic principle, working, characteristics and applications of strain gauge	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		transducers.		
	3 & 4	***PrO 3.*** *Select a temperature sensor based on its operating characteristic.* 2b. Explain the basic principle, working, characteristics and application of temperature sensors. 2c. Select a temperature sensor based on its operating characteristic.	Lecture With media	PPTs, black board
Unit– III Flow and Pressure Measurement	5	***PrO 2.*** *Perform the stress and strain analysis of industrial equipment.* 3a. Explain the basic principle, working, characteristics and application of flow sensors.	Lecture With media	PPTs, black board
	6	***PrO 4.*** *Explain the operating principle of flow restriction sensors.* 3b. Explain the basic principle, working, characteristics and application of pressure sensors.	Lecture With media	PPTs, black board
Unit – IV Measurement of Liquid Level & Humidity	7	***PrO 5.*** *Explain the operating principle of level sensors.* 5a. Explain the basic principle, working, characteristics and application of level sensors.	Lecture With media	PPTs, black board
		PrO 6. *Explain the operating principle of humidity sensors.* *5b.* Explain the basic principle, working, characteristics and application of humidity sensors.	Lecture With media	PPTs, black board
Unit – V Measurement of Electrical Parameters	8 & 9	***PrO 7.*** *Memorize single phase energy meter.* ***PrO 8.*** *Use DC ammeter and voltmeter for different ranges.* 7a. Explain the basic principle, working, characteristics and application of analog instruments.	Lecture With media	PPTs, black board
	10 & 11	***PrO 9.*** *Calibrate ammeter, voltmeter and wattmeter as per IS.* 7b. Explain the basic principle, working, characteristics and application of different bridge circuits. 7c. Calibrate ammeter, voltmeter and wattmeter.	Lecture With media	PPTs, black board
Unit – VI Instrument Transformer	12 & 13	***PrO 10.*** *Determine the different ratios and errors for potential transformer.* 8a. Explain the basic principle, working, characteristics and application of potential transformers. 8b. Explain the causes of errors and its solution in potential transformers.	Lecture With media	PPTs, black board
	14 &	***PrO 11.*** *Determine the different ratios and*	Lecture	PPTs, black

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
	15	*errors for current transformer.* 8c. Explain the basic principle, working, characteristics and application of current transformers. 8d. Explain the causes of errors and its solution in current transformers.	With media	board

XIII SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Electrical and Electronics Measurement and Instrumentation	Rajput, R. K.	S. Chand, , ISBN 978-9380386348
2	Electrical and Electronics Measurement and Instrumentation	Shawhney, A.K.	Dhanpat Rai Publisher, New Delhi

XIV SOFTWARE/LEARNING WEBSITES

i. www.isa.org
ii. http://nptel.iitm.ac.in/video.php?courseId=1062
iii. http://www.mywbut.com/syllabus.php?mode=SM&paper_id=57&dept_id=6

XV PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE243.01	**Semester III**	**Electrical Measurement and Industrial Instrumentation** **Mark 'H'=3** for high, **'M'**=2 for medium or **'L'=1'** for the relevant correlation of each PO or PSO, Competency or CO													
243.01	**Competency** Maintain different types of electrical instrumentation systems and transducers.	*3*	*2*	-	-	-	*1*	-	-	*2*	*1*	-	*1*	*2*	*1*
243.01.1	Memorize the different sensors, transducers and various measuring instruments.	3	-	1	1	-	2	-	-	2	1	-	2	2	1

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
243.01.2	Explain basic principle, working, characteristics and applications of the various measuring instruments and transducers.	3	-	2	-	-	2	-	-	2	2	-	2	2	-
243.01.3	Interpret the measured results and cause of any possible error.	3	2	-	2	-	2	-	-	2	2	-	2	2	2

XVI NAMES OF RESOURCE PERSONS

a) Prof. Mihir A. Bhatt

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS121.02 A: CREATIVITY, PROBLEM SOLVING AND INNOVATION

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
III	HS121.02 A	Creativity, Problem Solving and Innovation	02	02	--	--	30	70	100

II. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Creativity, Problem Solving and Innovation	06
2.	Questioning, Learning and Visualization	06
3.	Creative Thinking and Problem Solving	06
4.	Logic, Language and Reasoning	06
5.	Contemporary Issues and Practices in Creativity and Problem Solving	
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	**30**

III. Detailed Syllabus:

1.	Introduction to Creativity, Problem Solving and Innovation	06 Hours	20%
	Definitions of Creativity and Innovation, Need for Problem Solving and Innovation, Scope of Creativity in various Domains, Types and Styles of Thinking, Strategies to develop Creativity, Problem Solving and Innovation skills		
2.	Questioning, Learning and Visualization	6 Hours	20%
	Strategy and Methods of Questioning, Asking the Right Questions, Strategy of Learning and its Importance, Sources and Methods of Learning, Purpose and Value of Creativity Education in real life, Visualization strategies - Making thoughts Visible, Mind Mapping and Visualizing Thinking		
3.	Creative Thinking and Problem Solving	6 Hours	20%

	Creative Thinking and its need, Strategy of Thinking Fluency, Generating all Possibilities, SCAMPER Technique, Divergent Vs Convergent Thinking, Lateral Vs Vertical Thinking, Fusion of Ideas for Problem Solving, Applying strategies for Problem Solving		
4.	Logic, Language and Reasoning	6 Hours	20%
	Basic Concepts of Logic, Statement Vs Sentence, Premises Vs Conclusion, Concept of an Argument, Functions of Language: Informative, Expressive and Directive, Inductive Vs Deductive Reasoning, Critical Thinking & Creativity, Moral Reasoning		
5.	Contemporary Issues and Practices in Creativity and Problem Solving	6 Hours	20%
	Cognitive Research Trust Thinking for Creatively Solving Problems, Case Study on Contemporary Issues and Practices in Creativity and Problem Solving		

IV. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by Slides Presentations, Reading Material, Discussions, Case Studies, Puzzles, Ted Talks, Videos, Task-Based Learning, Projects, Assignments and various Individual and Interpersonal activities like, Critical reading, Group work, Independent and Collaborative Research, Presentations, etc.

V. Evaluation:

There will be no end semester university examinations. Students will be evaluated continuously in the form of internal as well as external evaluation. The evaluation is schemed as 30 marks for internal evaluation and 70 marks for external evaluation. The concerned teacher shall evaluate students distribute the marks (out of 30 as Internal and out of 70 as External) and submit them. There may also be a Computer Based Test at the end.

Evaluation Scheme

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Attendance	100 %	--	20
2	Individual Activity Participation	As stipulated by the Resource Person(s) in the Training		20
3	Group Activity Participation			20
4	Presentation			30
5	Feedback on Improvement			10
			Total	**100**

VI. Course Outcomes (COs)

At the end of the course, learners will be able to:

CO1	Demonstrate creativity in their day to day activities and academic output.
CO2	Solve personal, social and professional problems with a positive and an objective mindset.
CO3	Think creatively and work towards problem solving in a strategic way.
CO4	Initiate new and innovative practices in their chosen field of profession.
CO5	Give logical ideas, opinions, and solutions to problems.
CO6	Think critically over the situation and drawing conclusion.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	2	-	2	-	-
CO2	-	-	-	2	-	-	-	-	-	2	-	3	-	-
CO3	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO4	-	-	-	-	-	2	-	-	-	2	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO6	-	-	-	3	-	-	-	-	-	2	-	-	-	-

Correlation levels 1, 2 or 3 are as defined below:
1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

VII. Books / Reference Books / Reading

Text Books

1. R Keith Sawyer, ZigZag, The Surprising Path to Greater Creativity, Jossy-Bass Publication 2013
2. Michael Michalko, Crackling Creativity, The Secrets of Creative Genus, Ten Speed Press 2001

Reference Books

3. Michael Michalko, Thinker Toys, Second Edition, Random House Publication 2006
4. Edward De Beno, De Beno’s Thinking Course, Revised Edition, Pearson Publication 1994
5. Edward De Beno, Six Thinking Hats, Revised and Update Edition, Penguin Publication 1999
6. Tony Buzan, How to Mind Map, Thorsons Publication 2002
7. Scott Berkum, The Myths of Innovation, Expended and revised edition, Berkun Publication 2010
8. Tom Kelly and David Kelly, Creative confidence: Unleashing the creative Potential within Us all, William Collins Publication 2013
9. Ira Flatow, The all Laughed, Harper Publication 1992

10. Paul Sloane, Des Mac Hale& M.A. Di Spezio, The Ultimate Lateral & Critical Thinking Puzzle book, Sterling Publication 2002

Additional Readings

11. Keith Sawer, Group Genius, The Creative Power of Collaboration, Basic Books Publication 2007
12. Edward De Beno, Lateral Thinking, Creativity Step by Step, Penguin Publication 1973
13. Nancy Margulies with Nusa Mall, Mapping Inner Space, Crown House Publication 2002
14. Tom Kelly with Jonathan Littman,The Art of Innovation, Profile Publication 2001
15. Roger Von Oech,A Whack on the Side of the Head. Revised edition, Hachette Publication 1998
16. Roger Von Oech, A Kick in the Seat of the Head, William Morrow 1986
17. Jonah Lehrer, Imagine How Creativity Works, Canongate Books Publication 2012
18. James M Higgins, 101 Creative Problem Solving Techniques, New Management Publication1994
19. Soctt G Isaksen, K Brain Doval, Donald J Treffinger, Creative Approach to Problem Solving, Sage Publication 2000
20. Donald J Treffinger, scott G Isaksen, K Brain stead Dorval Creative Problem Solving An Introduction,Prufrock Press 2006
21. H Scott Fogler& Steven E. LeBlance, Strategies for Creative Problem Solving,Prentice Hall Publication 2008
22. Dave Gray, Sunni Brown and James Macanufo, Game Storming, O'reilly Publication 2010.
23. Howard Gardner, Creating minds,Basic Books Publication 1993
24. MihalyCsikzentmihalyi, Creativity–Flow and Psychology of Discovery and Invention,Harper Publication 1996
25. Martin Gerdner,W. H.,Ahal Insight, Freeman Publication 1978
26. Paul Sloane, Test Your Lateral Thinking IQ, Sterling Publication1994
27. Paul Sloane & Des Machale Intriguing, Lateral Thinking Puzzles, Sterling Publication 1996

Articles / Videos / Other Suggested Materials

- Internet Search based May TED talks and other sources for videos, slide shares, problems, etc

B. Tech. (Electrical Engineering) Programme

SYLLABI

(Semester – IV)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Course Code: **EE287**
Course Title: **MATLAB Programming**

UG Engineering Programme	Semester
Electrical Engineering	Fourth

I RATIONALE

MATLAB is considered as one of the most important tools and modern computer language, and this course enable the students to learn many of MATLAB commands and how to use them in programming to solve many problems in different mathematical subjects, specially in numerical analysis and other subjects which connected to computer oriented mathematics.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Creates valuable transferable skills, sets individuals apart in the job market, and can help accelerate professional growth.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1. ***Validate*** program for real world problems.
2. **Understand** the use of MATLAB software for numerical methods.
3. **Implement** solution for mathematical problems.
4. **Debug** and **Troubleshoot** MATLAB code.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			**Total Credits (L+T+P)**	**Examination Scheme**				
				Theory Marks		**Practical Marks**		**Total Marks**
L	**T**	**P**	**C**	**ESE**	PA	**ESE**	**PA**	
0	0	2	2	0	0	70	30*	100

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

(*): *Under the* ***Practical PA****, out of 30 marks,* ***10 marks*** *are for* ***Presentation*** *to facilitate integration of COs and the remaining 15 marks is converted from lab manual to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	**CO No.**	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	**Approx. Hrs. required**
1	1	Introduction To MATLAB basic Mathematical, trigonometric, Array and Matrix operations	02
2	1	Implement the various logical and Iterative loops in MATLAB programming.	02
3	2	Solve Linear equations with Gauss elimination & Gauss-Jordan elimination method using MATLAB programming.	02
4	2	Solve Linear equations with RLSE & Gauss Seidel method using MATLAB programming.	02
5	3	Computation of Matrices and Eigen values with Decomposition methods using MATLAB programming.	02
6	3	Computation of Matrices and Eigen values with Power methods using MATLAB programming.	02
7	1	Implementation of basic Curve Plotting Commands.	02
8	4	Computation of Lagrange Polynomial method and Newtons Polynomial method for interpolation of curve.	02
9	3	Solve nonlinear equation with Bisection method using MATLAB functions.	02
10	3	Solve nonlinear equation with Newton-Raphson method using MATLAB functions.	04
11	3	Solve ordinary differential equations with Euler's method using MALAB programming.	02
12	3	Solve ordinary differential equations with Runge-Kutta method using MALAB programming.	02
		Total	**26**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	**Equipment Name with Broad Specifications**	**PrO No.**
1	MATLAB Simulation software (version 2021a)	1,2,3,4,5,6,7,8,9,10,11,12

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Function as a team member.
3) Function as a team leader.
4) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII SPECIAL INSTRUCTIONAL STRATEGIES (if any)

i) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
j) ***'P' in item No. 4*** does not mean only the traditional practical method, but different types of teaching methods and media that are to be employed to develop the outcomes.
k) About ***15-20% of the simulation*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.

IX SUGGESTED STUDENT ACTIVITIES

Other than the laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different software used in industries.
b) Prepare 15 min. power point presentation on how to choose software for the particular application.

X SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Applied Numerical Methods Using MATLAB	Won Young Yang, Wenwu Cao Tae-Sang Chung, John Morris,	John Willey, 2005 ISBN: 9780471698333
2	Computer Oriented Numerical Methods	V. Rajaraman	PHI, 2006 ISBN: 9388028317

XXII SOFTWARE/LEARNING WEBSITES

1 http://www.nptel.ac.in/courses/122104018/node109.html
2 http://www.geos.ed.ac.uk/~yliu23/docs/lect_spline.pdf
3 https://mat.iitm.ac.in/home/sryedida/public_html/caimna/interpolation/lagrange.html

XI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	**PO 1** Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE287	**Semester IV**														
287	**Competency** *Creates valuable transferable skills, sets individuals apart in the job market, and can help accelerate professional growth*	***3***	***2***	***2***	***2***	***1***	-	-	-	-	-	-	***1***	***3***	-
287.1	Validate program for real world problems	3	3	2	-	2	-	-	-	-	-	-	-	3	2
287.2	Understand the use of MATLAB software for numerical methods	3	3	-	2	2	-	-	-	-	-	-	-	3	2
287.3	Implement solution for mathematical problems	3	2	2	2	-	-	-	-	1	-	-	-	3	3
287.4	Debug and Troubleshoot MATLAB code	-	3	3	2	2	-	-	-	-	-	-	-	2	-

XII NAMES OF RESOURCE PERSONS

a) Dr. Jigar Sarda

Course Code: **EE288 (University Elective)**
Course Title: **Maintenance of Household Apparatus**

UG Programme	Semester
All UG Program of CHARUSAT	Forth

I RATIONALE

All the houses needed major or minor electrical maintenances over period of time. Also, small maintenance needed for household apparatus. The major component of repairing cost is the labour/skill and minor component is cost of item to be replaces (Eg. Carbon brush in mixture machine, switch, capacitor in fan). If the students get the repairing/replacement skill, he/she can save money and time, most important is; he/she doesn't have to wait for repairing time taken by external person.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Repair the household apparatus with minimum cost.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) ***Safely Operate*** the electrical/electronic devices and safeguard their selves from electrical shock.
2) ***Repair*** *the domestic wiring.*
3) **Repair/maintain/troubleshoot** the *household electrical appliances.*
4) ***Develop*** small electronic project.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	**T**	**P**	**C**	**ESE**	PA	**ESE**	**PA**	
0	0	2	2	0	0*	70	30	100

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

(*): *Under the* ***Theory PA****, out of 30 marks,* ***10 marks*** *are for* ***micro-project assessment*** *to facilitate integration of COs and the remaining 20 marks is based on lab activities by assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Operate the multi-meter to measure the said electrical quantity.	02
2	1,2	Wiring the tube light, fan, Extension board (Power Socket), Staircase wiring (Two-way wiring).	02
3	1,2	Choose the correct rating of Fuse, MCB, ELCB. Connect the earthing pin wiring to the earth pit.	01 01
4	2	Prepare the drawing for wiring a newly built room, without any electrical wiring along with a bill of materials and specifications.	02
5	3	Solder and de-solder electronic components on PCB.	02
6	4	Develop simple mobile charger and DC voltage controller circuit.	02
7	3	Maintenance and Troubleshooting of Celling FAN.	02
8	3	Maintenance and Troubleshooting of Tube light & LED light.	02
9	3	Maintenance and Troubleshooting of Pump (Household Motor).	02
10	3	Maintenance and Troubleshooting of Water Purifier (RO).	02
11	3	Maintenance and Troubleshooting of Air-Conditioner.	02
12		Interpret the home electricity bill and can calculate the energy calculation of any equipment.	02
13	4	Choose the correct battery for the given application.	02
14			
15			
16			
		Total	**26**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Multi-meter	1
2	Multi-meter, Tube light set, extension board, wires.	2, 8
3	Fuse, MCB, ELCB	3
4	PCS, Soldering Iron, Soldering Wire.	5
5	230/9 V Transformer, Diodes, Capacitor	6
6	Ceiling fan, Multi-meter, capacitor	7
7	Electric Motor (Single Phase), Capacitor	9
8	RO System	10
9	Air Conditioner, Capacitor	11
10	Energy meter	12
11	Various types of batteries	13

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'P' in item No. 4*** does not mean only the traditional practical method, but different types of hands-on practices to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

IX SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding the evolution of arc quenching mechanism for different type of circuit breakers.
b) Prepare 15 min. power point presentation on how to choose the specific circuit breaker for the particular application.

X SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Electrical Design Estimation and Costing	Raina K.B. & Bhattacharya S.K.	New Age International Private Limited, Second Edition, 2018-19, ISBN : 978-8122443585
2	India Electrical Rules 1956 Hand book	Government of India and Vishal Singh	Kindle Edition
3	A Textbook of Electrical Technology in S.I. Units Volume III	B L Thareja	S Chand, 2007 ISBN : 978-8121924900.
4	National Building code of India Group 1 and Group 4, Bureau of Indian standard,		Government of India
5	Troubleshooting and Repairing Major Appliances	Eric Kleinert	McGraw Hill, 2012 ISBN : 978-0071770187

XI SOFTWARE/LEARNING WEBSITES

- https://www.instructables.com/id/How-to-solder/
- https://www.udemy.com/course/electronic-electronics-maintenance-electronic-devices-maintenance/

XII PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE288	**Semester IV**	**Fault Analysis and Switchgear** **Mark 'H'=3** for high, **'M'**=2 for medium or **'L'=1'** for the relevant correlation of each PO or PSO, Competency or CO													
288	**Competency** *Evaluate the performance of various circuit breakers employed*	***2***	***2***	***1***	***1***	**-**	***1***	**-**	**-**	**-**	**-**	**-**	**-**	***1***	**-**

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	for various applications in power system.														
288.1	Evaluate the symmetrical fault current and voltage at any location of power system.	3	2	-	1	-	3	-	-	-	-	-	-	2	-
288.2	Evaluate the unsymmetrical fault current and voltage at any location of power system with mathematical model and with software tools.	2	3	1	1	-	1	-	-	1	-	-	-	2	-
288.3	Interpret the phenomena of current interruption in circuit breaker.	2	3	1	1	-	1	-	-	-	-	-	-	1	-
288.4	Evaluate the different parameters for arc formation and arc interruption in circuit breaker.	1	1	3	1	1	-	-	-	2	-	1	-	1	-

XIII NAMES OF RESOURCE PERSONS

a) Dr. Nilaykumar A. Patel
b) ………………………………………………….

Course Code: **EE254**

Course Title: **ELECTRICAL POWER GENERATION AND ITS ECONOMICS**

UG Engineering Programme	Semester
Electrical Engineering	Fourth

I RATIONALE

As electrical power generation and economy is essential gradient for the industrial and all around development of any country This curriculum course structure is developed such that the student is able to evaluate the fundamentals of electromechanical energy conversion, concepts of electrical power generation, basic layout and function of conventional as well as non-conventional power generation technology, constructional & operational aspects of various power plant, basic knowledge of transmission line components, transmission line, various types of insulated cables and its selection criteria, economic aspects of power system operation, and required reactive power factor for most economical operation.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Evaluate the functioning and performance of various power generation plants and basic power system components**.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) Use the knowledge and describe the working of various power plants and various equipment of power generation plants.
2) Develop skill and Identify the key point to improve the performance of the power plants.
3) Aware with the new era of energy generation technology like wind, solar and fuel cell etc.
4) Analyze and select the proper cable for particular application.
5) Use the knowledge of basic power system structure and determine tariff means to reduce the electricity charges by improving the power factor.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	100
3	0	0	3	70	30	00	00	

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

V AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VI UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Introduction of renewable and non-renewable energy sources	**CO 1. *Use the knowledge and describe the working of various power plants and various equipment of power generation plants.*** 1a. Power and Energy, sources of renewable & non-renewable energy such as coal, water, nuclear, wind, solar, tidal	1.1 Power and Energy sources of energy such as coal, water, nuclear, wind, solar, tidal etc.
Unit– II Steam Power Plants	**CO 1. *Use the knowledge and describe the working of various power plants and various equipment of power generation plants.*** **CO 2. *Develop skill and Identify the key point to improve the performance of the power plants.*** 2a. Classification of steam power plants, layout of steam power plant, Component of steam power plants, Essential requirement for steam power plant, selection of site. 2b. Choice of steam condition, fuel	2.1 Classification of steam power plants 2.2 layout of steam power plant 2.3 Component of steam power plants 2.4 Essential requirement for steam power plant 2.5 selection of site 2.6 Choice of steam condition, fuel handling, Combustion

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	handling, Combustion equipments for steam boilers, Ash handling, electrostatic precipitator and dust collection, Boilers. 2c. steam turbines, pressure and velocity compounding, Steam condenser, feed water treatment, advantages and disadvantages of steam power plant	equipments for steam boilers, Ash handling, electrostatic precipitator and dust collection, Boilers, steam turbines, pressure and velocity compounding, Steam condenser, feed water treatment 2.7 advantages and disadvantages of steam power plant
Unit– III Hydro Electric Power Plant	**CO 1. *Use the knowledge and describe the working of various power plants and various equipment of power generation plants.*** **CO 2. *Develop skill and Identify the key point to improve the performance of the power plants.*** 3a. Layout and elements of hydro power plant, Classification of hydro power plant, Hydraulic turbines 3b. Application, advantages and disadvantages of hydro power plant, Site selection for hydro power plant.	3.1 Classification of hydro power plant 3.2 layout and elements of hydro power plant 3.3 Site selection for hydro power plant 3.4 advantages and disadvantages of hydro power plant 3.5 Application Hydraulic turbines
Unit-IV Nuclear Power Plant	**CO 1. *Use the knowledge and describe the working of various power plants and various equipment of power generation plants.*** **CO 2. *Develop skill and Identify the key point to improve the performance of the power plants.*** 4a. General aspect of nuclear engineering, layout of nuclear power plant 4b. Nuclear reactors, Main components of nuclear power plants, advantages and disadvantages of nuclear power plant	4.1 General aspect of nuclear engineering 4.2 layout of nuclear power plant 4.3 Main components of nuclear power plants 4.4 advantages and disadvantages of nuclear power plant
Unit – V Combined cycle power plant	**CO 1. *Use the knowledge and describe the working of various power plants and various equipment of power generation plants.*** **CO 2. *Develop skill and Identify the key point to improve the performance of the power plants.*** 5a. Various combined cycle power plant, working of open cycle, close cycle & combine cycle 5b. advantages and disadvantages,	5.1 Various combined cycle power plant 5.2 working of combined cycle power plant 5.3 advantages and disadvantages 5.4 comparison of various power plant

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	comparison of various power plant	
Unit – VI Wind Power Generation	**CO 1.** ***Use the knowledge and describe the working of various power plants and various equipment of power generation plants.*** **CO 2.** ***Develop skill and Identify the key point to improve the performance of the power plants.*** **CO 3.** ***Aware with the new era of energy generation technology like wind, solar and fuel cell etc.*** 6a. Nature and origin of wind, variables in wind energy conversion system, wind velocities and height from ground and site selection 6b. Types of wind energy system, wind turbine generator unit with battery storage facilities, solar wind hybrid 6c. Wind farm siting, Applications, Merits and demerits of wind energy	6.1 nature and origin of wind 6.2 variables in wind energy conversion system 6.3 Types of wind energy system 6.4 wind velocities and height from ground and site selection 6.5 wind turbine generator unit with battery storage facilities 6.6 solar wind hybrid, Wind farm siting
Unit – VII Solar Power Generation	**CO 1.** ***Use the knowledge and describe the working of various power plants and various equipment of power generation plants.*** **CO 2.** ***Develop skill and Identify the key point to improve the performance of the power plants.*** **CO 3.** ***Aware with the new era of energy generation technology like wind, solar and fuel cell etc.*** 7a. Solar energy routes and prospects, Types of solar thermal collectors 7b. comparison between conventional and solar thermal power plant, ratings of solar power plant heat transfer fluids 7c. solar parameters calculation 7d. Solar photovoltaic energy conversion 7e. solar pond and binary cycle solar thermal power plant, merits and limitations of solar energy conversion and utilization	7.1 Solar energy routes and prospects 7.2 Types of solar thermal collectors 7.3 solar photovoltaic cell conversion 7.4 comparison between conventional and solar thermal power plant 7.5 ratings of solar power plant heat transfer fluids 7.6 solar pond and binary cycle solar thermal power plant
Unit – VIII Other Renewable Energy Sources	**CO 1.** ***Use the knowledge and describe the working of various power plants and various equipment of power generation plants.*** **CO 2.** ***Develop skill and Identify the key point to improve the performance of the power plants.***	8.1 Fuel cell based power plant: Introduction, concept, types 8.2Electrochemical Reactions, Hydrogen, Oxygen Fuel cells.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	CO 3. *Aware with the new era of energy generation technology like wind, solar and fuel cell etc.* 8a. Layout of Fuel cell based power plant types, Electrochemical Reactions, Hydrogen, Oxygen Fuel cells 8b. Principles of MHD Power Generation, MHD Systems, Advantages of MHD Systems, Electrical conditions. 8c. Origin of geothermal resources, Geothermal energy conversion, utilization of geothermal energy, liquid dominated geothermal electric power plant, Applications. 8d. Ocean energy resources, advantages and limitations of ocean energy conversion, technologies, ocean thermal energy conversion, principle of OTEC, Tidal energy conversion	8.3 MHD Power Generation: Principles, MHD Systems, Advantages of MHD Systems, Electrical conditions 8.4 Geothermal energy: Applications, utilization of geothermal energy, origin of geothermal resources, liquid dominated geothermal electric power plant. 8.5 Ocean and tidal energy: Ocean energy resources, advantages and limitations of ocean energy conversion, technologies, ocean thermal energy conversion, principle of OTEC. Tidal energy conversion : High and low tides, tidal energy conversion
Unit – IX Insulated Cables	**CO 4. *Analyze and select the proper cable for particular application.*** 9a. Requirement of insulated cables, properties of conductor and insulating material for cables, construction of cables, types of cables, types of insulating material for cable. 9b. Insulation resistance of a single core cable 9c. capacitance of single and three core cable 9d. Electrostatic stresses in a single core cable, most economical conductor size in cable 9e. Capacitance grading and inter sheath grading 9f. Calculation of losses in cable and factors affecting it, current rating of a cable	9.1 Requirement of insulated cables 9.2 properties of conductor and insulating material for cables 9.3 construction of cables, types of cables, types of insulating material for cable 9.4 Insulation resistance of a single core cable 9.5 capacitance of single and three core cable 9.6 Electrostatic stresses in a single core cable 9.7 most economical conductor size in cable 9.8 Capacitance grading and inter sheath grading 9.9 Calculation of losses in cable and factors affecting

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
		it, current rating of a cable
Unit – X Components of Overhead Lines	**CO 5. *Use the knowledge of basic power system structure and determine tariff means to reduce the electricity charges by improving the power factor.*** 10.1 Basic power system structure, generation, transmission, substation, sub transmission, distribution, Complex power. 10.2 single phase transmission, three phase transmission. Conductors and line supports, Types of Insulators 10.3 potential distribution over a string of suspension insulators, mathematical expression for String efficiency 10.4 Methods of improving String efficiency, by using longer cross-arms, by insulator grading, by using guard rings.	10.1 Basic power system structure, generation, transmission, substation, sub-transmission, distribution, 10.2 Complex power 10.3 single phase transmission, three phase transmission 10.4 Conductors and line supports 10.5 Types of Insulators 10.6 potential distribution over a string of suspension insulators, String efficiency and methods to improve it, Examples on string efficiency
Unit – XI Mechanical Design of Transmission Lines (Sag-Tension Relationship)	**CO 5. *Use the knowledge of basic power system structure and determine tariff means to reduce the electricity charges by improving the power factor.*** 11.1 Sag-tension relation, calculation of sag at equal & unequal level support. 11.2 sag parameters calculation	11.1 Sag-tension relation and its example with different configurations
Unit – XII Economics of Power System	**CO 5. *Use the knowledge of basic power system structure and determine tariff means to reduce the electricity charges by improving the power factor.*** 12.1 Types of Loads in power system, effects of variable load on power system, load curves and load duration curves. 12.2 Important terms and factors related to power system economics of power system, importance of those factors. 12.3 calculation of load curve parameters 12.4 Load curves and selection of generating units, Cost of electrical energy and its expression 12.5 Define depreciation, methods of determining the depreciation	12.1 Loads in power system, effects of variable load on power system, load curves and load duration curves 12.2 Important terms and factors related to power system economics of power system, importance of those factors 12.3 Load curves and selection of generating units 12.4 Cost of electrical energy and its expression 12.5 methods of determining the

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	12.3 depreciation methods calculation 12.4 Define Tariff, Types of Tariffs 12.5 Calculation of tariff parameters	depreciation, examples 12.6 Types of Tariffs, Examples
Unit – XIII Power Factor Improvement in Power System	**CO 5. *Use the knowledge of basic power system structure and determine tariff means to reduce the electricity charges by improving the power factor.*** 13.1 Power factor basics, Causes of low power factor, disadvantages of low power factor 13.2 power factor improvement, Power factor improvement equipments 13.3 Power factor improvement equipment's, calculations of power factor correction, most economical power factor 13.4 power factor improvement calculation	13.1 Causes of low power factor, disadvantages of low power factor 13.2 power factor improvement, Power factor improvement equipments 13.3 calculations of power factor correction 13.4 most economical power factor

VII TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction	01	1	0	0	0	0	0	01
II	Steam Power Plants	06	1	3	2	2	2	0	10
III	Hydro Electric Power Plant	03	0	1	2	0	2	0	5
IV	Nuclear Power Plant	03	0	1	2	0	2	0	5
V	Combined cycle power plant	02	0	1	0	1	1	0	3
VI	Wind Power Generation	03	0	1	1	1	1	0	4
VII	Solar Power Generation	02	0	1	1	0	1	0	3
VIII	Other Renewable Energy Sources	03	0	2	1	1	1	0	5
IX	Insulated Cables	04	1	1	0	2	2	0	6
X	Components of Overhead Lines	08	1	2	2	3	4	0	12
XI	Mechanical Design of Transmission Lines (Sag-Tension	02	0	2	1	0	0	0	3

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
	Relationship)								
XII	Economics of Power System	05	0	1	1	2	4	0	8
XIII	Power Factor Improvement in Power System	03	0	1	1	1	2	0	5
Total		**45**	**04**	**17**	**14**	**13**	**22**	**0**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

VIII MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **Make the small scale module of different power generation plant.**
2) **Make the small scale module of integrated power plant.**
3) **Design the underground cable and select the parameters accordingly.**
4) **Make the power factor correction unit.**
5) **Design the smart meter for electricity billing.**

IX SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

I) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.

m) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
n) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
o) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

X SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

g) Do the survey of power generation state wise, national and internationally.
h) Prepare the poster of generation, transmission and distribution data.

XI COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Introduction of renewable and non-renewable energy sources **Unit– II** Steam Power Plants	1	1a. Power and Energy, sources of renewable & non-renewable energy such as coal, water, nuclear, wind, solar, tidal 2a. Classification of steam power plants, layout of steam power plant, Component of steam power plants, Essential requirement for steam power plant, selection of site. 2b. Choice of steam condition, fuel handling, Combustion equipments for steam boilers, Ash handling, electrostatic precipitator and dust collection, Boilers.	Lecture with media	PPTs, Video
Unit– II Steam Power Plants **Unit– III** Hydro Electric Power Plant	2	2c. steam turbines, pressure and velocity compounding, Steam condenser, feed water treatment, advantages and disadvantages of steam power plant. 3a. Layout and elements of hydro power plant, Classification of hydro power plant, Hydraulic turbines 3b. Application, advantages and disadvantages of hydro power plant, Site selection for hydro power plant.	Lecture With media	PPTs, Video

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit-IV Nuclear Power Plant	**3**	4a.General aspect of nuclear engineering, layout of nuclear power plant 4b.Nuclear reactors, Main components of nuclear power plants, advantages and disadvantages of nuclear power plant	Lecture with media	PPTs, black board, Video
Unit – V Combined cycle power plant	**4**	5a. Various combined cycle power plant, working of open cycle, close cycle & combine cycle 5b. advantages and disadvantages, comparison of various power plant 6a. Nature and origin of wind, variables in wind energy conversion system, wind velocities and height from ground and site selection	Lecture with media,	PPTs, Video
Unit – VI Wind Power Generation	**5**	6b. Types of wind energy system, wind turbine generator unit with battery storage facilities, solar wind hybrid 6c. Wind farm siting, Applications, Merits and demerits of wind energy	Lecture with media,	PPTs, Video
Unit – VII Solar Power Generation	**6**	7a. Solar energy routes and prospects, Types of solar thermal collectors 7b. comparison between conventional and solar thermal power plant, ratings of solar power plant heat transfer fluids 7c. solar parameters calculation	Lecture with media	PPTs
	7	7d. Solar photovoltaic energy conversion 7e. solar pond and binary cycle solar thermal power plant, merits and limitations of solar energy conversion and utilization	Lecture with media, Field visit	PPTs, Video
Unit – VIII Other Renewable Energy Sources	**8**	8a. Layout of Fuel cell based power plant types, Electrochemical Reactions, Hydrogen, Oxygen Fuel cells 8b. Principles of MHD Power Generation, MHD Systems, Advantages of MHD Systems, Electrical conditions	Lecture with media	PPTs
Unit – VIII Other Renewable Energy Sources	**9**	8c. Origin of geothermal resources, Geothermal energy conversion, utilization of geothermal energy, liquid dominated geothermal electric power plant, Applications. 8d. Ocean energy resources, advantages and limitations of ocean energy conversion, technologies, ocean thermal energy conversion, principle of OTEC, Tidal energy conversion	Lecture with media	PPTs
Unit – IX Insulated Cables	**10**	9a. Requirement of insulated cables, properties of conductor and insulating material for cables, construction of cables, types of cables, types of insulating material for cable. 9b. Insulation resistance of a single core cable 9c. capacitance of single and three core cable	Lecture with media	PPTs, black board
Unit – IX Insulated	**11**	9d. Electrostatic stresses in a single core cable, most economical conductor size in cable	Lecture with media	PPTs, black

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Cables		9e. Capacitance grading and inter sheath grading 9f. Calculation of losses in cable and factors affecting it, current rating of a cable		board
Unit – XI Mechanical Design of Transmission Lines (Sag-Tension Relationship)	**12**	10.1 Basic power system structure, generation, transmission, substation, sub transmission, distribution, Complex power. 10.2 single phase transmission, three phase transmission. Conductors and line supports, Types of Insulators 10.3 potential distribution over a string of suspension insulators, mathematical expression for String efficiency 10.4 Methods of improving String efficiency, by using longer cross-arms, by insulaor gradings, by using guard rings.	Lecture	black board
Unit– V Circuit Breakers	**13**	11.1 Sag-tension relation, calculation of sag at equal & unequal level support. 11.2 sag parameters calculation	Lecture	black board
Unit – XII Economics of Power System **Unit – XIII** Power Factor Improvement in Power System	**14**	Types of Loads in power system, effects of variable load on power system, load curves and load duration curves. 12.2 Important terms and factors related to power system economics of power system, importance of those factors. 12.3 calculation of load curve parameters 12.4 Load curves and selection of generating units, Cost of electrical energy and its expression 12.5 Define depreciation, methods of determining the depreciation 12.3 depreciation methods calculation 12.4 Define Tariff, Types of Tariffs 12.5 Calculation of tariff parameters	Lecture	black board
Unit – XIII Power Factor Improvement in Power System	**15**	13.1 Power factor basics, Causes of low power factor, disadvantages of low power factor 13.2 power factor improvement, Power factor improvement equipments 13.3 Power factor improvement equipment's, calculations of power factor correction, most economical power factor 13.4 power factor improvement calculation	Lecture	black board

XII SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Principles of Power system	V.K. Mehta	S. Chand & Company Ltd. ISBN: 978-81-219-2496-2
2	Energy Technology	S. Rao & Dr.	Khanna Publication, New Delhi.

S. No.	Title of Book	Author	Publication including ISBN
		B.B.Parulekar	ISBN: 978-81-7409-040-9
3	Renewable energy sources and conversion technology	N.K. Bansal	Tata McGraw Hill, New Delhi. ISBN: 0074600230, 9780074600238
4	A Course in electrical power	Soni & Bhatnagar	S. K. Kataria & Sons publications ISBN: 8188458538
5	Energy sources	G.D. Rai	Khanna Publication, New Delhi, 2019, ISBN: 978-8174090737

XIII SOFTWARE/LEARNING WEBSITES

1. Power System Generation, Transmission and Distribution:https://nptel.ac.in/courses/108102047
2. Power System Engineering: EE254 EPGE Syllabus EE 4th sem.docx

XIV PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs							POs and PSOs								
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Project management and finance	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.	**PSO3**
EE254	**Semester IV**				ELECTRICAL POWER GENERATION AND ITS ECONOMICS **Mark ‘H’=3** for high, **‘M’**=2 for medium or **‘L’=1’** for the relevant correlation of each PO or PSO, Competency or CO											
254	**Competency** *Evaluate the performance of various protection systems used in*	*3*	*3*	*3*	*2*	*1*	*1*	*3*	-	*1*	-	**-**	**-**	***3***	**-**	-

Course Code	Competency and COs	POs and PSOs														
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Project management and finance	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.	**PSO3**
	Industry															
254.1	Use the knowledge and describe the working of various power plants and various equipment of power generation plants.	3	2	2	1	1	1	2	-	-	1			3	1	-
254.2	Develop skill and Identify the key point to improve the performance of the power plants.	3	2	2	1	1	1	3	-	1	-			3	2	-
254.3	Aware with the new era of energy generation technology like wind, solar and fuel cell etc.	2	2	1	1	1	1	3	-	1	-			3	2	-
254.4	Analyze and select the proper cable for particular application.	2	2	1	2	1	1	-	-	-	-			2	1	-
254.5	Use the knowledge of basic power system structure and determine tariff means to reduce the electricity charges by improving the power factor.	3	2	1	1	1	1	-	-	1	1			2	-	-

XV NAMES OF RESOURCE PERSONS

a) **Ankur Patel**

Course Code: **EE246**
Course Title: **Microprocessor & Microcontroller**

UG Engineering Programme	Semester
Electrical Engineering	Fourth

I RATIONALE

With advancement in electrical and electronics industry all the electrical machine nowadays can be controlled with electronics devices. This type of electronics system requires controller or processor to think and act quickly. This curriculum course is developed such that the student is able to design a system to controller electrical devices as per the application's need.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Create an embedded system to solve real life problem**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) ***Understand*** *the basic building blocks & architecture of a microcontroller/ microprocessor*
2) ***Develop*** *a program using embedded C*
3) ***Demonstrate*** *use of Input, Output, Timers, Interrupts and serial communication.*
4) **Analyse**, **Evaluate** and **Integrate** peripheral devices
5) ***Design*** *and* ***create*** *hardware and software solution to real life problems.*

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme (Anna 2021)				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	
4	0	2	5	70	30*	25	25	150

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

(*): *Under the* ***Theory PA****, out of 30 marks,* ***15 marks*** *are for* ***micro-project assessment*** *to facilitate integration of Cos and the remaining 15 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Understand 8085 and Create Assembly programs for the given task.	4
2	3	Demonstrate Digital input and Digital Output example with simulation and hardware.	2
3	2	Understand C programming concept for embedded C in Keil Microvision IDE.	2
4	3	Demonstrate the use of Timer in different modes of operation and their use cases with simulation and hardware.	2
5	3	Demonstrate use of Counter in different modes of operation and their use cases with simulation and hardware.	2
6	3	Demonstration of External interrupt and it's use cases with simulation and hardware.	2
7	3	Demonstrate use of serial port of 8051 for serial communication.	2
8	4	Demonstrate interfacing a keypad.	2
9	4	Develop a system with 8051 and liquid crystal &7-Segment LED display to print given message on screen.	4
10	4	Develop system to measure temperature/pressure/moisture using ADC.	2
11	4	Demonstrate use of DAC with use case.	2
12	5	Create system to monitor and control temperature of room.	2
		Total	**28**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Dynalog 8085 kit	1
2	Keil uvison for C51	2,3,4,5,6,7,8,9,10,11,12
3	8051 Controller board	2,3,4,5,6,7,8,9,10,11,12
4	LEDs, Toggle Switch, Push Buttons, Connecting wires	3,4,5,6,12
5	Digital Signal Oscilloscope	3,4,5,6,10
6	4x4 Matrix Keypad	8
7	16x2 LCD, 7-Segment LED	9,12
8	ADC 0804 or 0808, DAC 0808	10,11,12
9	Temperature Sensor	10,12
10	Relay Card, Contactor	12

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I **Basic of 8085**	***CO 1. Understand the basic building blocks & architecture of a microcontroller/ microprocessor*** *PrO 2. Understand 8085 Dynalog Kit and Create Assembly programs for the given task.* 1a. Justify the need of Microprocessor for the given application 1b. Evaluate the working of the given microprocessor 1c. Create the program for the given application 1d. Choose the memory according to the given application	1.1 Introduction to Microprocessor, Block Diagram of 8085 and it's various registers. 1.2 Bus organization, Pin Diagram and its functions. 1.3 Assembly level programming. 1.4 Types of memory and memory interfacing

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit– II Introduction to Microcontroller	***CO 1. Understand the basic building blocks & architecture of a microcontroller/ microprocessor*** *PrO 3. Demonstrate Digital input and Digital Output example with simulation and hardware.* 2a. Determine the possibility of application for the given microcontroller 2b. Understand the working of the given microcontroller 2c. Understand the function of different pins for the hardware interfacing 2d. Understand the requirement of different type of register and their function 2e. Understand the operation of on chip memory 2f. Evaluate speed of execution for the given crystal 2g. Understand the different operation available through port	2.1 8051 Microcontroller family and its salient feature 2.2 Architectural block Diagram of INTEL 8051 2.3 Pin diagram and Pin Functions 2.4 General Purpose and Special Function Registers 2.5 Memory organization 2.6 Crystal Frequency & Crystal Connection and Speed of Execution 2.7 -PORT Structure
Unit– III 8051 Programming	***CO 2. Develop a program using embedded C*** *PrO 4. Understand C programming concept for embedded C in Keil Microvision IDE.* 3a. Differentiate operation of IDE and use specific module for the given situation 3b. Create the hardware to load microcontroller 3c. Create the program for the given application 3d. Differentiate and understand the need of assembly and C language for creating software as per the given application 3e. Understanding format used to write program in C language 3f. Justify the need of special keyword for assembly level language	3.1 Concept of Assembler, Editor, Linker, Loader, Debugger in IDE 3.2 Hardware Loader Circuit for ISP featured 8051. 3.3 Assembly Instruction Sets: Data Transfer & Addressing modes, Logical, Branching, Arithmetic & Bit wise Operation, 3.4 Comparison of Assembly and C Programming 3.5 Data type , Keywords, Various Operator , Control loops, Functions and Arrays in C 3.6 Special Keywords & Interrupt function in Keil

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
		Microvision IDE
Unit-IV On chip Hardware Modules/Peripherals in 8051	***CO 4. Demonstrate use of Input, Output, Timers, Interrupts and serial communication.*** *PrO 5. Demonstrate the use of Timer in different modes of operation and their use cases with simulation and hardware.* *PrO 6. Demonstrate use of Counter in different modes of operation and their use cases with simulation and hardware.* *PrO 7. Demonstration of External interrupt and it's use cases with simulation and hardware.* *PrO 8. Demonstrate use of serial port of 8051 for serial communication.* 4a. Evaluate the operation of Timer/Counter and use it to create different type of application 4b. Evaluate the operation of Interrupts and use it to create different type of application 4c. Evaluate the operation of UART and use it to create different type of application	4.1 Internal Architecture of Timer/Counter and Related C Programs with all modes 4.2 Basics of Interrupt: Mask-able/ Non mask-able Interrupt, Enabling the mask-able Interrupt, Interrupt vector table, interrupt latency time and interrupt priority, External Interrupt and Related C Programs 4.3 Basics of Serial communication Protocol, Difference Between Synchronous and Asynchronous Communication, Internal Architecture of Serial Communication module in 8051 (UART) and related C programs with all modes
Unit – V Interfacing Microcontroller & Applications	***CO 4. Analyse, Evaluate and Integrate peripheral devices*** *PrO 9. Demonstrate interfacing a keypad.* *PrO 10. Develop a system with 8051 and liquid crystal &7-Segment LED display to print given message on screen.*	

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	PrO 11. Develop system to measure temperature/pressure/moisture using ADC. *PrO 12. Demonstrate use of DAC with use case.* ***CO 5. Design and create hardware and software solution to real life problems.*** *PrO 13. Create system to monitor and control temperature of room.* 5a. Understand operation of relay and use it to control the given device 5b. Understanding memory interfacing to fulfill memory demand of the given application 5c. Preparing hardware and software to use keyboard/switch for the given application 5d. Preparing hardware and software to use LED/LCD display for the given application 5e. Preparing hardware and software to use ACD/DAC for the given application 5f. Evaluate operation of ADC using LM35 to measure temperature. 5g. Develop hardware and software to count the cars 5h. Develop hardware and software to control the DC motor	5.1 Interfacing: Hardware interfacing diagram for driving a relay by ULN 2803/Transistor ,solid state relay(162341), Controlling a Lamp using Opto-isolator(ILD74) 5.2 External Memory Interfacing (RAM and ROM) 5.3 Keyboard Interfacing (Key Debouncing Concept, Simple Key Interfacing Hardware & Software program, Matrix key board) 5.4 Displays Interfacing (LED, 7-Segment display and LCD display) 5.5 ADC & DAC interfacing (Serial and parallel output ADC/DAC). 5.6 Temperature measurement(LM35) with LM35 5.7 Counting Cars passing through passage/bottles on belt 5.8 Unidirectional and Bidirectional DC Motor Control with PWM using L293 and ILD74, Stepper Motor Drive with ULN2003.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Basic of 8085	10	6		2			4	12
II	Introduction to Microcontroller	10	8	3					11
III	8051 Programming	08		3		3	3		9
IV	On chip Hardware Modules/Peripherals in 8051	16	4	2			3	10	19
V	Interfacing Microcontroller & Applications	16	2	2	3	2	4	6	19
Total		60	**20**	**10**	**5**	**5**	**10**	**20**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **Protection of Motor:** Create hardware and software to monitor different parameter of motor and to control motor under any abnormal condition.
2) **Green house:** Create hardware and software to create green house, in which soil moisture, humidity, temperature are continuously monitored and based on this parameters sprinklers, irrigation and cooling system can be controlled.
3) **Fire safety system:** Create a hardware and software to detect fire and contolling fire by turning on water sprinklers to control fire.

4) **Autonomous car:** Prepare hardware and software for autonomus vehicle which can follow line.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

i) Visiting project expo organized by electrical or electronics department to understand application of microcontrollers used solve real life problems.
j) Prepare small application based on 8051, demonstrate the model with presentation/poster in front of class or at project expo.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Basic of 8085	1	***PrO 1.*** *Understand 8085 and Create Assembly programs for the given task.* 1d. Justify the need of Microprocessor for the given application 1e. Evaluate the working of the given microprocessor	Lecture with media	Black board, PPT
	2	***PrO 1.*** *Understand 8085 and Create Assembly programs for the given task.* 1f. Create the program for the given application	Lecture	Black board
	3	1g. Choose the memory according to the given application	Lecture	Black board
Unit– II Introduct	**3**	2a. Determine the possibility of application for the given microcontroller 2b. Understand the working of the given	Lecture	Black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
ion to Microcontroller		microcontroller		
	4	2c. Understand the function of different pins for the hardware interfacing 2d. Understand the requirement of different type of register and their function 2e. Understand the operation of on chip memory	Lecture with media	Black board, PPT
	5	2f. Evaluate speed of execution for the given crystal ***PrO 2.*** *Demonstrate Digital input and Digital Output example with simulation and hardware.* 2g. Understand the different operation available through port	Lecture with media, Demonstration	Black board, Software: Keil µvision
Unit– III 8051 Programming	**6**	3a. Differentiate operation of IDE and use specific module for the given problem 3b. Create the hardware to load microcontroller 3c. Create the program for the given application	Lecture with media, Demonstration	Black board, Software: Keil µvision
	7	***PrO 3.*** *Understand C programming concept for embedded C in Keil Microvision IDE.* 3d. Differentiate and understand the need of assembly and C language for creating software as per the given application 3e. Understanding format used to write program in C language 3f. Justify the need of special keyword for assembly level language	Lecture with media, Demonstration	Black board, Software: Keil µvision
Unit– IV On chip Hardware Modules/ Peripherals in 8051	**8** **9**	***PrO 4.*** *Demonstrate the use of Timer in different modes of operation and their use cases with simulation and hardware.* ***PrO 5.*** *Demonstrate use of Counter in different modes of operation and their use cases with simulation and hardware.* 4a. Evaluate the operation of Timer/Counter and use it to create different type of application	Lecture with media, Hardware and Software Demonstration	Black board, Software: Keil µvision
	10	***PrO 6.*** *Demonstration of External interrupt and it's use cases with simulation and hardware.* 4b. Evaluate the operation of Interrupts and use it to create different type of application	Lecture with media, Hardware and Software Demonstration	Black board, Software: Keil µvision
	11	***PrO 7.*** *Demonstrate use of serial port of 8051 for serial communication.* 4c. Evaluate the operation of UART and use it to create different type of application	Lecture with media, Hardware and Software Demonstration	Black board, Software: Keil µvision

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– V Interfacing Microcontroller & Applications	**12**	5a. Understand operation of relay and use it to control the given device 5b. Understanding memory interfacing to fulfill memory demand of the given application ***PrO 8.*** *Demonstrate interfacing a keypad.* 5c. Preparing hardware and software to use keyboard/switch for the given application	Lecture with media, Hardware and Software Demonstration	Black board, Software: Keil µvision
	13	***PrO 9.*** *Develop a system with 8051 and liquid crystal &7-Segment LED display to print given message on screen.* 5d. Preparing hardware and software to use LED/LCD display for the given application	Lecture with media, Hardware and Software Demonstration	Black board, Software: Keil µvision
	14	***PrO 10.*** *Develop system to measure temperature /pressure /moisture using ADC.* ***PrO 11.*** *Demonstrate use of DAC with use case.* 5e. Preparing hardware and software to use ACD/DAC for the given application 5f. Evaluate operation of ADC using LM35 to measure temperature.	Lecture with media, Hardware and Software Demonstration	Black board, Software: Keil µvision
	15	***PrO 12.*** *Create system to monitor and control temperature of room.* 5g. Develop hardware and software to count the cars 5h. Develop hardware and software to control the DC motor	Lecture with media, Hardware and Software Demonstration	Black board, Software: Keil µvision

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Microprocessor Architecture, Programming, and Applications with the 8085	Ramesh Gaonkar	Penram International Publishing (India) Ltd.
2	Exploring C for Microcontrollers	Jivan Parab, Vinod Shelake, Rajanish Kamat, Gourish Naik	Springer
3	The 8051 Microcontroller Architecture	Kenneth Ayala	West Publishing Company
4	The 8051 Microcontroller and Embedded Systems Using Assembly And C,	M A Mazidi, Janice Mazidi, Rolin Kinlay	Pearson Publication
5	Microcontrollers and PLCs	Sanjay Attri	Dhanpatrai& Co
6	MCS@51 Microcontroller		INTEL

S. No.	Title of Book	Author	Publication including ISBN
	Family User's Manual		

XV SOFTWARE/LEARNING WEBSITES

1. http://old.zntu.edu.ua/base/lection/rpf/lib/zhzh03/8051_tutorial.pdf
2. https://nptel.ac.in/courses/108105102

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE246	**Semester VI**	**Microprocessor & Microcontroller** Mark **'H'=3** for high, **'M'**=2 for medium or **'L'=1** for the relevant correlation of each PO or PSO, Competency or CO													
246	***Competency*** *Create an embedded system to solve real life problem*	***1***	***2***	***3***	***1***	***3***	***1***			***1***			***1***	***3***	***1***
246.1	*Understand the basic building blocks & architecture of a microcontroller/ microprocessor*	3	1										1		
246.2	*Develop a program using embedded C*	1	2	3	1	3				1			1		
246.3	*Demonstrate use of Input, Output, Timers, Interrupts and serial communication.*	1	2	1	1	3				1					
246.4	*Analyse, Evaluate and Integrate peripheral devices*	3	2	2	1	3				1				1	
246.5	*Design and create hardware and software solution to real life problems.*	1	3	3	1	3	1			1			1	3	1

XVII NAMES OF RESOURCE PERSONS

a) Jignesh S Patel

b) ……………………………………………………

Course Code: **EE247**
Course Title: Transformers And Induction Machines

UG Engineering Programme	Semester
Electrical Engineering	Fourth

I RATIONALE

The Transformers and Induction Machines are used in power Transmission and Distribution. The course is developed such that student would get the knowledge from basics to control schemes of these machines.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Evaluate the performance of various control methods used in field**.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) **Use** in depth knowledge about the construction details, operating principle, performance characteristics of Transformers and induction machines.
2) **Evaluate** the performance of the Transformers and Induction machines.
3) **Interpret** the suitable control technique for induction motors.
4) **Analyze** with the recent developments in transformers and induction machines.
5) **Choose** the type of testing method for Transformers and Induction machines.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	200
4	0	4	6	70	30*	50	50	

***Legends: L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

(*):** Under the **Theory PA**, out of 30 marks, **15 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs**.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the

students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	**CO No.**	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	**Approx. Hrs. required**
1	2	Open circuit and short circuit tests on single phase transformer	05
2	1	Determination of equivalent circuit parameters of transformer.	05
3	4	Load test on single-phase transformer and three phase transformers.	05
4	4	Parallel operation of single phase transformer and Three phase Transformers	05
5	5	Scott Connection of transformers.	05
6	5	No load and Blocked rotor test of single phase induction motor	05
7	1	Determination of equivalent circuit parameters and performance of single phase induction motor	05
8	2	No load and Blocked rotor test of ploy phase induction motor.	05
9	2	Load test of ploy phase induction motor	05
10	3	Speed control of wound rotor induction motor	05
11	3	Speed control of squirrel cage induction motor	05
12	3	Computation of circle diagram for induction motor	05
		Total	**60**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	**Equipment Name with Broad Specifications**	**PrO No.**
1	Single Phase Transformer, Three Phase Transformers, Single and Three Phase load, Voltage regulators (Variac), Meters,	1,2,3,4,5,6
2	Induction machine with prime-mover/load	7,8,9,10,11,12

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '***Apply***' and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain* *(to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Fundamentals, Testing and performance analysis of single phase transformers	**CO1. Use** in depth knowledge about the construction details, operating principle, performance characteristics of Transformers and induction machines **CO2. Evaluate** the performance of the Transformers and Induction machines **CO4. Analyze** with the recent developments in transformers and induction machines. **CO5.Choose** the type of testing method for Transformers and Induction machines. PrO1 . Open circuit and short circuit tests on single phase transformer PrO.2 Determination of equivalent circuit parameters of transformer. PrO3. Load test on single-phase transformer and three phase transformers. PrO 4. Parallel operation of single phase transformer and Three phase Transformers 1a. Justify the concept of static machines and fundamental component roles. 1b. Evaluate the performance of Single Phase Transformer 1c. Justify the output characteristics of Single phase Transformers 1d. Setup the circuit to determine output characteristics of single phase transformers. 1e. Obtain open circuit and short circuit characteristics of three phase transformer.	1.1 Concept of an ideal transformer, assumptions; ideal transformer at no load and on load 1.2 Phasor diagram, voltage current and power relations; Transformer on DC. 1.3 EMF equation 1.4 Voltage and current transformation ratio, Effects of voltage and frequency variations. 1.5 Constructional details of a practical single phase transformer, core, winding, insulation and cooling methods 1.6 Circuit model of a practical transformer incorporating an ideal transformer; 1.7 Analysis of a practical single phase transformer, exact and approximate equivalent circuits. 1.8 Phasor diagram, relation between referred variables and parameters, per unit system of representation 1.9 Open circuit and short circuit tests 1.10 Experimental determination of single

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain* *(to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
		phase transformer equivalent circuit parameters 1.11 Percentage regulation of a single phase transformer 1.12 Efficiency of a single phase transformer, maximum efficiency 1.13 All day efficiency 1.14 Sumpner's test for experimental determination of a single phase transformer efficiency 1.15 Auto transformer
Unit– II Fundamentals, Operation and performance of three phase transformers	**CO1.Use** in depth knowledge about the construction details, operating principle, performance characteristics of induction machines **CO2.Evaluate** the performance of the Transformers and Induction machines **CO4.Analyze** with the recent developments in transformers and induction machines. PrO.4 Parallel operation of single phase transformer and Three phase Transformers. PrO.5 Scott Connection of transformers. 2a. Determine the output from three phase transformer. 2b. Evaluate the performance of Three Phase Transformer	a. Bank of three single phase transformers b. voltage, current and kVA rating with different connections c. open Δ connection d. Three phase transformer as a single unit e. constructional features, three limb, five limb, core type and shell type constructions f. comparison with bank of three single phase transformers g. Three phase transformer connections h. polarity and terminal convention i. vector groups; Y/Y and Δ/Δ connections,

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain* *(to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
		connection and phasor diagram, per phase equivalent circuit based analysis j. Vector groups Y/Δ and Δ/Y connections. Connection and phasor diagrams, per phase equivalent circuit based analysis k. Y/Z and Δ/Z connections l. Connection and phasor diagrams m. voltage, current relations n. Phase conversion: 3 to 6 phase and 3 to 2 phase conversions (Scott Connections) o. Tap changing transformer and voltage control p. Parallel operation of single and three phase transformers. q. per phase equivalent circuit based analysis with equal and unequal no load voltages r. load sharing; Harmonics and switching transients in transformers effect of transformer connections, inrush current

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain* *(to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit-III Construction, operation, Testing and analysis of three phase induction machines.	**CO1.<u>Use</u>** in depth knowledge about the construction details, operating principle, performance characteristics of Transformers and induction machines **CO2.<u>Evaluate</u>** the performance of the Transformers and Induction machines **CO3.<u>Interpret</u>** the suitable control technique for induction motors. **CO4.<u>Analyze</u>** with the recent developments in transformers and induction machines. **CO5.<u>Choose</u>** the type of testing method for Transformers and Induction machines. PrO.8 No load and Blocked rotor test of ploy phase induction motor. PrO.9 Load test of ploy phase induction motor PrO.12 Computation of circle diagram for induction motor 3a.<u>Choose</u> the relevant operating parameters for load operation of three phase induction motor. 3b. <u>Evaluate</u> the performance of three phase induction motor for the specified condition.	a. Introduction, Classification of three phase induction motor, b. Production of three phase rotating magnetic field c. Locking of stator and rotor field d. Construction, Principle of operation, slip, Frequency of rotor current, Speed of rotor field e. Rotor EMF, Rotor current, Rotor reactance under starting and running condition f. Relation between torque and rotor power factor g. Torque equation, Torque under running condition, Torque-Speed curves and effect of change in rotor resistance h. Operating region, Starting torque, Full load torque, Breakdown torque, Condition for maximum torque i. Effect of change in supply voltage and frequency on torque and slip, Braking of induction motor j. Losses in induction machines k. Power flow diagram

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
		in a three phase induction machine l. air gap power, slip power, mechanical power; Torque, Synchronous watt, Efficiency m. Induction motor as generalized transformer n. Phasor diagram of induction motor o. Exact and approximate per phase equivalent circuit p. Induction motor no load test and blocked rotor test q. determining equivalent circuit parameters; losses and efficiency r. Concept of circle diagram, Series circuit and current locus, Construction of circle diagram s. Induction machine performance computation from circle diagram t. Effect of space harmonic fields, harmonics induction torques, Harmonic synchronous torques u. Crawling and Cogging, Deep bar and double cage induction motor (High torque motors)

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit-IV Starting and speed control of three phase induction machines	**CO3.Interpret** the suitable control technique for induction motors. **CO2.Evaluate** the performance of the Transformers and Induction machines PrO.10 Speed control of wound rotor induction motor PrO.11 Speed control of squirrel cage induction motor 4.1 Interpret the machine data with respect to the given power system. 4.2 Select the relevant control method for the specified application.	4.1 Starting of squirrel cage and slip ring induction motor 4.2 Direct on line starting 4.3 stator resistance and reactance based starting 4.4 rotor resistance based starting 4.5 Y/Δ and auto transformer based starting; examples 4.6 Speed control of induction motors by stator voltage variation and pole changing techniques 4.7 Variable frequency speed control of induction machines, v/f control method, slip power control method
Unit-V Single phase induction motor	**CO1.Use** in depth knowledge about the construction details, operating principle, performance characteristics of induction machines **CO2.Evaluate** the performance of the Transformers and Induction machines **CO4.Analyze** with the recent developments in transformers and induction machines. PrO.6 No load and Blocked rotor test of single phase induction motor PrO.7 Determination of equivalent circuit parameters and performance of single phase induction motor 5a. Choose the relevant operating parameters for load operation of single phase induction motor. 5b. Evaluate the performance of single phase induction motor for the specified condition.	5.1 General description of the motor and constructional features 5.2 Rotating and pulsating field, development of equivalent circuit based on double revolving field theory 5.3 starting of single phase induction motor, 5.4 Determination of parameters by test 5.5 performance analysis 5.6 Types of induction motor: split phase or resistance start, Capacitor-start, Capacitor start capacitorrun, Permanent capacitor, Shaded pole motor: Construction, Working, Starting and running performance

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
		5.7 Characteristics and Applications; selection of capacitor value for starting and running conditions 5.8 Selection capacitance values for maximum torque

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	**U Level**	**A Level**	**An Level**	**E Level**	**C Level**	**Total Marks**
I	Fundamentals, Testing and performance analysis of single phase transformers	15	2	6	2	6	0	0	16
II	Fundamentals, Operation and performance of three phase transformers	15	2	2	4	4	2	0	14
III	Construction, operation, Testing and analysis of three phase induction machines.	15	0	6	4	4	4	0	18
IV	Starting and speed control of three phase induction machines	9	0	4	4	2	4	0	14
V	Single phase induction motor	6	0	2	2	2	2	0	8
Total		**60**	**4**	**20**	**16**	**18**	**12**	**0**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom’s taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **OC/SC test on transformers to draw OCC and SCC**.
2) **No**-load and Blocked rotor test of Induction motors.
3) **Speed control of induction motor in variable torque condition**.
4) **Direct torque control of induction motor**.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different methods of voltage regulation.
b) Prepare 15 min. power point presentation on synchronizing of alternator.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Fundamentals, Testing and performance analysis of single phase transformers	1	PrO1. Open circuit and short circuit tests on single phase transformer 1a. Justify the concept of static machines and fundamental component roles. 1b. Evaluate the performance of Single Phase Transformer 1c. Setup the circuit to determine output characteristics of single phase transformers. 1d. Obtain open circuit and short circuit characteristics of three phase transformer.	Lecture with media	PPTs, black board
	2	PrO.2 Determination of equivalent circuit parameters of transformer. 2a. **Evaluate** the performance of Single Phase Transformer 2b. **Justify** the output characteristics of Single phase Transformers	Lecture With media	PPTs, black board
	3	*PrO3.* Load test on single-phase transformer and three phase transformers. 1a. Evaluate the performance of Single Phase Transformer 1b. Justify the output characteristics of Single phase Transformers 1c. Setup the circuit to determine output characteristics of single phase transformers.	Lecture With media	PPTs, black board
	4.1	*PrO 4.* Parallel operation of single phase transformer. 1a. Justify the concept of static machines and fundamental component roles. 1b. Evaluate the performance of Single Phase Transformer 1c. Justify the output characteristics of Single phase Transformers 1d. Setup the circuit to determine output characteristics of single phase transformers.	Lecture With media	PPTs, black board
Unit-2 Fundamentals, Operation and performa	**4.2**	PrO.4 Parallel operation of Three phase Transformers. 2a. Determine the output from three phase transformer.	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
nce of three phase transformers	**5**	PrO.5 Scott Connection of transformers. 2a. Evaluate the performance of Three Phase Transformer	Lecture With media	PPTs, black board
Unit-3 Construction, operation, Testing and analysis of three phase induction machines.	**6**	PrO.8 No load and Blocked rotor test of ploy phase induction motor. 3a. Choose the relevant operating parameters for load operation of three phase induction motor. 3c. Evaluate the performance of three phase induction motor for the specified condition.	Lecture With media	PPTs, black board
	7	PrO.9 Load test of ploy phase induction motor 3a. Choose the relevant operating parameters for load operation of three phase induction motor. 3c. Evaluate the performance of three phase induction motor for the specified condition.	Lecture With media	PPTs, black board
	8	PrO.12 Computation of circle diagram for induction motor 3a. Choose the relevant operating parameters for load operation of three phase induction motor. 3c. Evaluate the performance of three phase induction motor for the specified condition.	Lecture With media	PPTs, black board
Unit-4 Starting and speed control of three phase induction machines	**9**	PrO.10 Speed control of wound rotor induction motor 4.3 Interpret the machine data with respect to the given power system. 4.4 Select the relevant control method for the specified application.	Lecture With media	PPTs, black board
	10	PrO.11 Speed control of squirrel cage induction motor 4.5 Interpret the machine data with respect to the given power system. 4.6 Select the relevant control method for the specified application.	Lecture With media	PPTs, black board
Unit-5 Single phase	**11**	PrO.6 No load and Blocked rotor test of single phase induction motor	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
induction motor		3a. Choose the relevant operating parameters for load operation of single phase induction motor. 3c. Evaluate the performance of single phase induction motor for the specified condition.		
	12	***PrO.7*** Determination of equivalent circuit parameters and performance of single phase induction motor 3a. Choose the relevant operating parameters for load operation of single phase induction motor. 3c. Evaluate the performance of single phase induction motor for the specified condition.	Lecture With media	PPTs, black board

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
4	Theory and performance of electrical machines	J.B. Gupta	S.K. Kataria & Sons, ISBN: 978-93-5014-277-6
5	A textbook of electrical technology VOL II	B.L. Theraja	S. Chand, ISBN: 9788121924375
6	Electric Machinery Fundamentals	Stephen. J. Chapman	McGraw Hill, ISBN: 10. 978007352954

XV SOFTWARE/LEARNING WEBSITES

- http://nptel.ac.in/courses/ 108106072 /
  (video lectures Prof:D. Kastha, IIT Kharagpur)

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	**PO 1** Used of mathematical method	**PO 2** Analysis mechanical system	**PO 3** Used engineering soft.	**PO 4** Solve problem in machine	**PO 5** **Modern tool usage**	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work:	**PO 10** Communication	**PO11** Project management	**PO12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem
CO1	**Use** in depth knowledge about the construction details, operating principle, performance characteristics of Transformers and induction machines.	3	1	1	3	1				2		2	3	3	1
CO2	**Evaluate** the performance of the Transformers and Induction machines.	3	2	1	1	1			1	2		2	3	3	1
CO3	**Interpret** the suitable control technique for induction motors.	3	2	3	2	3			1	1		2	2	1	
CO4	**Analyze** with the recent developments in transformers and induction machines.	2		2	2	3				2				2	3
CO5	**Choose** the type of testing method for Transformers and Induction machines.	2		2	1	2							1	2	

XVII NAMES OF RESOURCE PERSONS

Prof. Pratik B Panchal

Course Code: **EE255.01**

Course Title: **Control Systems**

UG Engineering Programme	Semester
Control Systems	Fourth

I RATIONALE

Electrical engineers have to work on various electrical equipment with their dedicated control system. This course is developed such that student should be able to design, develop, implement and operate efficiently various controllers, compensators and closed loop control systems for different electrical applications for efficient performance.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Evaluate the performance of various control systems used for linear and no linear systems.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

Use the different methods to develop mathematical models for systems.
Evaluate *the performance of the system by means of stability.*
Evaluate *the performance of the system by means of* state space analysis.
Evaluate the performance of linear and nonlinear control system with time and frequency response analysis under *steady state and dynamic condition*.
Choose *the appropriate control scheme, controller, compensator for different applications***.**

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			**Total Credits (L+T+P)**	**Examination Scheme**				
				Theory Marks		**Practical Marks**		**Total Marks**
L	**T**	**P**	**C**	**ESE**	PA	**ESE**	**PA**	150
3	2	0	5	70	30	25	25	

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

(*): *Under the* ***Theory PA****, out of 40 marks,* ***20 marks*** *are for* ***micro-project assessment*** *to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	**CO No.**	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	**Approx. Hrs. required**
1	1	Set the value of controller gain for the effective performance of the feedback control system.	01
2	1	Determine the transfer function for the given SISO and MIMO system using block diagram reduction technique by using MATLAB software.	02
3	1	Determine the transfer function for the given SISO and MIMO system using Mason's gain Formula by using MATLAB software.	02
4	2	Determine the stability of the systems by using RH Criteria and Root locus with the help of standard procedure and compare the same with MATALB software.	02
5	3	Examine the controllability, observability for the systems with the help of MATLAB software.	02
6	3	Determine the Eigen values and Eigen vectors for the system using conventional and with the help of MATLAB software.	01
7	4	Determine the value of various transient & steady state response specifications for different type and order control systems with the help of MATALB software.	02
8	4	Determine the value of ζ for various 1st and 2nd order control system and compare the performance of the system with the help of graphical representation using MATLAB software.	02
9	4	Determine the most efficient values of P , D & I constants in P, PD ,PI and PID controllers for 2nd order control system and compare the performance of the system with the help of graphical representation using MATLAB software.	02
10	5	Examine the stability of the systems using BODE ,POLAR and NYQUIST plots with the help of MATLAB software.	02
11	5	Setup the relevant compensator for the given specific control system requirement using MATLB software.	02
		Total	**20**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	**Equipment Name with Broad Specifications**	**PrO No.**
1	MATLAB Simulation software (version 2021a)	1,2,3,4,5
2	Test kit for feedback control system.	1

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Introduction to control system.	**CO 1.** ***Use*** the different methods to develop mathematical models for systems. PrO 1. Set the value of controller gain for the effective performance of the feedback control system. *PrO 2.* Determine the transfer function for the given SISO and MIMO system using block diagram reduction technique by using MATLAB software. PrO 3. Determine the transfer function for the given SISO and MIMO system using Mason's gain Formula by using MATLAB software. Justify the need of protection scheme for the given situation. 1a. Evaluate the various control system various parameters.	1.1 Control system parameters 1.2 Types of control systems 1.3 Methods of finding transfer function of the system. 1.4 Analysis of Electrical, Mechanical and

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	1b. Justify the requirements of good control system. 1c. Determine the transfer function of the system using Block diagram reduction. 1d. Determine the transfer function of the system using Mason's Gain Formula. 1e. Evaluate the transfer function of electrical, mechanical and electromechanical system.	Electromechanical systems. 1.5 Need of servomechanism and servomotors
Unit– II Stability of Control Systems	**CO 2. *Evaluate*** *the performance of the system by means of stability.* PrO 4. Determine the stability of the systems by using RH Criteria and Root locus with the help of standard procedure and compare the same with MATALB software. 2a. Evaluate the various factors which are relevant for the stability. 2b. Evaluate the various methods for determining the stability of the system. 2c. Determine the stability of the higher order system with the help pf RH criteria. 2d. Choose the appropriate value of DC gain in order to achieve the stability condition. 2e. Evaluate the stability condition using root locus diagram method for stability condition.	2.1 Basic stability conditions 2.2 R-H Criteria for higher order control system. 2.3 Root locus construction and stability analysis. 2.4 Effect of adding poles and zeroes in the system
Unit– III State Space Analysis of Control Systems	**CO 3. Evaluate** *the performance of the system by means of* state space analysis. *PrO 5.* Examine the controllability, observability for the systems with the help of MATLAB software. *PrO 6.* Determine the Eigen values and Eigen vectors for the system using conventional and with the help of MATLAB software. 3a. Evaluate the various performance indices related to state space analysis. 3b. Determine the controllability & observability by means of Kalman's Test. 3c. Determine the Eigen values and Eigen vectors for the system.	3.1 Concept of state, state variable and state model 3.2 Concept of controllability and Observability, State – Transition Matrix 3.3 Computation of State Transition Matrix, Eigen Values and Eigen Vectors.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit-IV **Time response analysis**	**CO 4. *Evaluate*** the performance of linear and nonlinear control system with time and frequency response analysis under *steady state and dynamic condition*. *PrO 7.* Determine the value of various transient & steady state response specifications for different type and order control systems with the help of MATALB software. *PrO 8.* Determine the value of ζ for various 1^{st} and 2^{nd} order control system and compare the performance of the system with the help of graphical representation using MATLAB software. *PrO 9.* Determine the most efficient values of P , D & I constants in P, PD ,PI and PID controllers for 2^{nd} order control system and compare the performance of the system with the help of graphical representation using MATLAB software. *PrO 10.* Examine the stability of the systems using BODE, POLAR and NYQUIST plots with the help of MATLAB software 4a. Examine the various factors associated with steady state and transient state performance of the system. 4b. Choose the relevant values of ζ for the efficient performance of the system. 4c. Choose the relevant values of controller parameters for the efficient performance of the system. 4d. Examine the various factors associated with Time response analysis. 4e. Examine the various factors associated with frequency response analysis. 4f. Determine the stability of the system under frequency response approach.	4.1 Time response of first order and second order system 4.2 Damping ratio, natural un damped frequency, Transient response specifications 4.3 Steady state error and error constant. 4.4 Effect of integral and derivative control action on system. 4.5 Bode plot, Procedure for drawing bode plot and determination of gain margin, phase margin and stability 4.6 Polar plot Procedure for drawing bode plot and determination of gain margin, phase margin and stability
Unit – V **Frequency response analysis**	**CO 5. *Choose*** *the appropriate control scheme, controller, compensator for different applications*.	

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	PrO 11. Setup the relevant compensator for the given specific control system requirement using MATLB software 5a. Examine the the various factors associated with compensation. 5b. Examine the various compensators for efficient performance of the system. 5c. Determine the compensators for efficient performance of the system breaker. 5d. Prepare the specification of a compensator required for the given situation. 5e. Select the relevant compensator for the efficient performance of control system.	5.1 Phase lead compensation, Phase lag compensation, Phase lead lag compensation. 5.2 Selection of Compensator, Design of Phase Lag, Phase Lead and Phase Lag-Lead Compensator.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction to control Systems	10	02	04	04	04	02	0	16
II	Stability of Control Systems	06	02	02	02	00	04	0	10
III	State Space Analysis of Control Systems	06	02	02	02	00	04	0	10
IV	Time response analysis	08	02	02	02	02	06	02	16
V	Frequency response analysis	15	02	02	02	02	08	02	18
Total		**45**	**10**	**12**	**12**	**08**	**24**	**04**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **Mathematical modelling:** Collect the specifications of different types of electromagnetic relays used in different protection situations in different types of industries.
2) **Steady state and Dynamic Response :** Collect the specifications of different types of static relays used in different protection situations in different types of industries.
3) **Stability Analysis:** Prepare case studies of protections failures in different types of industries.
4) **Controllers and Compensators:** Prepare case studies of circuit breakers used in the protection schemes in different types of industries.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different control system used in industries.
b) Prepare 15 min. power point presentation on the implementation of closed loop control system.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Introduction to control system.	1	*PrO 1. Set the value of controller gain for the effective performance of the feedback control system.* *PrO 2.* Determine the transfer function for the given SISO and MIMO system using block diagram reduction technique by using MATLAB software. 1a. Evaluate the various control system various parameters. 1b. Justify the requirements of good control system. 1c. Determine the transfer function of the system using Mason's Gain Formula.	Lecture with media	PPTs, black board
	2	PrO 3. Determine the transfer function for the given SISO and MIMO system using Mason's gain Formula by using MATLAB software. Justify the need of protection scheme for the given situation. 1a. Determine the transfer function of the system using Block diagram reduction.	Lecture With media	PPTs, black board
	3	PrO 3. Determine the transfer function for the given SISO and MIMO system using Mason's gain Formula by using MATLAB software. Justify the need of protection scheme for the given situation. 1a. Determine the transfer function of the system using Mason's Gain Formula. 1b. Evaluate the transfer function of electrical, mechanical and electromechanical system.	Lecture with media Tutorial and Field visit	PPTs, black board
Unit– II Stability of Control Systems	**4**	**CO 2. *Evaluate*** *the performance of the system by means of stability.* PrO 4. Determine the stability of the systems by using RH Criteria and Root locus with the help of standard procedure and compare the same with MATALB software. 2a. Evaluate the various factors which are relevant for the stability.	Lecture with media, and Demonstration	Real samples of relays, PPTs, Videos

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		2b. Evaluate the various methods for determining the stability of the system.		
	5	2c. Evaluate the various methods for determining the stability of the system. 2d. Determine the stability of the higher order system with the help pf RH criteria. 2e. Choose the appropriate value of DC gain in order to achieve the stability condition.	Lecture with media, Tutorial	PPTs
	6	2e. Choose the appropriate value of DC gain in order to achieve the stability condition.	Lecture with media	PPTs
Unit– III State Space Analysis of Control Systems	7	**CO 3. Evaluate** *the performance of the system by means of* state space analysis. *PrO 7.* Examine the controllability, observability for the systems with the help of MATLAB software. 3a. Evaluate the various performance indices related to state space analysis. 3b. Determine the controllability & observability by means of Kalman's Test.	Lecture with media, Field visit	PPTs, Video
	8	*PrO 8.* Determine the Eigen values and Eigen vectors for the system using conventional and with the help of MATLAB software. 3c. Determine the transfer function of the system. 3d. Determine the Eigen values and Eigen vectors for the system.	Lecture with media, and Tutorial	PPTs
Unit– IV Time response analysis	9	**CO 4. *Evaluate*** the performance of linear and nonlinear control system with time and frequency response analysis under *steady state and dynamic condition*. *PrO 13.* Determine the value of various transient & steady state response specifications for different type and order control systems with the help of MATALB software. *PrO 14.* Determine the value of ζ for various 1^{st} and 2^{nd} order control system and compare the performance of the system with the help of graphical representation using MATLAB software. *PrO 15.* Determine the most efficient values of P , D & I constants in P, PD ,PI and PID controllers for 2^{nd} order control system and compare the performance of the system with the	Lecture with media, Tutorial	PPT, Video

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		help of graphical representation using MATLAB software. *PrO 16.* Examine the stability of the systems using BODE, POLAR and NYQUIST plots with the help of MATLAB software 4g. Examine the various factors associated with steady state and transient state performance of the system. 4h. Choose the relevant values of ζ for the efficient performance of the system. 4i. Choose the relevant values of controller parameters for the efficient performance of the system. 4j. Examine the various factors associated with Time response analysis. 4k. Examine the various factors associated with frequency response analysis. Determine the stability of the system under frequency response approach.		
	11		Lecture with media	PPTs, black board
	12		Lecture with media, case study	PPT, Pictures , black board
Unit– V Frequency response analysis	**13**	**CO 5.** ***Choose*** *the appropriate control scheme, controller, compensator for different applications*. *PrO 17.* Setup the relevant compensator for the given specific control system requirement using MATLB software 5f. Examine the the various factors associated with compensation. 5g. Examine the various compensators for efficient performance of the system. 5h. Determine the compensators for efficient performance of the system breaker. 5i. Prepare the specification of a compensator required for the given situation. Select the relevant compensator for the efficient performance of control system.	Lecture with media, Tutorial	PPTs, black board
	14		Lecture	PPTs, Video

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
			with media, Tutorial	
	15		Lecture with media, Tutorial	PPTs, Video

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Automatic control system	B.C. Kuo	Khanna Publishers, New Delhi, 2008, ISBN: 9788174092323
2	Modern control engineering.	K. Ogata	
3	Control System Engineering	J Nagrath & M Gopal	New Age International Publishers
4	Control Systems	A. Anand Kumar	PHI Learning Pvt. Ltd.
5	Automatic Control Systems	S. Hasan Saeed	Katson Books
6	Control Systems	Ashfaq Hussain, Haroon Ashfaq	Dhanpat Rai & Co.

XV SOFTWARE/LEARNING WEBSITES

1. http://nptel.iitg.ernet.in/courses/Elec_Engg/IIT%20Delhi/Control%20Systems.htm
2. http://nptel.ac.in/courses/108101037/
3. http://nptel.ac.in/courses/108102043/
4. http://nptel.ac.in/courses/108103007/

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs												
		PO 1 Used of mathematical method	PO 2 Analysis mechanical system	PO 3 Used engineering soft.	PO 4 Solve problem in machine	PO 5 The engineer and society	PO 6 Environment and sustainability	PO 7 Ethics	PO 8 Individual and team work:	PO 9 Communication	PO 10 Life-long learning	PSO1 Evaluate the performance of Electrical Machines and Power systems Power Systems	PSO2 Evaluate the performance of Electrical and Electronic Instrumentaton Systems	PSO3 Provide socially and ethically acceptable l technical solutions to complex electrical engineering problems
EE255.01	Semester VI	Protection and Switchgear Mark **'H'=3** for high, **'M'**=2 for medium or **'L'=1** for the relevant correlation of each PO or PSO, Competency or CO												
255	**Competency** *Evaluate the performance of*	3	2	2	2	1	1	1	-	2	1	3	-	1

Course Code	Competency and COs	POs and PSOs												
		PO 1 Used of mathematical method	**PO 2** Analysis mechanical system	**PO 3** Used engineering soft.	**PO 4** Solve problem in machine	**PO 5** The engineer and society	**PO 6** Environment and sustainability	**PO 7** Ethics	**PO 8** Individual and team work:	**PO 9** Communication	**PO 10** Life-long learning	**PSO1** Evaluate the performance of Electrical Machines and Power systems Power Systems	**PSO2** Evaluate the performance of Electrical and Electronic Instrumentaton Systems	**PSO3** Provide socially and ethically acceptable l technical solutions to complex electrical engineering problems
	various protection systems used in Industry													
255.01.1	Use the protection schemes for apparatus protection	3	2	1	1	1	1	1	-	1	1	1	-	1
255.01.2	Evaluate the performance of the electromagnetic relays	3	2	2	1	1	1	1	-	1	-	2	-	1
255.01.3	Interpret the suitable protection system for the apparatus protection	3	2	1	1	1	1	1	-	1	-	2	-	1
255.01.4	Evaluate the performance of the static and Numerical relays	3	1	1	2	1	1	1	-	1	-	1	-	-
255.01.5	Choose the type of circuit breaker for different applications	3	2	1	1	1	1	1	-	1	1	1	-	-

XVII NAMES OF RESOURCE PERSONS

a) ………………………………………………….

b) ……………………………………………………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS111.02 A: HUMAN VALUES AND PROFESSIONAL ETHICS

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
IV	HS111.02 A	Human Values & Professional Ethics	02	02	---	---	30	70	100

II. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Values and Ethics	05
2.	Elements and Principles of Values	08
3.	Applied Ethics	08
4.	Value, Ethics & Global Issues	05
5.	Contemporary Issues in Values and Ethics	04
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	**30**

III. Detailed Syllabus:

1.	Introduction to Values and Ethics	5 Hours	17%
	Need, Relevance and Significance of Values General, Concept and Meaning of Values and Ethics		
2.	Elements and Principles of Values	8 Hours	26%
	Universal & Personal Values, Social, Civic & Democratic Value		
3.	Applied Ethics	8 Hours	26%
	Universal Code of Ethics, Professional Ethics, Organizational Ethics, Ethical Leadership, Domain Specific Ethics		
4.	Value, Ethics & Global Issues	5 Hours	17%

	Cross-Cultural Issues, Role of Ethics & Values in Sustainability		
5.	Contemporary Issues in Values and Ethics	4 Hours	14%
	Case Studies, Presentations, Projects		

IV. Instruction, Method & Pedagogy:

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like case studies, critical reading, group work, independent and collaborative research, presentations, etc.

V. Evaluation:

The students will be evaluated continuously in the form of internal as well as external examinations. The evaluation (Practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation in the form of University Practical Examination.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Assignment	1	25	25
2	Project Work	1	25	25
3	Quiz	2	10 + 15	25
4	Presentations / Posters / Tasks	1	25	25
			Total	100*
5	Attendance and Class Participation			05
			Total	**30**

***To be converted to 25.**

External Evaluation

The University Practical examination will be of 70 marks and will test the value system of the students by asking them application based questions. There may also be a Computer Based Test at the end.

Sl. No.	Component	Number	Marks per incidence	Total Marks

1	Practical Exam / Viva Voce	01	70	70
			Total	70

There may also be a Computer Based Test at the end.

VI. Course Outcome (COs):

At the end of the course, learners will be able to:

At the end of the course, the students will be able to

CO1	Understand the concepts and mechanics of values and ethics.
CO2	Understand the significance of value and ethical inputs in and get motivated to apply them in their life and profession.
CO3	Understand the significance of value and ethical inputs in and get motivated to apply them in social, global and civic issues.
CO4	Develop their responsibility towards society.
CO5	Comprehend their own core values and adhere to those values at their workplace.
CO6	Practice Ethical Leadership.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	2	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	2	-	-	-	-	-	-	-	-
CO3	-	-	-	-	-	2	-	-	-	2	-	-	-	-
CO4	-	-	-	-	-	2	2	-	-	-	-	-	-	-
CO5	-	-	-	-	-	2	-	-	-	2	-	2	-	-
CO6	-	-	-	-	-	1	-	3	2	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

VII. Reference Books / Reading:

- Human Values and Ethics in Workplace, United Nations Settlement Program, 2006. (http://www.unwac.org/new_unwac/pdf/HVWSHE/Human%20Values%20&%20Ethics%20-%20Individual%20Guide.pdf).

- Ethics for Everyone, Arthur Dorbin, 2009. (http://arthurdobrin.files.wordpress.com/2008/08/ethics-for-everyone.pdf) .
- Values and Ethics for 21st Century, BBVA. (https://www.bbvaopenmind.com/wp-content/uploads/2013/10/Values-and-Ethics-for-the-21st-Century_BBVA.pdf)
- www.ethics.org

CE291: PYTHON PROGRAMMING SKILLS

B.TECH 4TH SEMESTER (Electrical)

INSTITUTE ELECTIVE – I

(For Minor Specialization in "Artificial Intelligence and Machine Learning)

Credits and Hours:

Teaching Scheme	Theory	Practical/Tutorial	Total	Credit
Hours/week	2	4	6	4
Marks	100	100	200	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Basics of Python	01
2.	Variables and Data Types	04
3.	Loops and Conditional Control	06
4.	Advanced Data Structures	04
5.	Functions	02
6.	Object Oriented Programming in python	08
7.	Reading and Writing Files	03
8.	Packages and Modular Programming	02

Total Hours (Theory): 30

Total Hours (Lab): 60

Total Hours: 90

B. Detailed Syllabus:

1. Basics of Python **01 Hour** **03%**

1.1 Introduction to Python

1.2 Installation and Environment Setup

1.3 Simple Python Program Practice

2. Variables and Data Types **04 Hours** **13%**

2.1 Variables, Operators, Precedence and Associativity

2.2 Data Types , Indentation, Comments, Reading Input, Print Output

2.3 Memory Optimization

2.4	Type Conversion		
3.	**Loops and Conditional Control**	**06 Hours**	**20%**
3.1	The if… Decision Control Flow Statement, The if…else Decision Control Flow Statement, The if…elif…else Decision Control Statement		
3.2	Nested if Statement, The while Loop, The for Loop,		
4.	**Advanced Data Structures**	**04 Hours**	**13%**
4.1	Lists- Basic List Operators, Replacing, Inserting, Removing an Element, Searching and Sorting Lists, Tuples.		
4.2	Dictionaries- Dictionary Literals, Adding and Removing Keys, Accessing and Replacing Values; Traversing Dictionaries.		
4.3	Sets, Strings, Files and their Libraries		
5.	**Functions**	**02 Hour**	**07%**
5.1	Defining a Function, Calling a Function		
5.2	Pass by Reference vs Value, Function Arguments, Required and Default Arguments, Keyword Arguments and Variable-Length Arguments		
5.3	The Anonymous Functions, The Return Statement		
5.4	Scope of Variables		
6.	**Object Oriented Programming in python**	**08 Hours**	**27%**
6.1	Object-Orientation Introduction, Class Definition, Class Objects, Instance Objects, Method Objects		
6.2	Class and Instance Variables, Inheritance, Private Variables, The Polymorphism		
7.	**Reading and Writing Files**	**03 Hour**	**10%**
7.1	Reading Files, Parsing and Processing Files		
7.2	File storage, File operations (delete, copy)		
8.	**Packages and Modular Programming**	**02 Hour**	**07%**
8.1	Executing modules as Scripts, Import and Use of Built-in Modules, Accessing Third Party Packages and Modules		
8.2	Creating Custom Modules and Packages, Importing and Using Custom Modules		

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams will be conducted as per pedagogy as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval.
- Surprise tests/Quizzes/Seminar will be conducted.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Practicals/Tutorials related to course content will be carried out in the laboratory.

D. Course Outcomes (COs):

At the end of the course the students will be able to:

CO1	Interpret the fundamental python syntax, semantics and fluent in the use of python control flow statements. (Understand)
CO2	Determine the methods to create and manipulate python programs by utilizing the data structures like lists, dictionaries, tuples and sets. (Apply)
CO3	Express proficiency in the handling of strings and functions. (Apply)
CO4	Articulate the Object-Oriented Programming concepts such as inheritance and polymorphism as used in Python. (Create)
CO5	Identify and use the commonly used operations involving file systems, and regular expressions (Create)

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	1	-	-	-	1	-	1	2	-	-	2	-	-
CO2	-	1	-	2	2	3	2	-	-	-	2	-	2	-
CO3	1	-	2	1	3	2	3	2	2	2	-	2	-	3
CO4	-	1	2	2	3	3	2	3	-	3	3	-	-	-
CO5	1	2	-	2	3	3	2	2	2	-	3	3	3	3

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

E. Recommended Study Material:

❖ Text Books:

1. Nagar, Sandeep, "Introduction to Python for Engineers and Scientists - Open Source Solutions for Numerical Computation", Apress, 2018.
2. Kenneth Lambert, "Fundamentals of Python: First Programs", ISBN-13: 978-1337560092, Cengage learning publishers, first edition, 2012.
3. Magnus Lie Hetland, "Beginning Python from Novice to Professional", Third Edition, Apress, 2017
4. Reema Thareja, "Python Programming using Problem Solving Approach", ISBN-13:978-0-19- 948017-3, Oxford University Press, 2017.

❖ Reference Books:

1. David Beazley, Brian K. Jones, "Python Cookbook", 3rd edition, OREILLY, 2016
2. Brett Slatkin, "Effective Python: 59 Specific Ways to Write Better Python", Novatec, 2016
3. Allen Downey, "Think Python: How to Think Like a Computer Scientist", Green Tea Press, 2015
4. Mark Lutz "Learning Python", 4th Edition, O"REILLY, 2016
5. Vamsi kurama, "Python programming: A modern approach", ISBN-978-93-325-8752- 6, Pearson, 2018.
6. W. Chun, "Core python programming", ISBN-13: 978-0132269933, Pearson, 2nd edition, 2016.

❖ Web Materials:

1. https://www.python.org/
2. http://www.diveintopython3.net/
3. https://developer.mozilla.org/en-US/docs/Learn/Server-side/Django
4. https://www.fullstackpython.com/django.html
5. https://codelabs.developers.google.com/
6. https://www.tutorialspoint.com/python/index.htm

❖ Software:

1. Python IDLE
2. Anaconda Python
3. PyCharm

ME291: BASICS OF ELECTRICAL VEHICLES

B.TECH 4TH SEMESTER (Electrical)

INSTITUTE ELECTIVE – I

(For Minor Specialization in “Electric Vehicle Systems”)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	00	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to Electric Vehicles	14
2.	Automotive Design	14
3.	Design of Electric and Hybrid Electric	15
4.	Regulation and Standardization of Vehicles	02

Total Hours (Theory): 45

Total Hours: 45

B. Detailed Syllabus:

1 Introduction to Electric Vehicles 14Hours 31.11%

1.1 I.C Engine powered Vehicles Vs Electrical Vehicles (EVs), How EV works, Types of EVs, Hybrid Electric Drive-train

1.2 Tractive effort in normal driving, Energy consumption Concept of Hybrid Electric Drive Trains, Architecture of Hybrid Electric Drive Trains, Series Hybrid Electric Drive Trains, Parallel hybrid electric drive trains, Electric Propulsion unit.

2 Automotive Design 14Hours 31.11%

2.1 Major parts of automobiles, Chassis, Frame, Body of automobile, Transmission system, Steering and Breaking systems, Suspension, Accessories.

3 Design of Electric and Hybrid Electric Vehicles 15Hours 33.34%

3.1 Series Hybrid Electric Drive Train Design: Operating patterns, control strategies, Sizing of major components, power rating of traction motor, power rating of engine/generator, design of PPS

3.2 Parallel Hybrid Electric Drive Train Design: Control strategies of parallel hybrid drive train, design of engine power capacity, design of electric motor drive capacity, transmission design, energy storage design

4 Regulation and Standardization of Vehicles 02Hours 4.44%

4.1 Motor vehicle act, registration of motor vehicles, driving license, control of traffic, insurance against third party, claims for compensation, traffic signs, central motor vehicle rules, vehicle safety standards and regulations, classification and definition of vehicles, enforcement of emission norms, duties of surveyor.

C. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1:	Understand needs and working of various systems in automobiles.
CO2:	Practically identify different automotive systems and subsystems.
CO3:	Practically identify different automotive components.
CO4:	Understand market and businesses of automobile industry.
CO5:	This course will give the students some awareness to recent trends and research areas in Automobiles.

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	-	-	-	-	-	-	-	-	-	-	1	-	-
CO2:	3	1	1	-	-	-	-	-	1	-	-	1	-	-
CO3:	3	1	1	-	-	-	-	-	1	-	-	2	-	-
CO4:	-	-	-	-	-	-	-	-	-	-	2	1	-	-

CO5:	1	1	1	1	-	-	-	-	1	-	-	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

D. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries 5 Marks weightage.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5 Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.
- Industrial Visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of assignment.

E. Recommended Study Material:

Text Books:

1. Singh Kirpal, "Automobile Engineering" Volume I & II, Standard Pub.& Dist.
2. Gupta K. M. "Automobile Engineering" Volume I & II, Umesh Pub.
3. Gupta R. B. "Automobile Engineering", Satya Prakashan.
4. Giri N. K., "Automobile Technology", Khanna Pub.

Reference Books:

1. Crouse W., “Automotive Mechanics”, Tata Mc Graw Hill.

2. Narang G. B. S. “Automobile Engineering”, Khanna Pub.

3. John Lowry and James Larminie , “Electric Vehicle Technology Explained”, Wiley

Web Material:

1. http://nptel.iitm.ac.in

2. http://www.sae.org/pubs/automotive/

3. https://www.springer.com/journal/12239/

Other Material:

1. International Journal of Automotive Technology (www.ijat.net/)

2. International Journal of Automotive Technology

ACADEMIC REGULATIONS & SYLLABUS

Faculty of Technology & Engineering

Bachelor of Technology Programme
(Electrical Engineering)

Education Campus – Changa, (ECC), hitherto a conglomerate of institutes of professional education in Engineering, Pharmacy, Computer Applications, Management, Applied Sciences, Physiotherapy and Nursing, is one of the choicest destinations by students. It has been transformed into **Charotar University of Science and Technology (CHARUSAT)** through an Act by Government of Gujarat. CHARUSAT is permitted to grant degrees under Section-22 of UGC- Govt. of India.

The journey of CHARUSAT started in the year 2000, with only 240 Students, 4 Programmes, one Institute and an investment of about Rs. 3 Crores (INR 30 million). At present there are seven different institutes falling under ambit of six different faculties. The programmes offered by these faculties range from undergraduate (UG) to Ph.D. degrees including M.Phil. These faculties, in all offer different programmes. A quick glimpse in as under:

Faculty	Institute	Programmes Offered
Faculty of Technology & Engineering	Chandubhai S. Patel Institute of Technology	B. Tech, M. Tech Ph.D.
	Devang Patel Institute of Advance Technology & Research	B. Tech
Faculty of Pharmacy	Ramanbhai Patel College of Pharmacy	B. Pharm, M. Pharm MPM, Ph.D. PGDCT/ PGDPT
Faculty of Management Studies	Indukaka Ipcowala Institute of Management	M.B.A., PGDM Ph.D. Dual Degree BBA+MBA
Faculty of Computer Applications	Smt. Chandaben Mohanbhai Patel Institute of Computer Applications	M.C.A/MCA (Lateral) M.Sc.(IT) Ph.D. Dual Degree BCA+MCA
Faculty of Applied Sciences	P. D. Patel Institute of Applied	M. Sc., M. Phil

	Sciences	Ph.D. Dual Degree B.Sc+M.Sc
Faculty of Medical Sciences	Ashok and Rita Institute of Physiotherapy	B.PT., M. PT Ph.D.
	Manikaka Topawala Institute of Nursing	B.Sc. , M.Sc. GNM, Ph.D.
	Charotar Institute of Paramedical Sciences	Ph.D. PGDHA, PGDMLT

The development and growth of the institutes have already led to an investment of over Rs.63 Crores (INR 630 Million). The future outlay is planned with an estimate of Rs. 250 Crores (INR 2500 Million).

The University is characterized by state-of-the-art infrastructural facilities, innovative teaching methods and highly learned faculty members. The University Campus sprawls over 100 acres of land and is Wi-Fi enabled. It is also recognized as the Greenest Campus of Gujarat.

CHARUSAT is privileged to have 350 core faculty members, educated and trained in IITs, IIMs and leading Indian Universities, and with long exposure to industry. It is also proud of its past students who are employed in prestigious national and multinational corporations.

From one college to the level of a forward-looking University, **CHARUSAT** has the vision of entering the club of premier Universities initially in the country and then globally. **High Moral Values like Honesty, Integrity and Transparency** which have been the foundation of ECC continue to anchor the functioning of **CHARUSAT**. Banking on the world class infrastructure and highly qualified and competent faculty, the University is expected to be catapulted into top 20 Universities in the coming five years. In order to align with the global requirements, the University has collaborated with internationally reputed organizations like Pennsylvania State University – USA, University at Alabama at Birmingham – USA, Northwick Park Institute –UK, ISRO, BARC, etc.

CHARUSAT has designed curricula for all its programmes in line with the current international practices and emerging requirements. Industrial Visits, Study Tours, Expert

Lectures and Interactive IT enabled Teaching Practice form an integral part of the unique **CHARUSAT** pedagogy.

The programmes are credit-based and have continuous evaluation as an important feature. The pedagogy is student-centred, augurs well for self-learning and motivation for enquiry and research, and contains innumerable unique features like:

- Participatory and interactive discussion-based classes.
- Sessions by visiting faculty members drawn from leading academic institutions and industry.
- Regular weekly seminars.
- Distinguished lecture series.
- Practical, field-based projects and assignments.
- Summer training in leading organizations under faculty supervision in relevant programmes.
- Industrial tours and visits.
- Extensive use of technology for learning.
- Final Placement through campus interviews.

Exploration in the field of knowledge through research and development and comprehensive industrial linkages will be a hallmark of the University, which will mould the students for global assignments through technology-based knowledge and critical skills.

The evaluation of the student is based on grading system. A student has to pursue his/her programme with diligence for scoring a good Cumulative Grade Point Average (CGPA) and for succeeding in the chosen profession and life.

CHARUSAT welcomes you for a Bright Future

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

ACADEMIC REGULATIONS

Bachelor of Technology (Electrical Engineering) Programme

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
Phone: 02697-265011, Email: info@charusat.ac.in
www.charusat.ac.in

Year – 2023-2024

CHARUSAT

FACULTY OF TECHNOLOGY AND ENGINEERING
ACADEMIC REGULATIONS
Bachelor of Technology Programmes

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

Choice based Credit System with Semester pattern of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a course works in the chosen subject of specialization and also complete a project/dissertation if any. Apart from the Programme Core courses, provision for choosing University level electives and Programme/Institutional level electives are available under the Choice based credit system.

2. Duration of Programme

Undergraduate programme (B. Tech.)

Minimum	8 semesters (4 academic years)
Maximum	12 semesters (6 academic years)

3. Eligibility for admissions

As enacted by Govt. of Gujarat from time to time.

4. Mode of admissions

As enacted by Govt. of Gujarat from time to time.

5. Programme structure and Credits

As per annexure – 1 attached

6. *Attendance*

6.1 All activities prescribed under these regulations and enlisted by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student regarding attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Principal.

6.2 Student's attendance in a course should be 80%.

7 *Course Evaluation*

7.1 The performance of every student in each course will be evaluated as follows:

7.1.1. Internal evaluation by the course faculty member(s) based on continuous assessment, the continuous assessment will be conducted by the respective department /institute.

7.1.2 Final end-semester examination by the University through written paper or practical test or oral test or presentation by the student or a combination of these.

7.1.3 The weightages of continuous assessment and End-semester university examination in overall assessment shall depend on individual course as approved by Academic Council through Board of Studies.

7.1.4 The performance of candidate in continuous assessment and in end-semester examination together (if applicable) shall be considered for deciding the final grade in a course.

7.1.5 In order to earn the credit in a course a student has to obtain grade other than FF.

7.2 Performance in continuous assessment and end-semester University Examination

7.2.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations (internal/external) are as follows

Minimum marks in University Exam per course	Minimum marks Overall per course
40%	45%

7.2.2 If a candidate obtains minimum required marks in each course but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

8 *Grading*

8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade **on** a ten-point scale as per the following scheme:

Table: Grading Scheme (UG)

Range of Marks (%)	≥80	**<80 ≥73**	**<73 ≥66**	**<66 ≥60**	**<60 ≥55**	**<55 ≥50**	**<50 ≥45**	**<45**
Corresponding Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his/her performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

(i) SGPA = $\Sigma C_i G_i / \Sigma C_i$ where C_i is the number of credits of course i
G_i is the Grade Point for the course i
and i = 1 to n, n = number of courses in the semester

(ii) CGPA = $\Sigma C_i G_i / \Sigma C_i$ where C_i is the number of credits of course i
G_i is the Grade Point for the course i
and i = 1 to n, n = number of courses of all semesters up to which CGPA is computed.

9 *Award of Class*

☞ The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Award of Class	CGPA Range
First Class with Distinction	CGPA ≥ 7.50
First class	7.49 ≥ CGPA ≥ 6.00
Second Class	5.99 ≥ CGPA ≥ 5.00
Pass Class	4.99 ≥ CGPA ≥ 4.50

Grade sheets of only the final semester shall indicate the class. In case of all the other semesters, it will simply indicate as Pass / Fail.

9.1 Maximum duration allowed for Completion of a programme

- Maximum duration to allow for completion of a particular programme shall not be more than twice the normal duration of the respective programme. For example, a 6-Semester programme should be completed within not more than 12 semesters.

10 Detention Criteria

- No student will be allowed to move further in next semester if CGPA is less than 3 at the end of an academic year.

- A student will not be allowed to move to third year if he/she has not cleared all the courses of second year.

- A student will not be allowed to move to fourth year if he/she has not cleared all the courses of third year.

11 Transcript:

- The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

Annexure – I

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)											
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN ELECTRICAL ENGINEERING											
Sem	Course Code	Course title	Teaching Scheme				Examination Scheme				Total
			Contact Hrs.			Credits	Theory		Practical		
			Th.	Pr.	Total		Int.	Ext.	Int.	Ext.	
TY Sem– V	EE351	Electrical Power Transmission and Distribution	3		3	3	30	70			100
	EE342	Synchronous and DC Machines	4	4	8	6	30	70	50	50	200
	EE353	Power Electronics and Drives – I	3	2	5	4	30	70	25	25	150
	EE344	Minor Project I		4	4	2			50	50	100
	EE345	Electrical Product Survey		2	2	2			50	50	100
	EE350	Summer Internship I				3			75	75	150
	HS131.02A	Communication and Soft Skill		2	2	2			30	70	100
	EE371-EE375	Advanced Microcontrollers/ VLSI Technology and Design/Optimization Techniques/Distributed Generation and Microgrid/Energy Conservation, Audit and Management	4	2	6	5	30	70	25	25	150
		TOTAL:	**14**	**16**	**30**	**27**	**120**	**280**	**305**	**345**	**1050**
TY Sem - VI	EE346	Digital Signal Processing	3	2	5	4	30	70	25	25	150
	EE347	Programmable Logic Controllers and Industrial Automation	2	4	6	4	30	70	50	50	200
	EE348	Power Electronics and Drives II	3	2	5	4	30	70	25	25	150
	EE349	Fault Analysis and Switchgear	4	2	6	5	30	70	25	25	150
	EE360	Minor Project II		4	4	2			50	50	100
	HS132.02A	Contributor Personality Development	2		2	2			30	70	100
	EE376-EE381	Special Electrical Machines and Applications/ Embedded Systems/Power Electronics Applications in Power System/High Voltage Engineering/Internet of Things/Hybrid and Electric Vehicles	3	2	5	4	30	70	25	25	150
		TOTAL:	**15**	**18**	**33**	**25**	**150**	**350**	**230**	**270**	**1000**

B. Tech. (Electrical Engineering) Programme

SYLLABI

(Semester – V)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Course Code: EE351

Course Title: **ELECTRICAL POWER TRANSMISSION AND DISTRIBUTION**

UG Engineering Programme	Semester
Electrical Engineering	Fifth

I RATIONALE

In today's time The electricity is generated in bulk at remote places using different type of generating station and transmitted to long distances, then distributed at various places as per consumer requirements. The transmission and distribution of electric power is a complex men made system which requires knowledge of different types of transmission lines and equipment's. Technicians are required to operate and maintain the power transmission and distribution system so that electrical energy is continuously available to the end users economically. So engineer or technician should able to work independently in the various area of transmission and distribution system. They can able to operate various control equipment's independently in normal and abnormal conditions. So at design and execution level skill development in student incorporate via this course.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course:

- **Design, Operate and Troubleshoot various transmission and distribution system.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) **Identify** and Calculate the different parameters of transmission line.
2) **Analyze** & calculate the performance of various types of transmission & distribution systems.
3) **Chose** the size and location of generating station, sub-station. with bus-bar arrangement.
4) **Design** and Connection schemes of A.C. & D.C. distribution system.
5) **Acquire** the knowledge of Power system planning for Generation, Transmission & Distribution System

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme (Anna 2022)				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	100
3	0	0	3	70	30*	0	0	

***Legends: L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

***(*):** Under the **Theory PA**, out of 30 marks, 30 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain	Topics and Sub-topics
Unit – I **Transmission Line Parameters**	**CO 1. Identify** and Calculate the different parameters of transmission line. **1a.** learn details of different parameters **1b.** calculate the different parameters of transmission line with different configuration of transmission line. **1c.** learn about importance of bundled conductor and its calculation.	1.1 Basics of Electric field. 1.2 Resistance of line, basic concepts of inductance and capacitance. 1.3 Flux linkages of an isolated current carrying conductor. 1.4 Inductance of a single phase two wire line, examples. 1.5 Flux linkages of one conductor in a group and inductance of composite conductor lines, examples. 1.6 Inductance of three phase line with unsymmetrical spacing and symmetrical spacing, 1.7 importance of transposition of tower, examples. Inductance of double circuit line, examples. 1.8 Inductance of bundled conductors and importance of bundled conductors, examples. 1.9 Electric field of a long straight conductor, potential difference between two conductors of a group of parallel conductors. 1.10 Capacitance of two wire

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain	Topics and Sub-topics
		line. Capacitance of three phase line with equal and unequal spacing, examples.
Unit– II **Characteristic and Performance of Transmission Line**	**CO 2. Analyze** & calculate the performance of various types of transmission & distribution systems. **2a.** analyze the general relation of a transmission line and its ABCD parameter. **2b.** analyze the performance of long transmission line and power transferred through transmission line. **2c.** acquire the knowledge regarding the surge impedance and surge impedance loading of transmission line.	2.1 Voltage regulation of short transmission line, example. 2.2 T and π representation of medium transmission line and their Phasor diagram, example. 2.3 Performance analysis of long transmission line and its ABCD parameter representation. 2.4 Surge impedance and surge impedance loading, 2.5 Equivalent T and π representation of long line, 2.6 Interpretation of long line equation, 2.7 Ferranti effect. Examples.
Unit– III **Design of Transmission system**	**CO 2. Analyze** & calculate the performance of various types of transmission & distribution systems. **2a.** State the need for EHV transmission **2b.** design Electrical side of EHV transmission line with tower. **2c.** gain the knowledge of RI and TVI. **2d.** Analyze transmission towers design	3.1 Requirement of Transmission lines, 3.2 selection of voltage for high-voltage transmission lines. 3.3 choice of conductors, spacing of conductor, 3.4 corona. 3.5 Surge impedance loading of transmission lines. 3.6 Electrical design of transmission lines, 3.7 Earth Wires and towers. 3.8 Transmission of Electrical Power at extra-high voltage. Design consideration of EHV lines. 3.9 Selection & spacing of conductors for EHV line. 3.10 Radio & television interference.
Unit-IV **Design of Power System**	**CO 3. Chose** the size and location of generating station, sub-station. with bus-bar arrangement.	4.1 Introduction, Selection of size and location of generation stations. 4.2 Selection of specification of

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain	Topics and Sub-topics
and Substation	**3a.** State the need for electrical substations. **3b.** Sketch the elevation layout of a typical 11/33/66/110 kV electrical substation with various switchgear and typical spacing between them and the ground level as well. **3c.** State the selection of the bus bar and their arrangement. **3d.** With sketches describe the various types of Earthing adapted for substations.	transmission lines. 4.3 Size and locations of substations. 4.4 Role of vector group in designing of sub-station. 4.5 Sub-station Grounding, Earthing, 4.6 step and touch potential. 4.7 Bus-bar arrangements in sub-station.
Unit – V **Design of Distribution system**	**CO 4. Design** and Connection schemes of A.C. & D.C. distribution system. **4a.** State the need for distribution system **4b.** With sketches describe the various connection schemes of the distribution system **4c**. Describe the Impact of wind power on the distribution system **4d.** Describe the measures to be adapted to take of the distributed generation in the distribution system **4e.** Solve simple numerical problems	5.1 Distribution system, requirements of distribution system, 5.2 design consideration of distribution system, 5.3 classification of distribution system. 5.4 Connection schemes of Distribution system. 5.5 D.C. distribution system, 5.6 Types of distribution system. 5.7 A.C. distribution system, A.C. distribution system calculations.
Unit-VI **Power System Planning**	**CO 5. Acquire** the knowledge of Power system planning for Generation, Transmission & Distribution System. **5a.** prepare flow chart of generation, transmission and distribution planning**.**	6.1 Introduction, Methods of Power system planning. 6.2 Forecasting load and energy requirements. 6.3 Generation Planning 6.4 Transmission System planning. 6.5 Distribution System Planning. 6.6 Reliability of electrical power

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain	Topics and Sub-topics
	5b. Acquire the knowledge about the reliable power system.	system, 6.7 Method of measuring power system reliability. 6.8 Trends in power system Planning in India.

VI TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Transmission Line Parameters	10	2	2	2	2	2	0	10
II	Characteristic and Performance of Transmission Line	10	2	2	2	2	2	0	10
III	Design of Transmission system	8	2	2	3	3	2	0	12
IV	Design of Power System and Substation	6	2	3	3	3	2	2	15
V	Design of Distribution system	8	2	3	3	3	2	2	15
VI	Power System Planning	3	2	2	2	0	0	2	08
Total		**45**	**12**	**14**	**15**	**13**	**10**	**06**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

VII SPECIAL INSTRUCTIONAL STRATEGIES (if any)

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.

b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.

c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.

VIII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding power system planning.
b) Prepare 15 min. power point presentation on different topics of syllabus.

IX COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Transmission Line Parameters	1-2-3	1a. understanding Basics of Electric field. 1b. study and calculation of different parameter. 1c. calculation of inductance and capacitances in various cases.	Lecture with media, Tutorial	PPTs, black board
Unit– II Characteristic and Performance of Transmission Line	4-5-6	2a. Learning performance of short, medium and long transmission line. 2b. Calculate voltage regulation and efficiency of system	Lecture With media Demonstration in software Tutorial	PPTs, black board
Unit– III Design of Transmission system	**7-8**	3a. Understand Requirement of Transmission lines. 3b. electrical design of EHV transmission line.	Lecture with media, Tutorial	PPTs, black board
Unit-IV Design of Power System and Substation	**9-10**	4a. Selection of size, location and different specification generating station, sub-station and transmission line. 4b. Basic of Earthing & grounding with details of step and touch potential.	Lecture with media, Case study	PPTs, black board
Unit – V Design of Distribution system	**11-12**	5a. Connection scheme of different distribution system 5b. AC & DC distribution study and calculation	Lecture with media, + PPT + Tutorial	PPTs, black board
Unit – VI Power System Planning	**13**	6a. Methods of generation, transmission and distribution system planning.	Lecture with media, + PPT	PPTs, black board

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Electrical Power System Design	M. V. Deshpande	Tata Mcgraw Hill
2	Principles of Power system	V.K.Mehta	S.Chand & Company Ltd.
3	A Course in electrical power	Soni & Bhattnagar	Dhanpat Rai & Sons.
4	Sub-station Design and Equipment	P.V. Gupta and P.S.Satnam,	Dhanpat Rai & Sons.
5	Handbook of Electrical Power Distribution	Ramamurthy University	Press (I) Pvt. Ltd
6	Power System Analysis and Design	B.R. Gupta	S.Chand Publications
7	Power system analysis	Hadi Saadat,	Tata McGraw Hill Publishing Company, New Delh
8	Modern Power System Analysis	D.P. Kothari & I. J. Nagrath,	Tata Mcgraw Hill
9	Power System Analysis	Grainger & Stevenson,	Tata Mcgraw Hill

XV SOFTWARE/LEARNING WEBSITES

1. http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-KANPUR/powersystem/ui/TOC.htm
2. http://en.wikipedia.org/wiki/Electric_power_transmission

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE351	**Semester VI**														
EE351	**Competency** Design, Operate and Troubleshoot various	*3*	*3*	*3*	*2*	*3*	*2*	*2*	-	*2*	*2*	*3*	*1*	*3*	*1*

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	transmission and distribution system.														
EE351.1	**Identify** and Calculate the different parameters of transmission line.	3	3	-	-	-	-	-	-	-	-	-	-	2	1
EE351.2	**Analyze** & calculate the performance of various types of transmission & distribution systems.	2	2	3	-	3	-	-	-	3	3	-	-	3	3
EE351.3	**Chose** the size and location of generating station, sub-station. with bus-bar arrangement.	-	-	3	3	3	3	-	2	3	2	-	-	3	3
EE351.4	**Design** and Connection schemes of A.C. & D.C. distribution system.	-	3	3	3	-	-	-	1	-	-	-	-	3	3
EE351.5	**Acquire** the knowledge of Power system planning for Generation, Transmission & Distribution System.	-	-	-	2	3	3	2	-	3	3	3	3	3	3

XVII NAMES OF RESOURCE PERSONS

a) Vibha Parmar
b) ……………………………………………………

Course Code: **EE342**
Course Title: **Synchronous & DC Machines**

UG Engineering Programme	Semester
Electrical Engineering	Fifth

I RATIONALE

The DC machines & synchronous machines are used power generation plants. The course is developed such that student would get the knowledge from basics to control schemes of these machines.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Evaluate the performance of various control methods used in field**.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) ***Use*** the control technique for electrical machines**.**
2) ***Evaluate*** *the performance of the synchronous and DC machines*
3) **Interpret** the suitable control technique for synchronous generator.
4) ***Evaluate*** the performance of the DC and synchronous generator.
5) ***Choose*** *the type of turbine for synchronous generator.*

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			**Total Credits (L+T+P)**	**Examination Scheme**				
				Theory Marks		**Practical Marks**		**Total Marks**
L	**T**	**P**	**C**	**ESE**	PA	**ESE**	**PA**	200
4	0	4	6	70	30*	50	50	

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

(*): *Under the* ***Theory PA****, out of 30 marks,* ***15 marks*** *are for* ***micro-project assessment*** *to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs****.***

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the

students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	To Study operating principle, working and cut model of DC machine.	04
2	2	To perform the magnetization (open circuit) characteristics and load test on a DC shunt generator to obtain the internal and external load characteristics.	04
3	3	To Perform the load test on a D.C. series generator to obtain the internal and external load characteristics.	04
4	3	To perform the speed control of DC series and DC shunt motor.	06
5	3	To find out the voltage regulation of three phase alternator by direct load test.	04
6	4	To find out voltage regulation of three phase alternator by Synchronous Impedance Method (EMF Method).	04
7	4	To find out voltage regulation of three phase alternator by Ampere Turn Method (MMF Method).	04
8	4	To find out voltage regulation of three phase alternator by ZPFC Method	06
9	4	To Perform slip test on salient pole alternator for determining direct axis synchronous reactance-Xd and quadrature axis synchronous reactance - Xq.	04
10	5	To perform parallel operation of two alternators and study method of synchronization of two alternators. .	06
11	5	To Obtain V Curve & Inverted V Curve of a Synchronous Motor.	04
12	5	To obtain constant excitation circle of synchronous motor.	04
13	1	TO study DC and AC machine windings.	06
		Total	**60**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	DC series/shunt/compound machine with prime-mover/load	1,2,3,4
2	Synchronous machine with prime-mover/load	5,6,7,8,9,10,11,12

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Fundamentals of DC Machines	**CO 1. *Able to recognize the electric and magnetic circuits of AC and DC machines.*** ***CO 2. Acquire the knowledge regarding the fundamentals, working and construction of Synchronous and DC machines.*** *PrO 1.* To Study operating principle, working and cut model of DC machine. PrO 13. To study DC and AC machine windings 1a. Justify the concept of rotating machines and fundamental component roles.	1.1 Classification of rotating electrical machines 1.2 Construction of DC machines, Comparison of AC and DC Generators 1.3 Principle of DC generator and motor, Comparison of motor and generator action, Elementary DC generator 1.4 essential parts, Types of DC machines 1.5 EMF equation, Voltage build up process, Causes of its failure 1.6 Critical resistance and speed, Armature reaction and its effect 1.7 Commutation in DC machines, Reactance voltage 1.8 methods of improving commutation, compensating winding

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
Unit– II Operation and performance Characteristics of DC Machine	**CO 4. *Aware with the performance characteristics and applications of Synchronous and DC machines*** ***CO 6. Understand the role and importance of DC and AC machine in power generation plants*** *PrO 2. To perform the magnetization (open circuit) characteristics and load test on a DC shunt generator to obtain the internal and external load characteristics.* *PrO3. To Perform the load test on a D.C. series generator to obtain the internal and external load characteristics.* *PrO 4. To perform the speed control of DC series and DC shunt motor.* 2a. Evaluate the performance of DC generator 2b. Evaluate the performance of DC motor 2c. Justify the output characteristics of DC machines 2d. Setup the circuit to determine output characteristics of DC machines.	2.1 Characteristics of separately excited DC machines (motor and generator), 2.2 Characteristics of shunt, series and compound machines (generators & motors) 2.3 Power flow diagram of DC machines 2.4 Losses in DC machines, Efficiency , condition for maximum efficiency 2.5 Voltage regulation in DC generator 2.6 Speed regulation in DC motor 2.7 Parallel operation of DC machines, Examples 2.8 Necessity of starter, types of starter 2.9 Speed control of DC shunt and series motors PV-Grid connected system design 2.10 Braking of DC motors, Examples 2.11 Applications of all types of DC generators and DC motors.
Unit– III Construction and Operation of Synchronous Generator	**CO 2. *Acquire the knowledge regarding the fundamentals, working and construction of Synchronous and DC machines.*** 3a. Determine the output from synchronous generator. 3b. Obtain open circuit and short circuit characteristics of synchronous generator.	3.1 Introduction, classification of synchronous machine 3.2 details of construction, advantages of rotating field type of construction 3.3 difference between salient and cylindrical alternator 3.4 damper winding, operating principle, Speed and Frequency, Synchronous speed 3.5 pitch factor, distribution factor and emf equation, emf waveform for

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
		concentrated and distributed winding, Examples 3.6 armature leakage reactance, synchronous reactance and impedance 3.7 armature reaction 3.8 Equivalent circuit, phasor diagram 3.9 Synchronous generator operating alone, operation of alternator on no load and on load 3.10 effect of variation in load on synchronous generator, Laboratory methods of determination of synchronous reactance 3.11 oc/sc test, determination of effective resistance of armature, Short circuit ratio 3.12 Losses and efficiency, power flow diagram, Examples,
Unit-IV Voltage Regulation, Power Equations and Synchronization of Alternators	**CO 4. Aware with the performance characteristics and applications of Synchronous and DC machines.** **CO 5. Create and test the synchronization of alternators to grid** *PrO 5.* To find out the voltage regulation of three phase alternator by direct load test. *PrO 6.* To find out voltage regulation of three phase alternator by Synchronous Impedance Method (EMF Method). *PrO 7.* To find out voltage regulation of three phase alternator by Ampere Turn Method (MMF Method). PrO 8. To find out voltage regulation of three phase alternator by ZPFC Method PrO 9. To Perform slip test on salient pole alternator for determining direct axis synchronous reactance-Xd and quadrature axis synchronous reactance – Xq PrO 10. To perform parallel operation of two	4.1 Voltage Regulation, methods of calculating voltage regulation, Direct load test 4.2 EMF method 4.3 MMF method 4.4 ZPFC method 4.5 Comparison of methods, examples 4.6 Active, Reactive and Complex Input and Output Power Equations for cylindrical synchronous generator 4.7 Maximum input and output power and torque developed, Power Angle Characteristics 4.8 Two reactance theory for

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	alternators and study method of synchronization of two alternators 4a. Choose the relevant operating parameters for load operation of alternator. 4b. Select the relevant control method for the specified application. 4c. Evaluate the performance of alternator for the specified condition.	salient pole synchronous machine, Phasor Diagram, Power Equations and Torque Angle Characteristics 4.9 Determination of Xd and Xq by slip test 4.10 Capability Curves for Synchronous Generators, Prime mover characteristics 4.11 Frequency Power and Voltage Reactive Power relations (F-P and Q-V), Power shared by two alternators 4.12 Necessity of parallel operation, condition required for parallel operation, the general procedure for synchronization of alternators 4.13 Infinite Bus, Synchronizing current, power, torque, Time period of Oscillations, effect of increasing the driving torque of one the alternator 4.14 effect of change in excitation of one of the alternator, load sharing between two alternators 4.15 operation of synchronous generator with infinite bus bar, operation of synchronous generator with other generator 4.16 various types of alternators based on prime movers, Factors affecting the size and efficiency of alternators
Unit – V Synchronous	**CO 3. *Acquire the practical exposure of rotating electrical machines.***	5.1 Introduction, construction, principle of

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Motor	PrO 11. To Obtain V Curve & Inverted V Curve of a Synchronous Motor. PrO 12. To obtain constant excitation circle of synchronous motor. 5a. Interpret the machine data with respect to the given power system. 5b. Determine machine charging control strategy. 5c. Select the method for machine control.	operation, Equivalent circuit and phasor diagram for cylindrical and salient rotor 5.2 synchronous motor with different excitation, starting methods of synchronous motor 5.3 power flow equations for synchronous motor, examples 5.4 steady state synchronous motor operation - effect of load change on synchronous motor 5.5 effect of field current change on synchronous motor, power flow diagram, examples 5.6 Different torques of synchronous motor, stability and maximum load angle 5.7 construction of V curves and inverted V curves, O curves 5.8 Synchronous condenser, synchronous phase modifier 5.9 Hunting 5.10 speed control of synchronous motor, merits, demerits and application 5.11 Comparison of three phase synchronous and induction motors

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Fundamentals of DC Machines	12	2	4	2	4	0	0	12
II	Operation and	12	2	2	4	4	2	0	14

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
	performance Characteristics of DC Machine								
III	Construction and Operation of Synchronous Generator	12	0	2	4	4	2	0	12
IV	Voltage Regulation, Power Equations and Synchronization of Alternators	14	0	4	4	4	6	0	18
V	Synchronous Motor	10	2	2	4	4	2	0	14
Total		**60**	**6**	**14**	**18**	**20**	**12**	**0**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **OC/SC test on alternator to draw OCC and SCC**.
2) **Calculation of synchronous impedance of synchronous machine.**
3) **Create setup of regenerative braking of DC motor.**
4) **Create a setup for load condition of synchronous motor for various power factor**.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

d) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
e) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
f) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
g) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different methods of voltage regulation.
b) Prepare 15 min. power point presentation on synchronizing of alternator.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Fundamentals of DC Machines	1	PrO1. To Study operating principle, working and cut model of DC machine. <u>1a. Justify</u> the concept of rotating machines and fundamental component roles.	Lecture with media	PPTs, black board
	2	PrO 13. To study DC and AC machine windings	Lecture With media	PPTs, black board
Unit II- Operation and performance Characteristics of DC Machine	**3**	*PrO 2. To perform the magnetization (open circuit) characteristics and load test on a DC shunt generator to obtain the internal and external load characteristics.* <u>2a. Evaluate</u> the performance of DC generator <u>2c. Justify</u> the output characteristics of DC machines <u>2d. Setup</u> the circuit to determine output characteristics of DC machines.	Lecture With media	PPTs, black board
	4	*PrO3. To Perform the load test on a D.C. series generator to obtain the internal and external load*	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		characteristics. 2b. Evaluate the performance of DC motor 2c. Justify the output characteristics of DC machines 2d. Setup the circuit to determine output characteristics of DC machines.		
	5	*PrO 4. To perform the speed control of DC series and DC shunt motor.* 2c. Justify the output characteristics of DC machines 2d. Setup the circuit to determine output characteristics of DC machines.	Lecture With media	PPTs, black board
Unit– IV - Voltage Regulation, Power Equations and Synchronization of Alternators	6	*PrO 5.* To find out the voltage regulation of three phase alternator by direct load test. 4a. Choose the relevant operating parameters for load operation of alternator.	Lecture With media	PPTs, black board
	7	*PrO 6.* To find out voltage regulation of three phase alternator by Synchronous Impedance Method (EMF Method) 4c. Evaluate the performance of alternator for the specified condition.	Lecture With media	PPTs, black board
	8	*PrO 7.* To find out voltage regulation of three phase alternator by Ampere Turn Method (MMF Method). 4c. Evaluate the performance of alternator for the specified condition.	Lecture With media	PPTs, black board
	9	PrO 8. To find out voltage regulation of three phase alternator by ZPFC Method 4c. Evaluate the performance of alternator for the specified condition.	Lecture With media	PPTs, black board
	11	PrO 9. To Perform slip test on salient pole alternator for determining direct axis synchronous reactance- Xd and quadrature axis synchronous reactance – Xq 4b. Select the relevant control method for the specified application.	Lecture With media	PPTs, black board
	12	PrO 10. To perform parallel operation of two alternators and study method of synchronization of two alternators 4b. Select the relevant control method for the specified application.	Lecture With media	PPTs, black board
Unit V- Synchronous Motor	13	PrO 11. To Obtain V Curve & Inverted V Curve of a Synchronous Motor. 5a. Select the method for machine control.	Lecture With media	PPTs, black board
	14	PrO 12. To obtain constant excitation circle of synchronous motor. 5b. Interpret the machine data with respect to the given power system.	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
	15	5c. Determine machine charging control strategy. Select (self)	Lecture With media	PPTs, black board

XVIII SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Theory and performance of electrical machines	J.B. Gupta	S.K. Kataria & Sons, ISBN: 978-93-5014-277-6
2	A textbook of electrical technology VOL II	B.L. Theraja	S. Chand, ISBN: 9788121924375
3	Electric Machinery Fundamentals	Stephen. J. Chapman	McGraw Hill, ISBN: 10. 978007352954

XIX SOFTWARE/LEARNING WEBSITES

- http://nptel.ac.in/courses/ 108106072 /

  (video lectures Prof:D. Kastha, IIT Kharagpur)

XX PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Used of mathematical method	PO 2 Analysis mechanical system	PO 3 Used engineering soft.	PO 4 Solve problem in machine	PO 5 **Modern tool usage**	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work:	PO 10 Communication	PO11 Project management	PO12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem
EE342	**Semester V**														
CO1	Able to recognize the electric and magnetic circuits of AC and DC machines.	*3*				*1*									

Course Code	Competency and COs	PO 1 Used of mathematical method	PO 2 Analysis mechanical system	PO 3 Used engineering soft.	PO 4 Solve problem in machine	**PO 5 Modern tool usage**	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work:	PO 10 Communication	PO11 Project management	PO12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem
CO2	Acquire the knowledge regarding the fundamentals, working and construction of Synchronous and DC machines.	*2*	*2*	*1*	*1*	*1*			*1*						
CO3	Acquire the practical exposure of rotating electrical machines.	*3*	*2*	*3*	*2*	*1*						*2*	*2*	*1*	
CO4	Aware with the performance characteristics and applications of Synchronous and DC machines.		*3*	*2*	*2*					*1*					
CO5	Create and test the synchronization of alternators to grid.	*2*	*3*	*2*	*3*	*2*							*1*		
CO6	Understand the role and importance of DC and AC machine in power generation plants.														*1*

XXI NAMES OF RESOURCE PERSONS

a) Prof. Pratik Mochi
b) …………………………………………………….

Course Code: **EE353**
Course Title: **POWER ELECTRONICS AND DRIVES-I**

UG Engineering Programme	**Semester**
Electrical Engineering	Fifth

I RATIONALE

1) Electrical engineers have to deal with various types of *semiconductor devices*, power electronics circuits and converters are used for different dc drives. This curriculum course structure is developed such that the student is able to evaluate the performance of the various *semiconductor devices,* types of power electronic converters and drives for different industry drive applications.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Evaluate the performance of various *SEMICONDUCTOR DEVICES*, TYPES OF Controlled rectifier, Chopper and DC DRIVES**.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1. *Evaluate the performance of the* various types of *SEMICONDUCTOR DEVICES and its use for different converters.*
2. *Evaluate the performance of the various types the Triggering, Commutation and Driver Circuit*
3. *Evaluate the performance of the* various types of Chopper for different application
4. *Evaluate the performance of the* various types of Controlled Rectifier for different application
5. *Evaluate the performance of the various types of DC Drives for different application*

1 TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			**Total Credits (L+T+P)**	**Examination Scheme**				
				Theory Marks		**Practical Marks**		**Total Marks**
L	**T**	**P**	**C**	**ESE**	PA	**ESE**	**PA**	
3	0	2	4	70	30*	25	25	150

Legends: L*-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

*(*): Under the **Theory PA**, out of 40 marks, **20 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

2 SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1,2	(A)To study the DC Gate control characteristics of a SCR	02
	1,2	(B)To study and plot the Anode Current Characteristics of a SCR.	02
2	3	A)To measure the holding current of a SCR.	02
	1,2	B)To Study the phase firing characteristics of a SCR.	02
3	3	To Study the V-I characteristics of a DIAC.	02
4	1,2	To Study application of DIAC as Sawtooth Wave Generator and Pulse Train Generator.	02
5	1,2	To Study the operation of UJT as Relaxation Oscillator.	02
6	1,2	To Study the single-phase Fully Controlled Bridge Rectifier.	02
7	1,2	A)To study the gate characteristics of a TRIAC.	02
	3,4	B)To study the Terminal characteristics of a TRIAC.	02
	1,2	C)To measure the Holding current of a Triac.	02
8	3	A)To study the application of a Triac as a Static Switch with DC control.	02
	1,2	B)To study the application of a Triac in control of AC with AC signal.	02
9	3	To Study the step down voltage commutated thyristorised chopper.	02
10	3	To Study and simulate the DC Drive in MATLAB	02
		Total	**30**

3 MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB Simulation software (version 2021a)	6
2	SCR characteristic kit with DC and AC gate control	1 (A), (B) and 2 (A), (B)
3	DIAC VI characteristic kit	3
4	DIAC as Sawtooth Wave Generator and Pulse Train Generator KIT	4
5	UJT as Relaxation Oscillator Circuit kit	5

S. No.	Equipment Name with Broad Specifications	PrO No.
6	single-phase Fully Controlled Bridge Rectifier kit with variac	6
7	TRIAC VI characteristic kit	7 (A), (B) , (C) and 8 (A), (B)
8	Step down voltage commutated thyristorised chopper KIT	9
9	DSO with isolation and its probes	4,5,6
10	Multimeter	1,2,3,7,8

4 AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

6) Follow safe practices.
7) Practice energy conservation.
8) Function as a team member.
9) Function as a team leader.
10) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

e) 'Responding Level' in 1st year
f) 'Valuing Level' in 2nd year
g) 'Organising Level' in 3rd year
h) 'Characterising Level' in 4th year.

5 UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Introduction	**CO 1. Evaluate the performance of the various types of SEMICONDUCTOR DEVICES and its use for different converters.** 1b. Justify the need of SEMICONDUCTOR DEVICES 1c. Justify the need of different converters	1.1 Introduction, 1.2 Scope and applications, Classification of power electronic converters

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain* *(to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit– II Power Semiconductor Devices	**CO 1. Evaluate the performance of the various types of SEMICONDUCTOR DEVICES and its use for different converters.** *PrO 1. (A)To study the DC Gate control characteristics of a SCR* *(B)To study and plot the Anode Current Characteristics of a SCR.* *PrO 2. A)To measure the holding current of a SCR.* *B)To Study the phase firing characteristics of a SCR.* *PrO 3. To Study the V-I characteristics of a DIAC.* *PrO 7. A) To study the gate characteristics of a TRIAC.* *B) To study the Terminal characteristics of a TRIAC.* *C)To measure the Holding current of a Triac.* *2a.* Evaluate the VI characteristics of SCR 2b. Evaluate the VI characteristics of DIAC 2c. Evaluate the VI characteristics of TRIAC	2.1 Diode: Classification and its application 2.2 SCR: VI characteristics, Switching characteristics, Gate characteristics, Ratings, Effect of dv/dt and di/dt, Snubber circuit and its design 2.3 Thyristor Family: Triac, DIAC, LASCR, SUS, SCS 2.4 Gate commutated devices: GTO, MOSFET and IGBT
Unit– III Triggering, Commutation and Driver Circuit	**CO 2. Evaluate the performance of the various types the Triggering, Commutation and Driver Circuit** PrO 4. To Study application of DIAC as Sawtooth Wave Generator and Pulse Train Generator. PrO 5. To Study the operation of UJT as Relaxation Oscillator. 2a. Evaluate the application of DIAC as Sawtooth Wave Generator and Pulse Train Generator. 2b. Evaluate the operation of UJT as Relaxation Oscillator.	3.1 Triggering circuits: Resistance triggering circuit, Resistance capacitance trigger circuit, DIAC trigger circuit, UJT based trigger circuit; Micro-controller based Trigger circuit 3.2 Commutation circuits: Line commutation, Load commutation, Forced commutation, External pulse commutation 3.3 Driver Circuit: Optocoupler and pulse transformer based SCR driver circuit, Gate drive circuit for power MOSFET, Driver circuit for IGBT and BJT
Unit– IV DC-DC	**CO 3 Evaluate the performance of the various types of Chopper for different**	4.1 Introduction, Principle of chopper operation,

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Converters (Choppers)	**application** *PrO 9. To Study the step down voltage commutated thyristorised chopper.* *3a. Evaluate the performance of step down voltage commutated thyristorised chopper*	4.2 Control strategies, 4.3 Step down (buck) converter, 4.4 Step up(boost) converter 4.5 Step up down converter, 4.6 Cuk DCDC converter, Chopper configuration
Unit-V AC-DC Converters (Controlled Rectifier)	**CO 4. Evaluate the performance of the various types of Controlled Rectifier for different application** *PrO 6.* To Study the single-phase Fully Controlled Bridge Rectifier *4a. Evaluate the performance of single-phase Fully Controlled Bridge Rectifier*	5.1 Introduction, Principle of phase control, 5.2 Single phase half wave converter with different loads, 5.3 Single phase full wave converter with different loads, 5.4 Single phase semi converter, 5.5 Three phase half wave converter with different loads, 5.6 Three phase full converter, 5.7 Three phase semi converter with different loads, 5.8 Dual converter, 5.9 Multi-pulse Rectifier
Unit – VI DC Motor Drives	**CO 5. Evaluate the performance of the various types of DC Drives for different application** *PrO 10.* To Study and simulate the DC Drive in MATLAB 5d. Understand the operation of DC Drive in different mode condition. 5e. Justify the need of *DC Drives for different application*	6.1 Concept of Electric Drive, 6.2 Basic DC Machine Equations and its characteristics, 6.3 Control of DC separately exited motor from single phase and three phase controlled rectifier, 6.4 Rectifier control of DC series motor, 6.5 Chopper control of separately exited DC motor, 6.6 Chopper control of DC series motor, 6.7 Multi quadrant operation of, 6.8 DC separately excited motor

1 TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction	1.5	0	1	1	2	1	0	05
II	Power Semiconductor	13	0	1	1	6	4	0	12

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
	Devices								
III	Triggering, Commutation and Driver Circuit	7.5	0	1	4	4	1	0	10
IV	DC-DC Converters (Choppers)	7.5	0	2	2	4	6	0	14
V	AC-DC Converters (Controlled Rectifier)	7.5	1	2	2	4	6	0	15
VI	DC Motor Drives	08	0	2	2	6	4	0	14
Total		**45**	**1**	**9**	**12**	**26**	**22**	**0**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

2 MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

5) **Rectifier:** Simulate the rectifier for particular one application.
6) **Chopper:** Design and Develop the hardware of Chopper.
7) **Triggering circuit:** Design and simulate the SCT triggering circuit
8) **DC drive**: Design the simulation of DC drives in MATLAB.

3 SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

h) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.

i) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
j) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
k) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

4 SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different Semiconductor switches used in converters.
b) Prepare 15 min. power point presentation on how to choose the particular converter for the particular application.

5 COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Introduction	1	*1a. Justify the need of SEMICONDUCTOR DEVICES* *1b. Justify the need of different converters*	Lecture with media	PPTs, black board, MATLAB Software
Unit– II Power Semiconductor Devices	**6**	*PrO 1. (A)To study the DC Gate control characteristics of a SCR* *(B)To study and plot the Anode Current Characteristics of a SCR.* *PrO 2. A)To measure the holding current of a SCR.* *B)To Study the phase firing characteristics of a SCR.* *PrO 3. To Study the V-I characteristics of a DIAC.* *PrO 7. A) To study the gate characteristics of a TRIAC.* *B) To study the Terminal characteristics of a TRIAC.* *C)To measure the Holding current of a Triac.* *2a.* Evaluate the VI characteristics of SCR 2b. Evaluate the VI characteristics of DIAC 2c. Evaluate the VI characteristics of TRIAC	Lecture with media, and Demonstration	PPTs, black board, MATLAB Software, Practical demonstration
Unit– III Triggering,	**8**	*PrO 4. To Study application of DIAC as Sawtooth Wave Generator and Pulse Train Generator.* *PrO 5. To Study the operation of UJT as*	Lecture with media,	PPTs, black

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Commutation and Driver Circuit		*Relaxation Oscillator.* 2a. Evaluate the application of DIAC as Sawtooth Wave Generator and Pulse Train Generator. 2b. Evaluate the operation of UJT as Relaxation Oscillator.	Field visit	board, MATLAB Software, Practical demonstration
Unit– IV DC-DC Converters (Choppers)	**10**	*PrO 9. To Study the step down voltage commutated thyristorised chopper.* 4a. *3a. Evaluate the performance of step down voltage commutated thyristorised chopper*	Lecture with media, Tutorial	PPTs, black board, Practical demonstration
Unit-V AC-DC Converters (Controlled Rectifier)	**15**	*PrO 6.* To Study the single-phase Fully Controlled Bridge Rectifier 4a. Evaluate the performance of single-phase Fully Controlled Bridge Rectifier	Lecture with media, Tutorial	PPTs, black board, MATLAB Software, Practical demonstration
Unit – VI DC Motor Drives		*PrO 10.* To Study and simulate the DC Drive in MATLAB 5a. Understand the operation of DC Drive in different mode condition. 5b. Justify the need of *DC Drives for different application*	Lecture with media, Tutorial	PPTs, black board, MATLAB Software

XXII SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Power Electronics	Dr.P.S.Bimbhra	Khanna publishers
2	Power electronics	M D Singh and K B K hanchandani,	Tata Mcgraw-Hill Publication
3	Power electronics handbook	by M.H. Rashid	Academic press
4	High power converters and AC drives	IEEE press	IEEE press
5	Power Electronics Converters, Applications and design	Mohan, Undeland, Robbins	Wiley publication

XXIII SOFTWARE/LEARNING WEBSITES

1. http://www.nptel.ac.in/courses/108105066/ (Notes from IIT Kharagpur)
2. http://www.nptel.ac.in/courses/108101038/1

(Videos and Notes by Prof. B.G. Fernandes, IIT Bombay)

3. http://www.nptel.ac.in/courses/108108035/1
4. http://www.nptel.ac.in/courses/108108036/

(Videos and Notes by Prof. L. Umanand, IIT Bangalore)

5. http://www.nptel.ac.in/courses/108102046/
6. http://www.nptel.ac.in/courses/108108077/

(Videos and Notes by Prof. K. Gopakumar, IISC, Bangalore)

XXIV PO-COMPETENCY-CO MAPPING

Course Code	**Competency and COs**	**PO 1** Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE353	**Semester VII**														
353	**Competency** • Evaluate the performance of various SEMICONDUCTOR DEVICES, TYPES OF Controlled rectifier, Chopper and DC DRIVES	***3***	***2***	***2***	***2***	***1***	**-**	***1***	**-**	**-**	**-**	**-**	***1***	***3***	**-**
353.1	Evaluate the performance of the various types of SEMICONDUCTOR DEVICES and its use for different converters.	*3*	*2*	*1*	-	*2*	-	-	-	-	*1*	-	-	-	-
353.2	Evaluate the performance of the various types the Triggering, Commutation and Driver Circuit	3	3	2	1	2	-	-	-	-	-	-	-	2	-
353.3	Evaluate the performance of the various types of Chopper for different application	3	3	3	1	2	-	-	-	-	-	-	-	2	-
353.4	Evaluate the performance of the various types of Controlled Rectifier for different application	3	1	3	1	2	-	-	-	-	-	-	-	2	-
353.5	Evaluate the performance of the various types of DC Drives for different application	3	1	-	-	-	-	-	-	-	-	-	-	-	-

XXV NAMES OF RESOURCE PERSONS

a) Dr. Dharmesh Dabhi

Course Code: **EE344**

Course Title: **Minor Project-I**

UG Engineering Programme	**Semester**
Electrical Engineering	Fifth

I RATIONALE

This course enables the students to apply some of the knowledge and/or skills developed during the programme to new problem for which there are number of engineering solutions. This course include planning of the tasks which are to be completed within the time allocated, and in turn, helps to develop ability to plan, , use, monitor and control resources optimally and economically. By studying this course abilities like creativity, imitativeness and performance qualities are also developed in students. Leadership development and supervision skills are also integrated objectives of learning this course.

II COMPETENCY

The course should be taught and implemented with the aim to develop the required course outcomes (COs) so that students will acquire following competency needed by the industry:

- **Find the solution of a real-life problem**.

III COURSE OUTCOMES (COs)

Depending upon the nature of the projects undertaken, the following could be some of the minor course outcomes that could be attained, although, in case of some projects few of the following course outcomes may not be applicable.

Apply knowledge and skills learned from project to real-world problems.

Solve engineering problems by the team work and work with other discipline.

Use experience related to professional and ethical issues in the work environment.

Apply impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context in order to provide employability and making entrepreneur.

Implement the skills related engineering on contemporary issues related with engineering in general with advanced tools and technologies.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			**Total Credits (L+T+P)**	**Examination Scheme**				
				Theory Marks		**Practical Marks**		**Total Marks**
L	**T**	**P**	**C**	**ESE**	PA	**ESE**	**PA**	100
0	0	04	02	00	00	50	00	

***Legends: L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

V COURSE DETAILS

i. Minor Project showcases that, a student during the fifth semester will able to apply the engineering knowledge to a practical problem.

ii. Students can carry out their project work in-house at institute/ industry/ research organization/govt. organization.

iii. Project should be related to the Product/prototype design and development or Experimentation, Simulations and evaluation of any system etc.

iv. A student is required to carry out the project work related to Electrical Engineering or some interdisciplinary work under the guidance of a faculty member and the supervisor of the concerned industry/institute/organization.

v. The minor project work provides student an opportunity to develop something on their own or group of maximum 3 students. Final decision related to the number of students in one group is based on internal supervisor recommendation.

vi. Project will be evaluated at least twice during the semester by internal faculty as a part of continuous evaluation and final evaluation will be carried out by external examiner at the end of semester.

vii. Student should submit the project report at the end of the semester duly signed by project supervisor and head of department. Report comprising literature review, objective, methodology, scope of the project work undertaken, outcome of project and conclusions.

viii. The students are required to identify their problem and they are required to follow all the rules and instruction issued by department/industry.

VI PROJECT REPORT

At the end of the semester student should prepare project report. Guideline of the Project report is as follow

a) Certificate
b) Acknowledgements
c) Abstract (in one paragraph not more than 150 words)
d) Content Page
e) Chapter-1 Introduction and background of the Problem/Task
f) Chapter-2 Literature Survey for Problem/Task Identification and Definition,
g) Chapter-3 Proposed Detailed Methodology of solving the problem/completing the task
h) Chapter-4 Methodology actually followed to solve the problem/complete the task
i) Chapter-5 The solution, its development and its implementation/execution (with details including drawings, software, used for or developed for the solution etc. in detail). In case of survey/experiment type of project this chapter will include data analysis, information presentation and analysis.
j) Chapter-6 Final conclusions and recommendations if any for future projects.

Note:

a) The report should contain as many diagrams and figures, charts etc. as relevant for the project.
b) Originality of the report (written in own words) would be given more importance rather than quality of printing and use of glossy paper or multi-colour printing

VII ASSESSMENT OF PROJECT WORK

Assessment of Project work also has two components, first is Progressive Assessment, while another is End of the Term Assessment.

i. Progressive Assessment Criteria

a) Project guide or the group of internal faculty are supposed to carry out this assessment. It is a continuous process, during which for developing desired qualities in the students, faculty should orally give informal feedback to students about their performance and interpersonal behaviour while guiding them on their project work every week.

S. No.	Criteria	Marks
	First Progressive Assessment at the end of 4th week	
1	**Project Proposal** for work to be completed at the end of semester: Including problem definition and methodology adapted, literature review adopted (in case of working models prepared then quality of workman ship, appropriateness of raw materials/components/ technology being used, functioning of the prototype etc. has to be also assessed)	**10**
	General Behaviour (initiative, resourcefulness, reasoning ability, imagination/creativity, self-reliance)	**Only written feed-back/suggestions**
	Second Progressive Assessment at the end of 12th week	
2	**Execution of Plan during semester** (marks to be also given based on ability to collect relevant information, ability to follow correct procedure, manipulative skills, ability to observe, record & interpret, ingenuity in the use of material and equipment, target achievement) In case of working models, quality of workman ship (including accuracy in dimensions, shape, tolerance limits), appropriateness of raw materials/components/ technology being used, functioning of the prototype, cost effectiveness, marketability, modernity etc. has to be also assessed	**5**
3	**Log book** (for work done during semester- detailed and regular entry would be basis of marks)	**5**
4	**General Behaviour** (persistence, interest, confidence, problem solving ability, decision making ability, initiative to act, team spirit, sharing of material etc., participation in discussions, completion of individual responsibilities, leadership)	**Only written feed-back./suggestions**
	Third Progressive Assessment at the end of 14th week	
5	**Portfolio for Self-learning and reflection** (for work done in semester- marks based on amount of reflection and completion of the portfolio)	**10**
6	**Project Report including documentation -** for work done in semester (marks based on: clarity in presentation and organization; styles and	**5**

S. No.	Criteria	Marks
	language; quality of diagrams, drawings and graphs; accuracy of conclusion drawn; citing of cross references; suggestion for further research/project work)	
7	**Presentation** (presentation skills including communication skills to be assessed by observing presentations and asking questions during presentation for work done during semester)	**5**
8	**Defence** (ability to defend the methods/materials used and technical knowledge, and involvement of individual to be assessed by asking questions during presentation and viva/voce for work done during semester)	**10**
Total		**50**

Note: Oral feedback on general behaviour may also be given whenever relevant/ required during day to day guidance and supervision.

ii. END SEMESTER EXAMINATION

b) Criteria of Marks for ESE for Capstone Project Execution.

S. No.	Description	Marks
1	**Problem Identification/Project Title** (Innovation /Utility of the Project for industry/ User/Academia) marks to be also given based on (i) Accuracy or specificity of the scope and (ii) Appropriateness of the work with reference to desired course outcomes.	**5**
2	**Industrial Survey and Literature Review** (marks to be given based on extent/volume and quality of the survey of Industry / Society / Institutes/Literature/Internet for Problem Identification and possible solutions)	**5**
3	**Project Proposal** (In case of working models, detailed design and planning of fabrication/assembly of the prototype has to be also assessed). Marks to be given also based on appropriateness, flexibility, detail and clarity in methods/planning.	**10**
4	**Final Report including documentation**. (marks based on: clarity in presentation and organization; styles and language; quality of diagrams, drawings and graphs; accuracy of conclusion drawn; citing of cross references; suggestion for further research/project work)	**10**
5	**Presentation** (presentation skills including communication skills to be assessed by observing presentations and asking questions during presentation)	**10**
6	**Defence** (ability to defend the methods/materials used and technical knowledge, and involvement of individual to be assessed by asking questions during presentation and viva/voce)	**10**
Total		**50**

VIII Guidelines for Portfolio Preparation and assessment

The main purpose of portfolio preparation is learning based on self-assessment and portfolio is not to be used for assessment in traditional sense.

a) Each student has to prepare his/her portfolio separately. However, he/she can discuss with the group members about certain issues on which he/she wants to write in the portfolio.

b) Whatever is written inside the portfolio is never to be used for assessment, because if teachers start giving marks based on whatever is written in the portfolio, then students would hesitate in true self-assessment and would not openly describe their own mistakes or shortcomings.

c) Some marks are allocated for portfolio, these marks are to be given based on how sincerely portfolio has been prepared and not based on what strengths and weaknesses of the students are mentioned in the portfolio.

d) Portfolio has to be returned back to the students after assessing it (assessment is only to see that whether portfolio is completed properly or not) by teachers. Because student is the real owner of the portfolio.

e) Students mainly learn during portfolio preparation, but they can further learn if they read it after a gap. And hence they are supposed to keep the portfolios with them even after completion of the diploma because it is record of their own experiences (it is like diary some people write about their personal experiences), because they can read it again after some time and can revise their learning (about their own qualities)

IX PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	**PO 1** Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE344	**Semester V**														
344	**Competency** *Find the solution of a real-life problem.*	***3***	***2***	***2***	***2***	***2***	***2***	***1***	***1***	***2***	***2***	***2***	***1***	***2***	***2***
344.1	Apply knowledge and skills learned from project to real-world problems.	3	-	1	-	-	1	-	1	-	1	1	1	-	1
344.2	Solve engineering problems by the team work and work with other discipline.	3	3	2	2	1	2	-	-	3	2	1	2	2	-
344.3	Use experience related to professional and ethical issues in the work environment.	3	2	1	-	-	-	-	3	1	-	2	-	-	1
344.	Apply impact of	3	-	-	3	3	3	-	-	2	1	3	2	3	3

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
4	engineering solutions, developed in a project, in a global, economic, environmental, and societal context in order to provide employability and making entrepreneur.														
344.5	Implement the skills related engineering on contemporary issues related with engineering in general with advanced tools and technologies.	3	1	3	2	3	3	2	-	-	2	2	3	3	2

X NAMES OF RESOURCE PERSONS

a) Prof. Mahammadsoaib Saiyad

Course Code: **EE345**
Course Title: **Electrical Product Survey**

UG Engineering Programme	**Semester**
Electrical Engineering	Fifth

I RATIONALE

Objective of this course is to understand the electrical and electronics products and its market statistics. Students get aware about the product, Manufacturing method, consumer requirement and perception, product satisfaction, consumer experience (good and bad), and intent to purchase behavior.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- Specification, market survey, identification of different products like wire, cable, and switchgear can be done.
- To understand various components and its behavior.
- To make them industry ready.
- To get familiar with modern tools and technologies

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1. *To apply knowledge and skills learned from project to real-world engineering problems.*
2. *To function in a team work with teammates from other disciplines.*
3. *To use experience related to professional and ethical issues in the work environment.*
4. *To explain the impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context.*
5. *To demonstrates knowledge of contemporary issues related with engineering in general with the use new tools and technologies.*

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			**Total Credits (L+T+P)**	**Examination Scheme**				
				Theory Marks		**Practical Marks**		**Total Marks**
L	**T**	**P**	**C**	**ESE**	PA	**ESE**	**PA**	
0	0	2	2	00	00	50	50	100

***Legends: L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

***(*):** Under the **Theory PA**, out of 30 marks, **15 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The practical outcomes (PrOs) , which are created for particular electrical product, predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Electrical Product selected by student group	-

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Note: The course evaluation is based on the presentation of students. No theory exam is involved. However, following instructions are given to students.

- Electrical product survey has 2 credit subject and 2 hours per week.
- Students can find his/her electrical product.
- The student may take the guidance from department faculty members.
- Product must be from Electrical bag round.
- The student shall submit product presentations to the Coordinator for the evaluation of the product survey.
- After the completion of product survey, a soft copy of "presentation" must be submitted to the Coordinator by the end of the semester.
- The students must make a group of 3 or 4 and submit their proposal to departmental coordinator.
- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision. If the revised presentation is still unsatisfactory the student shall be requested to repeat the project

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given

to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.

d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different methods of voltage regulation.
b) Prepare 15 min. power point presentation on synchronizing of alternator.

XIII COURSE PLAN

There is no theory part for which course plan is needed.

XXVI SUGGESTED LEARNING RESOURCES

The user manual and product booklet of various industries.

XXVII SOFTWARE/LEARNING WEBSITES

- NPTEL online courses on particular electrical product might be helpful.

XXVIII PO-COMPETENCY-CO MAPPING

Course Code	**Competency and COs**	**PO 1** Used of mathematical method	**PO 2** Analysis mechanical system	**PO 3** Used engineering soft.	**PO 4** Solve problem in machine	**PO 5 Modern tool usage**	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work:	**PO 10** Communication	**PO11** Project management	**PO12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem
EE345	**Semester V**														
CO1	To apply knowledge and skills learned from project to real-world engineering problems.	*3*	*2*											*2*	
CO2	To function in a team work with teammates from other disciplines	1		1	1	2				3					1
CO3	To use experience related to professional and ethical issues in the work environment	1					1		2						1
CO4	To explain the impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context	1						2		1		2		1	1
CO5	To demonstrates knowledge of contemporary issues related with engineering in general with the use new tools and technologies.	3	1		1								1		2

XXIX NAMES OF RESOURCE PERSONS

a) Pratik Mochi

b) ………………………………………………….

Course Code: **EE350**
Course Title: **Summer Internship-I**

UG Engineering Programme	**Semester**
Electrical Engineering	Fifth

I RATIONALE

This course offers the opportunity for the young students to acquire on job the skills, knowledge, attitudes, and perceptions along with the experience needed to constitute a professional identity. This course includes planning of the tasks which are to be completed within the time allocated, and in turn, helps to develop ability to plan, use, monitor and control resources optimally and economically. By studying this course ability like creativity, imitativeness and performance qualities are also developed in students. Also give an insight into the working of the real organizations. Leadership development and supervision skills are also integrated objectives of learning this course. To appreciate the linkages among different functions and departments. This course helps the students in exploring career opportunities in their areas of interest. By attending this course student get deeper understanding in specific functional areas.

II COMPETENCY

The course should be taught and implemented with the aim to develop the required course outcomes (COs) so that students will acquire following competency needed by the industry:

- **Develop professional habit, attitude and ethics related to the industrial working environment**.

III COURSE OUTCOMES (COs)

On successful completion of this course, the student will be able to:

CO1: Integrate class room theory with real-world engineering problems of industry/organization.

CO2: Develop work habits, attitudes and ethics for professional environment.

CO3: Develop communication, interpersonal and other critical skills.

CO4: Develop technical report writing and presentation skill.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			**Total Credits (L+T+P)**	**Examination Scheme (Anna 2021)**			
				Theory Marks		**Practical Marks**	**Total Marks**
L	**T**	**P**	**C**	**ESE**	PA	**ESE**	
0	0	0	3	00	00	150	150

***Legends: L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

V Course Details: Instructional Method and Pedagogy

Format of Summer Internship Report:

At the start of the semester student should prepare project report. Guideline of the Project report is as follow

The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- Chapter-1: Overview of company: Detail of the company including Location and spread of the company, Number of employees, engineers, technicians, administrators in the company, Divisions of the company, Administrative tree (if available), Customer profile and market share, details of process or services, various department and role of each department, detail of your department.
- Chapter-2: Overview of Internship Experience: In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out.
- Chapter 3: Observations: In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization. Present your observations.
- Chapter 4: Learning Outcomes: In the last section, summarize the summer internship activities, learning outcomes, contributions and intellectual benefits. If

this is your second summer internship, compare the first and second summer internships and your preferences.

- References: List any source you have used in the document including books, articles and web sites in a consistent format.
- Appendices: If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it and waste tons of paper.

Note:

a) The report should contain as many diagrams and figures, charts etc. as relevant for the internship.
b) Originality of the report (written in own words) would be given more importance rather than quality of printing and use of glossy paper or multi-colour printing

VI ASSESSMENT OF Summer Internship

Assessment of summer internship is done at the beginning of term by the internal examiner and external examiner after successful completion of summer internship

A) Assessment of summer internship by Internal Examiner.

S. No.	Description	Marks
1	**Final Report including documentation**. (marks based on: clarity in presentation and organization; styles and language; quality of diagrams, drawings and graphs; accuracy of conclusion drawn; citing of cross references; suggestion for further research/project work)	**25**
2	**Presentation** (presentation skills including communication skills to be assessed by observing presentations and asking questions during presentation)	**25**
3	**Viva** (ability to give answers the methods/materials used and technical knowledge, and involvement of individual to be assessed by asking questions during presentation and viva/voce)	**25**
	Total	**75**

B) Assessment of summer internship by External Examiner.

S. No.	Description	Marks
1	**Final Report including documentation**. (marks based on: clarity in presentation and organization; styles and language; quality of diagrams,	**25**

S. No.	Description	Marks
	drawings and graphs; accuracy of conclusion drawn; citing of cross references; suggestion for further research/project work)	
2	**Presentation** (presentation skills including communication skills to be assessed by observing presentations and asking questions during presentation)	**25**
3	**Viva** (ability to give answers the methods/materials used and technical knowledge, and involvement of individual to be assessed by asking questions during presentation and viva/voce)	**25**
Total		**75**

VII ASSESSMENT OF Summer Internship

Course Code	**Competency and COs**	**PO 1** Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE350	**Semester V**														
350	**Competency** The course should be taught and implemented with the aim to develop the required course outcomes (COs) so that students will acquire following competency needed by the industry:	***3***	***2***	***2***	***2***	***2***	***2***	***1***	***1***	***2***	***2***	***2***	***1***	***2***	***2***
350.1	Solve engineering problems by working in a team with across other disciplines.	3	-	1	-	-	1	-	1	-	1	1	1	-	1
350.2	Use experience related to professional and ethical issues in the work environment	3	3	2	2	1	2	-	-	3	2	1	2	2	-
350.3	Use experience related to professional and ethical issues in the work environment.	3	2	1	-	-	-	-	3	1	-	2	-	-	1
350.4	Examine the impact of engineering	3	-	-	3	3	3	-	-	2	1	3	2	3	3

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	solutions, developed in a project, in a global, economic, environmental, and societal context.														
350.5	Evaluate contemporary issues related with engineering in general by adopting new tools and technologies	3	1	3	2	3	3	2	-	-	2	2	3	3	2

XI NAMES OF RESOURCE PERSONS

A) Mahammadsoaib Saiyad

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS131.02A: COMMUNICATION AND SOFT SKILLS

I. Credits and Schemes

Sem	Course Code	Credits	Teaching Scheme	Evaluation Scheme				
			Contact Hours/Week	Theory		Practical		Total
				Internal	External	Internal	External	
V	HS131.02 A	02	02	--	--	30	70	100

II. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	An Introduction to Communication	06
2.	Cross-cultural Communication and Globalization	03
3.	Communication for Career Building	10
4.	Group Dynamics and Soft Skills	05
5.	Effective Presentation Strategies	04
6.	Contemporary Issues in Communication and Soft Skills	02
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

III. Detailed Syllabus:

1.	An Introduction to Communication	06 Hours	20%
	Basics of Communication: Origin, Concept, Process, Levels, Principles and Barriers; Applications of Communication; Rhetoric in Professional Communication; Importance of Ethos, Logos, and Pathos in Communication		
2.	Cross-cultural Communication and Globalization	03 Hours	10%
	Basic Concepts: Culture, Globalization and Cross-cultural Communication; Social and People Skills; Communicating with People of Different Cultures; Conflicts in Cross-cultural Communication and Tactics / techniques to resolve them; Persuasive Communication		
3.	Communication for Career Building	10 Hours	33%

	Cover Letters and Resume; E-mail and Report; Types of Resume; Concept and Rationale of Group Discussion Skills and Aspects assessed in Group Discussion; Concept and Rationale of Personal Interview; Types of Personal Interview; Writing Statement of Purpose		
4.	Group Dynamics and Soft Skills	05 Hours	17%
	An Introduction to Group Dynamics and Soft Skills; Groups and their Structures; Roles and Functions of Members in Groups; Conflict Management; Aptitude and Attitude; Various Intelligences; Developing an Open Mindset		
5.	Effective Presentation Strategies	04 Hours	14%
	Designing Appealing Presentation; Audience Analysis and Supporting Material; Presentation Mechanics and Presentation Process; Managing Yourself during Q and A Session; Fundamentals of Persuasion		
6.	Contemporary Issues in Communication and Soft Skills	02 Hours	06%
	Trends and Practices in Communication, Case Studies		

IV. Pedagogy

Teaching will be facilitated by reading material, discussions, task-based learning, projects, assignments and interpersonal activities like group work, independent and collaborative study projects and presentations, etc.

V. Evaluation

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Assignment / Tutorial	01	25	
2	Project	01	25	
3	Quiz	02	25	
4	Case Study and Presentation	01	25	100*
5	Attendance and Class Participation			05
			Total	30

*** Marks will be converted into 25**

External Evaluation

The University Examination will be for 70 marks and will be based on practical (computer based) tests and a viva-voce. There may also be a Computer Based Test at the end.

Details are:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Practical / Viva	01	70	70
			Total	70

VI. Course Outcomes (COs)

At the end of the course, the students will be able to

CO1	Gain thorough understanding and proficiency in various Professional Communication Skills.
CO2	Develop awareness and competence in cross-cultural communication in their personal, academic and professional environments.
CO3	Develop business writing and presentation skills to succeed in career.
CO4	Develop soft skills to stand out and take their career to the next level.
CO5	Develop various intelligences and open Mindset to function in multi-disciplinary and cross-cultural work environment.
CO6	Practice new trends in communication in multiple perspectives at personal, professional, and social level.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	2	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	2	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO6	-	-	-	-	-	-	-	-	-	3	-	3	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

VII Preference

It will be preferred that students take globally recognized tests by Cambridge, British Council or any other equivalent agency. Equivalent marks shall be considered in the University Exam.

VIII Recommended Study Material:

❖ **Text book:**

1. Koneru, A. Professional Communication, Tata McGrow Hill Education Private Limited
2. Disanza, J.R. &Legge, N. Business and Professional Communication, Pearson Education
3. Raman, M & Singh, P. Business Communication, Oxford University Press

❖ **Reference book:**

1. Disanza, J.R. &Legge, N. Business and Professional Communication, Pearson Education
2. Anandamurugan, A. Placement Interviews – Skills for Success, Tata McGrow Hill Education Private Limited

❖ **Web material:**

1. https://www.coursera.org/learn/careerdevelopment
2. https://www.futurelearn.com/courses/writing-applications
3. https://www.futurelearn.com/courses/workplace-englis

Course Code: **EE371**
Course Title: **Advanced Microcontrollers**

UG Engineering Programme	**Semester**
Electrical Engineering	Fifth

I RATIONALE

Controllers are an essential part of modern electronic systems, providing the processing power and intelligence required to control and monitor various functions. They are widely used in a variety of industries, from automotive and aerospace to medical and consumer electronics, to control everything from engines and motors to temperature and lighting. Controllers offer a compact and cost-effective solution for a wide range of applications, with built-in peripherals, such as timers, ADCs, and DACs, that make it easy to interface with sensors and actuators. With the ever-increasing demand for automation and connectivity, the importance of controllers in modern systems is only expected to grow in the coming years.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Ability to design, program, and integrate microcontrollers into electronic systems for the purpose of controlling and monitoring various functions.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) *Develop software for microcontrollers.*
2) *Understand the architecture of advanced microcontrollers*
3) *Design and implement conventional timer-based systems and PCA based systems.*
4) *Design and implement ADC and DAC-based systems*
5) *Analyse and Understand communication protocol.*

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			**Total Credits (L+T+P)**	**Examination Scheme (Anna 2021)**				
				Theory Marks		**Practical Marks**		**Total Marks**
L	**T**	**P**	**C**	**ESE**	PA	**ESE**	**PA**	
4	0	2	5	70	30*	25	25	150

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

***(*):** Under the **Theory PA**, out of 30 marks, 15 **marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 15 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1,2	Understand operation of CrossBar, SFR paging using C-programming and debugging on keil microvision platform.	4
2	1,2	Configure the internal oscillator and PLL for various SYSCLK.	4
3	1,3	Develop C program for timer T0 and T1 in various modes of operations.	2
4	1,3	Develop C program for timer T2, T3 and T4 in various modes of operation.	4
5	1,4	Develop C program for ADC0 in various mode of operation (Cover Single ended & Differential mode with various start of conversion method).	4
6	1,4	Develop C program for ADC2 in various mode of operation (Cover Single ended & Differential mode with various start of conversion method).	2
7	1,4	Develop C program for using window comparator mode of ADC0.	2
8	1,4	Develop C program for on chip analog comparator.	2
9	1,4	Develop C program for DAC and generating signals.	2
10	1,3	Develop C program for various PCA mode of operations.	4
		Total	

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	C8051f120 development board	1,2,3,4,5,6,7,8,9,10
2	Software Tool: Keil uvision v4.0 and above,	1,2,3,4,5,6,7,8,9,10
3	Silicon lab Flash Programming Utilities	1,2,3,4,5,6,7,8,9,10
4	Digital Signal Oscilloscope , Digital Multi Meter	1,2,3,4,5,6,7,8,9,10

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Embedded C Language Programming for Microcontroller	**CO 1.** ***Develop software for microcontrollers*** 1a. Choose appropriate programming language for the given scenario. 1b. Evaluate operation of super loop. 1c. Analyse the round robin concept and implement concept in the given application. 1d. Understand working of IDE platform and use them for developing and simulating code for the given application.	1.1 Comparison of Assembly & High Level Programming 1.2 Concept of Super loop/Infinite loop used in Embedded C Programs 1.3 Concept of Round Robin Execution in Real Time Operating System based Programming Environment. 1.4 Introduction to Keil Microvision IDE & Related debugging Techniques, Basic Working of various Control Loops, Variables: Global & Local, Various types of Variables & it's Memory Scope, Mathematical & Logical

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
		Operators, Array, Structure & Union, Pointers and its use in programming. Various C library and its use in programming.
Unit– II Introduction to c8051f120 Microcontroller	**CO 1. *Develop software for microcontrollers*** ***CO 2. Understand the architecture of advanced microcontrollers*** *PrO.1 Understand operation of CrossBar, SFR paging using C-programming and debugging on keil microvision platform.* *PrO.2 Configure the internal oscillator and PLL for various SYSCLK.* 2a. Justify the need of different microcontrollers and select the appropriate microcontroller for the given embedded system. 2b. Evaluate the importance of WATCHDOG timer and use them in the hardware debugging. 2c. Choose the appropriate speed of operation for the given application. 2d. Understand Memory management. 2e. Justify the need of multifunctional pin and select appropriate function for the given application.	2.1 Comparison of 8051 & CIP-51 core Microcontrollers, Peripherals available in c8051f120 2.2 Microcontroller Watchdog Timer & it's Reset codes 2.3 System Clock configuration with & without Internal PLL 2.4 SFR & SFR Paging 2.5 Concept of Crossbar & Crossbar Decoder, Internal Structure & Configuration of Port in Open drain & Push pull mode with & without Weak pull up disabled.
Unit– III Timers in c8051f120 Microcontroller	**CO 1. *Develop software for microcontrollers*** **CO 3. *Design and implement conventional timer-based systems and PCA based systems*.** *PrO.3 Develop C program for timer T0 and T1 in various modes of operations.* *PrO.4 Develop C program for timer T2, T3 and T4 in various modes of operation.* 3a. Understand and Evaluate operation of timer and use them in appropriate mode in the given application. 3b. Justify need of multiple timer and use them in appropriate mode in the given application.	3.1 Block diagram & Operation of Timer0 & 1 with Associated SFR, Calculation of input clock frequency for various SYSCLKOUT, Various Modes of Timer & Associated Programs With & Without Interrupt enabled. 3.2 Block diagram & Operation of Timer2,3 & 4 with Associated SFR.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
		Various modes of Operations & Associated Programs.
Unit-IV Analog Peripherals of c8051f120 Microcontroller	**CO 1.** ***Develop software for microcontrollers*** **CO 4.** ***Design and implement ADC and DAC-based systems.*** *PrO.5 Develop C program for ADC0 in various mode of operation (Cover Single ended & Differential mode with various start of conversion method).* *PrO.6 Develop C program for ADC2 in various mode of operation (Cover Single ended & Differential mode with various start of conversion method).* *PrO.7 Develop C program for using window comparator mode of ADC0.* *PrO.8 Develop C program for on chip Analog comparator.* *PrO.9 Develop C program for DAC and generating signals.* 4a. Understand and Program on chip ADC for the given application. 4b. Understand and Program on chip DAC for the given application. 4c. Understand and Program on chip Comparator for the given application.	4.1 Block diagram & Working of 12 bit ADC0 & 8 bit ADC2 A/D Converters, Start of Conversion using Various modes, ADC in Window Comparator Mode, Temperature sensing using Onchip Temperature Sensor, Programs Related to ADC Operation 4.2 Digital to Analog Converter in c8051f120 with various mode of conversion, Generation of waveforms using DAC 4.3 On-chip Comparator.
Unit – V PCA and Application	**CO 1.** ***Develop software for microcontrollers*** **CO 3.** ***Design and implement conventional timer-based systems and PCA based systems.*** *PrO.10 Develop C program for various PCA mode of operations.* 5a. Justify need of PCA and use them in the given application.	5.1 Block diagram of PCA module, PCA Timer in Capture Compare, High Speed output, Frequency Output & 8/16 Bit Pulse Width Modulation Mode.: Associated Register & Working, Associated Programs.
Unit – VI Communication Protocol	**CO 1.** ***Develop software for microcontrollers*** **CO 5.** ***Analyse and Understand communication protocol.*** 6a. Justify need of different communication protocol and select appropriate protocol for the given situation.	6.1 SPI, SMBUS and I2C, UART Communication

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit	Unit Title	Teaching	Distribution of Theory Marks

No.		Hours	R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Embedded C Language Programming for Microcontroller	12	1	2	2	3	4	2	14
II	Introduction to c8051f120 Microcontroller	08	1	1	2	3	1	1	9
III	Timers in c8051f120 Microcontroller	12	1	2	3	3	3	2	14
IV	Analog Peripherals of c8051f120 Microcontroller	12	1	1	3	3	3	3	14
V	PCA and Application	10	1	2	2	3	2	2	12
VI	Communication Protocol	06	1	2	2	2	0	0	7
Total			**6**	**10**	**14**	**17**	**13**	**10**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **Inverter:** Use knowledge of power electronics and microcontrollers to implement hase inverter.
2) **Buck-Boost converter:** Implement buck boost converter to control DC voltage of system.

3) **3-ph Inverter:** Use knowledge of power electronics and microcontrollers to implement three phase inverter.
4) **Over current protection:** Implement current monitoring system using current sensor and ADC to prevent faulty operation of equipment under abnormal conditions.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different protection schemes used in industries.
b) Prepare small application based on 8051, demonstrate the model with presentation/poster in front of class or at project expo.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Embedded C Language Programming for Microcontroller	1	1a. Choose appropriate programming language for the given scenario. 1b. Evaluate operation of super loop.	Lecture with media	PPTs, black board, Computer Demonstration
	2	1c. Analyse the round robin concept and implement concept in the given application.	Lecture with media	PPTs, black board, Computer Demonstration

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
	3	1d. Understand working of IDE platform and use them for developing and simulating code for the given application.	Lecture with media	PPTs, black board, Computer Demonstration
Unit– II Introduction to c8051f120 Microcontroller	4	PrO.1 Understand operation of CrossBar, SFR paging using C-programming and debugging on keil microvision platform. 2a. Justify the need of different microcontrollers and select the appropriate microcontroller for the given embedded system. 2b. Evaluate the importance of WATCHDOG timer and use them in the hardware debugging.	Lecture with media	PPTs, black board, Computer Demonstration
	5	PrO.2 Configure the internal oscillator and PLL for various SYSCLK. 2c. Choose the appropriate speed of operation for the given application. 2d. Understand Memory management. 2e. Justify the need of multifunctional pin and select appropriate function for the given application.	Lecture with media	PPTs, black board, Computer Demonstration
Unit– III Timers in c8051f120 Microcontroller	6	PrO.3 Develop C program for timer T0 and T1 in various modes of operations. 3a. Understand and Evaluate operation of timer and use them in appropriate mode in the given application.	Lecture with media	PPTs, black board, Computer Demonstration
	7	PrO.3 Develop C program for timer T0 and T1 in various modes of operations. PrO.4 Develop C program for timer T2, T3 and T4 in various modes of operation. 3a. Understand and Evaluate operation of timer and use them in appropriate mode in the given application. 3b. Justify need of multiple timer and use them in appropriate mode in the given application.	Lecture with media	PPTs, black board, Computer Demonstration
	8	PrO.4 Develop C program for timer T2, T3 and T4 in various modes of operation. 3b. Justify need of multiple timer and use them in appropriate mode in the given application.	Lecture with media	PPTs, black board, Computer Demonstration
Unit-IV Analog Peripherals of c8051f120 Microcontroller	9	PrO.5 Develop C program for ADC0 in various mode of operation (Cover Single ended & Differential mode with various start of conversion method). PrO.6 Develop C program for ADC2 in various mode of operation (Cover Single ended & Differential mode with various start of conversion method). 4a. Understand and Program on chip ADC for	Lecture with media	PPTs, black board, Computer Demonstration

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		the given application.		
	10	PrO.7 Develop C program for using window comparator mode of ADC0. PrO.8 Develop C program for on chip Analog comparator. 4a. Understand and Program on chip ADC for the given application. 4b. Understand and Program on chip DAC for the given application.	Lecture with media	PPTs, black board, Computer Demonstration
	11	PrO.9 Develop C program for DAC and generating signals. 4b. Understand and Program on chip Comparator for the given application.	Lecture with media	PPTs, black board, Computer Demonstration
Unit – V PCA and Application	12	PrO.10 Develop C program for various PCA mode of operations. 5a. Justify need of PCA and use them in the given application.	Lecture with media	PPTs, black board, Computer Demonstration
	13	PrO.10 Develop C program for various PCA mode of operations. 5a. Justify need of PCA and use them in the given application.	Lecture with media	PPTs, black board, Computer Demonstration
	14	PrO.10 Develop C program for various PCA mode of operations. 5a. Justify need of PCA and use them in the given application.	Lecture with media	PPTs, black board, Computer Demonstration
Unit – VI Communication Protocol	14	6a. Justify need of different communication protocol and select appropriate protocol for the given situation.	Lecture with media	PPTs, black board
	15	6a. Justify need of different communication protocol and select appropriate protocol for the given situation.	Lecture with media	PPTs, black board

XXX SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Embedded System Design using C8051	Han-Way Huang	Cengage Engineering 9788131512241
2	Introduction to Embedded Systems	Shibu K V	Tata McGraw Hill 9780070678798

S. No.	Title of Book	Author	Publication including ISBN
3	Microprocessors & Interfacing	Douglas Hall	Tata McGraw Hill 9781283188982
4	The 8051 Microcontroller and Embedded Systems using Assembly and C	Muhammad Ali Mazidi	Pearson 9780199681273
5	Exploring C for Microcontrollers: A Hands on Approch	Jivan S. Parab, Vinod G. Shelake	Springer 9781402060663

XXXI SOFTWARE/LEARNING WEBSITES

1. http://www.keil.com/dd/docs/datashts/cast/cast_c8051.pdf (Data sheet of C8051)
2. https://www.silabs.com/documents/public/data-sheets/C8051F12x-13x.pdf
3. https://www.silabs.com/documents/public/user-guides/C8051F12x-DK.pdf

XXXII PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE371	**Semester V**	**Advanced Microcontrollers** **Mark 'H'=3** for high, **'M'**=2 for medium or **'L'=1'** for the relevant correlation of each PO or PSO, Competency or CO													
371	***Competency:*** *Ability to design, program, and integrate microcontrollers into electronic systems for the purpose of controlling and monitoring various functions*	***1***	***2***	***3***	***1***	***3***	***1***			***1***			***1***	***3***	***1***
371.1	*Develop software for microcontrollers.*	1	2	3	3	3	1			1		1	1	1	1
371.2	*Understand the architecture of advanced microcontrollers*	2	1	1	1					1			1	1	
371.3	*Design and implement conventional timer-*	1	2	3	2	2				1			1	3	

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	based systems and PCA based systems.														
371.4	*Design and implement ADC and DAC-based systems*	1	2	3	2	2				1			1	3	
371.5	*Analyse and Understand communication protocol.*	1	2	3	2	2				1			1	3	

XXXIII NAMES OF RESOURCE PERSONS

a) Jignesh S Patel

b) ………………………………………………….

Course Code: **EE372**
Course Title: VLSI Design and Technology

UG Engineering Programme	Semester
Electrical Engineering	Fifth

I RATIONALE

This syllabus has been designed to make the students know about the fundamental IC fabrication technology, NMOS and CMOS design of digital circuits. This subject aims to give the basic design idea of digital circuits using MOS technology and different parameter associated with CMOS technologies. The course curriculum is supported by all of the VLSI/Electronics industries such as Cadence, Arrow, Einfochips, Sankalp semiconductors, Nvidia, ST Microelectronics etc.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- To learn IC fabrication process and different parameter associated with it.
- To learn the detail operation of MOS transitions and inverters
- Design the digital circuits using different MOS technologies.
- To learn advance technology in IC fabrication

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an <u>integrated performance</u> of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

CO-1) Identify the various IC Fabrication methods. To Understand the VLSI design flow and get aware about the trends in VLSI technology.
CO-2) Analyses the operation of MOS transistor. Design and Evaluate the MOS inverter with different load technology.
CO-3) Design and Analyze combination circuits and sequential circuits using CMOS and NMOS Technology that realize specified digital functions.
CO-4) To understand and design the dynamic logic using the CMOS technologies. To understand and evaluate the Domino logic circuits.
CO-5) Explain and understand the advance technologies in MOS fabrication and compare it with the current CMOS technologies.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)	Total Credits (L+T+P)	Examination Scheme		
		Theory Marks	Practical Marks	Total Marks

L	T	P	C	ESE	PA	ESE	PA	150
4	0	2	5	70	30	25	25	

***Legends: L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

***(*):** Under the **Theory PA**, out of 30 marks, **15 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 15 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	**1**	Introduction to Xilinx ISE suite Tools.	02
2	**2,3**	. **Introduction to switch level modeling.** . Write a Verilog module using switch-level modeling to implement the following: CMOS Inverter, CMOS NAND Gate, CMOS NOR gate, CMOS AND,CMOS OR	04
3	**3**	A. Implement CMOS 2X1 MUX using switch level modeling B. Implement CMOS EXOR gate and also implement X=AB+C using CMOS	04
4	**3**	A. Implement CMOS 4X1 MUX using "Tran" switches B. Implement Pseudo NMOS NOR gate..	04
5	**3**	Implement Full adder using transistor level modeling.	02
6	**1**	A. Introduction to LTSpice Tool. B. LTSpice Simulation of AND, OR, NOT gate.	04
7	**3**	LTSpice Simulation of NAND and NOR gate using CMOS.	02
8	**2**	Obtain V-I characteristics of NMOS/PMOS MOSFET using LT Spice (DC Analysis).	02
9	**2**	LTSpice Simulation of NOT gate using NAND & NOR with CMOS Technology.	02
10	**3**	LTSpice Simulation of Flip-flops using NAND gates.	04
		Total	**30**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Xilinx 14.9	1,2,3,4,5
2	LT Spice tool	6,7,8,9,10

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Introduction and fabrication of MOSFET	CO-1) Identify the various IC Fabrication methods. To Understand the VLSI design flow and get aware about the trends in VLSI technology. Pro1: Introduction to Xilinx ISE suite Tool. Pro6: **A. Introduction to LTSpice Tool.** 1a.Introduction to switch level modeling in Xilinx and basic simulation in LT Spice	1. **Introduction and fabrication of MOSFE** 1.1VLSI Design Flow 1.2Design Hierarchy 1.3Design Methodology 1.4 NMOS, PMOS and CMOS fabrication process.
MOS Transistor	CO-2)Analyses the operation of MOS transistor. Design and Evaluate the MOS inverter with different load technology. PRO2: Write a Verilog module using switch-level modeling to implement the following: CMOS Inverter, CMOS NAND Gate, CMOS NOR gate, CMOS AND, CMOS OR PRO8: Obtain V-I characteristics of	**2. MOS Transistor** 2.1 Metal Oxide Semiconductor (MOS) structure. 2.2 The MOS system under external bias. 2.3 Structure and operation of MOS transistor 2.4 MOSFET current-voltage characteristics.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	NMOS/PMOS MOSFET using LT Spice (DC Analysis). 1b. Design CMOS basic gates in Xilinx with its test bench. 1c. Obtain V-I characteristics of MOSFET and obtain net list in LT spice	2.5 MOSFET scaling & Small-geometry effects, 2.6 MOSFET capacitance
MOS Inverters: static characteristic and switching characteristic	CO-2) Analyses the operation of MOS transistor. Design and Evaluate the MOS inverter with different load technology. PRO9: LTSpice Simulation of NOT gate using NAND & NOR with CMOS Technology. 1d. Explain the design of NOT gate using NAND and NOR gates	**3. MOS Inverters: static characteristic and switching characteristic** 3.1 Introduction and Resistive load inverter 3.2 Inverter with n-type MOSFET load (Enhancement & Depletion type MOSFET load) 3.3 CMOS inverter 3.4 Definition and calculation of delay times 3.5 Inverter design with delay constraints 3.6 Estimation of interconnect parasitic and calculation of interconnect delay 3.7 Switching power dissipation of CMOS inverters
MOS combinational circuits	CO-3) Design and Analyze combination circuits and sequential circuits using CMOS and NMOS Technology that realize specified digital functions. PRO10 LTSpice Simulation of Flip-flops using NAND gates 1f. Design various flip-flop using CMOS technology.	**4. MOS combinational circuits** 4.1 Introduction 4.2 MOS logic circuit with Pseudo NMOS loads 4.3 CMOS logic circuit 4.4 Complex logic circuits 4.5 CMOS Transmission gate
MOS sequential logic circuits	CO-3) Design and Analyze combination circuits and sequential circuits using CMOS and NMOS Technology that realiz specified digital functions. PRO10 LTSpice Simulation of Flip-flops using NAND gates 1f Design various flip-flop using CMC technology.	**5. MOS sequential logic circuits** 5.1 Introduction 5.2 Behavior of Bi-stable elements 5.3 The S-R latch circuit 5.4 Clocked latch and Flip-flop circuits 5.5 CMOS D-latch and edge triggered Flip-flop

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
MOS dynamic logic circuits	CO-4) To understand and design the dynamic logic using the CMOS technologies. To understand and evaluate the Domino logic implementation.	**6.MOS dynamic logic circuits** 6.1 Principles of pass transistor circuits 6.2 Synchronous Dynamic Circuit Techniques 6.3 CMOS Dynamic Circuit Techniques

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction and fabrication of MOSFET	05	0	2	4	4	0	0	8
II	MOS Transistor	08	0	2	4	4	2	0	12
III	MOS Inverters:static characteristic and switching characteristic	16	0	4	4	4	4	0	20
IV	MOS combinational Circuits	06	0	0	0	2	2	6	10
V	MOS sequential logic Circuits	04	0	0	4	0	2	4	10
VI	MOS dynamic logic Circuits	04	2	0	0	4	2	2	10
Total		**45**	**4**	**10**	**14**	**16**	**12**	**12**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) Implement the combinational circuit using Xilinx software and Spartan 3E board.
2) Implement the sequential circuit using Xilinx software and Spartan 3E board.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- Prepare specifications of some electronic devices.
- Undertake micro-projects in groups (Max. Limit-4)
- Give seminars on any relevant topic

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Introduction and fabrication of MOSFET	**1**	Pro1: Introduction to Xilinx ISE suite Tool. 1a. Introduction to switch level modeling in Xilinx and basic simulation in LT Spice	Lecture with media	PPTs, black board
	9,10	Pro6: **A. Introduction to LTSpice Tool.** 1a. Introduction to switch level modeling in Xilinx and basic simulation in LT Spice	Lecture With media	PPTs, black board
Unit– II MOS Transistor	**2,3**	PRO2: Write a Verilog module using switch- level modeling to implement the following: CMOS Inverter, CMOS NAND Gate, CMOS NOR gate, CMOS AND, CMOS OR	Lecture with media, and Demonstration	Real samples of relays, PPTs, Videos
	12	PRO8: Obtain V-I characteristics of NMOS/PMOS MOSFET using LT Spice (DC Analysis). 1b. Design CMOS basic gates in Xilinx with its test bench. 1c. Obtain V_I characteristics of MOSFET and obtain net list in LT spice.	Lecture with media, and Demonstration	Real samples of relays, PPTs, Videos
Unit– III MOS Inverters: static characteristic and switching characteristic	**13**	PRO9: LTSpice Simulation of NOT gate using NAND & NOR with CMOS Technology. 1d. Explain the design of NOT gate using NAND and NOR gates	Lecture with media, Field visit	PPTs, Video
Unit– IV MOS combinational circuits	**4,5**	PRO3 A. Implement CMOS 2X1 MUX using switch level modeling B. Implement CMOS EXOR gate and also implement X=AB+C using CMOS Design various combinational circuit using CMOS and Pseudo NMOS technology.	Lecture with media, Tutorial	PPT, Video
	6,7	PRO4 A. Implement CMOS 4X1 MUX using “Tran” switches B. Implement Pseudo NMOS NOR gate. 1e. Design various combinational circuit using CMOS and Pseudo NMOS technology.	Lecture with media	PPT, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
	8	PRO5 Implement Full adder using transistor level modeling. 1e. Design various combinational circuit using CMOS and Pseudo NMOS technology.	Lecture with media, case study	PPT, Pictures, black board
	11	PRO7 LTSpice Simulation of NAND and NOR gate using CMOS. 1e. Design various combinational circuit using CMOS and Pseudo NMOS technology.	Lecture with media, case study	PPT, Pictures, black board
Unit– V MOS sequential logic circuits	15	PRO10 LTSpice Simulation of Flip-flopsusing NAND gates 1f. Design various flip-flop using CMOS technology.	Lecture with media, case study	PPT, Pictures, black board
MOS dynamic logic circuits	14	Principles of pass transistor circuits Explain Voltage Bootstrapping, Synchronous Dynamic Circuit Techniques Explain CMOS Dynamic Circuit Techniques , High performance Dynamic CMOS circuits	Lecture with media,	PPT, black board

XXXIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	CMOS digital integrated circuits Analysis and Design	Sung-Mo-Kang, Yusuf Leblebici, Chulwoo Kim	Tata McGrawhill ISBN-13: 978-93-5316-509-3 ISBN-10: 93-5316-509-1
2	Basic VLSI Design	Douglas Pucknell	Pearson Publication ISBN-13: 978-0724801053 ISBN-10: 0724801057
3	Modern VLSI Design	Wayne Wolf	Person Education ISBN-13: 978-0134186047 ISBN-10:0134186

XXXV SOFTWARE/LEARNING WEBSITES

- www.edaplayground.com
- www.chipverify.com
- www.nptel.iitm.ac.in

XXXVI PO-COMPETENCY-CO MAPPING

Course Code	**Competency and Cos**	**PO 1** Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Project management and finance	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.	**PSO 3**
						POs and PSOs										
EE372	**Semester V**					VLSI Design and Technology **Mark** ‘**H’=3** for high, ‘**M**’=2 for medium or **‘L’=1’** for the relevant correlation of each PO or PSO, Competency or CO										
372	**Competency** *Evaluate the performance of various analog and digital circuits used in Industrial applications.*	3	1	3	2	2	0	0	0	0	0	-	-	3	1	-
372.1	Identify the various IC Fabrication methods. To Understand the VLSI design flow and get aware about the trends in VLSI technology	3	1	1	-	3	-	-	-	-	3	3	-	1	1	-
372.2	Analyses th operation of MOS transistor. Design and Evaluate the MOS inverter with different load technology	2	3	3	3	2	1	-	-	1	2	3	-	3	1	-
373.3	Design and Analy combination circui and sequenti circuits using CMC and NMC Technology th realize specifie digital functions	3	3	3	3	3	1	-	1	2	1	3	1	3	1	-
374.4	To understand ar design the dynam logic using the CMOS technologies. T understand ar evaluate the Domino logic circuits.	3	1	1	1	-	-	-	-	2	2	-	2	2	2	-
375.5	Explain ar understand th advance technologies in	3	1	1	1	-	-	-	-	1	1	-	3	2	2	-

	MOS fabrication and compare it with the current CMOS technologies															

XXXVII NAMES OF RESOURCE PERSONS

a) **Ankur H Patel**

Course Code: **EE373**

Course Title: **OPTIMIZATION TECHNIQUES (PROGRAMME ELECTIVE -1)**

UG Engineering Programme	Semester
Electrical Engineering	Fifth

I RATIONALE

Electrical engineers have to deal with various types of optimization problem used in many applications. This curriculum course structure is developed such that the student is able to understand and apply the basic classical optimization methods to solve the basic power system problems.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

Understand and analyze the classical optimization techniques to solve the power system problem.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1. Develop a fundamental understanding of concepts of various classical methods
2. Understand and formulate the linear programming problem to solve the power system linear problems.
3. Understand and formulate the unconstrained programming problem to solve the power system unconstrained problems.
4. Understand and formulate the Nonlinear programming problem to solve the power system Nonlinear problems.
5. The students will understand different methods of unconstrained and least squares-based optimization for solving real world problem.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	
4	0	2	5	70	30*	25	25	150

***Legends: L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

***(*):** Under the **Theory PA**, out of 40 marks, **20 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	3,4	Formulation of optimization problem	05
2	3,4	Solution of Linear Programming Problem by graphical method	05
3	3,4	Kuhn-Tucker method	05
4	1,2	Nonlinear programming method	05
5	1,2	Use of Penalty function method	05
6	1,2	Particle swarm optimization to solve constrained and unconstrained optimization problems.	05
		Total	**30**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Matlab optimization solver	1 to 6
2	GAMS Software	1 to 6

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Fundamentals of Optimization	**CO 1. Develop a fundamental understanding of concepts of various classical methods** PrO 1 Formulation of optimization problem 1a. *Understand the Convexity* 1b. *Understand the* Rates of convergence. 1c. *Understand* the Feasibility and optimality,	a. Introduction, Feasibility and optimality, b. Convexity, c. constraints, Rates of convergence.
Unit– II Linear Programming	**CO 2. Understand and formulate the linear programming problem to solve the power system linear problems.** PrO 1 Solution of Linear Programming Problem by graphical method PrO 5 Use of Penalty function method 2a *Understand and apply the* Formulation of Linear programming problem 2b. *Understand and analyze the* Graphical method 2c. *Understand and apply the* Simplex method 2d. *Understand and apply the* Big M method	2.1 Introduction, Formulation of Linear programming problem, 2.2 Graphical method, Simplex method, Basic solution, 2.3 Basic feasible solution, Simplex algorithm, Big M method (Penalty method), 2.4 Two phase method, Special cases in simplex method application. Economic load dispatch using Linear programming method.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit– III Unconstrained optimization	**CO 3. Understand and formulate the unconstrained programming problem to solve the power system unconstrained problems.** **CO 5. The students will understand different methods of unconstrained and least squares-based optimization for solving real world problem.** PrO 3 Kuhn-Tucker method PrO 6 Particle swarm optimization to solve constrained and unconstrained optimization problems. *3a. Understand and analyze the Optimality conditions,* *3b. Understand and analyze the Newton's method for minimization,* *3c. Understand and analyze the Quasi-Newton method* *3d Understand and analyze the Line search methods, Steepest-Descent method,*	3.1 Introduction, Optimality conditions, Newton's method for minimization, 3.2 Line search methods, Steepest-Descent method, 3.3 Quasi-Newton method.
Unit– IV **Nonlinear Programming**	**CO 4. Understand and formulate the Nonlinear programming problem to solve the power system Nonlinear problems.** PrO 4 Nonlinear programming method PrO 5 Use of Penalty function method *4a. Understand and apply the Optimality conditions for constrained problems, Kuhn Tucker conditions,* *4b. Understand and apply the* Penalty function method, Barrier method, *4c. Understand and analyze the* Computing the Lagrange multipliers, Sequential quadratic programming, *4d Understand and analyze the* Interior point method. Active and reactive power	4.1 Optimality conditions for constrained problems, Kuhn Tucker conditions, 4.2 Penalty function method, Barrier method, 4.3 The Lagrange multipliers and the Lagrangian function, Sensitivity analysis, 4.4 Computing the Lagrange multipliers, Sequential quadratic programming, 4.5 Interior point method. Active and reactive power optimization using NLP.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	optimization using NLP.	
Unit– V **Least squares based methods**	**CO 5. The students will understand different methods of unconstrained and least squares-based optimization for solving real world problem.** *5a. Understand and apply the* LSQ fit *5b. Evaluate the* of unknown coefficients *5c. Understand and analyze the* Approximating the shifted waveform by a power series *5d. Understand and analyze the* multi-variable series LSQ technique, Derivation of the multi variable series.	1.1 Introduction, Integral LSQ fit, Basic assumptions, 1.2 Determination of unknown coefficients, Power series LSQ fit, 1.3 Assumptions, Shifted waveform, Approximating the shifted waveform by a power series, 1.4 Multi-variable series LSQ technique, Derivation of the multi variable series.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Fundamentals of Optimization	05	0	2	2	0	1	0	05
II	Linear Programming	12	0	1	1	4	5	3	14
III	Unconstrained optimization	13	2	1	2	2	5	3	15
IV	Nonlinear Programming	15	2	2	3	3	6	2	18
V	Least squares-based methods	15	1	3	3	4	5	2	18
Total		60	**5**	**9**	**11**	**13**	**22**	**10**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) Formulate the Economic load dispatch non liner problem in MATLAB and GAMS.
2) Formulate the Unit commitment problem in MATLAB and GAMS.
3) Formulate the Mathematical nonlinear and linear constrained problem and solved using the classical optimization techniques using MATLAB and GAMS.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different Semiconductor switches used in converters.
b) Prepare 15 min. power point presentation on how to choose the particular converter for the particular application.

XIII COURSE PLAN

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Fundamentals of Optimization	PrO 1 Formulation of optimization problem *1a. Understand the* *Convexity* *1b. Understand the* Rates of convergence. *1c. Understand* the Feasibility and optimality,	1.1 Introduction, Feasibility and optimality, 1.2 Convexity, 1.3 constraints, Rates of convergence.
Unit– II Linear Programming	PrO 1 Solution of Linear Programming Problem by graphical method PrO 5 Use of Penalty function method *2a Understand and apply the* Formulation of Linear programming problem 2b. *Understand and analyze the* Graphical method 2c. *Understand and apply the* Simplex method 2d. *Understand and apply the* Big M method	2.1 Introduction, Formulation of Linear programming problem, 2.2 Graphical method, Simplex method, Basic solution, 2.3 Basic feasible solution, Simplex algorithm, Big M method (Penalty method), 2.4 Two phase method, Special cases in simplex method application. Economic load dispatch using Linear programming method.
Unit– III Unconstrained optimization	PrO 3 Kuhn-Tucker method PrO 6 Particle swarm optimization to solve constrained and unconstrained optimization problems. *3a. Understand and analyze the Optimality conditions,* *3b. Understand and analyze the Newton's method for minimization,* *3c. Understand and analyze the Quasi-Newton method* *3d Understand and analyze the Line search methods, Steepest-Descent method,*	3.1 Introduction, Optimality conditions, Newton's method for minimization, 3.2 Line search methods, Steepest-Descent method, 3.3 Quasi-Newton method.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit– IV Nonlinear Programming	PrO 4 Nonlinear programming method PrO 5 Use of Penalty function method *4a. Understand and apply the Optimality conditions for constrained problems, Kuhn Tucker conditions,* *4b. Understand and apply the* Penalty function method, Barrier method, *4c. Understand and analyze the* Computing the Lagrange multipliers, Sequential quadratic programming, *4d Understand and analyze the* Interior point method. Active and reactive power optimization using NLP.	4.1 Optimality conditions for constrained problems, Kuhn Tucker conditions, 4.2 Penalty function method, Barrier method, 4.3 The Lagrange multipliers and the Lagrangian function, Sensitivity analysis, 4.4 Computing the Lagrange multipliers, Sequential quadratic programming, 4.5 Interior point method. Active and reactive power optimization using NLP.
Unit– V Least squares based methods	*5a. Understand and apply the* LSQ fit *5b. Evaluate the* of unknown coefficients *5c. Understand and analyze the* Approximating the shifted waveform by a power series *5d. Understand and analyze the* multi-variable series LSQ technique, Derivation of the multi variable series.	5.1 Introduction, Integral LSQ fit, Basic assumptions, 5.2 Determination of unknown coefficients, Power series LSQ fit, 5.3 Assumptions, Shifted waveform, Approximating the shifted waveform by a power series, 5.4 Multi-variable series LSQ technique, Derivation of the multi variable series.

XXXVIII SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Electric power applications of Fuzzy systems	M. E. El-Hawary	IEEE press, 2006
2	Artificial intelligence and intelligent systems	N.P.Padhy	Oxford University Press, 2005
3	Soft Computing Techniques and its Applications in Electrical Engineering	Chaturvedi	Springer

S. No.	Title of Book	Author	Publication including ISBN
4	Optimization Principles	Narayan Rao	John Wiley & Sons
5	Optimization for engineering design	K. Deb	PHI

XXXIX SOFTWARE/LEARNING WEBSITES

1. https://nptel.ac.in/courses/111/105/111105100/
2. https://nptel.ac.in/courses/105/108/105108127/

XL PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE373	**Semester V**														
373	**Competency** • Understand and analyze the classical optimization techniques to solve the power system problem.	***3***	***2***	***2***	***2***	***1***	**-**	***1***	**-**	**-**	**-**	**-**	***1***	***3***	**-**
373.1	Develop a fundamental understanding of concepts of various classical methods	***3***	-	-	1	-	-	-	-	-	2	**-**	**-**	-	**-**
373.2	Understand and formulate the linear programming problem to solve the power system linear problems.	3	2	3	-	2	-	-	-	1	-	-	1	1	3
373.3	Understand and formulate the unconstrained programming problem to solve the power system unconstrained problems.	3	3	2	-	2	-	-	-	1	-	-	1	1	3
373.4	Understand and formulate the Nonlinear programming problem to solve the power system Nonlinear problems.	3	3	2	-	2	-	-	-	1	-	-	1	1	3
373.5	The students will understand different	3	3	2	-	2	-	-	-	1	-	-	1	1	3

Course Code	Competency and COs	**PO 1** Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	methods of unconstrained and least squares-based optimization for solving real world problem.														

XLI NAMES OF RESOURCE PERSONS

a) Dr. Dharmesh Dabhi

Course Code: **EE374**
Course Title: **Distributed Generation and Microgrid**

UG Engineering Programme	Semester
Electrical Engineering	Fifth

I RATIONALE

The micro grid can help measure and reduce energy consumption and costs. One of the visions for the micro grid is that consumers will be able to monitor their energy consumption in real time, or at least near real time, via the Web. This curriculum course structure is developed such that the student is able to understand the basic concepts, components, architecture, various measurement technologies of micro grid and the importance of distributed generation in power system.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Evaluate the performance of various distributed generator, energy storage equipment and converters which are used in industry.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) ***Understand*** the basic difference between conventional and non-conventional energy resources**.**
2) ***Analyze*** the concept of distributed generation and installation.
3) **Design** the grid integration system with conventional and non-conventional energy sources.
4) ***Evaluate*** the performance of the converters in micro grid.
5) ***Analyze*** power quality issues and control operation of micro grid**.**

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	**T**	**P**	**C**	**ESE**	PA	**ESE**	**PA**	
4	0	2	5	70	30*	25	25	150

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

(*): *Under the* ***Theory PA****, out of 30 marks,* ***10 marks*** *are for* ***micro-project assessment*** *to facilitate integration of COs and the remaining 20 marks is converted from 2 tests to be taken*

during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Prepare the review report of implementing non-conventional energy resources in Micro grid.	04
2	2	*Design* the Active Distribution System with Distributed Generation.	02
3	2	*Design* the Passive cell balancing for a Lithium-ion battery pack.	04
4	2	*Analyze* the DC-DC converter that can be used to maintain a constant load voltage when drawing power from an ultra-capacitor.	02
5	2	*Analyze* the charging and discharging behavior of supercapacitor.	04
6	3	*Design* the 250-kW Grid connected PV array.	04
7	4	*Design* of DC microgrid.	04
8	5	*Analyze* the islanded operation of a remote- microgrid model using droop controllers.	04
		Total	**28**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB Simulation software (version 2021a)	1,2,3,4,5,6,7, 8

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year

d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Introduction	**CO 1.** ***Understand*** *the basic difference between conventional and non-conventional energy resources*. *PrO 2. Prepare the review report of implementing non-conventional energy resources in Micro grid.* 1a. *Prepare* the report on energy crises 1b. *Prepare* the review report on non-conventional energy sources	1.1 Conventional power generation: advantages and disadvantages 1.2 Energy crises 1.3 Non-conventional energy (NCE) resources: review of Solar PV, Wind Energy systems, Fuel Cells, micro turbines, biomass, and tidal sources.
Unit– II Basic of Distributed Generation	**CO 2. *Analyze*** the concept of distributed generation and installation. *Pro 2. Design* the Active Distribution System with Distributed Generation. *Pro 3. Design* the Passive cell balancing for a Lithium-ion battery pack. *Pro 4. Analyze* the DC-DC converter that can be used to maintain a constant load voltage when drawing power from an ultra-capacitor. *Pro 5. Analyze* the charging and discharging behavior of supercapacitor. 2a. Understand the concept of distributed generations. 2b. Choose the correct standards for interconnecting the distributed generation for the specified application. 2c. Analyze the security issues in	2.1 Concept of distributed generations, topologies, selection of sources 2.2 Regulatory standards/framework, Standards for interconnecting Distributed resources to electric power systems: IEEE 1547. 2.3 DG installation classes, security issues in DG implementations 2.4 Energy storage elements: Batteries, ultra-capacitors, flywheels.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	DG implementations. 2d. Design the energy storage elements.	
Unit– III **Impact of Grid Integration**	**CO 3. Design** the grid integration system with conventional and non-conventional energy sources. *Pro 6. Design* the 250-kW Grid connected PV array. 3a. Determine the requirements for grid interconnection for the given distributed generation. 3b. Evaluate the operational parameters for grid integration. 3c. Evaluate the impact of grid integration with NCE sources on existing power system. 3d. Understand the islanding issues of grid integration.	3.1 Requirements for grid interconnection, 3.2 Limits on operational parameters: voltage, frequency, THD 3.3 Response to grid abnormal operating conditions 3.4 Islanding issues. 3.5 Impact of grid integration with NCE sources on existing power system: reliability, stability and power quality issues.
Unit-IV **Basic of Microgrid**	**CO 4. *Evaluate*** the performance of the converters in micro grid. *Pro 7. Design* of DC microgrid. 4a. Design the microgrid for the specified application. 4b. Evaluate the performance of the microgrid drivers for the specified condition. 4c. Evaluate the performance of the converters for the specified application.	4.1 Concept and definition of microgrid, 4.2 Microgrid drivers and benefits, 4.3 Review of sources of microgrids, 4.4 Typical structure and configuration of a microgrid, 4.5 AC and DC microgrids, 4.6 Power Electronics interfaces in DC and AC microgrids
Unit – V **Control and Operation of Microgrid**	**CO 5. *Analyze*** power quality issues and control operation of micro grid. *Pro 8. Analyze* the islanded operation of a remote- microgrid model using droop controllers. 5a. Evaluate the different controls of grid connected microgrid. 5b. Evaluate the different controls of islanded microgrid. 5c. Design the active and reactive control of microgrid. 5d. Design the communication	5.1 Modes of operation and control of microgrid: grid connected and islanded mode, 5.2 Active and reactive power control, protection issues, 5.3 Anti-islanding schemes: passive, active and communication based techniques 5.4 Microgrid communication infrastructure 5.5 Power quality issues in microgrids, regulatory standards, Microgrid economics 5.6 Introduction to smart microgrids.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain* *(to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	infrastructure for microgrid. 5e. Evaluate the power quality issues in microgrid.	

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	**Unit Title**	**Teaching Hours**	**Distribution of Theory Marks**						
			R Level	**U Level**	**A Level**	**An Level**	**E Level**	**C Level**	**Total Marks**
I	Introduction	08	4	4	2	2	0	0	12
II	Basic of Distributed Generation	12	4	4	4	2	0	0	14
III	Impact of Grid Integration	16	0	4	4	2	2	4	16
IV	Basic of Microgrid	08	2	2	2	2	0	2	10
V	Control and Operation of Microgrid	16	0	4	4	2	4	4	18
Total		**60**	**10**	**18**	**16**	**10**	**6**	**10**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **Converters:** Collect the specifications of different types of converters manufacturing by different types of industries.
2) **Battery:** Collect the specifications of different types of battery used in different Electric vehicle by different types of industries.
3) **Renewable generator:** Prepare case studies on power quality issues in renewable energy integration and microgrid.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different protection schemes used in industries.
b) Prepare 15 min. power point presentation on how to choose relay for the particular application.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Introduction	1	**Pro 1.** *<u>Prepare</u> the review report of implementing non-conventional energy resources in Micro grid.* 1a. <u>Prepare</u> the report on energy crises.	Lecture with media	PPTs, black board
	2	*1b.* <u>Prepare</u> the review report on non-conventional energy sources.	Lecture With media	PPTs, black board
	3	*1b.* <u>Prepare</u> the review report on non-conventional energy sources.	Lecture with media,	PPTs, black board
Unit– II Basic of Distributed	**4**	**Pro 2**. *<u>Design</u>* the Active Distribution System with Distributed Generation. 2a. <u>Understand</u> the concept of distributed generations.	Lecture with media, Tutorial	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Generation	**5**	2b. Choose the correct standards for interconnecting the distributed generation for the specified application. 2c. Analyze the security issues in DG implementations.	Lecture with media, Tutorial	PPTs, black board
	6	**Pro 3**. *Design* the Passive cell balancing for a Lithium-ion battery pack. **Pro 4**. *Analyze* the DC-DC converter that can be used to maintain a constant load voltage when drawing power from an ultra-capacitor. **Pro 5**. *Analyze* the charging and discharging behavior of supercapacitor. 2d. Design the energy storage elements.	Lecture with media	PPTs, black board
Unit– III Impact of Grid Integration		3a. Determine the requirements for grid interconnection for the given distributed generation. 3b. Evaluate the operational parameters for grid integration.	Lecture with media,	PPTs, black board
	8	3c. Evaluate the impact of grid integration with NCE sources on existing power system. 3d. Understand the islanding issues of grid integration.	Lecture with media	PPTs, black board
	9	**Pro 6.** *Design* the 250-kW Grid connected PV array.	Lecture with media, + PPT	PPTs, black board
Unit-IV Basic of Microgrid	**10**	**Pro 7**. *Design* of DC microgrid. 4a. Design the microgrid for the specified application.	Lecture with media, Tutorial	PPT, black board
	11	4b. Evaluate the performance of the microgrid drivers for the specified condition.	Lecture with media	PPTs, black board
	12	4c. Evaluate the performance of the converters for the specified application.	Lecture with media, case study	PPT, black board
Unit – V Control and Operation of Microgrid	**13**	*Pro 8. Analyze* the islanded operation of a remote- microgrid model using droop controllers. 5a. Evaluate the different controls of grid connected microgrid. 5b. Evaluate the different controls of islanded microgrid.	Lecture with media	PPTs, black board
	14	5c. Design the active and reactive control of microgrid. 5d. Design the communication infrastructure for microgrid.	Lecture with media, Tutorial	PPTs, black board
	15	5e. Evaluate the power quality issues in microgrid.	Lecture with media, Tutorial	PPTs, black board, Video

XLII SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Microgrids: Architectures and Control	N. Hatziargyriou	Wiley-IEEE Press, 2014. ISBN: 9781118720677
2	Distributed Generation-Induction and Permanent Magnet Generators	Loi Lei Lai, Tze Fun Chan	Wiley-IEEE Press, 2007. ISBN: 978-0-470-06208-1
3	Voltage Source Converters in Power Systems: Modeling, Control and Applications	Amirnaser Yezdani, Reza Iravani	Wiley-IEEE Press, 2009 ISBN: 978-0-470-52156-4
4	Renewable Energy Sources	N. Twidell, A. D. Weir	University press, 2001, ISBN: 9781315766416
5	Fuel Cell Systems	James Larminie , Andrew Dicks	Wiley, 2000 ISBN: 9780470848579

XLIII SOFTWARE/LEARNING WEBSITES

1. DC Microgrid and Control System: - https://nptel.ac.in/courses/108/107/108107143/

PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE374	**Semester V**														
374	**Competency** Evaluate the performance of various distributed generator, energy storage equipment and converters which are used in industry	***3***	***2***	***2***	***2***	***1***	**-**	***1***	**-**	**-**	**-**	**-**	***1***	***3***	**-**
374.1	Understand the basic difference between	3	2	2	-	2	-	2	-	-	1	-	-	3	-

Course Code	Competency and COs	**PO 1** Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	conventional and non-conventional energy resources.														
374.2	Analyze the concept of distributed generation and installation	3	2	1	-	-	-	-	-	1	-	-	-	2	-
374.3	Design the grid integration system with conventional and non-conventional energy sources	3	2	3	3	2	-	-	-		-	-	-	2	2
374.4	Evaluate the performance of the converters in micro grid	3	3	1	2	-	-	2	-	1	-	-	-	2	2
374.5	Analyze power quality issues and control operation of micro grid	3	-	2	-	2	-	-	-	1	2	2	-	-	3

XLIV NAMES OF RESOURCE PERSONS

c) Dr. Jigar Sarda

Course Code: **EE375**
Course Title: ENERGY CONSERVATION, AUDIT AND MANAGEMENT (PE IV)

UG Engineering Programme	Semester
Electrical Engineering	Fifth

I RATIONALE

The pressure of technological development in all sectors on renewable energy sources has led to growing the cost of energy around the world. Efficient and judicious use of available energy sources would lead to the easing of such pressures and decrease in the operating costs of the organizations and industries. Thus it is necessary to conserve the energy. Also essential theoretical knowledge and practical skills about the concept of energy conservation is to be provided through different approaches. The process of energy audit will help to identify the various possible avenues in which savings of energy can be effectively adopted.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Perform Electrical Energy Audit for Conservation of the Energy.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) ***Use*** the different energy management tools and techniques.
2) ***Perform*** *the* Energy Conservation opportunities in different area.
3) **Interpret** the suitable energy storage system.
4) ***Perform*** *the Electrical Energy Audit of HT and LT Consumer.*
5) ***Choose*** *the energy efficient technology in electrical systems.*

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	
4	0	2	5	70	30*	25	25	150

***Legends: L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

***(*):** Under the **Theory PA**, out of 30 marks, 30 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Study energy management and conservation techniques	02
2	1	Use different energy audit tools to perform audit	02
3	2	Perform energy conservation opportunity in domestic user appliances	02
4	2	Perform energy conservation opportunity in Industrial user appliances	02
5	4	Identify star labeled electrical apparatus and compare the data for various star ratings.	02
6	4	Compare power consumption of different types of light, FTL & LED.	02
7	4	Perform Bill analysis of HT consumers and suggest suitable corrective measures for conservation and reduction of it's energy bill.	02
8	4	Perform energy audit of HT consumer	02
9	5	Determine the reduction in power consumption by replacement of lightning system in class room / laboratory.	02
10	5	Determine the reduction in power consumption by replacement of Fans and AC in class room / laboratory.	02
11	5	Prepare an energy audit report.(phase I)	02
12	5	Prepare an energy audit report.(phase II)	02
		Total	**22**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Ammeter MI Type: AC/DC 0-5-10 Amp	1
2	Voltmeter MI Type: AC/DC 0-150/300V,0-250/500V	2
3	Wattmeter: Three Phase double element 5/10Amp,250/500V	3
4	Wattmeter: Single Phase double element 2.5/5Amp,200/400V	4
5	Automatic Power factor controller (APFC)	5
6	Lux Meter	6
7	Clip on meter (amp, volts) digital/analog	7
8	FTL,CFL,LED of different ratings	8
9	Power analyzer	9
10	Thermography meter	10
11	Electric regulators, Electronics ballast	11

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Electrical Energy Conservation	**CO 1. *Use*** the different energy management tools and techniques. *PrO 3.* Study energy management and conservation techniques *PrO 4.* *Use different energy audit tools to perform audit* 1d. Interpret the given energy conservation clause 1e. Explain the specified BEE roles 1f. Describe the techniques for energy conservation in given electric machine	1.1 Introduction to energy science and energy technology, various forms of energy. Law of conservation of energy. Primary and Secondary energy, Commercial and Non-commercial energy, Renewable and Non-Renewable energy. Carbon credit. 1.2 Energy conservation act-2001

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain* *(to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
		and its features, key definitions as outlined in the Act, Schemes of BEE under the energy conservation Act, 2001. 1.3 National Mission for Enhanced Energy Efficiency. 1.4 Energy scenario of India, Energy conservation and its importance. 1.5 Load Management and Maximum demand control. 1.6 Potential energy conservation opportunities in: (1) HVAC System (2) Lighting systems (3) Motors and Transformers
Unit– II Energy Management	**CO 2.** ***Perform*** *the* Energy Conservation opportunities in different area. **CO 1. Use** the different energy management tools and techniques. PrO 3. Perform energy conservation opportunity in domestic user appliances PrO 4. Perform energy conservation opportunity in Industrial user appliances	2.1 Definition and objective of energy management, Responsibilities and Duties of energy manager, Management tools for effective Implementation like 5S, Kaizen, Small group Activity, TPM, TQM & ISO 50001. 2.2 Electrical System Optimization: Electricity rate Tariff, Key to reduction in electrical energy consumption, methods to improve plant power factor, Load Management, Conduction Loss, Switching Loss, Magnetic Loss. 2.3 Cogeneration system 2.4 Intelligent buildings 2.5 Energy planning, Energy staffing, Energy Organization, Energy Requirement, Energy Costing, Energy Budgeting, Energy Monitoring, Energy consciousness, Energy Management Professionals, Environment pollution due to

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
		energy use.
Unit– III **Energy storage system**	**CO 3.** **Interpret** the suitable energy storage system.	3.1 Energy storage: Demand for energy storage, Classification of Energy storage systems. 3.2 Mechanical energy storage: Pumped hydro, Compressed Air, Flywheels. 3.3 Electrochemical energy storage: secondary batteries, Flow batteries. 3.4 Chemical energy storage: Hydrogen, Synthetic natural gas. 3.5 Electrical and magnetic energy storage systems.
Unit-IV **Energy Audit**	**CO 4.** ***Perform*** *the Electrical Energy Audit of HT and LT Consumer.* *PrO 5.* *dentify star labeled electrical apparatus and compare the data for various star ratings.* *PrO 6.* *compare power consumption of different types of light, FTL & LED.* *PrO 7.* *Perform Bill analysis of HT consumers and suggest suitable corrective measures for conservation and reduction of it's energy bill.* *PrO 8.* *Perform Bill analysis of HT consumers and suggest suitable corrective measures for conservation and reduction of it's energy bill.* *PrO 9.* *Perform energy audit of HT consumer*	4.1 Introduction, Definition, Need for Energy Audit, 4.2 Types and Procedures for energy audit. Preliminary energy audit, detailed energy audit, Industrial Audit approaches. 4.3 Procedure for energy audit and equipment's required. 4.4 Energy Audit Report, Benchmarking, 4.5 Electricity Billing for LT and HT consumer.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain* *(to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	4a. Suggest relevant instrument for the specified energy audit with justification 4b. Develop the energy flow diagram of the given apparatus 4c. Develop questionnaire for the energy audit of the given facility 4d. Prepare the energy audit report for the given apparatus.	
Unit – V Energy Efficient Technologies In Electrical Systems	**CO 5.** ***Choose*** *the energy efficient technology in electrical systems.*	5.1 Automatic Power Factor Controllers 5.2 Energy Efficient Motors. 5.3 Variable Speed Drives and variable frequency Drives. 5.4 Soft Starter. 5.5 Energy Efficient Transformers. 5.6 Electronic Ballasts. 5.7 Energy Efficient Lighting. 5.8 Electrical Furnaces.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	**Unit Title**	**Teaching Hours**	**Distribution of Theory Marks**						
			R Level	**U Level**	**A Level**	**An Level**	**E Level**	**C Level**	**Total Marks**
I	Electrical Energy Conservation	13	3	5	5	2	0	0	15
II	Energy Management	15	3	3	5	3	2	2	18
III	Energy storage system	13	2	2	2	2	2	0	10
IV	Energy Audit	11	0	3	4	4	2	2	15
V	Energy Efficient Technologies In Electrical Systems	08	2	3	5	0	2	0	12
Total		**60**	**06**	**17**	**17**	**18**	**06**	**06**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **Star label calculation:** choose the different appliances for star label calculation.
2) **Load Calculation:** Collect the specifications of different load in university campus.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding audit methodology of different utility used in industries.
b) Prepare 15 min. power point presentation on different topics of energy audit, conservation and management.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Electrical Energy Conservation	1-2-3	1a. Identify different methods used for energy conservation. 1b. Use different methods to perform of energy conservation. 1c. Explain energy management concept.	Lecture with media	PPTs, black board
Unit– II Energy Management	4-5-6	2a. Calculate and find costing of energy conservation project. 2b. Do Case study of energy conservation. 2c. Improve power factor in electrical system. 2d. Optimize solution steps for conservation & management	Lecture With media Demonstration	PPTs, black board
Unit– III Energy storage system	**7-8-9**	3a. Choose the efficient method for energy storage. 3b. Select method as per applications.	Lecture with media, Tutorial	PPTs, black board
Unit-IV Energy Audit	**10-11**	4a. Describe concept of energy audit in detail. 4b. Use different tools of electrical energy audit. 4c. Perform energy audit in CHARUSAT campus.(3 month)	Lecture with media, Case study	PPTs, black board
Unit – V Energy Efficient Technologies In Electrical Systems	**12-13**	5c. Study different technology available in market. 5d. Select energy efficient electrical equipment's in industries.	Lecture with media, + PPT	PPTs, black board

XLV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Guide Book no 1 to 4 for national Certification	Bureau of energy efficiency (BEE)	Bureau of energy efficiency

S. No.	Title of Book	Author	Publication including ISBN
	Examination for Energy Manager & Auditor		
2	Art of reading Electricity bill	Talware Yogendra	Danyatavyaorakashan
3	Energy Engineering and Management	Chakrabarti ,Amlan	PHI Learning Private Limited
4	Energy Audit in Electrical Utilities	Rajiv Shankar	Viva books Pvt.
5	Non – conventional energy sources	G. D. Rai	G. D. Rai
6	Industrial Energy Conservation Techniques	K. Nagabhusan Raju	Atlantic Publishers & Distributors (P) Ltd.

XLVI SOFTWARE/LEARNING WEBSITES

1. www.energymanagertraining.com
2. www.bee-india.gov.in

XLVII PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE375	Semester V														
EE375	**Competency Perform Electrical Energy Audit for Conservation of the Energy.**	***3***	***3***	***3***	***2***	***3***	***2***	***2***	***-***	***2***	***2***	***3***	***1***	***3***	***1***
EE375.1	***Use*** the different energy management	2	3	3	2	3	-	2	-	3	3	2	2	3	2

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	tools and techniques														
EE375.2	***Perform*** *the* Energy Conservation opportunities in different area.	2	3	-	-	-	-	-	-	-	-	-	3	3	-
EE375.3	**Interpret** the suitable energy storage system.	1	-	1	-	-	2	-	-	-	-	-	-	1	-
EE375.4	***Perform*** *the Electrical Energy Audit of HT and LT Consumer.*	3	-	3	-	3	-	-	3	3	3	3	3	3	3
EE375.5	***Choose*** *the energy efficient technology in electrical systems.*	2	-	-	3	-	2	-	-	-	3	-	3	3	3

XLVIII NAMES OF RESOURCE PERSONS

a) Vibha Parmar

b) ………………………………………………….

B. Tech. (Electrical Engineering) Programme

SYLLABI

(Semester – VI)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Course Code: **EE346**
Course Title: **Digital Signal Processing**

UG Engineering Programme	Semester
Electrical Engineering	Sixth

I RATIONALE

Computer and Programmable device can process discrete information and discrete time systems can be implemented inside. The electrical engineer need to apply signal processing algorithm and design the discrete systems for products like, Numerical protection relays, close loop power electronic controller, digital twin, measuring and analysis instruments etc. Development of a real time system covers various steps such as, developing discrete mathematical model/ set of equation, verify its stability and remodifying the operational characteristics by inculcating several signal processing algorithm and discrete compensators. So before programming the microcontroller or similar programmable discrete device, the system designer has to develop and test the algorithm in software. At-least, the basic signal processing algorithms and digital system design can be taught, leaving the capacity of individual to think out of box, correlate the learning and known things to find and develop new innovative algorithm.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- Test, Verify and implement Signal processing algorithms in MATLAB and similar software: The design-development and Research industries hire 'Domain Knowledge Software Engineer' for developing/implementing and testing the research hypothesis and algorithms.
- Design, implementation and Testing of Discrete Digital Systems.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) **Understand** basic Signal Processing concepts and **Find** source of error and problem associated in Signal Processing, **Select** Processor and Peripherals for signal Processing application.
2) **Design** and **realize/implement** Discrete system from time domain equation and S domain transfer function.
3) **Design** and **realize/implement** Digital systems like Digital Compensator and Digital Filters.
4) **Understand** the theory and Mathematics of various signal processing terms, algorithms and Transforms. **Apply** it and **interpret** the result, **simulate** and **implement** it.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
3	0	2	4	70	30*	25	25	

***Legends: L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

***(*):** Under the **Theory PA**, out of 30 marks, **20 marks** are for **attendance**, **micro-project assessment** and other **assignment** if given any, to facilitate integration of COs and the remaining 10 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	*Identify the presence/occurrence of Aliasing Phenomenon in MATLAB environment and recommend appropriate solution in terms choice of sampling frequency and specification of antialiasing filter.*	01
2	4	*Code various Mathematical equations & Operation in MATLAB, that can be useful in implementing signal processing equations/transforms/ algorithms. Also determine numbers of operations, time and memory requirement.* *(Example equations: Various statistics terms, Feature extracting terms, Integration, Differentiation, Shifting, Convolution, Correlation, DFT, Continuous Wavelet Transform etc.)*	04
3	2,3	*Realize the given electrical and Mechanical engineering systems on digital platform and Verify the response.*	04
4	2,3,4	*Design a digital close loop system and analyze-verify the response after incorporating the Digital Compensator (P, PI, PID, Lag Lead Compensator).*	02
5	3	*Design various IIR and FIR Filters with Hand calculation and Excel, Python and MATLAB.*	03
6	3,4	*Simulate and Verify the operation of various IIR and FIR filters in MATLAB, dSPACE and Microcontroller.*	03
7	3,4	*Design and Simulate Mutli-Resolution Wavelet Filter bank and Wavelet Packet Filter banks in MATLAB.*	02
8	2,3,4	*Implement Signal Processing algorithms for power system and Power Electronics applications in MATLAB, dSPACE Real time digital simulator and Microcontroller.*	04

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
		(Special interest are Fourier and Wavelet based algorithm for protection, event detection and power quality assessment)	
9	2,3,4	*Test- Verify and choose various concepts/properties related to Signal Processing algorithms for and in given application.* *The algorithms to be considered are: Feature extraction criteria, statistical equations, correlation, DFT, S Transform, Wavelet, IIR/FIR Filters.* *Various example of failure of algorithm, in giving desired output will generate insight about selection of algorithm. Few of the selection criteria needs to be covered are time resolution, frequency resolution, correct frequency domain representation, delays introduced in filtering for removal of particular harmonic, response in presence of random signal, decaying dc and non-integer multiple harmonics)*	03
10	3	*Simulate Signal processing algorithms on image in MATLAB.*	04
		Total	**30**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

Sr. No.	Equipment Name with Broad Specifications	PrO No.
1	*MATLAB Simulation software (version higher than 2016a), Python (An Open source compiler), Microsoft Excel or Libre Calculator (An Open source software)*	All (1 to 10)
2	*dSPACE Real Time Digital Simulator*	1, 2,4,5,7
3	*Fast Microcontroller with ADC and DAC (Like Texas DSP, ARM M4)*	6,8
4	*Signal Generator and Digital Storage Oscilloscope, Harmonic Analyzer*	6,8
5	*Converter Drive /Grid connected converter*	8
6	*Camera interface*	10

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Test – verify result by simulation before believing the algorithm. Develop the boundary with in which algorithm will success.
2) Think out of box for new invention of algorithm. Anything learned/unlearned can be useful for new research at any point of research and development. Learn and make habit of noting down the conclusions and mathematics involved.
3) Follow ethics and predict about the prospective damage by a wrong algorithm. Don’t hide bad side of your algorithm from yourself and others.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I ***Introduction to Signal Processing***	***CO 1.*** *Understand basic Signal Processing concepts and Find source of error and problem associated in Signal Processing, select Processor and Peripherals for signal Processing application.* ***CO 4.*** *Understand the theory and Mathematics of various signal processing terms, algorithms and Transforms. Apply it and interpret the result, simulate and implement it.* *PrO 5. Identify the presence/occurrence of Aliasing Phenomenon in MATLAB environment and suggest and apply appropriate solution in terms choice of sampling frequency and specification of antialiasing filter.* *PrO 6. Code various Mathematical equations & Operation in MATLAB, that can be useful in implementing signal processing equations/transforms/ algorithms. Also determine numbers of operations, time and memory requirement.* *PrO 9. Test- Verify and choose various concepts/properties related to Signal Processing algorithms for and in given application.* *PrO 10. Simulate Signal processing algorithms on image in MATLAB.*	1.1 *Basic block diagram of Digital processing system,* 1.2 *Sampling and Sampling Theorem for Reconstruction of signal in Digital domain, Aliasing and Solution, Re-construction Filter,* 1.3 *Errors and noise,* 1.4 *Mathematical and graphical representation of various signals and basic operations over it,* 1.5 *Various Discrete systems and their characterization, Convolution and Correlation.* 1.6 *Introduction to DSP Processor.*

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	1a. Justify the need of proper sampling rate and decide it for given case. 1b. Identify source of error and Quantify various error in processing the signal on a given hardware. 1c. Design Antialiasing Analog Filter. 1d. Code basic signal processing math like signal statistics, Integration, Differentiation, Interpolation, Time Shifting Convolution, correlation etc.	
Unit– II **Z Transform and Realization of Discrete System**	**CO 2**. Design and realize/implement Discrete system from time domain equation and S domain transfer function. **CO 3**. Design and realize/implement Digital systems like Digital Compensator and Digital Filters. PrO 3. Realize the given electrical and Mechanical engineering systems on digital platform and Verify the response. PrO 4. Design a digital close loop system and observe-verify the response after incorporating the Digital Compensator (P, PI, PID, Lag Lead Compensator). 2a. Devise Discrete systems from continuous equations or system. 2b. Design digital compensator for a close loop system. 2c. Determine the stability of Discrete System.	2.1 Definition and ROC for Z Transform, 2.2 S plane to Z plane mapping using various techniques like impulse invariant and Bilinear Transformation, 2.3 Inverse Z Transform, 2.4 Realization of IIR& FIR Systems (Direct, Cascade and Parallel form Realization), 2.5 Discrete Integrator and Differentiator
Unit– III **Fourier Analysis**	**CO 4**. Understand the theory and Mathematics of various signal processing terms, algorithms and Transforms. Apply it and interpret the result, simulate and implement it. PrO 8. Implement Signal Processing algorithms for power system and Power Electronics applications in MATLAB, dSPACE Real time digital simulator and Microcontroller. PrO 9. Test- Verify and choose various concepts/properties related to Signal Processing algorithms for and in given application.	3.1 Continuous time and Discrete time Fourier Series (CFS and DFS) and their properties, 3.2 Discrete Time Fourier Transform(DTFT), 3.3 Discrete Fourier Transform(DFT), 3.4 Computational Advantage of Fast Fourier Transform(FFT) Over DFT, Bit Reversed ordering, 3.5 Decimation in time (DIT)

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	PrO 10. Simulate Signal processing algorithms on image in MATLAB. 3a. Code the Discrete Fourier Transform, draw spectrogram/harmonic spectrum, of signal and calculate numbers of operation required in DFT Operation. 3b. Determine the Sampling frequency for extraction of harmonic components. 3c. Write equation of each stage of FFT for given total N point sample set (Using DIT and DIF butterfly diagram, means without deriving the equations), and determine numbers of operation required. 3d. Implement the DFT, STFT and FFT on Micro-controller 3e. Judge the performance of Fourier Transform for power system signals.	& Decimation in frequency (DIF) Algorithm for implementation of FFT.
Unit-IV **Filter Design**	**CO 3**. Design and realize/implement Digital systems like Digital Compensator and Digital Filters. PrO 5. Design various IIR and FIR Filters with Hand calculation and Excel, Python and MATLAB. PrO 6. Simulate and Verify the operation of various IIR and FIR filters in MATLAB, dSPACE and Microcontroller. PrO 7. Design and Simulate Multi-resolution Wavelet Filter bank and Wavelet Packet Filter banks in MATLAB. PrO 8. Implement Signal Processing algorithms for power system applications in MATLAB, dSPACE Real time digital simulator and Microcontroller. PrO 9. Test- Verify and choose various concepts/properties related to Signal Processing algorithms for and in given application. PrO 10. Simulate Signal processing algorithms on image in MATLAB. 4a. Select the specification of filter for filtering the signal.	4.1 Introduction to Digital Filters and their classification, Applications, 4.2 Design of IIR and FIR Filters using various methods, their comparison. 4.3 Introduction to adaptive Filtering concepts. 4.4 Multi-rate Signal Processing: Concept and Applications, decimation, Interpolation, Sampling rate conversion by rational factor, Multi-rate filter banks.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	4b. Design different types of IIR Filter for given specification. *4c. Design different types of FIR filter for given specification.* *4d. Implement the IIR and FIR Filter on Microcontroller and dSPACE Real Time Simulator.* *4e. Decide if application really requires adaptive filtering.* *4f. Implement Multi rate Filters.*	

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction to Signal Processing	10	1	3	2	2	2	0	10
II	Z Transform and Realization of Discrete System	10	2	4	10	2	2	0	20
III	Fourier Analysis	10	2	4	10	2	2	0	20
IV	Filter Design	15	10	4	12	1	1	0	30
Total		**45**	**15**	**15**	**34**	**7**	**7**	**0**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. It should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

Just to give idea, A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here:

1) **Object identification**: Review various object identification methods. Write Matlab code and C code that is based on related mathematics or algorithms (Don't use ready libraries available in software) and verify the result for various images and videos.
2) **Posture identification**: Identify, alarm and announce who is sleeping (or degree on concentration and involvement) in class (where lectures are conducted) . Use camera and code image processing algorithms for the same. (CO 2,4)
3) **Harmonic analyzer**: code STFT and wavelet in microcontroller and give display on computer screen or GLCD.(co 3)
4) **Wavelet based Power Quality measurement**: Download research papers for Wavelet based Power quality indices and implement it in microcontroller. (CO 3,4)
5) **Impedance Relay:** Use Fourier transform (STFT) to find fundamental component of voltage current and implement the Impedance relay on Microcontroller. (CO 4)
6) **VHDL/Verilog implementation of Specialized DSP algorithm**: Select the algorithm and code on FPGA. The algorithm selected should be much more complex to be executed by microcontroller in real time and hence the use of FPGA should be justified. (CO 1,2,3,4)

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

The teacher can use following strategies, to accelerate the attainment of the various types of learning outcomes in this course:

a) Tutorials that need hand calculation and testing in Software.
b) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance may to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) The teacher can suggest several research paper to read and refer. This can be assign to specially to evaluate the capacity of individual, to understand topics related to signal processing but not in syllabus. The capacity can be judge by the way or quality of efforts put, the simulation developed to verify the algorithm and brain storming done and presented to evaluator in form of presentation.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Make a report on new signal processing algorithms, their application in Power Engineering and their benefit over classical Research publication. You may refer IEEE Transaction on signal processing and similar high impacting journals for the same.
b) Identify a hypothetical non existing as well as existing phenomenon which can fail certain signal processing algorithm.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Introduction to Signal Processing	1	*PrO 1. Identify the presence/occurrence of Aliasing Phenomenon in MATLAB environment and give appropriate solution in terms choice of sampling frequency and specification of antialiasing filter. (1 hour)* *1a. <u>Get</u> overall idea of signal Processing hardware.* *1b. <u>Know</u> Role of Signal Processing in various engineering domain, to explore the application of this field, which can help student to identify and plan his/her dream signal processing related job.* *1c. <u>Justify</u> the need of proper sampling rate and calculate/Decide it.* *1d. <u>Design</u> Antialiasing Analog Filter and Reconstruction Filter.*	*Lecture + Experimentation*	*Mixture of black board and ICT*
	2	*PrO 2. <u>Code</u> various Mathematical equations & Operation in MATLAB, that can be useful in implementing signal processing equations/transforms/ algorithms. Also <u>determine</u> numbers of operations, time and memory requirement. (2 Hours)* 1e. *<u>Determine</u> source of problem and <u>quantify</u> various error involved in processing the signal on a given hardware.*	*Lecture + Experimentation + Solved/Unsolved Tutorials from Resource Person.*	*Mixture of black board and ICT*
	3	**PrO2. Continued. For 2 Hours. The outcomes will be achieved by end of this week.** *2a. <u>Devise</u> Discrete Systems from continuous equations and systems.*	*Lecture + Experimentation + Solved/Unsolved Tutorials from Resource Person.*	*Mixture of black board and ICT*
Unit– II Z Transform and Realization of Discrete	**4**	*PrO 3. <u>Realize</u> the Given electrical and Mechanical engineering systems on digital platform and Verify the response. (2 Hours)* *2b. <u>Design</u> digital compensator for a close loop system.*	*Lecture + Experimentation + Solved/Unsolved Tutorials from Resource Person.*	*Mixture of black board and ICT*

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
System	5	**PrO3. (Continued from last lab session for 2 Hours). Outcome will be achieved by end of this week.** *PrO 4. Design a digital close loop system and observe-verify the response after incorporating the Digital Compensator (P, PI, PID, Lag Lead Compensator).* *2c. Determine the stability of Discrete System.*	*Lecture + Experimentation + Solved/Unsolved Tutorials from Resource Person.*	*Mixture of black board and ICT*
	6	*PrO 8. implement Signal Processing algorithms for power system and Power Electronics applications in MATLAB, dSPACE Real time digital simulator and Microcontroller.* **(Part 2: The DFT FFT based application for Power system)** *3a. Code the Discrete Fourier Transform, draw spectrogram and harmonic spectrum, and calculate numbers of operation required.* *3b. Determine the Sampling frequency for extraction of harmonic components using DFT.*	*Lecture + Experimentation*	*Mixture of black board and ICT*
Unit– III Fourier Analysis	7	**Note: Lab hours will be utilized in this week for lecture.** *3c. Draw out equation of each stage of FFT for given total N point sample set (Using DIT and DIF butterfly diagram, means without deriving the equations), and determine numbers of operation required.* *3d. Judge the performance of Fourier Transform for given signal (e.g for power system signals) .* *3e. Decide if FT based algorithm can be used in required application or not.* *3f. Find out time domain information from frequency domain information.*	*Lecture only.*	*Mixture of black board and ICT*
	8	**PrO8. Part 1: Statistical equation, correlation and other such algorithm based applications (Continued from week 6 for 1 hour)** *3a. Find out time domain information from frequency domain information.* *3b. Implement the DFT, STFT and FFT on Micro Controller*	*Lecture + Experimentation.*	*Mixture of black board and ICT*
	9	**Note: Lab hours in this week will be utilized for lecture.** *4a. Select the specification of filter for filtering the signal.* *4b. Design different types of IIR Filter for given specification.*	*Lecture only.*	*Mixture of black board and ICT*

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– IV Filter Design	**10**	**Note: Lab hours in this week will be utilized for lecture.** **4b. continued…. Outcome will be achieved by end of this week.**	*Lecture + Experimentation*	*Mixture of black board and ICT*
	11	**Note: 1 Hour of lecture will be utilized in laboratory during this week.** *PrO 5. Design various IIR and FIR Filters with Hand calculation and Excel, Python and MATLAB.* **(Part 1: Design of IIR Filters: 3 Hours)** *4c. Design different types of FIR filter for given specification.*	*Lecture*	*Mixture of black board and ICT*
	12	*PrO 6. Simulate and Verify the operation of various IIR and FIR filters in MATLAB, dSPACE and Microcontroller.* *4d. Implement the IIR and FIR Filter on Microcontroller and dSPACE Real Time Simulator.*	*Lecture + Experimentation*	*Mixture of black board and ICT*
	13	*PrO 7. Design and Simulate Multi-resolution Wavelet Filter bank and Wavelet Packet Filter banks in MATLAB.* *4e. Decide if application really requires adaptive filtering.* *4f. Implement Multi rate Filters.*	*Lecture + Experimentation*	*Mixture of black board and ICT*
	14	**PrO8. Part 3: Wavelet based algorithm for power system application (Continued from week 8, for 1 hour). The outcome of PrO 8 will be achieved by end of this week.** **4f. continued.** **The outcome of 4f will be achieved by end of this week.**	*Lecture + Experimentation + Resource Person's Material*	*Mixture of black board and ICT*
	15	**Note: Lecture hours in this week will be utilized for lab.** *PrO 9. Test-Verify and choose various concepts/properties related to Signal Processing algorithms for and in given application.* *PrO 10. Simulate Signal processing algorithms on image in MATLAB.*	*Experimentation only.*	*Mixture of black board and ICT*

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	*Digital Signal Processing*	*Anand Kumar*	*PHI Learning Private Ltd. ISBN-978-81-203-5071-7*
2	*"Real Time Digital*	*Udayashankara*	*Prentice-Hall Of India Pvt Limited, 2010*

S. No.	Title of Book	Author	Publication including ISBN
	Signal Processing: Fundamentals, Algorithms and Implementation using TMS Processor"		*ISBN-10* ***8120340493 :*** *ISBN-13 : 978-8120340497*
3	*Digital Signal Processing*	*Ramesh Babu*	*SCITECH Publications* *ISBN-10* ***818371630X :*** *ISBN-13* ***978-8183716307 :***
4	*Digital Signal Processing Principles", Algorithms and Application*	*John Proakis, Dimitris G Manolakis*	*PHI, 3rd Edition (2000).* *ISBN-10* ***9788131710005 :*** *ISBN-13 : 978-8131710005*

XV SOFTWARE/LEARNING WEBSITES

1. *nptel video cource by prof.s.c. duttaroy, iitdelhi*
   http://nptel.iitm.ac.in/video.php?subjectid=117102060
2. *DSP Video Lectures by Prof. E. Ambikairajah:*
   http://onlinevideolecture.com/?course_id=373
3. A beginner's guide to digital signal processing
   http://www.analog.com/en/processorsdsp/processors/beginners_guide_to_dsp/fca.html

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE 346	**Semester VI**	**Protection and Switchgear** **Mark 'H'=3** for high, **'M'**=2 for medium or **'L'=1'** for the relevant correlation of each PO or PSO, Competency or CO													
346.1	*Understand basic Signal Processing concepts, Find source of error and problem associated in Signal Processing, Select Processor and Peripherals for signal Processing*	***2***	***2***	***1***	-	-	-	-	***1***	-	***2***	-	-	***1***	-

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	application.														
346.2	Design and realize/implement Discrete system from time domain equation and S domain transfer function.	2	3	2	2	3	-	-	1	-	2	-	1	3	-
346.3	Design and realize/implement Digital systems like Digital Compensator and Digital Filters.	2	3	3	2	3	-	-	1	-	2	-	1	3	-
346.4	Understand the theory and Mathematics of various signal processing terms, algorithms and Transforms. Apply it and interpret the result, simulate and implement it.	2	3	3	3	3	-	-	2	2	3	2	2	-	3

XVII NAMES OF RESOURCE PERSONS

a) Jivanadhar A. Joshi

Course Code: **EE347**
Course Title: **Programmable Logic Controllers and Industrial Automation**

UG Engineering Programme	Semester
Electrical Engineering	Sixth

I RATIONALE

In modern industry all machines are connected with each other and they operate automatically. This automated system requires PLCs, HMI and SCADA system to operate satisfactorily. This curriculum course structure is developed such that the student is able to understand basic functions of PLCs, Sensor and Actuators; and can design small scale automated systems.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- Understand and Evaluate the operation of automated systems in industry

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) ***Enhance*** skills in basics of PLC
2) ***Know*** the performance and classification of Input and Output Devices.
3) ***Understand*** different Programming language used in PLC
4) ***Design*** any automate control system for any machinery
5) ***Program*** PLC and apply their knowledge in the field of automation and control

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme (Anna 2021)				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	
2	0	4	4	70	30	50	50	200

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

(*): *Under the* ***Theory PA****, out of 30 marks,* ***15 marks*** *are for* ***micro-project assessment*** *to facilitate integration of Cos and the remaining 15 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Understand basics of PLC and SCADA	4
2	2	Understand different programming language used to program PLC and merits and demerits of each language.	6
3	3	Gain knowledge of sensors and actuators.	4
4	4,5	Implement logic gates in Ladder language and on PLC (AND, OR, NOT, EX-OR, EX-NOR, NAND, NOR)	6
5	4,5	Design and implement motor latch and unlatch Ladder on PLC (with and without using instruction)	4
6	4,5	Design and implement Liquid level control of tank using PLC	4
7	4,5	Design and implement flashing of Lamp on PLC.	4
8	4,5	Design and implement traffic light control sequence using ladder logic on PLC.	4
9	4,5	Design and implement flashing of Lamp 10 times on PLC.	4
10	4,5	Design and implement Pattern of Lamp based on ADC (battery level indicator).	4
11	4,5	Design and implement flashing of LED with adjustable ON time	4
12	4,5	Interfacing of RTD with PLC.	4
13	4,5	Implement SCADA screens with PLC programs.	8
		Total	**60**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Schneider Modicon M340 (PLC Controller) with its digital and analog peripherals(DDM16025,AMI0800, ART0414,AMO0410)	1,3,4,5,6,7,8, 9,10,11,12,13
2	Unity pro XL (any version)	2, 4,5,6,7,8, 9,10,11,12,13
3	Push Button, Toggle Switch	4,5,6,7,8, 9,10,11,12,13
4	LEDs	4,5,6,7,8, 9,10,11,12,13
5	10K POT with Digital Meter	10,11
6	Programmable Digital Voltmeter	10,11
7	RTD	12

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Basics of Control	**CO 1. *Enhance* skills in basics of PLC** 1a. Differentiate control scheme 1b. Evaluate the control strategy for the specified system 1c. Differentiate Relay logic and PLC logic 1d. Differentiate Analog , Automation and Sequential Logic 1e. Understand evolution in control strategy	1.1 Introduction to Sequential/ Logic Control 1.2 control strategy, control philosophy and control algorithm, 1.3 Difference between Relay logic control and PLC 1.4 Difference between Analog/ Automatic and Sequential/ Logic Control 1.5 Evolution of Control System
Unit– II Introduction to Programmable Logic Control (PLC)	**CO 1. *Enhance* skills in basics of PLC** *PrO 7. Understand basics of PLC and SCADA* 2a. Evaluate performance of PLC for given Application 2b. Understand need of PLC 2c. Evaluate operation of given PLC 2d. Justify need of PLC for given application	2.1 Definition of PLC, introduction to standard (IEC61131) used for PLC and basics of process automation 2.2 History and Evolution of PLC

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
		2.3 Block Diagram of PLC 2.4 Advantages and Disadvantages of PLC
Unit– III **Input and Output Devices**	**CO 2. *Know the performance and classification of Input and Output Devices.*** *PrO 3. Gain knowledge of sensors and actuators* 3a. Evaluate operation of sensors for given application. 3b. Select the transmitter wiring under given situation. 3c. Evaluate operation of output device for given application. 3d. Evaluate operation of control valve for given application.	3.1 A brief overview of sensors and its types, Overview of Transmitter 3.2 difference between two wire Transmitter and four wire transmitter 3.3 output devices connected to PLC 3.4 Control Valves and its classification
Unit-IV **PLC input and Output Module**	**CO 2. *Know the performance and classification of Input and Output Devices.*** *PrO 1. Understand basics of PLC and SCADA* *PrO 3. Gain knowledge of sensors and actuators* *4a.* Select the power supply for the given application *4b.* Select the appropriate input and output module for given application	4.1 Power supply circuit/Wiring Diagram, Selection of power supply 4.2 Function of input and output modules, Classification and block diagram of Input and output Modules(AC input module, Discrete/Digital input module ,Special input module cards TTL Logic, Relay and Triac output module)and selection of cards/modules
Unit – V **PLC Operation**	***CO 5. Program PLC and apply their knowledge in the field of automation and control*** *PrO 1. Understand basics of PLC and SCADA* 5a. Understand type of memory used in the given scenario 5b. Prepare addressing table for the given application 5c. Determine the operation of PLC in the	5.1 PLC memory types and its mapping, PLC Register basics 5.2 Addressing: Internal and External 5.3 PLC Scan Cycle and

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	given situation	response time
Unit – VI PLC Programming	***CO 3. Understand different Programming language used in PLC*** ***CO 5.Program PLC and apply their knowledge in the field of automation and control*** *PrO 4. Implement logic gates in Ladder language and on PLC (AND, OR, NOT, EX-OR, EX-NOR, NAND, NOR)* *PrO 5. Design and implement motor latch and unlatch Ladder on PLC (with and without using instruction)* *PrO 6. Design and implement Liquid level control of tank using PLC* *PrO 7. Design and implement flashing of Lamp on PLC.* *PrO 8. Design and implement traffic light control sequence using ladder logic on PLC.* *PrO 9. Design and implement flashing of Lamp 10 times on PLC.* *PrO 10. Design and implement Pattern of Lamp based on ADC (battery level indicator).* *PrO 11. Design and implement flashing of LED with adjustable ON time* *PrO 12. Interfacing of RTD with PLC.* *6a. Understand types of programming language available and use any of the language to implement the given application* *6b. Evaluate operation of basic block of ladder logic for the given program* *6c. Design* ladder logic program for the specified application.	6.1 Types of programming languages 6.2 Various PLC Instruction: NO, NC and output contacts and coils, Set, Reset, Timer (along with types), Counters (its types), PID, Logical, Arithmetic, Data handling, skip, MCR, jump, Bit operations and move instructions 6.3 Programming related to above instructions
Unit – VII Designing Automation System	***CO 4. Design any automated control system for any machinery*** ***CO 5. Program PLC and apply their knowledge in the field of automation and control*** *PrO 13.* Implement SCADA screens with PLC	

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	programs. 7a. Select PLC for the given application/machinery. 7b. Design and Prepare detailed document for the given application/machinery. 7c. Implement automated system for the given application/machinery.	7.1 Selection of PLC 7.2 Documentation: System architecture, Piping and Instrumentation Diagram (P&ID) (Flow sheet symbols as per standard ISA S5.1 1984(R1992)), General Arrangement (GA) Drawing, Wiring Diagram, I/O Listing and program flowchart, Debugging (Simulation) 7.3 Commissioning, troubleshooting and maintenance of PLC system

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	**Unit Title**	**Teaching Hours**	**Distribution of Theory Marks**						
			R Level	**U Level**	**A Level**	**An Level**	**E Level**	**C Level**	**Total Marks**
I	Basics of Control	02	3	2	0	0	0	0	5
II	Introduction to Programmable Logic Control (PLC)	02	3	2	0	0	0	0	5
III	Input and Output Devices	03	2	3	1	1	0	0	7
IV	PLC input and Output Module	04	3	2	1	3	0	0	9
V	PLC Operation	03	1	0	2	2	2	0	7
VI	PLC Programming	10	2	5	2	4	6	4	23
VII	Designing Automation System	06	0	0	1	4	6	3	14
Total		**30**	**14**	**14**	**7**	**14**	**14**	**7**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **Temperature Monitoring System:** Implement Temperature control of liquid in tank using PLC.
2) **Speed control of Induction Motor:** Control speed of induction motor using VFD. PLC will act as remote unit commanding/controlling VFD to control speed of motor.
3) **Automatic bottle filler:** Implement bottle filler using PLC. Where, each bottle will position automatically under dispenser and will be filled automatically without any human intervention.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical

evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different protection schemes used in industries.
b) Prepare 15 min. power point presentation on how to implement control strategy for the machinery in particular industry.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Basics of Control	1	1a. Differentiate control scheme 1b. Evaluate the control strategy for the specified system 1c. Differentiate Relay logic and PLC logic 1d. Differentiate Analog , Automation and Sequential Logic 1e. Understand evolution in control strategy	Lecture with media	PPTs, black board
Unit– II Introduction to Programmable Logic Control (PLC)	**2**	*PrO 1. Understand basics of PLC and SCADA* 2a. Evaluate performance of PLC for given Application 2b. Understand need of PLC 2c. Evaluate operation of given PLC 2d. Justify need of PLC for given application	Lecture with media	PPTs, black board
Unit– III Input and Output Devices	**3**	*PrO 3. Gain knowledge of sensors and actuators* 3a. Evaluate operation of sensors for given application. 3b. Select the transmitter wiring under given situation. 3c. Evaluate operation of output device for given application.	Lecture with media	PPTs, black board
	4	*PrO 3. Gain knowledge of sensors and actuators* 3d. Evaluate operation of control valve for given application.	Lecture with media	PPTs, black board
Unit-IV PLC input and Output Module	**4**	*PrO 1. Understand basics of PLC and SCADA* 4a. Select the power supply for the given application	Lecture with media	PPTs, black board
	5	*PrO 1. Understand basics of PLC and SCADA* *PrO 3. Gain knowledge of sensors and actuators* 4a. Select the power supply for the given application 4b. Select the appropriate input and output module for given application	Lecture with media	PPTs, black board
	6	*PrO 3. Gain knowledge of sensors and actuators* 4b. Select the appropriate input and output module for given application	Lecture with media	PPTs, black board
Unit – V PLC	**6**	*PrO 1. Understand basics of PLC and SCADA*	Lecture with media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Operation		5a. Understand type of memory used in the given scenario		
	7	*PrO 1. Understand basics of PLC and SCADA* 5a. Prepare addressing table for the given application 5b. Determine the operation of PLC in the given situation	Lecture with media	PPTs, black board
Unit – VI PLC Programming	**8**	*PrO 4. Implement logic gates in Ladder language and on PLC (AND, OR, NOT, EX-OR, EX-NOR, NAND, NOR)* 6a. Understand types of programming language available and use any of the language to implement the given application *6b.* Evaluate operation of basic block of ladder logic for the given program	Lecture with media	PPTs, black board
	9	*PrO 5. Design and implement motor latch and unlatch Ladder on PLC (with and without using instruction)* *PrO 6. Design and implement Liquid level control of tank using PLC* 6b. Evaluate operation of basic block of ladder logic for the given program *6c.* Design ladder logic program for the specified application.	Lecture with media	PPTs, black board
	10	*PrO 7. Design and implement flashing of Lamp on PLC.* *PrO 8. Design and implement traffic light control sequence using ladder logic on PLC.* 6b. Evaluate operation of basic block of ladder logic for the given program *6c.* Design ladder logic program for the specified application.	Lecture with media	PPTs, black board
	11	*PrO 9. Design and implement flashing of Lamp 10 times on PLC.* *PrO 10. Design and implement Pattern of Lamp based on ADC (battery level indicator).* 6b. Evaluate operation of basic block of ladder logic for the given program *6c.* Design ladder logic program for the specified application.	Lecture with media	PPTs, black board
	12	*PrO 11. Design and implement flashing of LED with adjustable ON time* *PrO 12. Interfacing of RTD with PLC.* 6b. Evaluate operation of basic block of ladder logic for the given program *6c.* Design ladder logic program for the specified application.	Lecture with media	PPTs, black board
Unit – VII	**13**	*PrO 13. Implement SCADA screens with PLC*	Lecture	PPTs, black

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Designing Automation System		*programs.* 7a. Select PLC for the given application/machinery.	with media	board
	14	*PrO 13. Implement SCADA screens with PLC programs.* 7b. Design and Prepare detailed document for the given application/machinery.	Lecture with media	PPTs, black board
	15	*PrO 13. Implement SCADA screens with PLC programs.* 7c. Implement automated system for the given application/machinery.	Lecture with media	PPTs, black board

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Programmable Logic Controllers	John W. Webb and Ronald A. Reis	PHI Publication 978-8120323087
2	Programmable Logic Controllers	W. Bolton	Newnes 978-0750681124
3	Programmable Logic Controllers & their Engineering Application	A.J. Crispin	McGraw Hill 978-0077093174
4	Programmable Logic Controllers	Thomas A. Hughes	ISA 978-1556178993
5	Programmable Logic Controllers	John R. Hackworth	Pearson Education 978-0130607188
6	Exploring Programmable Logic Controllers with Applications	Pradeep Kumar Srivastava	BPB Publications 978-8176569330
7	Programmable Logic Control Principles and Applications	NIIT	PHI Publication 978-8120325258

XV SOFTWARE/LEARNING WEBSITES

2. http://coep.vlab.co.in/?sub=33&brch=97
3. http://www.plcdev.com/book/export/html/9
4. http://www.plcmanual.com/
5. http://nptel.ac.in/courses/108105062
6. http://nptel.ac.in/courses/108105088
7. http://nptel.ac.in/courses/108105063
8. www.siemensplcweb.com
9. http://www.ab.com/programmablecontrol/plc

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE347	**Semester VI**	**Programmable Logic Controllers and Industrial Automation** **Mark 'H'=3** for high, '**M**'=2 for medium or **'L'=1'** for the relevant correlation of each PO or PSO, Competency or CO													
347	*Understand and Evaluate the operation of automated systems in industry*	***1***	***2***	***3***	***1***	***3***				***1***			***1***	***3***	***1***
347.1	***Enhance*** skills in basics of PLC	3													
347.2	***Know*** the performance and classification of Input and Output Devices.	3												1	
347.3	***Understand*** different Programming language used in PLC	2													
347.4	***Design*** any automate control system for any machinery		3	3	3	2				2				3	
347.5	***Program*** PLC and apply their knowledge in the field of automation and control		3	3	3	2				2				3	

XVII NAMES OF RESOURCE PERSONS

a) Jignesh S Patel
b) ………………………………………………….

Course Code: **EE348**
Course Title: **POWER ELECTRONICS AND DRIVES-II**

UG Engineering Programme	Semester
Electrical Engineering	sixth

I RATIONALE

Electrical engineers have to deal with various types of *semiconductor devices*, power electronics circuits and converters are used for different dc drives. This curriculum course structure is developed such that the student is able to evaluate the performance of the various *semiconductor devices,* types of power electronic converters and drives for different industry drive applications.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Evaluate the performance of various *SEMICONDUCTOR DEVICES*, TYPES OF POWER ELECTRONICS CIRCUITS AND AC DRIVES**.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1. *Evaluate the performance of the* various types of INVERTERS for different application
2. *Evaluate the performance of the* various types of power SUPPLIES for different application
3. *Evaluate the performance of the various types of AC Voltage controllers for different application*
4. *Evaluate the performance of the various types of Cycloconverters for different application*
5. *Evaluate the performance of the various types of AC Drives for different application*

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	
3	0	2	4	70	30*	25	25	150

***Legends: L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

***(*):** Under the **Theory PA**, out of 40 marks, **20 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1,2	Create the simulation of a single phase voltage source inverter and Current source inverter for different load condition using MATLAB software.	02
2	1,2	Create the simulation of 180°, 120˚ and 150 mode of conduction for 3-Φ Voltage source Inverter using MATLAB software.	04
3	3	Analyze the waveform of 180°, 120˚ and 150 mode of conduction for 3-Φ Voltage source Inverter using MATLAB	02
4	1,2	Create the simulation of Unipolar and Bi- polar Sine PWM (SPWM) technique using MATLAB software.	02
5	3	Analyze the waveform of Unipolar and Bi- polar Sine PWM (SPWM) technique in MATLAB.	02
6	1,2	Create the simulation of half bridge topology of SMPS using MATLAB software.	02
7	1,2	Create the simulation of full bridge topology of SMPS using MATLAB software.	02
8	1,2	Create the simulation of uninterruptible power supply (UPS) using MATLAB software.	02
9	1,2	Create the simulation of different types of AC voltage controller using MATLAB software.	04
10	3,4	Evaluate the Performance of different types of AC voltage controller for different load condition and firing angle delay using MATLAB software.	02
11	1,2	Create the simulation of step-up and step-down single- phase Cycloconverters using MATLAB software.	04
12	3	Analyze the Performance of the operation of variable frequency drive on hardware.	02
		Total	**30**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB Simulation software (version 2021a)	1,2,3,4,5,6,7,8,9,10,11
2	variable frequency drive hardware kit	12
3	DSO with isolation and its probes	12
4	Multimeter	12

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Inverters	**CO 1.*Evaluate the performance of the various types of INVERTERS for different application*** *PrO 1. Create the simulation of a single phase voltage source inverter and Current source inverter for different load condition using MATLAB software.* *PrO 2. Create the simulation of 180°, 120° and 150 mode of conduction for 3-Φ Voltage source Inverter using MATLAB software.* *PrO 3. Analyze the waveform of 180°, 120°*	1.1 Introduction and revision: Overview of switches 1.2 Introduction to inverters Voltage Source Inverter : Single-phase half bridge and full bridge inverter with different loads 1.3 Three-phase 180 and 120 degree mode inverter 1.4 Fourier analysis of Single-phase and Three-phase inverter 1.5 Output Voltage control in

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	and 150 mode of conduction for 3-Φ Voltage source Inverter using MATLAB. *PrO 4. Create the simulation of Unipolar and Bi- polar Sine PWM (SPWM) technique using MATLAB software.* *PrO 5. Analyze the waveform of Unipolar and Bi- polar Sine PWM (SPWM) technique in MATLAB.* 1a. Justify the need of three phase and single phase voltage source inverter 1b. Justify the need of three phase and single phase current source inverter. 1c. Evaluate the operation of inverters for different load condition. 1d. Determine switches used for inverters 1e. Choose Harmonic reduction techniques in inverter	inverters 1.6 Harmonic reduction techniques in inverter output voltage 1.7 Current Source Inverter: Concept of CSI, CSI with different loads 1.8 Load and Self-commutated inverter: Single-phase Series Inverter and Parallel Inverter
Unit– II Power Supplies	**CO 2.** *Evaluate the performance of the various types of power SUPPLIES for different application* *PrO 6. Create the simulation of half bridge topology of SMPS using MATLAB software.* *PrO 7. Create the simulation of full bridge topology of SMPS using MATLAB software.* *PrO 8. Create the simulation of uninterruptible power supply (UPS) using MATLAB software.* *2a.* Justify the need of SMPS 2b.Justify the need of UPS 2c. Evaluate the operation of different SMPS	2.1 DC Power Supplies: Switched-Mode DC Power Supplies, 2.2 Half bridge converter, Full bridge converter, Fly back converter, Push-pull converter 2.3 AC Power Supplies: Uninterruptible Power Supplies (UPS) , Batteries used in UPS 2.4 Static switches and Solid State relays
Unit– III AC Voltage controllers	**CO 3** ***Evaluate the performance of the various types of AC Voltage controllers for different application*** *PrO 9. Create the simulation of different types of AC voltage controller using MATLAB software* *PrO 10. Evaluate the Performance of different types of AC voltage controller for different load condition and firing angle delay using MATLAB software* *3a. Evaluate the performance of different types of AC Voltage controllers for different load condition.*	3.1 Voltage control techniques: Principle of voltage controllers, Phase control, Integral Cycle control, 3.2 Single-phase Voltage controllers with R and RL load Sequence control of AC voltage controllers: Two-stage and Multistage sequence control of voltage controllers

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	3b. Justify the need of AC Voltage controllers	
Unit-IV Cycloconverters	**CO 4.*Evaluate the performance of the various types of Cycloconverters for different application*** *PrO 11.* Create the simulation of step-up and step-down single- phase Cycloconverters using MATLAB software 4a. Choose the appropriate cycloconverter control strategy as per required application 4b. Evaluate the performance of different types of cycloconverter for different load condition. 4c. . Justify the need of cycloconverter	4.1 Introduction: Principle and Applications of Cycloconverters, Types of Cycloconverters, 4.2 Single-phase to single-phase Cycloconverters: Bridge-type Cycloconverters, Midpoint Cycloconverters, 4.3 Three-phase Half-wave Cycloconverters: Three-phase to Single-phase, Threephase to Three-phase, Output voltage Equation of a Cycloconverter
Unit – V AC Drives	**CO 5. *Evaluate the performance of the various types of AC Drives for different application*** *PrO 12.* Analyze the Performance of the operation of variable frequency drive on hardware. 5a. Understand the operation of VFD Drive in different mode condition. 5b. Justify the need of *AC Drives for different application*	5.1 Basic machine equations: Analysis and Performance of AC motors, 5.2 Speed Control of three-phase Squirrel-cage Induction Motors (SCIMs): Stator Voltage Control, Stator Frequency Control, Stator Voltage and Frequency Control, Stator Current Control, 5.3 Speed Control of three-phase Wound Rotor Induction Motors (WRIMs): Static Rotor Resistance Control, Slip-energy recovery Control.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Inverters	15	0	3	3	10	7	0	23
II	Power Supplies	05	0	1	1	4	2	0	8
III	AC Voltage controllers	06	0	1	4	4	1	0	10
IV	Cycloconverters	07	0	2	2	4	6	0	14

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
V	AC Drives	12	1	2	2	4	6	0	15
Total		**45**	**1**	**9**	**12**	**26**	**22**	**0**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **Inverter:** Simulate the inverter for particular one application.
2) **Power supply:** Design and Develop the hardware of full wave SMPS.
3) **Ac voltage controller:** Design and Develop the hardware of **Ac voltage controller**
4) **Cycloconverter:** Design and Develop the hardware of **Cycloconverter.**
5) **Ac drive**: Design the simulation of AC drives in MATLAB.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given

to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.

d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different Semiconductor switches used in converters.
b) Prepare 15 min. power point presentation on how to choose the particular converter for the particular application.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Inverters	1	**PrO 1.** *Create the simulation of a single phase voltage source inverter and Current source inverter for different load condition using MATLAB software.* **PrO 2.** *Create the simulation of 180°, 120˚ and 150 mode of conduction for 3-Φ Voltage source Inverter using MATLAB software.* **PrO 3.** *Analyze the waveform of 180°, 120˚ and 150 mode of conduction for 3-Φ Voltage source Inverter using MATLAB* **PrO 4.** *Create the simulation of Unipolar and Bi- polar Sine PWM (SPWM) technique using MATLAB software.* **PrO 5.** *Analyze the waveform of Unipolar and Bi- polar Sine PWM (SPWM) technique in MATLAB.* 1a. Justify the need of three phase and single phase voltage source inverter 1b. Justify the need of three phase and single phase current source inverter. 1c. Evaluate the operation of inverters for different load condition. 1d. Determine switches used for inverters 1e. Choose Harmonic reduction techniques in inverter	Lecture with media	PPTs, black board, MATLAB Software

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– II Power Supplies	**6**	*PrO 6. Create the simulation of half bridge topology of SMPS using MATLAB software.* *PrO 7. Create the simulation of full bridge topology of SMPS using MATLAB software.* *PrO 8. Create the simulation of uninterruptible power supply (UPS) using MATLAB software.* *2a.* Justify the need of SMPS 2b.Justify the need of UPS 2c. Evaluate the operation of different SMPS	Lecture with media, and Demonstration	PPTs, black board, MATLAB Software
Unit– III AC Voltage controllers	**8**	*PrO 9. Create the simulation of different types of AC voltage controller using MATLAB software* *PrO 10. Evaluate the Performance of different types of AC voltage controller for different load condition and firing angle delay using MATLAB software* *3a. Evaluate the performance of different types of AC Voltage controllers for different load condition.* *3b.* Justify the need of *AC Voltage controllers*	Lecture with media, Field visit	PPTs, black board, MATLAB Software
Unit-IV Cycloconverters	**10**	*PrO 11.* Create the simulation of step-up and step-down single- phase Cycloconverters using MATLAB software 4a. Choose the appropriate cycloconverter control strategy as per required application 4b. Evaluate the performance of different types of cycloconverter for different load condition. 4c. . Justify the need	Lecture with media, Tutorial	PPTs, black board, MATLAB Software
Unit – V AC Drives	**15**	*PrO 12.* Analyze the Performance of the operation of variable frequency drive on hardware. *5a. Understand the operation of VFD Drive in different mode condition.* 5b. Justify the need of *AC Drives for different application*	Lecture with media, Tutorial	PPTs, black board, MATLAB Software, Practical demonstration

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1		Dr.P.S.Bimbhra	Khanna publishers

S. No.	Title of Book	Author	Publication including ISBN
	Power Electronics		
2	Power electronics	M D Singh and K B K Khanchandani,	Tata Mcgraw-Hill Publication
3	Power electronics handbook	by M.H. Rashid	Academic press
4	High power converters and AC drives	IEEE press	IEEE press
5	Power Electronics Converters, Applications and design	Mohan, Undeland, Robbins	Wiley publication

XV SOFTWARE/LEARNING WEBSITES

1). http://nptel.ac.in/courses/108101038/
2. http://cusp.umn.edu/power_electronics.php
3. http://ecee.colorado.edu/~ecen5797/notes.html

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE348	Semester VII														
348	**Competency** Evaluate the performance of various SEMICONDUCTOR DEVICES, TYPES OF POWER ELECTRONICS CIRCUITS AND AC DRIVES.	3	2	2	2	1	-	1	-	-	-	-	1	3	-

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
348.1	Evaluate the performance of the various types of INVERTERS for different application	3	2	1	-	2	-	-	-	-	1	-	-	-	-
348.2	Evaluate the performance of the various types of power SUPPLIES for different application	3	3	2	1	2	-	-	-	-	-	-	-	2	-
348.3	Evaluate the performance of the various types of AC Voltage controllers for different application	3	3	3	1	2	-	-	-	-	-	-	-	2	-
348.4	Evaluate the performance of the various types of Cycloconverters for different application	3	1	3	1	2	-	-	-	-	-	-	-	2	-
348.5	Evaluate the performance of the various types of AC Drives for different application	3	1	-	-	-	-	-	-	-	-	-	-	-	-

XVII NAMES OF RESOURCE PERSONS

a) Dr. Dharmesh Dabhi

Course Code: **EE349**
Course Title: **Fault Analysis and Switchgear**

UG Engineering Programme	Semester
Electrical Engineering	Sixth

I RATIONALE

Electrical engineers have to deal with various types of circuit breakers. Different types of circuit breakers are used for various applications for different voltage levels. This curriculum course structure is developed such that the student is able to evaluate the performance of the circuit breaker and calculate the different types of fault current at any location of power system.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Evaluate the performance of various circuit breakers employed for various applications in power system.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) ***Evaluate*** the symmetrical fault current and voltage at any location of power system.
2) ***Evaluate*** *the un*symmetrical fault current and voltage at any location of power system with mathematical model and with software tools.
3) **Interpret** the *phenomena of current interruption in circuit breaker.*
4) ***Evaluate*** the different parameters for arc formation and arc interruption in circuit breaker.
5) ***Choose*** *the type of circuit breaker for different applications***.**

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	
4	0	2	5	70	30*	25	25	150

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

(*): *Under the* ***Theory PA****, out of 30 marks,* ***10 marks*** *are for* ***micro-project assessment*** *to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken*

during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Formulate the mathematical model for the given power system network.	02
2	1	Find the power system symmetrical fault current for the given power system network.	02
3	2	Find the sequence impedances (Positive, Negative and Zero-sequence) of the three-phase transformer.	02
4	2	Find the sequence impedances (Positive, Negative and Zero-sequence) of the three-phase Synchronous machine.	02
5	2	Verification of sequence network for the given connections of 3-phase transformer.	02
6	2	Develop the mathematical model of symmetrical components for the given power system network.	02
7	2	Find the power system unsymmetrical fault current for the given power system network.	02
8	2	Simulate the power system unsymmetrical fault current for the given power system network.	02
9	3	Classify the fault clearing process and problems associated with fault clearing process for the given conditions.	02
10	4	Evaluate the arc extinction process for the given circuit breaker.	02
11	5	Compare the performance of air break, air blast and SF6 for the given data.	02
12	5	Evaluate the applications of vacuum and direct current circuit breaker for the given voltage levels.	02
		Total	**24**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Power World Simulation software	2,7,8
2	Air break circuit breaker	11
3	Test rig of fault analysis	2
4	Test rig of 3 single-phase transformer.	3,5
5	Test rig of three phase synchronous generator.	4

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Per Unit System	**CO 1. *Evaluate* the symmetrical fault current and voltage at any location of power system.** *PrO 1.* Formulate the mathematical model for the given power system network. 1a. Develop the impedance and reactance diagram for the given power system network. 1b. Determine the per unit values of reactances of equipment for the given base.	1.1 Single phase solution to three phase system, one line diagram, impedance 1.2 and reactance diagram 1.3 Per unit system and examples

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit– II Symmetrical Fault Analysis	**CO 1. *Evaluate* the symmetrical fault current and voltage at any location of power system.** PrO 2. Find the power system symmetrical fault current for the given power system network. 2a. Determine the making current of the circuit breaker for the specified fault current levels. 2b. Determine the 3-phase short circuit current for the given power system network. 2c. Choose the correct value of synchronous machine reactance for fault current calculation for the given condition.	2.1 sudden short circuit on R-L series circuit 2.2 3 – phase short circuit current calculation, examples. 2.3 Sub-transient, transient and steady state model of synchronous machine
Unit– III Symmetrical Components	**CO 2. *Evaluate* the unsymmetrical fault current and voltage at any location of power system with mathematical model and with software tools.** PrO 3. Find the sequence impedances (Positive, Negative and Zero-sequence) of the three-phase transformer. PrO 4. Develop the mathematical model of symmetrical components for the given power system network. PrO 5. Find the sequence impedances (Positive, Negative and Zero-sequence) of the three-phase Synchronous machine. PrO 6. Verification of sequence network for the given connections of 3-phase transformer. 3a. Find the symmetrical components for the given phase values. 3b. Develop sequence networks form the given one line diagram of power system. 3c. Justify the phase shift to be consider for the specified transformer connection for unsymmetrical fault calculation.	3.1 Symmetrical Component transformation and its examples. 3.2 Sequence impedance and sequence network of various equipment of power system. 3.3 Phase shift in star delta Transformers.
Unit-IV Unsymmetrical Fault Analysis	**CO 2. *Evaluate* the unsymmetrical fault current and voltage at any location of power system with mathematical model and with software tools.**	

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	PrO 7. Find the power system unsymmetrical fault current for the given power system network. PrO 8. Simulate the power system unsymmetrical fault current for the given power system network. 4a. Formulate the fault current equation for the given type of fault for the specified condition. 4b. Evaluate the unsymmetrical fault current and voltage the given power system network for the specified condition.	4.1 Types of unsymmetrical faults and its analysis 4.2 Examples on unsymmetrical fault analysis
Unit – V **Fundaments of Fault Clearing, Switching Phenomena and Circuit Breaker (CB) Ratings.**	**CO 3. Interpret the *phenomena of current interruption in circuit breaker.*** PrO 9. Classify the fault clearing process and problems associated with fault clearing process for the given conditions. 5a. Interpret the data with respect to the given interruption method. 5b. Determine RV and RRRV for the specified condition of the given type circuit breaker. 5c. Prepare the specification of a circuit breaker required for the given situation. 5d. Determine the rating of the circuit breaker for the given conditions.	5.1 Current interruption in AC circuit breaker. 5.2 Transient recovery voltage, rate of rise of restriking voltage and Examples. Different factors affecting TRV. 5.3 Problems associated with circuit breaker for different types of faults, applications and configurations. 5.4 Resistance switching, current chopping and interruption of capacitive current 5.5 Rating and selection of circuit breakers
Unit – VI **Arc Extinction Process in Circuit Breaker**	***CO 4. Evaluate* the different parameters for arc formation and arc interruption in circuit breaker.** **PrO 10.** Evaluate the arc extinction process for the given circuit breaker. 6a. Interpret the various modes of ionization	6.1 Ionization and Deionization of gases 6.2 Formation of electric arc

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	and deionization of gases under the specified conditions. 6b. Interpret the phenomena of arc formation and arc extinction in the given condition.	and methods of arc extinction. 6.3 Arc interruption theories.
Unit – VII Types of Circuit Breakers	***CO 5. Choose the type of circuit breaker for different applications.*** PrO 11. Compare the performance of air break, air blast and SF6 for the given data. PrO 12. Evaluate the applications of vacuum and direct current circuit breaker for the given voltage levels. 7a. Select the relevant type of CB for the minimum cost for the given situation.	7.1 Types of circuit breakers. 7.2 Constructional details and principals of operations.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Per Unit System	04	0	0	6	0	0	0	06
II	Symmetrical Fault Analysis	08	0	7	0	7	0	0	14
III	Symmetrical Components	10	0	7	0	4	0	0	11
IV	Unsymmetrical Fault Analysis	10	0	0	0	4	7	0	11
V	Fundaments of Fault Clearing, Switching Phenomena and Circuit Breaker (CB) Ratings.	10	0	2	7	4	0	0	13
VI	Arc Extinction Process in Circuit Breaker	05	0	0	5	0	0	0	5
VII	Types of Circuit Breakers	13	0	4	0	0	6	0	10
Total		**60**	**0**	**20**	**18**	**19**	**13**	**0**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **Per Unit System:** Collect the nameplate rating of different types of equipment (Power transformer, Distribution Transformer, alternators etc) and compare the pu value of each type of equipment.
2) **Symmetrical components:** Google the different protective relays and list the relays work based on symmetrical components principals.
3) **Fault Clearing Process:** Develop MATLAB model which shows the waveform of voltage at time of contact separation.
4) **Circuit Breakers:** Prepare case studies of circuit breakers used in the protection schemes in different types of industries.
5) Undertake the survey in CHARUSAT campus and submit a report on different type of circuit breaker used with their rating.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given

to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.

d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding the evolution of arc quenching mechanism for different type of circuit breakers.
b) Prepare 15 min. power point presentation on how to choose the specific circuit breaker for the particular application.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Per Unit System	1	1a. Develop the impedance and reactance diagram for the given power system network. 1b. Determine the per unit values of reactances of equipment for the given base.	Lecture with media, field visit	PPTs, black board
Unit– II Symmetrical Fault Analysis	2	**PrO 1.** Formulate the mathematical model for the given power system network. 2a. Determine the making current of the circuit breaker for the specified fault current levels.	Lecture With media	PPTs, black board
	3	**PrO 2.** Find the power system symmetrical fault current for the given power system network. 2b. Determine the 3-phase short circuit current for the given power system network. 2c. Choose the correct value of synchronous machine reactance for fault current calculation for the given condition.	Lecture with media Tutorial	PPTs, black board
Unit– III Symmetrical Components	**4**	**PrO 3.** Find the sequence impedances (Positive, Negative and Zero-sequence) of the three-phase transformer. **PrO 4.** Develop the mathematical model of symmetrical components for the given power system network. **PrO 5.** Find the sequence impedances (Positive, Negative and Zero-sequence) of the three-phase Synchronous machine.	Lecture with media,	PPTs, Videos

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
	5	**PrO 6.** Verification of sequence network for the given connections of 3-phase transformer. 3a. Develop sequence networks form the given one line diagram of power system.	Lecture with media, Tutorial and Demonstration	PPTs
	6	3b. Justify the phase shift to be consider for the specified transformer connection for unsymmetrical fault calculation.	Lecture with media	PPTs
Unit– IV Unsymmetrical Fault Analysis	7	**PrO 7.** Find the power system unsymmetrical fault current for the given power system network. 4a. Formulate the fault current equation for the given type of fault for the specified condition.	Lecture with media, Field visit	PPTs, Video
	8	**PrO 8.** Simulate the power system unsymmetrical fault current for the given power system network. 4b. Evaluate the unsymmetrical fault current and voltage the given power system network for the specified condition.	Lecture with media, and Tutorial	PPTs
Unit – V Fundaments of Fault Clearing, Switching Phenomena and Circuit Breaker (CB) Ratings.	9	5a. Interpret the data with respect to the given interruption method.	Lecture with media, + PPT	PPTs, Video
	10	**PrO 9.** Classify the fault clearing process and problems associated with fault clearing process for the given conditions. 5b. Determine RV and RRRV for the specified condition of the given type circuit breaker. 5c. Prepare the specification of a circuit breaker required for the given situation.	Lecture with media, Tutorial	PPT, Video
	11	5d. Determine the rating of the circuit breaker for the given conditions.	Lecture with media	PPTs, black board
Unit – VI Arc Extinction Process in Circuit Breaker	12	**PrO 10.** Evaluate the arc extinction process for the given circuit breaker. 6a. Interpret the various modes of ionization and deionization of gases under the specified conditions. (self-learn) 6b. Interpret the phenomena of arc formation and arc extinction in the given condition.	Lecture with media, case study	PPT, Pictures , black board
Unit – VII Types of Circuit	13	**PrO 11.** Compare the performance of air break, air blast and SF6 for the given data.	Lecture	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Breakers		7a. Select the relevant type of CB for the minimum cost for the given situation.	with media, Tutorial	
	14	**PrO 12.** Evaluate the applications of vacuum and direct current circuit breaker for the given voltage levels. 7a. Select the relevant type of CB for the minimum cost for the given situation.	Lecture with media, Tutorial	PPTs, Video
	15	7a. Select the relevant type of CB for the minimum cost for the given situation.	Lecture with media, Tutorial	PPTs, Video

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Power System Analysis	Hadi Saadat	PSA Publishing , Third Edition, ISBN : 9780984543809, 9780984543823
2	Principles of Power system	V. K. Mehta , Rohit Mehta	S. Chand & Company Ltd, 2005 ISBN: 8121924960, 9788121924962
3	Modern Power System Analysis	D.P. Kothari & I. J. Nagrath	McGraw Hill Education; 5th edition ISBN: 9789354600968
4	Power System Analysis	Grainger & Stevenson	McGraw Hill Education,2017 ISBN: 9780070612938
5	Power System Analysis and Design	Glover and Sharma	Nelson Engineering; 4th edition ISBN: 9780534548841
6	Analysis of Faulted Power System	P.M. Anderson	IEEE Press, 1995 ISBN: 9780780311459
7	Computer Techniques In Power System Analysis	M A Pai and Chatterjee	McGraw Hill Education; 3rd edition (1 July 2017) ISBN: 978-9332901131
8	J&P Switchgear	R T Lythall	J&P Limited London, 1963 Identifier: in.ernet.dli.2015.474861 ISBN: -

XV SOFTWARE/LEARNING WEBSITES

- https://nptel.ac.in/courses/108105067
- https://nptel.ac.in/courses/117105140

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE349	**Semester VI**	**Fault Analysis and Switchgear** Mark '**H'=3** for high, '**M**'=2 for medium or **'L'=1'** for the relevant correlation of each PO or PSO, Competency or CO													
349	**Competency** *Evaluate the performance of various circuit breakers employed for various applications in power system.*	***3***	***2***	***1***	***2***	***1***	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
349.1	Evaluate the symmetrical fault current and voltage at any location of power system.	3	3	1	2	2	-	-	-	-	-	-	-	-	-
349.2	Evaluate the unsymmetrical fault current and voltage at any location of power system with mathematical model and with software tools.	3	3	2	2	2	-	-	-	-	-	-	-	-	-
349.3	Interpret the phenomena of current interruption in circuit breaker.	3	1	-	1	-	-	-	-	-	-	-	-	-	-
349.4	Evaluate the different parameters for arc formation and arc interruption in circuit breaker.	3	1	-	1	-	-	-	-	-	-	-	-	-	-
349.5	Choose the type of circuit breaker for different applications.	3	1	2	2	-	2	-	-	-	-	-	-	-	-

XVII NAMES OF RESOURCE PERSONS

a) Dr. Nilaykumar A. Patel
b) ……………………………………………………

Course Code: **EE360**
Course Title: **Minor Project-II**

UG Engineering Programme	Semester
Electrical Engineering	Sixth

I RATIONALE

This course enables the students to apply some of the knowledge and/or skills developed during the programme to new problem for which there are number of engineering solutions. This course include planning of the tasks which are to be completed within the time allocated, and in turn, helps to develop ability to plan, , use, monitor and control resources optimally and economically. By studying this course abilities like creativity, imitativeness and performance qualities are also developed in students. Leadership development and supervision skills are also integrated objectives of learning this course.

II COMPETENCY

The course should be taught and implemented with the aim to develop the required course outcomes (COs) so that students will acquire following competency needed by the industry:

- **Find the solution of a real-life problem**.

III COURSE OUTCOMES (COs)

Depending upon the nature of the projects undertaken, the following could be some of the minor course outcomes that could be attained, although, in case of some projects few of the following course outcomes may not be applicable.

Apply knowledge and skills learned from project to real-world problems.
Solve engineering problems by the team work and work with other discipline.
Use experience related to professional and ethical issues in the work environment.
Apply impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context in order to provide employability and making entrepreneur.
Implement the skills related engineering on contemporary issues related with engineering in general with advanced tools and technologies.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	
0	0	04	02	00	00	50	00	100

***Legends: L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

V COURSE DETAILS

i. Minor Project showcases that, a student during the fifth semester will able to apply the engineering knowledge to a practical problem.

ii. Students can carry out their project work in-house at institute/ industry/ research organization/govt. organization.

iii. Project should be related to the Product/prototype design and development or Experimentation, Simulations and evaluation of any system etc.

iv. A student is required to carry out the project work related to Electrical Engineering or some interdisciplinary work under the guidance of a faculty member and the supervisor of the concerned industry/institute/organization.

v. The minor project work provides student an opportunity to develop something on their own or group of maximum 3 students. Final decision related to the number of students in one group is based on internal supervisor recommendation.

vi. Project will be evaluated at least twice during the semester by internal faculty as a part of continuous evaluation and final evaluation will be carried out by external examiner at the end of semester.

vii. Student should submit the project report at the end of the semester duly signed by project supervisor and head of department. Report comprising literature review, objective, methodology, scope of the project work undertaken, outcome of project and conclusions.

viii. The students are required to identify their problem and they are required to follow all the rules and instruction issued by department/industry.

VI PROJECT REPORT

At the end of the semester student should prepare project report. Guideline of the Project report is as follow

a) Certificate
b) Acknowledgements
c) Abstract (in one paragraph not more than 150 words)
e) Content Page
f) Chapter-1 Introduction and background of the Problem/Task
g) Chapter-2 Literature Survey for Problem/Task Identification and Definition,
h) Chapter-3 Proposed Detailed Methodology of solving the problem/completing the task
d) Chapter-4 Methodology actually followed to solve the problem/complete the task
e) Chapter-5 The solution, its development and its implementation/execution (with details including drawings, software, used for or developed for the solution etc. in detail). In case of survey/experiment type of project this chapter will include data analysis, information presentation and analysis.
f) Chapter-6 Final conclusions and recommendations if any for future projects.

Note:

a) The report should contain as many diagrams and figures, charts etc. as relevant for the project.
b) Originality of the report (written in own words) would be given more importance rather than quality of printing and use of glossy paper or multi-colour printing

VII ASSESSMENT OF PROJECT WORK

Assessment of Project work also has two components, first is Progressive Assessment, while another is End of the Term Assessment.

i. Progressive Assessment Criteria

a) Project guide or the group of internal faculty are supposed to carry out this assessment. It is a continuous process, during which for developing desired qualities in the students, faculty should orally give informal feedback to students about their performance and interpersonal behaviour while guiding them on their project work every week.

S. No.	Criteria	Marks
	First Progressive Assessment at the end of 4th week	
1	**Project Proposal** for work to be completed at the end of semester: Including problem definition and methodology adapted, literature review adopted (in case of working models prepared then quality of workman ship, appropriateness of raw materials/components/ technology being used, functioning of the prototype etc. has to be also assessed)	**10**
	General Behaviour (initiative, resourcefulness, reasoning ability, imagination/creativity, self-reliance)	**Only written feed-back/suggestions**
	Second Progressive Assessment at the end of 12th week	
2	**Execution of Plan during semester** (marks to be also given based on ability to collect relevant information, ability to follow correct procedure, manipulative skills, ability to observe, record & interpret, ingenuity in the use of material and equipment, target achievement) In case of working models, quality of workman ship (including accuracy in dimensions, shape, tolerance limits), appropriateness of raw materials/components/ technology being used, functioning of the prototype, cost effectiveness, marketability, modernity etc. has to be also assessed	**5**
3	**Log book** (for work done during semester- detailed and regular entry would be basis of marks)	**5**
4	**General Behaviour** (persistence, interest, confidence, problem solving ability, decision making ability, initiative to act, team spirit, sharing of material etc., participation in discussions, completion of individual responsibilities, leadership)	**Only written feed-back./suggestions**
	Third Progressive Assessment at the end of 14th week	
5	**Portfolio for Self-learning and reflection** (for work done in semester- marks based on amount of reflection and completion of the portfolio)	**10**
6	**Project Report including documentation -** for work done in semester (marks based on: clarity in presentation and organization; styles and	**5**

S. No.	Criteria	Marks
	language; quality of diagrams, drawings and graphs; accuracy of conclusion drawn; citing of cross references; suggestion for further research/project work)	
7	**Presentation** (presentation skills including communication skills to be assessed by observing presentations and asking questions during presentation for work done during semester)	**5**
8	**Defence** (ability to defend the methods/materials used and technical knowledge, and involvement of individual to be assessed by asking questions during presentation and viva/voce for work done during semester)	**10**
	Total	**50**

Note: Oral feedback on general behaviour may also be given whenever relevant/ required during day to day guidance and supervision.

ii. END SEMESTER EXAMINATION

b) Criteria of Marks for ESE for Capstone Project Execution.

S. No.	Description	Marks
7	**Problem Identification/Project Title** (Innovation /Utility of the Project for industry/ User/Academia) marks to be also given based on (i) Accuracy or specificity of the scope and (ii) Appropriateness of the work with reference to desired course outcomes.	**5**
8	**Industrial Survey and Literature Review** (marks to be given based on extent/volume and quality of the survey of Industry / Society / Institutes/Literature/Internet for Problem Identification and possible solutions)	**5**
9	**Project Proposal** (In case of working models, detailed design and planning of fabrication/assembly of the prototype has to be also assessed). Marks to be given also based on appropriateness, flexibility, detail and clarity in methods/planning.	**10**
10	**Final Report including documentation**. (marks based on: clarity in presentation and organization; styles and language; quality of diagrams, drawings and graphs; accuracy of conclusion drawn; citing of cross references; suggestion for further research/project work)	**10**
11	**Presentation** (presentation skills including communication skills to be assessed by observing presentations and asking questions during presentation)	**10**
12	**Defence** (ability to defend the methods/materials used and technical knowledge, and involvement of individual to be assessed by asking questions during presentation and viva/voce)	**10**
	Total	**50**

XII Guidelines for Portfolio Preparation and assessment

The main purpose of portfolio preparation is learning based on self-assessment and portfolio is not to be used for assessment in traditional sense.

a) Each student has to prepare his/her portfolio separately. However, he/she can discuss with the group members about certain issues on which he/she wants to write in the portfolio.

b) Whatever is written inside the portfolio is never to be used for assessment, because if teachers start giving marks based on whatever is written in the portfolio, then students would hesitate in true self-assessment and would not openly describe their own mistakes or shortcomings.

c) Some marks are allocated for portfolio, these marks are to be given based on how sincerely portfolio has been prepared and not based on what strengths and weaknesses of the students are mentioned in the portfolio.

d) Portfolio has to be returned back to the students after assessing it (assessment is only to see that whether portfolio is completed properly or not) by teachers. Because student is the real owner of the portfolio.

e) Students mainly learn during portfolio preparation, but they can further learn if they read it after a gap. And hence they are supposed to keep the portfolios with them even after completion of the diploma because it is record of their own experiences (it is like diary some people write about their personal experiences), because they can read it again after some time and can revise their learning (about their own qualities)

XIII PO-COMPETENCY-CO MAPPING

Course Code	**Competency and COs**	**PO 1** Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE360	**Semester VI**														
360	**Competency** *Find the solution of a real-life problem.*	***3***	***2***	***2***	***2***	***2***	***2***	***1***	***1***	***2***	***2***	***2***	***1***	***2***	***2***
360.1	Apply knowledge and skills learned from project to real-world problems.	3	-	1	-	-	1	-	1	-	1	1	1	-	1
360.2	Solve engineering problems by the team work and work with other discipline.	3	3	2	2	1	2	-	-	3	2	1	2	2	-
360.3	Use experience related to professional and ethical issues in the work environment.	3	2	1	-	-	-	-	3	1	-	2	-	-	1
360.	Apply impact of	3	-	-	3	3	3	-	-	2	1	3	2	3	3

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
4	engineering solutions, developed in a project, in a global, economic, environmental, and societal context in order to provide employability and making entrepreneur.														
360.5	Implement the skills related engineering on contemporary issues related with engineering in general with advanced tools and technologies.	3	1	3	2	3	3	2	-	-	2	2	3	3	2

XIV NAMES OF RESOURCE PERSONS

d) Prof. Mahammadsoaib Saiyad

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS132.02 A: CONTRIBUTORY PERSONALITY DEVELOPMENT

I. Credits and Schemes:

Sem.	Course Code	Course Name	Credits	Teaching Scheme Contact Hours/Week	Evaluation Scheme				Total
					Theory		Practical		
					Internal	External	Internal	External	
VI	HS133.02 A	CONTRIBUTORY PERSONALITY DEVELOPMENT	02	02	--	--	30	70	100

II. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Concept of Personality	06
2.	Soft Skills and Personality Development	08
3.	Developing Contributory Personality	06
4.	Life skills and Personality Development	06
5.	Contemporary Issues in CPD	04
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

III. Detailed Syllabus:

1.	Concept of Personality	06 Hours	20%
	Meaning of Personality, Types of Personality, Factors contributing to Personality, Personality Traits, Personality Profiling		
2.	Soft Skills and Personality Development	08 Hours	26%
	Positive Thinking and Mind Set, Leadership, Assertiveness and Negotiation Skills, Self-Management, Interpersonal Skills, Being a Team Player		
3.	Developing Contributory Personality	06 Hours	20%

	Concept of Contributory Personality, Characteristics of a Contributor, The Contributor's Vision of Success & Career, The Scope of Contribution in a field, Embarking on the Journey to Contributor ship, Developing Contributor Personality, Reviewing Some Contributors Personalities		
4.	Life skills and Personality Development	06 Hours	20%
	Concept of life skills, Self-awareness, Empathy, Decision Making, Problem Solving		
5.	Contemporary Issues in CPD	04 Hours	14%
	Contemporary Trends and Practices in Contributory Personality Development, Case Study & Presentations		

IV. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by personality profiling, reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like case studies (Great Personality) , critical reading, group work, independent and collaborative research, presentations etc.

V. Evaluation:

The students will be evaluated continuously in the form of internal as well as external examinations. The practical evaluation is schemed as 30 marks for internal evaluation and 70 marks for external evaluation in the form of University examination

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Components	Number	Marks per incidence	Total Marks
1	Assignments	2	25	
2	Project work	2	25	
3	Personality Profiling and Presentation	1	25	
4	Quiz	2	25	100*
5	Attendance and Classroom participation			05
			Total	30

*** Total Marks will be converted to 25**

External Evaluation

The University Exam will be for 70 marks and will be based on practical computer based tests and a viva-voce. It will test the contributory personality aspects and their applications. There may also be a Computer Based Test at the end.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Practical Exam / Viva	01	70	70
			Total	70

VI. Course Outcomes (COs)

At the end of the course, learners will be able to:

CO1	Identify one's individual personality strengths and challenges.
CO2	Develop more assertive and optimist attitude towards work and life.
CO3	Develop quintessential soft skills to groom one's personality.
CO4	Identify traits of contributor personality.
CO5	Contribute to self, society, nation, and globe.
CO6	Develop skills of global citizenship to perform societal responsibilities.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO5	-	-	-	-	-	-	2	-	2	2	-	-	-	-
CO6	-	-	-	-	-	2		-	-	2	-	3	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

VII. Preference

It will be preferred that at the end of the course, students take an official Personality Profiling test like MBTI through CHARUSAT. Equivalent marks shall be considered in the University Exam.

VIII. Recommended Study Material:

- **Text book:**

    1. Personality Development & Soft Skills, Oxford University Press

2. Soft Skills, Bookboon
3. Personality Development, Swami Vivekananda; Advaita Ashrama

- **Reference book:**

1. Contributor Personality Program Workbook (Volume 1,2),
2. Contributor Personality Program ActivGuide, Illumine Knowledge Pvt. Ltd

- **Web material:**

4. https://www.coursera.org/learn/wharton-succcess
5. https://www.coursera.org/learn/personality-types-at-work
6. https://www.coursera.org/learn/self-awareness

Course Code: **EE376**
Course Title: **Special Electrical Machines And Applications**

UG Engineering Programme	Semester
Electrical Engineering	Sixth

I RATIONALE

The special electrical machines are widely used in power industries. The course is developed such that student would get the knowledge from basics to control schemes of these machines.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Evaluate the performance of various control methods used in field**.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) ***Use*** the control technique for special electrical machines**.**
2) ***Evaluate*** *the performance of the* special electrical machines.
3) **Interpret** the suitable control technique for special electrical machines.
4) ***Evaluate*** the performance of the special electrical machines.
5) ***Choose*** *the type of turbine for induction generator.*

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	
3	0	2	4	70	30*	25	25	150

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

(*): *Under the* ***Theory PA****, out of 30 marks,* ***15 marks*** *are for* ***micro-project assessment*** *to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs****.***

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the

students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	2	To Perform the Direct Load Test On Auto Synchronous Motor.	04
2	2	To Perform the Constant Excitation Test On Auto Synchronous Motor.	04
3	3	To Observe a Change in Output Voltage with Change in Rotor Position of 3 Phase Induction Regulator.	04
4	3	To Perform the Speed Control and Power Factor Control on Commutator Motor.	04
5	1	Study of a Stepper Motor.	02
6	1	Study of a Switched Reluctance Motor and Synchronous Reluctance Motor.	02
7	1	Study of a Permanent Magnet Synchronous Motor.	02
8	4	Demonstration of an Induction Generator.	04
9	4	To Understand the Behavior of Induction Generators Considering the Impact of a Load.	02
10	4	To Understand the Working of a Doubly Fed Induction Generator.	02
		Total	**30**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Commutator machine with load	4
2	Auto Synchronous machine with prime-mover/load	1,2
3	Induction machine with prime-mover/load	8,9
4	Induction voltage regulator	3

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year

b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Introduction of Special Electrical Machines	**CO 1. Recognize & understand the performance, operation and application of various AC & DC special machines.** PrO 1. To Perform the Direct Load Test On Auto Synchronous Motor. PrO 2. To Perform the Constant Excitation Test On Auto Synchronous Motor. PrO 3. To Observe a Change in Output Voltage with Change in Rotor Position of 3 Phase Induction Regulator. PrO 4. To Perform the Speed Control and Power Factor Control on Commutator Motor 1a. Justify the concept of special machines and fundamental component roles.	1.1 Rosenberg Generator 1.2 Dynamotors, Printed Circuit Board Motor 1.3 Special Induction Machines: Synchronous Induction Motor 1.4 Induction Voltage Regulator 1.5 Linear Induction Motor 1.6 Special Transformers: Induction Heating Transformer 1.7 Welding Transformers, Pulse Transformer 1.8 Schrage Motor 1.9 Repulsion Motor 1.10 Hysteresis Motor 1.11 Universal Motor, DC-AC Servomotor 1.12 Applications, Examples
Unit– II Special Synchronous Machines	**CO 3. Acquire the knowledge regarding the fundamentals of permanent magnet, motors/generators, switched reluctance motor, linear motors and stepper motors.** PrO 5. Study of a Stepper Motor. PrO 6. Study of a Switched Reluctance Motor and Synchronous Reluctance Motor. 2a. Evaluate the performance of stepper motor. 2b. Evaluate the performance of SRM.	2.1 Stepper Motor 2.2 Switched Reluctance Motor 2.3 Linear Synchronous Motor 2.4 Applications, Examples
Unit– III Permanent	**CO 2. Define various permanent magnet material used for manufacturing of special**	3.1 Basics of permanent magnet property

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Magnet Machines	**machine.** PrO 7. Study of a Permanent Magnet Synchronous Motor. 3a. Evaluate the performance of PMSM.	3.2 Unique features of Permanent Magnet (PM) Machines 3.3 Permanent Magnet Material & Characteristics 3.4 Permanent magnet DC motor (PMDC) 3.5 Permanent magnet brushless DC motor (PMBLDC) : construction, operation, control, advantages, dis-advantages, applications 3.6 permanent magnet synchronous motor(PMSM), Applications
Unit-IV Wind Turbine Generators	**CO 4. *Understand the types and operation of wind turbine generators and evaluate case studies based on renewable energy sources.*** *PrO 8. Demonstration of an Induction Generator.* *PrO 9. To Understand the Behavior of Induction Generators Considering the Impact of a Load.* *Pro 10. To Understand the Working of a Doubly Fed Induction Generator.* 4a. Choose the relevant operating parameters for load operation of induction generator.	4.1 Introduction, Types of induction generator (line excited and self-excited) 4.2 Introduction of fixed speed induction generator 4.3 Introduction of doubly fed induction generator 4.4 Working of Induction generator: Principle of Operation, Reactive power requirement 4.5 voltage build up process, Load and Power factor control, Effect of capacitor 4.6 Characteristics of induction generator, Advantages, Applications, Examples
Unit – V Modelling & Application	**CO 5. *To understand the modeling of machines and reference frames for machine control.*** 5a. Interpret the machine data with respect to the given power system. 5b. Select the method for machine control.	5.1 Synchronous Reference Frames: dq Axis Theory 5.2 Conversions: abc to αβ, αβ to dq, dq to αβ, αβ to abc 5.3 Space Vector Pulse Width Modulation 5.4 discrete and analog

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
		implementation, Applications
Unit VI Case Study	**CO 4. *Understand the types and operation of wind turbine generators and evaluate case studies based on renewable energy sources.*** 6a. Evaluate performance of Wind generator application	6.1 Case Study: Grid Connected Renewable Energy Source

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction of Special Electrical Machines	13	2	4	4	4	0	0	16
II	Special Synchronous Machines	05	0	2	2	2	2	0	08
III	Permanent Magnet Machines	07	0	2	4	4	2	0	12
IV	Wind Turbine Generators	06	0	2	2	2	6	0	12
V	Modelling & Applications	09	2	2	4	4	2	2	16
VI	Case Study	05	0	0	0	2	4	0	06
Total		**45**	**4**	**12**	**16**	**18**	**16**	**2**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **Load test on induction generator**.
2) **Calculation of step angle of stepper motor for various applications.**
3) **Create setup of regenerative braking of auto synchronous motor.**
4) **Create a setup for load condition of auto synchronous motor for various power factor**.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different methods of voltage regulation.
b) Prepare 15 min. power point presentation on synchronizing of alternator.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Introduction of	1	*PrO 1. To Perform the Direct Load Test On Auto Synchronous Motor.*	Lecture with media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Special Electrical Machines	2	*PrO 2. To Perform the Constant Excitation Test On Auto Synchronous Motor.*	Lecture With media	PPTs, black board
	3	*PrO 3. To Observe a Change in Output Voltage with Change in Rotor Position of 3 Phase Induction Regulator.*	Lecture with media	PPTs, black board
	4	*PrO 4. To Perform the Speed Control and Power Factor Control on Commutator Motor* *1a. Justify the concept of special machines and fundamental component roles.*	Lecture With media	PPTs, black board
Unit II- Special Synchronous Machines	**5**	*PrO 5. Study of a Stepper Motor.* *2a. Evaluate the performance of stepper motor.*	Lecture With media	PPTs, black board
	6	PrO 6. Study of a Switched Reluctance Motor and Synchronous Reluctance Motor. 2b. Evaluate the performance of SRM.	Lecture With media	PPTs, black board
Unit– III - Permanent Magnet Machines	**7**	PrO 7. Study of a Permanent Magnet Synchronous Motor.	Lecture With media	PPTs, black board
	8	3a. Evaluate the performance of PMSM.	Lecture With media	PPTs, black board
Unit IV- Wind Turbine Generators	**9**	*PrO 8. Demonstration of an Induction Generator*	Lecture With media	PPTs, black board
	10	*PrO 9. To Understand the Behavior of Induction Generators Considering the Impact of a Load.*	Lecture With media	PPTs, black board
	11	*Pro 10. To Understand the Working of a Doubly Fed Induction Generator.*	Lecture With media	PPTs, black board
	12	*4a.* Choose the relevant operating parameters for load operation of induction generator.	Lecture With media	PPTs, black board
Unit V - Modelling & Applications	**13**	5a. Interpret the machine data with respect to the given power system.	Lecture With media	PPTs, black board
	14	5b. Select the method for machine control.	Lecture With media	PPTs, black board
Unit VI- Case study	**15**	6a. Evaluate performance of Wind generator application	Lecture With media	PPTs, black board

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Theory and performance of electrical machines	J.B. Gupta	S.K. Kataria & Sons, ISBN: 978-93-5014-277-6
2	A textbook of electrical	B.L. Theraja	S. Chand, ISBN: 9788121924375

S. No.	Title of Book	Author	Publication including ISBN
	technology VOL II		
3	Special Electrical Machines	K. Venkataratnam	University press, ISBN: 13. 978-8173716317

XV SOFTWARE/LEARNING WEBSITES

- http://www.academia.edu/9885014/special_electrical_machines_nptel_notes

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Used of mathematical method	PO 2 Analysis mechanical system	PO 3 Used engineering soft.	PO 4 Solve problem in machine	PO 5 **Modern tool usage**	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work:	PO 10 Communication	PO11 Project management	PO12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem
EE376	**Semester VI**														
CO1	Recognize & understand the performance, operation and application of various AC & DC special machines.	3	-	1	2	-	-	-	-	1	2	3	1	-	-
CO2	Define various permanent magnet material used for manufacturing of special machine.	3	1	2	2	2	-	1	-	-	1	-	-	2	-
CO3	Acquire the knowledge regarding the fundamentals of permanent magnet, motors/generators, switched reluctance motor, linear motors and stepper motors.	3	3	3	1	2	2	1	-	-	3	2	1	2	-
CO4	Understand the types and operation of wind	3	3	3	1	2	2	3	-	2	2	2	3	3	3

Course Code	Competency and COs	PO 1 Used of mathematical method	PO 2 Analysis mechanical system	PO 3 Used engineering soft.	PO 4 Solve problem in machine	**PO 5 Modern tool usage**	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work:	PO 10 Communication	PO11 Project management	PO12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem
	turbine generators and evaluate case studies based on renewable energy sources.														
CO5	To understand the modeling of machines and reference frames for machine control.	3	-	2	3	3	-	-	2	2	3	1	2	3	3

XVII NAMES OF RESOURCE PERSONS

a) Pratik Mochi
b) ………………………………………………….

Course Code: **EE377**
Course Title: **Embedded Systems**

UG Engineering Programme	Semester
Electrical Engineering	Sixth

I RATIONALE

Looking to a large set of skills required by an Embedded System Development Engineer, there should be a course serving as Fast/Crash Course bridging the requirement like, introducing the various concepts of Embedded system and inculcating the most basic skills set just in a single course.

The Embedded software engineer and Embedded Hardware Engineer has to work in close collaboration in many or most of the project targeting Embedded system development for an application. While the Embedded Hardware Engineer read and understand the datasheets of various chips and components involved in design, integrate/interface them together and finally generate the professional Printed Circuit Board (PCB), the Embedded software engineer write wide varieties of firmware for an Embedded application. The course does not focus on skills required for an Embedded hardware engineer.

For most of application, computationally fast microcontrollers with rich set of on chip peripherals support and Memory are enough (ARM series Microcontroller are one of the popular choice of Embedded system designers). Due to several reasons, the Field Programmable Gate Array (FPGA) are selected as main or supporting chip. The Embedded software engineer needs to generate hardware inside such chips using hardware description language, like VHDL or Verilog. The course covers FPGA programming in VHDL language. It starts with developing some basic hardware block and it is extended for developing several communication hardware modules/protocols as well.

Embedded C coding for Microcontroller is already covered in a separate course and it is considered a pre-requisite for this course. But those course are not covering 32-bit microcontroller programming. The memory mapped registers/SFR in 32-bit microcontroller are generally accessed using pointer-structure-union mechanism. Writing and understanding ready driver is an art and needs some practice. often, The Embedded Software Engineer need to write Embedded Application firmware using market available Real Time Operating System (RTOS). The course teaches to develop small RTOS after discussing the basic concepts and to write code using RTOS.

The computer on board/chip (like raspberry Pi, Jetson Nano and other such boards) has revolutionaries the way of programming. Even one can program the system without knowing much architectural details of microcontroller. Specially these devices are gaining attention for systems involving some AI/ML algorithms and Internet connectivity. Such topics are also covered here briefly.

Such a course can be helpful to student to understand Embedded system programming, with- out spending much time for it. The industry may find this subject useful, for example if they want someone to offer a role requiring regular discussion with team of embedded system software developers and some time to get involved in their process to modify it. The course is not target to make a full stack embedded system engineer (Hardware + software both) but it helps and work out much for it.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- Understand given Embedded Firmware/HDL coding and Explain/Present to a team working on it. Remodify it for new requirement.
- Write the Firmware/HDL coding for target Embedded system and Debug/Correct it.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) Use the ready driver to configure the on chip peripherals and develop driver as a C callable function, find the bug in driver function.

2) Write/Develop VHDL program for required architecture. Test and Debug it.

3) Develop Small Real Time Operating System(RTOS) and Write code for Embedded application using RTOS.

4) Apply and Test Artificial Intelligence (AI) and Machine Learning (ML) algorithms for an Embedded Application implemented on Computer on Board Devices.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			**Total Credits (L+T+P)**	**Examination Scheme**				
				Theory Marks		**Practical Marks**		**Total Marks**
L	**T**	**P**	**C**	**ESE**	PA	**ESE**	**PA**	150
3	0	2	4	70	30*	25	25	

***Legends: L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

***(*):** Under the **Theory PA**, out of 30 marks, **20 marks** are for **attendance**, **micro-project assessment** and other **assignment** if given any, to facilitate integration of COs and the remaining 10 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	**Approx. Hrs. required**
1	1	*Develop memory map registers in header file using pointer, structure and union for Peripheral registers in ARM M4 microcontroller.*	01
2	2	*Develop/Write basic codes for FPGA in VHDL as in follows, and also write test bench program for the same.* *a. Combinational Circuits like Digital Gates and Digital Circuits based on Digital Gates, SR Non Clocked Flip Flop* *b. Sequential Clock based Circuits*	03
3	2	*Learn the following architecture in an Unknown Microcontroller, and develop/Write the VHDL code to implement the given architecture of peripheral in FPGA.* *a. GPIO* *b. SPI,I2C, UART, USART, CAN communication*	05
4	1	*Learn the following architecture in an Unknown Microcontroller, Develop/Write C driver for it* *a. SPI, I2C,* *b. CAN communication*	02
5	2	*Learn the following architecture in an Unknown Microcontroller, Develop/Write VHDL code to implement the given architecture of peripheral in FPGA.* *a. UP, Down, Up-Down Timer/Counter* *b. Watchdog Timer* *c. Pulse Width Modulator* *d. Capture Compare Unit*	05
6	1	*Learn the following architecture in an Unknown Microcontroller, Develop/Write C driver for it.* *a. UP, Down, Up-Down Timer/Counter* *b. Watchdog Timer* *c. Pulse Width Modulator* *d. Capture Compare Unit*	02
7	4	*Write AI and ML codes in Prolog, Python and C, and Test the result and speed of execution on given Computer On chip/board device and make (Develop interfacing and Write code) it communicating it to main controller/FPGA chip in Embedded board.*	08
8	3	*Test the small Real Time Operating system (developed during theory hours or market available with support of documentation) for various functionality.*	04
		Total	**30**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

Sr. No.	Equipment Name with Broad Specifications	PrO No.
1	Computer On chip Device (Raspberry Pi, Jetson Nano, Raspberry Pi Pico or similar)	7
2	*Data Set for AI and Machine Learning*	7

Sr. No.	Equipment Name with Broad Specifications	PrO No.
3	*FPGA kit.*	2,3,5
4	*ARM or similar 32 bit Fast Microcontroller Development boards*	1,4
5	*Unknown Microcontroller (The microcontroller which is not covered in any offered course)*	1,4,7
6	*Digital Storage Oscilloscope(DSO), Signal Generator*	3,4,8

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when he/she performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Understand the specification, requirements and device a plan of action for developing an Embedded Application/product.
2) Lead the team, distribute the task, and finally integrate them all to gether, for developing Embedded system firmware

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I ***Introduction***	Co1. Use the ready driver to configure the on chip peripherals and develop driver as a C callable function, find the bug in driver function. *Pro1. Develop memory map registers in header file using pointer, structure and union for Peripheral registers in ARM M4 microcontroller.* *Pro4. Learn the following architecture in an*	*1.1 Peripherals available in Microcontroller, their working, modes and Controlling and configuration Registers generally available in (Various)Microcontroller (GPIO, Watchdog, Timer, ADC, DAC, Compare-capture, Pulse width Modulator),* *1.2 Developing Driver using*

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	Unknown Microcontroller, Develop/Write C driver for it *a. SPI, I2C,* *b. CAN communication* *Pro6. Learn the following architecture in an Unknown Microcontroller, Develop/Write C driver for it.* *a. UP, Down, Up-Down Timer/Counter* *b. Watchdog Timer* *c. Pulse Width Modulator* *d. Capture Compare Unit* *1a. Understand the Architecture of peripheral from datasheet and write down driver to configure and use the peripheral.* *1b. Understand and use ready Device driver.*	*pointer, structure, Union in C and Using ready driver.*
Unit– II Communication Protocol and Device driver.	Co1. Use the ready driver to configure the on chip peripherals and develop driver as a C callable function, find the bug in driver function. *Pro3. Learn the following architecture in an Unknown Microcontroller, and develop/Write the VHDL code to implement the given architecture of peripheral in FPGA.* *a. GPIO* *b. SPI,I2C, UART, USART, CAN communication* *Pro4. Learn the following architecture in an Unknown Microcontroller, Develop/Write C driver for it* *a. SPI, I2C,* *b. CAN communication* *2a. identify the problems with communication and provide the solution.* *2b. Understand and use Device driver for various communication Protocols and find bug if any.*	*2.1 Working of SPI, I2C bus, CAN bus,* *2.2 Device driver development in C*

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
Unit– III **FPGA Programming in VHDL**	Co2. <u>Write/Develop</u> VHDL program for required architecture. <u>Test</u> and <u>Debug</u> it. *Pro3. <u>Learn</u> the following architecture in an Unknown Microcontroller, and <u>develop/Write</u> the VHDL code to implement the given architecture of peripheral in FPGA.* *a. GPIO* *b. SPI, I2C, UART, USART, CAN communication* *Pro5. <u>Learn</u> the following architecture in an Unknown Microcontroller, <u>Develop/Write</u> VHDL code to implement the given architecture of peripheral in FPGA.* *a. UP, Down, Up-Down Timer/Counter* *b. Watchdog Timer* *c. Pulse Width Modulator* *d. Capture Compare Unit* 3a. <u>Understand</u> the given VHDL program. 3b. <u>Develop</u> VHDL program for given architecture. 3c. <u>Develop</u> VHDL Program for interfacing external chip with VHDL.	*3.1 History of VHDL, Introduction to VHDL and meaning of Hardware abstraction, General Architecture of a VHDL Program* *3.2 Entity Declaration, Architecture body (Structural, Behavioural and Mix style of Modelling),* *3.3 Lexical Elements(comments, identifiers, numbers, characters, strings, bit strings),* *3.4 Data types (Integer datatypes, Arrays, Records),* *3.5 Object declaration,* *3.6 Signal Vs Variable,* *3.7 Programming OR, AND and XOR Gate in VHDL* *3.8 Loop statements, Wait statement,* *3.9 Libraries,* *3.10 Programming exercises (Like Adder, comparator, ALU, Latch, D and JK Flip flop, Register and shift Registers, Counter, Finite state Machine),* *3.11 Conditional and selected signal Assignment,* *3.12 Writing functions and procedures,* *3.13 Structural modelling,* *3.14 Test benches, Role of RTL engineer in Embedded industries,* *3.15 Implementation of communication protocols on FPGA, Interfacing ADC with FPGA*

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit-IV **Fundamental of Real Time Operating System**	Co3. Develop Small Real Time Operating System(RTOS) and Write code for Embedded application using RTOS. *Pro8. Test the small Real Time Operating system (developed during theory hours or market available with support of documentation) for various functionality.* 4a. Select the Operating system as per requirement. 4b. Understand basic concepts and Components of RTOS and develop small Real Time Operating system. 4c. Write and Debug RTOS code for required Embedded System Application.	*4.1 Basics of OS and RTOS: Code Execution in Operating system with simplest Round robin Scheduling, Round robin with interrupts, Multi Processing with multi core and Muti tasking, Types of Operating System (Real Time, Non Real Time and Soft Real time), Selection of RTOS for an Embedded system,* *4.2 Architectural Diagram and functioning of Operating System, Kernel space and user space, Task, Thread and Process State of Task (Ready, running, suspended, deleted),* *4.3 Various Task scheduling algorithms(Various multi-tasking system),* *4.4 Task Communication and related programs, Task synchronization, Message Passing, mailbox, signalling, semaphore, Mutex.* *4.5 Problems (like Deadlock, live lock, starvation),*
Unit-V **Expert Systems implementation on computer On chip**	Co4. Apply and Test Artificial Intelligence (AI) and Machine Learning (ML) algorithms for an Embedded Application implemented on Computer on Board Devices. *Pro7. Write AI and ML codes in Prolog, Python and C, and Test the result and speed of execution on given Computer On chip/board device and make (Develop interfacing and Write code) it communicating it to main controller/FPGA chip in Embedded board.*	*5.1 Introduction to several computers on chip boards, Architecture,* *5.2 Accessing peripherals using python,* *5.3 Python implementation of AI and Machine learning on Computer on chip device* *5.4 Communication of result*

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	5a. Select Embedded board for required AI ML algorithm and target Embedded application. 5b. Write python code to access pins of computer on board device. 5c. Implement AI/ML algorithm on computer on board device and develop interfacing scheme and code to communicate the result to main controller/FPGA chip.	

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction	06	1	3	5	0	0	1	10
II	Communication Protocol and Device driver.	09	2	3	8	2	0	0	15
III	FPGA Programming in VHDL	09	1	3	2	3	1	5	15
IV	Fundamental of Real Time Operating System	09	2	4	6	1	0	2	15
V	Expert Systems implementation on computer On chip	12	4	2	2	2	2	3	15
Total		**45**	**10**	**15**	**23**	**8**	**3**	**11**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. It should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

Just to give idea, A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here:

1) **Smart Class Room:** The machine learning and AI algorithm learned in unit IV, can be used to identify an object or a gesture. The algorithm can be trained for particular application like detecting a particular student in class, observing his learning effort and evaluating its daily, weekly and semester performance. Report can be sent to student for his/her betterment. The prototype design can be re investigated and re design to increase the speed of response. More function that make classroom teaching smart and help teaching learning and evaluation process can be incorporated.
2) **VHDL/Verilog implementation of Specialized DSP algorithm**: Select the algorithm and code on FPGA. The algorithm selected should be much more complex to be executed by microcontroller in real time and hence the use of FPGA should be justified. (CO 1,2,3,4)
3) **VHDL implementation of Space Vector modulation**: The space Vector modulation is one of the popular modulation technique use to generate the Gate driving signal in converter application. The VHDL implementation can be useful to maintain the minimum precision required for duty cycle as high frequency PWM modulators can be implemented on FPGA platform.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

The teacher can use following strategies, to accelerate the attainment of the various types of learning outcomes in this course:

a) Tutorials that need hand calculation and testing in Software.
b) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance may to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) The teacher can suggest several research paper to read and refer. This can be assign to specially to evaluate the capacity of individual, to understand topics related to signal processing but not in syllabus. The capacity can be judge by the way or quality of efforts put, the simulation developed to verify the algorithm and brain storming done and presented to evaluator in form of presentation.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical

evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Make a report by literature review on new Embedded boards
b) Identify the possible reduction in mathematical burden required in particular AI/ML algorithm, modify the code and test verify the speed for new algorithm.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Introduction	1	*Pro1. <u>Develop</u> memory map registers in header file using pointer, structure and union for Peripheral registers in ARM M4 microcontroller.* *1a. <u>Understand</u> the Architecture of peripheral from datasheet and <u>write</u> down driver to configure and <u>use</u> the peripheral.* *1b. <u>Understand</u> and <u>use</u> ready Device driver.*	*Lecture Only (Experiment 1, will be covered during class room exercise)*	*Mixture of black board and ICT*
Unit– II Communication Protocol and Device driver.	**2**	*2a. <u>identify</u> the problems with communication and <u>provide</u> the solution.*	*Lecture Only. The Lab hours will also be utilized for theory.*	*Mixture of black board and ICT*
	3	*Pro4. <u>Learn</u> the following architecture in an Unknown Microcontroller, <u>Develop</u>/<u>Write</u> C driver for it* *a. SPI, I2C,* *b. CAN communication* *2b. <u>Understand</u> and <u>use</u> Device driver for various communication Protocols and <u>find</u> bug if any.*	*Lecture + Experimentation*	*Mixture of black board and ICT*
Unit– III FPGA Programming in VHDL	**4**	*Pro2. <u>Develop/Write</u> basic codes for FPGA in VHDL as in follows, and also write test bench program for the same.* *a. Combinational Circuits like Digital Gates and Digital Circuits based on Digital Gates, SR Non Clocked Flip Flop* *b. Sequential Clock based Circuits* 3a. <u>Develop</u> VHDL program for given architecture.	*Lecture + Experimentation* *One of the lecture of this week will be utilized for experiment along with regular two-hour lab in a week.*	*Mixture of black board and ICT*
	5	*Pro3. <u>Learn</u> the following architecture in an Unknown Microcontroller, and <u>develop/Write</u> the VHDL code to implement the given architecture of peripheral in FPGA.*	*Experimentation only. (The lecture hours of this week will be utilized as lab hours along with regular slot*	*Mixture of black board and ICT*

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		a. GPIO *b. SPI,I2C, UART, USART, CAN communication* 3b. <u>Develop</u> VHDL program for given architecture.	*of lab)*	
	6	*Pro5. <u>Learn</u> the following architecture in an Unknown Microcontroller, <u>Develop/Write</u> VHDL code to implement the given architecture of peripheral in FPGA.* *a. UP, Down, Up-Down Timer/Counter* *b. Watchdog Timer* *c. Pulse Width Modulator* *d. Capture Compare Unit* *3a. <u>Understand</u> the given VHDL program, and <u>modify</u> it as per requirement.* 3b. <u>Develop</u> VHDL program for given architecture.	*Experimentation only. (The lecture hours of this week will be utilized as lab hours along with regular slot of lab)*	*Mixture of black board and ICT*
	7	*Pro6. <u>Learn</u> the following architecture in an Unknown Microcontroller, <u>Develop/Write</u> C driver for it.* *a. UP, Down, Up-Down Timer/Counter* *b. Watchdog Timer* *c. Pulse Width Modulator* *d. Capture Compare Unit* 3c. <u>Develop</u> VHDL Program for interfacing external chip with FPGA.	*Lecture + Experimentation*	*Mixture of black board and ICT*
	8	3d. <u>Develop</u> VHDL Program for interfacing external chip with FPGA.	*Lecture Only. The Lab hours will also be utilized for theory.*	*Mixture of black board and ICT*
Unit– V **Expert Systems implementation on computer On chip**	**9**	*Pro7. <u>Write</u> AI and ML codes in Prolog, Python and C, and <u>Test</u> the result and speed of execution on given Computer On chip/board device and <u>make</u> (<u>Develop</u> interfacing and <u>Write</u> code) it communicating it to main controller/FPGA chip in Embedded board.* 5a. <u>Select</u> Embedded board for required AI ML algorithm and target Embedded	*Lecture + Experimentation*	*Mixture of black board and ICT*

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		application. 5b. Write python code to access pins of computer on board device.		
	10	*Pro7. Write AI and ML codes in Prolog, Python and C, and Test the result and speed of execution on given Computer On chip/board device and make (Develop interfacing and Write code) it communicating it to main controller/FPGA chip in Embedded board.* 5c. Briefly understand various AI/ML Algorithm and the parameters related to it and parameters to be passed to python AI/ML function and Apply it in coding. 5d. Implement AI/ML algorithm on computer on board device and develop interfacing scheme and write code to communicate the result to main controller/FPGA chip.	*Lecture + Experimentation*	*Mixture of black board and ICT*
	11	*Pro7. Write AI and ML codes in Prolog, Python and C, and Test the result and speed of execution on given Computer On chip/board device and make (Develop interfacing and Write code) it communicating it to main controller/FPGA chip in Embedded board.* 5c. Briefly understand various AI/ML Algorithm and the parameters related to it and parameters to be passed to python AI/ML function and Apply it in coding. 5d. Implement AI/ML algorithm on computer on board device and develop interfacing scheme and write code to communicate the result to main controller/FPGA chip.	*Lecture + Experimentation*	*Mixture of black board and ICT*
	12	*Pro7. Write AI and ML codes in Prolog, Python and C, and Test the result and speed of execution on given Computer On chip/board device and make (Develop interfacing and Write code) it*	*Lecture + Experimentation*	*Mixture of black board and ICT*

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		communicating it to main controller/FPGA chip in Embedded board. 5c. Briefly understand various AI/ML Algorithm and the parameters related to it and parameters to be passed to python AI/ML function and Apply it in coding. 5d. Implement AI/ML algorithm on computer on board device and develop interfacing scheme and write code to communicate the result to main controller/FPGA chip.		
Unit– IV **Fundamental of Real Time Operating System**	**13**	*4a. Select the Operating system as per requirement.*	*Lecture Only.*	*Mixture of black board and ICT*
	14	*4b. Develop Real Time small Operating system and use it for developing an Embedded application.*	*Lecture + Experimentation in class room*	*Mixture of black board and ICT*
	15	*Pro8. Test the small Real Time Operating system (developed during theory hours or market available with support of documentation) for various functionality.* *4b. Develop Real Time small Operating system and use it for developing an Embedded application.*	*Lecture of 1 hour + Experimentation in class room and lab location for 4 hours*	*Mixture of black board and ICT*

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	*Introduction to Embedded system*	*Shibu K V*	*MacGraw Hill Education (India) Pvt Ltd Thirteen Reprint* *ISBN (13 digit): 978-0-07-014589-4* *ISBN (10 digit):0-07-014589-X*
2	*Embedded Systems*	*Raj Kamal*	*Tata – MGHill, Second Edition*

S. No.	Title of Book	Author	Publication including ISBN
	Architecture, programming and design		*ISBN 13: 978-0-07-066764-8* *ISBN 10: 0-07-066764-0*
3	*Embedded System Architecture: A Comprehensive Guide for Engineers and Programmers*	*Tammy Noergaard*	*Elsevier, Newnes* *ISBN: 0-7506-7792-9*
4	*The Definitive Guide to ARM cortex M3 and M4 Processor,*	*Joseph Yiu*	*Elsevier, Newnes* *Third Edition* *ISBN-13: 978-0-12-408082-9*
5	*Real Time Operating Systems for ARM cortex-M Microcontrollers,.*	*Jonathan W. Valvano*	*Volume 3, 4th Edition* *ISBN-13: 978-1466468863* *ISBN-10: 1466468866*
6	*ARM System-on-chip Architecture*	*Steve furber*	*Pearson* *ISBN-13: 9780201675191*
7	*Machine Learning using Python*	*Manaranjan Pradhan, U Dinesh Kumar*	*Wiley,* *ISBN: 978-81-265-7990-7* *ISBN: 978-265-8855-8(ebk)*
8	*Artificial Intelligence*	*Elaine Rich, Kevin Knight, Shivashankar B Nair*	*Third Edition, TATA MACGRAW HILL* *ISBN 13: 978-0-07-008770-5* *ISBN 10: 0-07-008770-9*
9	*VLSI Design*	*A. Albert Raj, T. Latha*	*PHI Learning* *ISBN-978-81-203-3431-1*
10	*Circuit Design and simulation with VHDL.*	*Volnei A. Pedroni*	*MIT Press* *ISBN-10: 0262014335* *ISBN-13: 978-0262014335*

XV SOFTWARE/LEARNING WEBSITES

1. Free RTOS online Documentation: https://www.freertos.org/
2. *NPTEL Courses by Prof. Santanu Chaudhary, IIT Delhi*
   https://nptel.ac.in/courses/108102045

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE 377	**Semester VI**	**Protection and Switchgear** Mark '**H'=3** for high, '**M**'=2 for medium or **'L'=1'** for the relevant correlation of each PO or PSO, Competency or CO													
377.1	*<u>Use</u> the ready driver to configure the on chip peripherals and develop driver as a C callable function, <u>find</u> the bug in driver function.*	***1***	***2***	-	-	-	-	-	-	-	-	-	-	-	-
377.2	*<u>Write/Develop</u> VHDL program for required architecture. <u>Test and Debug</u> it.*	**1**	**3**	**3**	**2**	**1**	-	-	-	**2**	**2**	**2**	**1**	**2**	-
377.3	*Develop Small Real Time Operating System(RTOS) and <u>Write</u> code for Embedded application using RTOS.*	**1**	**3**	**3**	**2**	**1**	-	-	-	**2**	**2**	**2**	-	**2**	-
377.4	*<u>Apply and Test</u> Artificial Intelligence (AI) and Machine Learning (ML) algorithms for an Embedded Application implemented on Computer on Board Devices.*	**1**	**3**	**3**	**2**	**1**	-	-	-	**2**	**2**	**2**	**1**	**2**	**2**

XVII NAMES OF RESOURCE PERSONS

a) Jivanadhar A. Joshi

Course Code: **EE378**

Course Title: **POWER ELECTRONICS APPLICATIONS IN POWER SYSTEM**

UG Engineering Programme	**Semester**
Electrical Engineering	six

I RATIONALE

Electrical engineers have to deal with various types of electrical and electronics *circuit components, its application, functions and basic electrical circuit theorem* used in many applications. This curriculum course structure is developed such that the student is able to understand and apply the basic principles and laws of electrical and electronics engineering with emphasis on the analysis and application to simple practical engineering problems.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

Understand and analyze the Modeling the FACTS and HVDC system to solve the power system problem.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an <u>integrated performance</u> of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1. To learn the operating principle of FACTS DEVICES.
2. Develop the skill to model the FACTS devices to solve the power system problems
3. To learn the operating principle of HVDC system .
4. Develop the skill to model the different types of HVDC system to solve the power system problems
5. To address the underlying concepts of power quality, its improvement and standards

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			**Total Credits (L+T+P)**	**Examination Scheme**				
				Theory Marks		**Practical Marks**		**Total Marks**
L	**T**	**P**	**C**	**ESE**	PA	**ESE**	**PA**	150
3	0	2	4	70	30*	25	25	

***Legends: L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

***(*):** Under the **Theory PA**, out of 40 marks, **20 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	3,4	To study and simulate the operation of Multi-Pulse Converter Using MATLAB Simulation.	03
2	3,4	To study and analyze the operation of 6 pulse HVDC system using MATLAB Simulation in Power System.	03
3	3,4	To study and analyze the operation of 6 pulse HVDC system using Hardware model.	03
4	1,2	To study and analyze the operation of Static VAr Compensator (SVC) using MATLAB Simulation in Power System.	03
5	1,2	To Study and Analyze the Operation of 3-Phase 3-Level Diode Clamped Multi-Level Inverter Using Hardware Model.	03
6	1,2	To Study and analyze the operation of Static Synchronous Compensator (STATCOM) using MATLAB Simulation in Power System.	03
7	1,2	To Study and analyze the operation of Thyristor Controlled Series Capacitor (TCSC) using MATLAB Simulation in Power System.	03
8	1,2	To Study and analyze the operation of VSC-based FACTS Controllers using Hardware model.	03
9	1,2	Comparison of SVC and STATCOM in Static Voltage Stability Margin Enhancement using PSAT Software.	03
10	1,2	To Study and Observe the effect of Voltage Variations on the Performance of DVR.	03
		Total	**30**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	HVDC demo kit	1,2,3
2	DSO	1 to 10
3	FACTS demo kit with all FACTS Controller	3
4	Multimeter	3,4
5	Matlab simulink	1 to 10

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I FACTS Controllers	**CO 1.To learn the operating principle of FACTS DEVICES.** **CO 2. Develop the skill to model the FACTS devices to solve the power system problems** PrO 4 To study and analyze the operation of Static VAr Compensator (SVC) using MATLAB Simulation in Power System. PrO 5 To Study and Analyze the Operation of 3-Phase 3-Level Diode Clamped Multi-Level Inverter Using Hardware Model. PrO 6 To Study and analyze the operation of Static Synchronous	1.1 Need of FACTS devices 1.2 classification of FACTS devices, operating principle of SVC 1.3 V-I characteristic of SVC 1.4 Advantage of slope in V-I characteristic, SVC applications for transient and voltage stability improvement, mitigation of SSR 1.5 Advantages of TCSC, different mode of operation of TCSC, different modeling concepts of TCSC, 1.6 Operating principle of STATCOM, SSSC and UPFC and their applications for power

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	Compensator (STATCOM) using MATLAB Simulation in Power System. PrO 7 To Study and analyze the operation of Thyristor Controlled Series Capacitor (TCSC) using MATLAB Simulation in Power System. PrO 8 To Study and analyze the operation of VSC-based FACTS Controllers using Hardware model. PrO 9 Comparison of SVC and STATCOM in Static Voltage Stability Margin Enhancement using PSAT Software. PrO 10 To Study and Observe the effect of Voltage Variations on the Performance of DVR. 1a. *Analyse the operation of SVC* 1b. *Analyse the operation of TCSC* 1c. *Analyse the operation of STATCOM* 1d. *Analyse the operation of SSSC and UPFC*	system performance improvement, 1.7 Power flow solution with SVC, TCSC, SSSC and UPFC
Unit– II HVDC Transmissi	**CO 3. To learn the operating principle of HVDC system .** **CO 4. Develop the skill to model the different types of HVDC system to solve the power system problems** PrO 1 To study and simulate the operation of Multi-Pulse Converter Using MATLAB Simulation. PrO 2 To study and analyze the operation of 6 pulse HVDC system using MATLAB Simulation in Power System. PrO 3 To study and analyze the operation of 6 pulse HVDC system using Hardware model. *2a Analyse the operation of 6-pulse rectifier* 2b. *Analyse the operation of 6-*	2.1 Introduction, equipment required for HVDC systems, 2.2 Comparison of AC and DC Transmission, Limitations of HVDC transmission lines, 2.3 Reliability of HVDC systems, comparison of HVDC link with EHVAC link, HVDC system configuration and components, 2.4 Fundamental equations in HVDC system, HVDC links, converter theory and performance equation, valve characteristic, converter circuits, 2.5 Converter transformer rating, multi bridge converters, abnormal operation of HVDC system, control of HVDC

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	pulse inverter 2c. *Analyse the operation of bilateral HVDC system*	system, harmonics and filters. Influence of AC system strength on AC/DC system interaction, response to AC and DC system faults
Unit– III Power Quality problems & Standards	**CO 5. To address the underlying concepts of power quality, its improvement and standards** 3a. Understand the basic terms of power quality. 3b. Understand the standards of power quality	3.1 Introduction: Definition & Reasons of Occurrence of following Voltage Dip, Brief voltage increases, Brief voltage interruption, Transients, Voltage Notches, Flickers, Distortion, Un-balance. Power Quality Indices, 3.2 Limits of Harmonic Distortion according to IEEE, IEC, EN and NORSOK limits. 3.3 Brief Introduction of Power quality Standards: IEC 61000-2-5,IEC 61000-2-1, IEC 1159 (Categories of Power quality variation according to IEEE 1159 standard with their relevant Spectral content, Duration of occurrence & Magnitude)

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	FACTS Controllers	30	3	3	6	9	10	4	35
II	HVDC Transmission	20	2	3	5	6	6	2	24
III	Power Quality problems & Standards	10	2	1	3	2	3	0	11
Total		60	**7**	**7**	**14**	**17**	**19**	**6**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) Design the any FACTS controller and Simulink in MATLAB.
2) Design the 12 Pulse, 18 pulse and 24 pulse controller and Simulink in MATLAB.
3) Determine the power quality of incoming power in terms of all parameters using power quality devices.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different Semiconductor switches used in converters.
b) Prepare 15 min. power point presentation on how to choose the particular converter for the particular application.

XIII COURSE PLAN

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I FACTS Controllers	PrO 4 To study and analyze the operation of Static VAr Compensator (SVC) using MATLAB Simulation in Power System. PrO 5 To Study and Analyze the Operation of 3-Phase 3-Level Diode Clamped Multi-Level Inverter Using Hardware Model. PrO 6 To Study and analyze the operation of Static Synchronous Compensator (STATCOM) using MATLAB Simulation in Power System. PrO 7 To Study and analyze the operation of Thyristor Controlled Series Capacitor (TCSC) using MATLAB Simulation in Power System. PrO 8 To Study and analyze the operation of VSC-based FACTS Controllers using Hardware model. PrO 9 Comparison of SVC and STATCOM in Static Voltage Stability Margin Enhancement using PSAT Software. PrO 10 To Study and Observe the effect of Voltage Variations on the Performance of DVR. 1a. *Analyse the operation of SVC* 1b. *Analyse the operation of TCSC* 1c. *Analyse the operation of STATCOM* 1d. *Analyse the operation of SSSC and UPFC*	1.1 Need of FACTS devices 1.2 classification of FACTS devices, operating principle of SVC 1.3 V-I characteristic of SVC 1.4 Advantage of slope in V-I characteristic, SVC applications for transient and voltage stability improvement, mitigation of SSR 1.5 Advantages of TCSC, different mode of operation of TCSC, different modeling concepts of TCSC, 1.6 Operating principle of STATCOM, SSSC and UPFC and their applications for power system performance improvement, 1.7 Power flow solution with SVC, TCSC, SSSC and UPFC
Unit– II HVDC Transmissi	PrO 1 To study and simulate the operation of Multi-Pulse Converter Using MATLAB	2.1 Introduction, equipment required for HVDC systems, 2.2 Comparison of AC and DC

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain* *(to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	Simulation. PrO 2 To study and analyze the operation of 6 pulse HVDC system using MATLAB Simulation in Power System. PrO 3 To study and analyze the operation of 6 pulse HVDC system using Hardware model. *2a Analyse the operation of 6-pulse rectifier* 2b. *Analyse the operation of 6-pulse inverter* 2c. *Analyse the operation of bilateral HVDC system*	Transmission, Limitations of HVDC transmission lines, 2.3 Reliability of HVDC systems, comparison of HVDC link with EHVAC link, HVDC system configuration and components, 2.4 Fundamental equations in HVDC system, HVDC links, converter theory and performance equation, valve characteristic, converter circuits, 2.5 Converter transformer rating, multi bridge converters, abnormal operation of HVDC system, control of HVDC system, harmonics and filters. Influence of AC system strength on AC/DC system interaction, response to AC and DC system faults
Unit– III Power Quality problems & Standards	3a. Understand the basic terms of power quality. 3b. Understand the standards of power quality	3.1 Introduction: Definition & Reasons of Occurrence of following Voltage Dip, Brief voltage increases, Brief voltage interruption, Transients, Voltage Notches, Flickers, Distortion, Un-balance. Power Quality Indices, 3.2 Limits of Harmonic Distortion according to IEEE, IEC, EN and NORSOK limits. 3.3 Brief Introduction of Power quality Standards: IEC 61000-2-5,IEC 61000-2-1, IEC 1159 (Categories of Power quality variation according to IEEE 1159 standard with their relevant Spectral content, Duration of occurrence & Magnitude)

XLIX SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Direct Current Transmission,	E. W. Kimbark	Vol 1, Wiley-Inderscience, 1971
2	Power System Stability and Control	P. Kundur	McGraw Hill, 1994.
3	HVDC Transimission,	S. Kamakshaiah and V. Kamaraju	McGraw Hill, 2011
4	,HVDC Power Transmission Systems: Technology and System Interactions,	K.R. Padiyar	New Age International
5	FACTS: Modeling and Simulation in Power Networks,	E. Acha	Wiley 2004
6	Reactive Power Control in Electric Systems,	T.J.E Miller	Wiley India Edition, 1982
7	Thyristor-based FACTS controllers for Electrical Transmission systems	R. Mohan Mathur and R. K. Varma	IEEE Press, 2002

XIV SOFTWARE/LEARNING WEBSITES

1. http://nptel.ac.in/courses/108108077/
2. http://nptel.ac.in/courses/108102046/
3. http://nptel.ac.in/courses/108104011/

XV PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE378	Semester I														
378	**Competency** •Understand and analyze the Modeling the FACTS and HVDC system to solve the power system problem.	3	2	2	2	1	-	1	-	-	-	-	1	3	-
378.1	To learn the operating principle of FACTS	3	3	-	-	-	-	-	-	2	-	-	3	3	-

Course Code	Competency and COs	**PO 1** Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	DEVICES.														
378.2	Develop the skill to model the FACTS devices to solve the power system problems	2	1	3	-	3	-	-	-	-	-	-	-	2	-
378.3	To learn the operating principle of HVDC system .	3	2	2	3	2	-	3	3	3	3	-	2	3	-
378.4	Develop the skill to model the different types of HVDC system to solve the power system problems	2	2	-	-	1	-	-	-	2	-	-	-	1	-
378.5	To address the underlying concepts of power quality, its improvement and standards	3	-	-	-	-	-	-	-	-	-	-	2	1	-

XVI NAMES OF RESOURCE PERSONS

a) Dr. Dharmesh Dabhi

Course Code: **EE379**
Course Title: **High Voltage Enginering**

UG Engineering Programme	Semester
Electrical Engineering	Sixth

I RATIONALE

The electrical stresses experienced by the insulations will reduce the life expectancy of high voltage equipment. Hence, a thorough understanding of the fundamental properties of the materials and their failure mechanisms during service is essential for appropriate and optimal design. This curriculum course structure is developed such that the student is able to use the knowledge of different insulation systems and perform the high voltage tests on electrical equipment under the impression of different heavy electrical stress.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- Perform the different high voltage tests on electrical equipment to access the performance parameters.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) ***Explain*** *the breakdown mechanisms involved in different dielectrics.*
2) ***Use*** *the various generation and measurement techniques of high voltages and current.*
3) ***Perform*** *the insulation coordination studies.*
4) ***Perform*** *the destructive tests on electrical equipment.*
5) ***Perform*** *the non-destructive tests on electrical equipment.*

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			**Total Credits (L+T+P)**	**Examination Scheme**				
				Theory Marks		**Practical Marks**		**Total Marks**
L	**T**	**P**	**C**	**ESE**	PA	**ESE**	**PA**	
3	0	2	4	70	30*	25	25	150

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

(*): *Under the* ***Theory PA****, out of 30 marks,* ***15 marks*** *are for* ***micro-project assessment*** *to facilitate integration of COs and the remaining 15 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Create a detailed report for the different types of insulating materials.	02
2	2	Create a detailed report for the generation and measurement of high voltage and current in high voltage laboratory.	02
3	2	Examine the impact of certain conditions on the breakdown voltage using sphere gap assembly.	04
4	2	Examine the breakdown characteristic for the given rod-rod, rod-plane and plane-plane electrode configurations.	04
5	3	Examine the function of horn gap lightning arrestor.	02
6	4	Determine the dielectric strength of the given transformer oil.	02
7	4	Determine the dielectric strength of the given solid dielectric materials.	02
8	4	Determine the power frequency flashover voltage of the given insulator.	02
9	4	Perform the impulse voltage test on the given insulator.	04
10	5	Measure the dielectric loss factor and dc resistivity of the given transformer oil.	02
11	5	Trace the equipotential lines and electric filed lines in the given electrode systems using electrolytic tank.	02
12	5	Examine the localized discharge activity for a given insulator using COMSOL Multiphysics.	02
		Total	**30**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Test rig for sphere gap & rod gap assembly: • 100kV AC/140kV DC high voltage tester • 65mm, 100mm & 150mm spherical electrodes • Earth electrode, rod electrodes and plane electrodes	3,4,8,10
2	Horn gap lightning arrestor	5
3	0-100 kV motorized oil BDV test kit	6
4	30kV AC/40kV DC high voltage tester	7
5	5-stage, 1.5kJ, 150kV Marx Impulse voltage generator	9
6	Oil dissipation factor and dc resistivity meter	11
7	Electrolytic Tank	12
8	COMSOL Multiphysics Software	13

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Function as a team member.
3) Function as a team leader.
4) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Breakdown mechanisms in gaseous, liquid and solid dielectrics	**CO 1.** ***Explain*** *the breakdown mechanisms involved in different dielectrics.* *PrO 8. Create a detailed report for the different types of insulating materials.* 1a. Justify the requirement of insulation for the given electrical equipment. 1b. Memorize the different types of insulating materials. 1c. Explain the ionization/conduction mechanisms involved in dielectric breakdown. 1d. Evaluate the dielectric strength of liquid dielectrics. 1e. Evaluate the dielectric strength of solid dielectrics. 1f. Evaluate effect of impurities over the performance of the insulation.	1.1 Need of insulation 1.2 Different types of insulation 1.3 Nature and causes of insulation failures 1.4 Ionization processes 1.5 Breakdown theories

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit– II Generation of High Voltages and Currents	**CO 2.** *Use the various generation and measurement techniques of high voltages and current.* *PrO 2.* Create a detailed report for the generation and measurement of high voltage and current in high voltage laboratory. 2a. Choose the relevant high voltage generation technique. 2b. Determine the circuit parameters to obtain the standard wave shape of test voltage as per the Indian or international standards. 2c. Evaluate the performance of generating methods for the specified conditions.	2.1 Different methods to generate high DC, AC and impulse voltages and currents. 2.2 Operating principles and working of various high DC voltage generating techniques. 2.3 Operating principles and working of various high AC power frequency voltage generating methods. 2.4 Operating principle and working of high frequency high voltage AC generation technique. 2.5 Impulse voltage wave shape and Marx impulse voltage generator. 2.6 Operating principle and working of high impulse current generator.
Unit– III Measurement of High Voltages and Currents	**CO 2.** ***Explain*** *the various generation and measurement techniques of high voltages and current.* PrO 2. Create a detailed report for the generation and measurement of high voltage and current in high voltage laboratory. PrO 3. Examine the impact of certain conditions on the breakdown voltage using sphere gap assembly. PrO 4. Examine the breakdown characteristic for the given rod-rod, rod-plane and plane-plane electrode configurations.	3.1 Different methods to measure high DC, AC and impulse voltages and currents. 3.2 Operating principles and working of various high DC voltage measurement techniques. 3.3 Operating principles and

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	3a. Choose the relevant high voltage and current measurement technique. 3b. Evaluate the performance of high voltage and current measurement methods for the specified conditions.	working of various high AC voltage measurement techniques. 3.4 Operating principles and working of various impulse and high frequency voltage measurement techniques. 3.5 Factors influencing the high voltage measurement using sphere gap assembly. 3.6 Operating principles and working of various high current measurement techniques.
Unit-IV Overvoltage phenomenon and Insulation coordination in Electric Power System	**CO 3. *Perform insulation coordination studies.*** *PrO 5.* <u>Examine the function of horn gap lightning arrestor.</u> 4a. <u>Understand the causes of overvoltage phenomena in electric power system.</u> 4b. <u>Perform</u> insulation coordination studies to determine basic impulse insulation level. 4c. <u>Explain</u> the function of lightning arrestors. 4d. <u>Select</u> the relevant lightning arrestor for <u>the specified</u> application.	4.1 Natural cause of overvoltage – Lightning phenomena 4.2 Overvoltage due to switching surges 4.3 Overvoltage due to system faults and other specified conditions 4.4 Types of lightning arrestors 4.5 Insulation coordination 4.6 Basic impulse insulation level
Unit – V Destructive Testing of Electrical Equipment	**CO 4. *<u>Perform</u>*** *the destructive tests on electrical equipments.* *PrO 6.* <u>Determine the dielectric strength of the given transformer oil.</u> *PrO 7.* <u>Determine the dielectric strength of the given solid dielectric materials.</u> *PrO 8.* <u>Determine the power frequency flashover voltage of the given insulator.</u> *PrO 9.* <u>Perform the impulse voltage test on</u>	

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	the given insulator. 5a. Perform power frequency flashover voltage test on electrical equipment under the specified test conditions. 5b. Perform power frequency withstand voltage test on electrical equipment under the specified test conditions. 5c. Perform impulse voltage test on electrical equipment under the specified test conditions. 5d. Perform short circuit test on circuit breakers. 5e. Perform synthetic test on circuit breaker.	5.1 Power frequency flashover voltage test 5.2 Power frequency voltage withstand test 5.3 Impulse flashover voltage test 5.4 Testing of insulators, bushings, capacitors, transformers, circuit breakers. 5.5 Short circuit test of circuit breakers. 5.6 Synthetic test of circuit breakers.
Unit – VI Non-destructive Testing of Electrical Equipment	**CO 5. *Perform*** *the non-destructive tests on electrical equipments.* *PrO 10.* Measure the dielectric loss factor and dc resistivity of the given transformer oil. *PrO 11.* Trace the equipotential lines and electric filed lines in the given electrode systems using electrolytic tank. *PrO 12.* Examine the localized discharge activity for a given insulator using COMSOL Multiphysics. 6a. Measure the dissipation factor and unknown capacitance of the given insulation. 6b. Measure the dc resistivity of the given insulating liquid. 6c. Perform partial discharge test. 6d. Perform dissolved gas analysis test on transformer oil. 6e. Perform sweep frequency response analysis of electrical equipment. 6f. Predict remaining life assessment of the given high voltage equipment.	6.1 Determination of dissipation factor and unknown capacitance of insulation. 6.2 Determination of DC resistivity of insulation liquid. 6.3 Partial discharge test of electrical equipments. 6.4 Dissolved gas analysis of transformer oil. 6.5 Sweep frequency response analysis of electrical equipment

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Breakdown mechanisms in gaseous, liquid and solid dielectrics	12	2	2	2	4	2	0	12
II	Generation of High Voltages and Currents	12	2	2	4	4	4	2	18
III	Measurement of High Voltages and Currents	08	0	0	4	2	2	0	08
IV	Overvoltage phenomenon and Insulation coordination in Electric Power System	02	2	2	0	2	2	0	08
V	Destructive Testing of Electrical Equipment	09	0	2	4	4	4	0	14
VI	Non-destructive Testing of Electrical Equipment	02	0	2	2	2	2	2	10
Total		**45**	**06**	**10**	**16**	**18**	**16**	**4**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) Undertake the survey and submit a report on different type of insulating materials.
2) Create a prototype model of voltage multiplier circuit for HVDC generation.
3) Create a prototype model of multistage impulse voltage generator.
4) Create a prototype model of tesla coil.
5) Create a prototype model of Van de Graff generator.
6) Examine the equipotential and electric field lines between plane-plane, plane-convex and convex-convex electrode configuration using finite element method magnetics (FEMM) software.
7) Perform electric field and electric potential distribution for the given insulator with AC, DC and impulse voltage using COMSOL Multiphysics software.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different insulations used in industries.
b) Prepare 15 min. power point presentation on how to choose insulation for the particular application.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Breakdown mechanisms in gaseous, liquid and solid dielectrics	1	**PrO 6.** *Create a detailed report for the different types of insulating materials.* 1a. Justify the requirement of insulation for the given electrical equipment. 1b. Memorize the different types of insulating materials. ***(self-learn)*** 1c. Explain the ionization/conduction mechanisms involved in dielectric breakdown.	Lecture with media	PPTs, black board
	2	1d. Explain the ionization/conduction mechanisms involved in dielectric breakdown.	Lecture With media	PPTs, black board
	3	1e. Explain the ionization/conduction mechanisms involved in dielectric breakdown.	Lecture With media	PPTs, black board
	4	1f. Evaluate the dielectric strength of liquid dielectrics. 1g. Evaluate the dielectric strength of solid dielectrics. 1h. Evaluate effect of impurities over the performance of the insulation.	Lecture With media	PPTs, black board
Unit– II Generation of High Voltages and Currents	**5**	**PrO 2.** Create a detailed report for the generation and measurement of high voltage and current in high voltage laboratory. 2a. Choose the relevant high voltage generation technique. 2b. Analyze the high voltage and current generation techniques.	Lecture With media	PPTs, black board
	6	2b. Analyze the high voltage and current generation techniques.	Lecture With media	PPTs, black board
	7	2b. Analyze the high voltage and current generation techniques. 2c. Determine the circuit parameters to obtain the standard wave shape of test voltage as per the Indian or international standards.	Lecture With media	PPTs, black board
	8	2c. Determine the circuit parameters to obtain the standard wave shape of test voltage as per the Indian or international standards. 2d. Evaluate the performance of generating methods for the specified conditions.	Lecture With media	PPTs, black board
Unit III Measurement of High Voltages and Currents	**9**	**PrO 2.** *Create a detailed report for the generation and measurement of high voltage and current in high voltage laboratory.* **PrO 3.** *Examine the impact of certain conditions on the breakdown voltage using sphere gap assembly.*	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		PrO 4. *Examine the breakdown characteristic for the given rod-rod, rod-plane and plane-plane electrode configurations.* 3a. Choose the relevant high voltage and current measurement technique. 3b. Evaluate the performance of high voltage and current measurement methods for the specified conditions.		
	10	3b. Evaluate the performance of high voltage and current measurement methods for the specified conditions.	Lecture With media	PPTs, black board
	11	3b. Evaluate the performance of high voltage and current measurement methods for the specified conditions.	Lecture With media	PPTs, black board
Unit-IV Overvoltage phenomenon and Insulation coordination in Electric Power System **Unit – V Destructive Testing of Electrical Equipment**	**12**	**PrO 5.** Examine the function of horn gap lightning arrestor. 4a. Understand the causes of overvoltage phenomena in electric power system. 4b. Perform insulation coordination studies to determine basic impulse insulation level. 4c. Explain the function of lightning arrestors. 4d. Select the relevant lightning arrestor for the specified application. *PrO 6.* Determine the dielectric strength of the given transformer oil. *PrO 7.* Determine the dielectric strength of the given solid dielectric materials. *PrO 8.* Determine the power frequency flashover voltage of the given insulator. *PrO 9.* Perform the impulse voltage test on the given insulator. 5a. Perform power frequency flashover voltage test on electrical equipments under the specified test conditions.	Lecture With media	PPTs, black board
	13	5a. Perform power frequency flashover voltage test on electrical equipments under the specified test conditions. 5b. Perform power frequency withstand voltage test on electrical equipments under the specified test conditions.	Lecture with media	PPTs, black board
	14	5c. Perform impulse voltage test on electrical equipment under the specified test conditions. 5d. Perform short circuit test on circuit breakers.	Lecture with media	PPTs, black board
Unit – VI Non-	**15**	5e. Perform synthetic test on circuit breaker.	Lecture with media,	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
destructive Testing of Electrical Equipment		*PrO 10.* Measure the dielectric loss factor and dc resistivity of the given transformer oil. *PrO 11.* Trace the equipotential lines and electric filed lines in the given electrode systems using electrolytic tank. *PrO 12.* Examine the localized discharge activity for a given insulator using COMSOL Multiphysics. 6a. Measure the dissipation factor and unknown capacitance of the given insulation. 6b. Measure the dc resistivity of the given insulating liquid. 6c. Perform partial discharge test. 6d. Perform dissolved gas analysis test on transformer oil. 6e. Perform sweep frequency response analysis of electrical equipment. 6f. Predict remaining life assessment of the given high voltage equipment.	Tutorial	

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
10	High Voltage Engineering	Naidu, M.S., Kamaraju, V.	McGraw-Hill, USA, ISBN 0-07-462286-2
11	High Voltage Engineering	Wadhwa, C.L.	New Age International (P) Ltd., New Delhi, ISBN: 9788122430905

XV SOFTWARE/LEARNING WEBSITES

i. http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-KANPU/HighVoltageEngg/ui/TOC.htm
ii. http://www.mv.helsinki.fi/tpaulin/Text.hveng.htm
iii. http://ocw.mit.edu/courses/physics/8-02-electricity-and-magnetism-spring-2002/video-lectures/lecture-6-high-voltage-breakdown-and-lightning.htm
iv. http://www.sayedsaad.com/High_voltage/index_solids.htm

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs											
		PO 1 Used of mathematical method	PO 2 Analysis mechanical system	PO 3 Used engineering soft.	PO 4 Solve problem in machine	PO 5 The engineer and society	PO 6 Environment and sustainability	PO 7 Ethics	PO 8 Individual and team work:	PO 9 Communication	PO 10 Life-long learning	PSO1 Evaluate the performance of Electrical Machines and Power systems Power Systems	PSO2 Evaluate the performance of Electrical and Electronic Instrumentation Systems	PSO3 Provide socially and ethically acceptable l technical solutions to complex electrical engineering problems

Course Code	Competency and COs	POs and PSOs												
		PO 1 Used of mathematical method	**PO 2** Analysis mechanical system	**PO 3** Used engineering soft.	**PO 4** Solve problem in machine	**PO 5** The engineer and society	**PO 6** Environment and sustainability	**PO 7** Ethics	**PO 8** Individual and team work:	**PO 9** Communication	**PO 10** Life-long learning	**PSO1** Evaluate the performance of Electrical Machines and Power systems Power Systems	**PSO2** Evaluate the performance of Electrical and Electronic Instrumentation Systems	**PSO3** Provide socially and ethically acceptable l technical solutions to complex electrical engineering problems
EE379	**Semester VI**	**High Voltage Engineering** **Mark 'H'=3** for high, '**M**'=2 for medium or **'L'=1'** for the relevant correlation of each PO or PSO, Competency or CO												
379	**Competency** Perform the different high voltage tests on electrical equipments to access the performance parameters.	3	2	2	2	1	1	1	-	2	1	3	-	1
379.1	**Explain the breakdown mechanisms involved in different dielectrics.**	3	2	1	1	1	1	1	-	1	1	1	-	1
379.2	**Explain the various generation and measurement techniques of high voltages and current.**	3	2	2	1	1	1	1	-	1	-	2	-	1
379.3	**Perform the insulation coordination studies.**	3	2	1	1	1	1	1	-	1	-	2	-	1
379.4	**Perform the destructive tests on electrical equipments.**	3	1	1	2	1	1	1	-	1	-	1	-	-
379.5	**Perform the non-destructive tests on electrical equipments.**	3	2	1	1	1	1	1	-	1	1	1	-	-

XVII NAMES OF RESOURCE PERSONS

a) Dr. Mihir A. Bhatt

EE380: INTERNET OF THINGS
6th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Course Objectives:

As electrical power generation and economy is essential gradient for the industrial and all around development of any country, the objectives of the course are:

1. To understand the embedded concepts and embedded system Architecture.
2. To Understand the Architectural Overview of IoT.
3. To Understand the IoT Reference Architecture and Real World Design Constraints.
4. To create protection scheme for three phase power transformer with CT and relay connection and settings.
5. To determine whether existing applications to the cloud makes technical and business sense.
6. To analyze and compare the long-term costs of cloud services.
7. To focus on the application of IOT in Electrical Engineering.

D. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction To Embedded Concepts	07
2	Iot Architecture And Protocols	11
3	Cloud Architecture And Computing	09
4	Computer Networks And Internet	06

5	Internet of Things: Sensors and Actuators	06
6	Preparing Iot Projects	06

Total hours (Theory) : 45 Hrs
Total hours (Lab) : 30 Hrs
Total hours : 75 Hrs

E. Detailed Syllabus:

1 Introduction To Embedded Concepts 15.00% 07 Hrs

Introduction to embedded systems, Application Areas, Categories of embedded systems,
Overview of embedded system architecture, Specialties of embedded systems, recent trends in embedded systems, Architecture of embedded systems, Hardware architecture, Software architecture, Application Software, Communication Software.

2 Iot Architecture And Protocols 25.00% 11 Hrs

Overview: IoT-An Architectural Overview– Building an architecture, Main design principles
and needed capabilities, An IoT architecture outline, standards considerations. M2M and IoT Technology Fundamentals- Devices and gateways, Local and wide area networking, Data management, Business processes in IoT, Everything as a Service(XaaS), M2M and IoT Analytics, Knowledge Management
Reference Architecture: IoT Architecture-State of the Art – Introduction, State of the art, Reference Model and architecture, IoT reference Model - IoT Reference Architecture-Introduction, Functional View, Information View, Deployment and Operational View.

3 Cloud Architecture And Computing 20.00% 09 Hrs

Cloud Architecture Basics: The Cloud -Hype cycle-metaphorical interpretation-cloud architecture standards and interoperability- Cloud types; IaaS, PaaS, SaaS. Benefits and challenges of cloud computing, public, private clouds community cloud, role of virtualization in enabling the cloud.
End to End Design : Requirement analysis: strategic alignment and architecture development cycle-strategic impact-Risk impact-financial impact-Business criteria-technical criteria-cloud opportunities –evaluation criteria and weight-End to end design-content delivery networks-capacity planning-security architecture and design
Cloud Application Architectures: Development environments for service development; Amazon, Azure, Google App-cloud platform in industry
How To Move Application Into The Cloud: Web Application Design- Machine Image Design-privacy design –Database management.

4 Computer Networks And Internet 13.33% 6 Hrs

Internet - The network edge- The network core- Delay , Loss and Throughput in packet switched networks- protocol layers and their service models- TCP/IP protocol architecture- Frame relay networks- ATM networks- protocol architecture- ATM logical connections ATM cell.
Congestion Control In Data Networks And Internet: Effects of congestion – Congestion control - Traffic management - Congestion control in packet switching networks – Frame relay- Congestion control -TCP flow control and TCP congestion control.

5 Internet Of Things: Sensors and Actuators 13.33% 6 Hrs

Introduction: Internet of Things Promises–Definition– Scope–Sensors for IoT Applications– Structure of IoT– IoT Map Device
Sensors: Linear and angular displacement sensors: resistance sensor – induction displacement sensor – digital optical displacement sensor – pneumatic sensors. Speed and flow rate sensors: electromagnetic sensors – fluid flow sensor – thermal flow sensor. Force sensors: piezoelectric sensors – strain gauge sensor – magnetic flux sensor – inductive pressure sensor – capacitive pressure sensor. Temperature sensors: electrical – thermal expansion – optical.

6 Preparing Iot Projects 13.33% 6 Hrs

Creating the sensor project - Preparing Raspberry Pi - Clayster libraries - Hardware- Interacting with the hardware - Interfacing the hardware- Internal representation of sensor values - Persisting data - External representation of sensor values - Exporting sensor data - Creating the actuator project- Hardware - Interfacing the hardware - Creating a controller - Representing sensor values - Parsing sensor data - Calculating control states - Creating a camera - Hardware -Accessing the serial port on Raspberry Pi - Interfacing the hardware - Creating persistent default settings - Adding configurable properties - Persisting the settings
- Working with the current settings -Initializing the camera

F. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	20	10	20	20	10

G. Course Outcomes (Learning Outcomes):

Upon successful completion of this course, a student would be able to

1. Enhance their skills in basics of embedded systems. (1)
2. Know the protocols of IOT and its types of protocols. (1,5, PSO1)

3. Understand applications different sensors in internet of things and in electrical engineering and programming in embedded C. (2,5, PSO1)
4. Know the all communication protocols and Design and troubleshoot hardware circuits based on different sensors. (2,3,4,5,9, PSO1,2)

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	2	-	-	-	2	-	-	-	-	-	-	-	2	-
CO3	-	3	-	-	2	-	-	-	-	-	-	-	3	-
CO4	-	2	3	2	3	-	-	-	3	-	-	-	3	2

H. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The visit of CHARUSAT MEDICAL Campus HT control room and nearby substation will clear the concept of relay and relay setting and student will realiase the actual application of relay.

- In lab session, the student will calculate the relay setting for particular application, will set the calculated value in relay and by performing the practical, realize the operation of relay under pre-determined conditions.

I. Recommended Study Material:

Reference Book:

1. Designing the Internet of Things ,by Adrian McEwen & Hakim Cassimally , John Wiley & Sons ltd.
2. From machine to machine to the Internet of Things, by Jan Holler, vlasios Tsiatsis, Catherine mulligan, Elsevier.
3. Internet of Things, Converting Technologies for Smart Environments and Integrated Ecosystems, by Ovidiu Vermesan and Peter fries, River Publishers.

Web Material:

https://nptel.ac.in/courses/106/105/106105166/ (IIT Kharagpur)

https://nptel.ac.in/courses/108108123/ (IISC Bangalore)

https://nptel.ac.in/courses/106105195/# (IIT Kharagpur)

J. List of Experiments:

Sr. No.	Title
1	Using GPIO and Timer: Using inbuilt timer of ESP, blink LED at an interval of one second.
2	Using UART for Serial Print: Print a statement on Serial Terminal
3	Using ADC for analog sensing: Use POT as an analog input to ESP8266 and print its value on serial terminal
4	Using I2C communication: Use I2C pins for communication with LCD and display characters on it.
5	Using PWM: Use GPIO to change the brightness of LED using pulse width modulation
6	Use POT as an analog input to ESP8266 and use this value to change the brightness of LED using pulse and print value on serial terminal also.
7	Using ESP as Access Point: Use ESP module as access point and connect other device to it using WiFi
8	Using ESP as Station: Use ESP module as station and connect to a router (or access point).

9	Plotting data on plot.ly: Take analog input from ESP and pass that data to www.thingspeak.com and prepare an online graph.
10	Using IoT protocol: Demonstrate simple publish subscribe mechanism of MQTT protocol using MQTT protocol
11	Using OneWire interface: Interface DS18b20 with ESP module using onewire interface
12	Expanding GPIO: Interface ESP with MCP23008 to expand GPIOs using I2C protocol
13	Develop offline Webserver to control GPIO: Demonstrate offline webserver using HTML webpage which can be accessed from web browser and through which LED can be toggled
14	Control GPIO from an Android App: Use an android application to control the GPIO using HTTP GET requests.
15	EEPROM Clear, Write and Read in ESP
16	TELNET (TCP) in ESP

K. **Course Assignments:**

	% weightge (Approx.)
Midterm exams	**20**
Final exams	**70**
Assignments	**10**

L. **Course Assessments & Course Outcomes matrix (LOs or CLOs)**

Assessment Methods	**COs or LOs**
1st Midterm exam	CO No: 1, 2
2nd Midterm exam	CO No: 3,4
Final exam	CO No: 1 to 4

M. Teaching Methods & Learning Activities: (tick the applicable activities)

Telling/Explaining	✓	Collaborating	
Questioning	✓	Experiments	✓
Reading	✓	Oral Presentations/Reports	
Demonstrating	✓	Web Searching	✓
Problem Solving	✓		

N. **Assessment Methods (Formal & Informal)** (tick the applicable activities)

Test/Exam	✓
Quiz	✓

O. Student Work load: (Total 135 Hours)

Course Readings	20 hrs	Exams/Quizzes	10 hrs
Problem Solving	30 hrs	Lectures	45 hrs
Lab Work	30 hrs		

Course Code: **EE381**

Course Title: **Hybrid and Electric Vehicles**

UG Engineering Programme	**Semester**
Hybrid and Electric Vehicles	Sixth

I RATIONALE

To lower the greenhouse gases and promote the ecofriendly vehicles to the urban infrastructure, the EVs will dominate the future mobility. Electrical engineer has to develop the EVs which ecofriendly, high performance and according to the industry standards. This course is developed such that students should be able to analyze, evaluate, design, develop, implement the EV, HEV and PHEV and coordinate the operation of EVs in smart grid.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Analyze the performance of Hybrid and Electric vehicles and its effect on power system grid.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

Identify *the* various terminologies associated with EV, HEV and PHEV.

Analyze the performance of motors used in modern electrical vehicles.

Analyze the *various chargers, charging schemes and batteries used for EV applications.*

Evaluate the factors associated grid when integrating EVs in the smart grid.

Examine *the various market policies and standards associated with EVs.*

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			**Total Credits (L+T+P)**	**Examination Scheme**				
				Theory Marks		**Practical Marks**		**Total Marks**
L	**T**	**P**	**C**	**ESE**	**PA**	**ESE**	**PA**	
3	2	0	4	70	30	25	25	150

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

(*): *Under the* ***Theory PA****, out of 40 marks,* ***20 marks*** *are for* ***micro-project assessment*** *to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken*

during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	**CO No.**	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	**Approx. Hrs. required**
1	1	Examine the specifications of MG ZS EV, also make a brief report on various systems in the car.	02
2	1	Examine the performance of differential gear drive for vehicular applications.	02
3	2	Examine the performance of DC motor drive model available in MATLAB Simulink.	02
4	2	Examine the performance of Induction motor drive model available in MATLAB Simulink.	02
5	2	Examine the performance of PMBLDC motor drive model available in MATLAB Simulink.	02
6	2	Examine the performance of PMSM motor drive model available in MATLAB Simulink.	02
7	2	Prepare the open loop control & closed loop control PMBLDC drive with the help of hall sensor based controller in MATLAB Simulink.	02
8	2	Prepare the open loop & closed loop control PMBLDC drive with the help of back emf based controller in MATLAB Simulink.	02
9	2	Prepare the open loop & closed loop control PMSM motor drive with bidirectional control in MATLAB Simulink.	02
10	2	Examine the performance of BMS model available in MATLAB Simulink.	02
11	3	Examine the performance of charger.	02
12	4	Examine the effect of EV charging/discharging and its effects on grid based on reactive power compensation model available in MATLAB Simulink.	02
13	4	Examine the effect of EV charging/discharging and its effects on grid based on frequency regulations model available in MATLAB Simulink.	02
		Total	**26**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	**Equipment Name with Broad Specifications**	**PrO No.**
1	AC & DC power supply	2,10
2	MATLAB Simulation software (version 2021a)	3 to 9, 11,12

S. No.	Equipment Name with Broad Specifications	PrO No.
3	MG ZS EV	1
4	Differential gear setup	2
5	Off board DC charger	10

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Introduction of Electric Vehicle	**CO 1.** ***Identify*** *the* various terminologies associated with EV, HEV and PHEV. PrO 1. Examine the specifications of MG ZS EV, also make a brief report on various systems in the car. PrO 2. Examine the performance of differential gear drive for vehicular applications. 1a. Describe the History of Electric Vehicles	1.1 History of Electric Vehicles (EVs), EV and the environment 1.2 EV challenges, Pure EV, Hybrid EV, PHEVs, Fuel cell EV 1.3 EV Technologies: motor drive technology, energy source technology. 1.4 Battery charging technology.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	from gasoline to PHEVs. 1b. State the challenges in implementing the EVs in the world. 1c. List the Pure EV, Hybrid EV, PHEVs, Fuel cell EV. 1d. Discuss Four quadrant operation of an electric drive. 1e. Explain Various energy sources for EV applications. 1f. List Various batteries used for EVs and its behavior.	
Unit– II Electric Motor Drives for EV	**CO 2. Analyze** the performance of motors used in modern electrical vehicles. PrO 3. Examine the performance of DC motor drive model available in MATLAB Simulink. PrO 4. Examine the performance of Induction motor drive model available in MATLAB Simulink. PrO 5. Examine the performance of PMBLDC motor drive model available in MATLAB Simulink. PrO 6. Examine the performance of PMSM motor drive model available in MATLAB Simulink. PrO 7. Prepare the open loop control & closed loop control PMBLDC drive with the help of hall sensor based controller in MATLAB Simulink. PrO 8. Prepare the open loop & closed loop control PMBLDC drive with the help of back emf based controller in MATLAB Simulink. PrO 9. Prepare the open loop & closed loop control PMSM motor drive with bidirectional control in MATLAB Simulink. 2a. Explain Block diagram of an electric drive system and explain each block. 2b. Explain the working of DC-DC converter. 2c. Examine the control of DC motor 2d. Compare the various methods for speed braking of DC motor and	2.1 DC Motor Drive: system configuration, structure, operation, DC-DC converter, control of DC motor, regenerative braking. 2.2 Induction Motor Drive: system configuration, structure, operation, Inverters, control of induction motor 2.3 PMBLDC Motor Drive: PM Materials, system configuration, structure, operation, inverters, control of PMBLDC motor, Comparison of various motor drives for EV application. 2.4 PMSM Motor Drive: system configuration, structure, operation, inverters, control of PMSM motor, Comparison of various motor drives for EV application.

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	explain regenerative braking. 2e. Examine the control of induction motor drive. 2f. List out various PM materials. 2g. Explain PMBLDC motor construction and working principle. 2h. Examine PMBLDC motor and its working under open loop and closed loop configurations. 2i. Explain PMSM motor construction and working principle. 2j. Examine PMSM motor and its working under open loop and closed loop configurations.	
Unit– III **EV Charging Strategy**	**CO 3. Analyze** the various chargers, charging schemes and batteries used for EV applications. *PrO 10.* Examine the performance of BMS model available in MATLAB Simulink. *PrO 11.* Examine the performance of charger. 3a. Identify EV charging requirements. 3b. List various charging methods for EVs. 3c. Examine the control strategies for residential, commercial & photovoltaic charging. 3d. Identify the effect of charging and its effect on Power Demand on Grid, Voltage Profile and frequency regulations. 3e. Examine the concept of Battery Management System and its functionality.	3.1 Categorizing EV charging 3.2 EV Charging Approaches: dumb charging, smart charging coordination, PV Based PEV Charging Facilities, 3.3 Control Strategies: residential photovoltaic charging, Power Demand on Grid, Voltage Profile.
Unit-IV **Vehicle to Grid**	**CO 4. Evaluate** the factors associated grid when integrating EVs in the smart grid. *PrO 12.* Examine the effect of EV charging/discharging and its effects on grid based on reactive power compensation model available in MATLAB Simulink. *PrO 13.* Examine the effect of EV charging/discharging and its effects on grid based on frequency regulations	4.1 Introduction to V2G and G2V concept 4.2 Communication and Control 4.3 Impact of EV Charging On Electrical Distribution Systems 4.4 Concept of V2G and G2V: Definition, Benefits, Risks, Grid Integration, EV Supply Equipment Design And Power Levels

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	model available in MATLAB Simulink. 4a. Discuss the concept of V2G and G2V. concept. 4b. Analyze Impact of EV Charging On Electrical Distribution Systems. 4c. List various factors associated with Grid Integration, EV Supply Equipment Design and Power Levels. 4d. Analyze the Charging Station requirement and its physical connection to the Grid. 4e. Identify various V2X systems and its future.	4.5 Charging Station Environment, Physical Connection to the Grid, 4.6 V2X systems and its future.
Unit – V EV Business Trend & Market Policy	**CO 5.** ***Examine*** *the various market policies and standards associated w0ith EVs.* 5a. Explain the concept of centralized charging control. 5b. Describe the role of EVs in electricity market and EV in smart grid. 5c. Prepare the algorithm for EV charging scheduling, data structure and V2X control.	5.1 Introduction to centralized charging control. 5.2 Electricity market, EV Aggregator, Algorithm for EV charging scheduling, data structure and requirement.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction of Electric Vehicle	06	04	04	02	-	-	-	10
II	Electric Motor Drives for EV	15	10	05	05	08	-	02	30
III	EV Charging Strategy	12	04	02	04	-	-	04	14
IV	Vehicle to Grid	09	02	04	04	-	-	-	10
V	EV Business Trend & Market Policy	03	06	-	-	-	-	-	06
Total		**45**	**06**						**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **Electric motors used for EVs:** Collect the specifications of different types of electric motors used by different vehicle manufacturers.
2) **Batteries used for EVs:** Collect the specifications of different types of batteries used by different vehicle manufacturers.
3) **BMS used for EVs:** Collect the specifications of different types of BMS techniques used by different vehicle manufacturers.
4) **Chargers used for EVs:** Collect the specifications of different types of chargers manufactured by different manufacturers.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different polices for EVs in different countries.
b) Prepare 15 min. power point presentation on the implementation of EVs in different contries.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Introduction of Electric Vehicle	1	**CO 1. *Identify*** *the* various terminologies associated with EV, HEV and PHEV. PrO 1. Examine the specifications of MG ZS EV, also make a brief report on various systems in the car. 1a. Describe the History of Electric Vehicles from gasoline to PHEVs. 1b. State the challenges in implementing the EVs in the world. 1c. List the Pure EV, Hybrid EV, PHEVs, Fuel cell EV.	Lecture with media	PPTs, black board
	2	**CO 1. *Identify*** *the* various terminologies associated with EV, HEV and PHEV. PrO 2. Determine the Mesh current for DC resistive circuits and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink. 1d. Discuss Four quadrant operation of an electric drive. 1e. Explain Various energy sources for EV applications. 1f. List Various batteries used for EVs and its behavior.	Lecture With media	PPTs, black board
Unit– II Electric Motor Drives for EV	**3**	**CO 2. Analyze** the performance of motors used in modern electrical vehicles. PrO 3. Examine the performance of DC motor drive model available in MATLAB Simulink. PrO 4. Examine the performance of Induction motor drive model available in MATLAB Simulink. 2a. Explain Block diagram of an electric drive	Lecture with media, and Demonstration	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		system and explain each block. 2b. Explain the working of DC-DC converter.		
	4	**CO 2. Analyze** the performance of motors used in modern electrical vehicles. PrO 5. Examine the performance of PMBLDC motor drive model available in MATLAB Simulink. 2c. Examine the control of DC motor 2d. Compare the various methods for speed braking of DC motor and explain regenerative braking.	Lecture with media, and Demonstration	PPTs, black board
	5	**CO 2. Analyze** the performance of motors used in modern electrical vehicles. PrO 6. Examine the performance of PMSM motor drive model available in MATLAB Simulink. 2e. Examine the control of induction motor drive. 2f. List out various PM materials.	Lecture with media, and Demonstration	PPTs, black board
	6	**CO 2. Analyze** the performance of motors used in modern electrical vehicles. PrO 7. Prepare the open loop control & closed loop control PMBLDC drive with the help of hall sensor based controller in MATLAB Simulink. 2g. Explain PMBLDC motor construction and working principle. 2h. Examine PMBLDC motor and its working under open loop and closed loop configurations.	Lecture with media, and Demonstration	PPTs, black board
	7	**CO 2. Analyze** the performance of motors used in modern electrical vehicles. PrO 8. Prepare the open loop & closed loop control PMBLDC drive with the help of back emf based controller in MATLAB Simulink. PrO 9. Prepare the open loop & closed loop control PMSM motor drive with bidirectional control in MATLAB Simulink.	Lecture with media, and Demonstration	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		2i. <u>Explain</u> PMSM motor construction and working principle. 2j. <u>Examine</u> PMSM motor and its working under open loop and closed loop configurations.		
Unit– III EV Charging Strategy	**8**	**CO 3.** **<u>Analyze</u>** the various chargers, charging schemes and batteries used for EV applications. *PrO 10.* <u>Examine</u> the performance of BMS model available in MATLAB Simulink. *3a. <u>Identify</u> EV charging requirements.* *3b. <u>List</u> various charging methods for EVs.*	Lecture with media, Field visit	PPTs, Video
	9	**CO 3.** **<u>Analyze</u>** the various chargers, charging schemes and batteries used for EV applications. *PrO 10.* <u>Examine</u> the performance of BMS model available in MATLAB Simulink. 3c. <u>Examine</u> the control strategies for residential, commercial & photovoltaic charging.	Lecture with media, and Tutorial	PPTs
	10	**CO 3.** **<u>Analyze</u>** the various chargers, charging schemes and batteries used for EV applications. *PrO 11.* <u>Examine</u> the performance of charger. 3d. <u>Identify</u> the effect of charging and its effect on Power Demand on Grid, Voltage Profile and frequency regulations.	Lecture with media, and Tutorial	PPTs
	11	**CO 3.** **<u>Analyze</u>** the various chargers, charging schemes and batteries used for EV applications. *PrO 11.* <u>Examine</u> the performance of charger. 3e. Examine the concept of Battery Management System and its functionality.	Lecture with media, and Tutorial	PPTs
Unit– IV Vehicle to Grid	**12**	**CO 4.** **<u>Evaluate</u>** the factors associated grid when integrating EVs in the smart grid. *PrO 12.* <u>Examine</u> the effect of EV charging/discharging and its effects on grid	Lecture with media, and Tutorial	PPTs

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		based on reactive power compensation model available in MATLAB Simulink. 4a. Discuss the concept of V2G and G2V. concept. 4b. Analyze Impact of EV Charging On Electrical Distribution Systems.		
	13	**CO 4. Evaluate** the factors associated grid when integrating EVs in the smart grid. *PrO 12.* Examine the effect of EV charging/discharging and its effects on grid based on frequency regulations model available in MATLAB Simulink. 4c. List various factors associated with Grid Integration, EV Supply Equipment Design and Power Levels. 4d. Analyze the Charging Station requirement and its physical connection to the Grid.	Lecture with media, and Tutorial	PPTs
	14	**CO 4. Evaluate** the factors associated grid when integrating EVs in the smart grid. *PrO 12.* Examine the effect of EV charging/discharging and its effects on grid based on frequency regulations model available in MATLAB Simulink. 4d. Analyze the Charging Station requirement and its physical connection to the Grid. 4e. Identify various V2X systems and its future.	Lecture with media, and Tutorial	PPTs
Unit– V EV Business Trend & Market Policy	15	**CO 5. *Examine*** *the various market policies and standards associated w0ith EVs.* *5a.* Explain the concept of centralized charging control. 5b. Describe the role of EVs in electricity market and EV in smart grid. 5c. Prepare the algorithm for EV charging scheduling, data structure and V2X control.	Lecture with media, and Tutorial	PPT

XIV SUGGESTED LEARNING RESOURCES

Sr. No.	Title of Book	Author	Publication including ISBN
1.	Plug in Electric Vehicles in Smart Grids: Charging Strategy	Sumedha Rajakaruna, Farhad Shahnia, Arindam Ghosh	Springer
2.	Smart Energy Grid Engineering	Hossam A Gabbar	ELSEVIER
3.	Electric Vehicle Technology Explained	James Larminie & John Lowry, Wiley	
4.	Electric Vehicle Machines & Drives Design, Analysis & Application	K.T.Chau,	IEEE press, Wiley
5.	Influences of Electric Vehicles on Power System and Key Technologies of Vehicle to Grid	Canbing Li, Yijia Cao, Yonghong Kuang & Bin Zhou,	Springer

XV SOFTWARE/LEARNING WEBSITES

1. https://www.edx.org/course/electric-cars-introduction (Electric Cars: Introduction)
2. https://www.edx.org/course/electric-cars-technology-0 (Electric Cars: Technology)
3. https://www.edx.org/course/electric-cars-business (Electric Cars: Business)
4. https://www.edx.org/course/electric-cars-policy (Electric Cars: Policy)

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs		POs and PSOs												
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE381	**Semester VI**		**Protection and Switchgear** **Mark 'H'=3** for high, '**M**'=2 for medium or **'L'=1'** for the relevant correlation of each PO or PSO, Competency or CO												
381	**Competency** To lower the greenhouse gases	*3*	*3*	*3*	*2*	*2*	-	-	-	-	-	-	*2*	*3*	*1*

Course Code	Competency and COs			POs and PSOs											
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	and promote the ecofriendly vehicles to the urban infrastructure, the EVs will dominate the future mobility. Electrical engineer has to develop the EVs which ecofriendly, high performance and according to the industry standards. This course is developed such that students should be able to analyze, evaluate, design, develop, implement the EV, HEV and PHEV and coordinate the operation of EVs in smart grid.														
CO1	Identify the various terminologies associated with EV, HEV and PHEV.	3	-	-	-	3	2	3	-	-	-	-	2	2	-
CO2	Analyze the performance of motors used in modern electrical vehicles.	3	1	1	2	1	-	-	-	-	-	-	2	1	1
CO3	Analyze the various chargers, charging schemes and batteries used for EV applications.	3	2	3	3	3	3	3	-	2	2	2	2	3	3
CO4	Evaluate the factors associated grid when integrating EVs in the smart grid.	3	2	2	3	2	-	-	-	-	-	2	2	2	3
CO5	Examine the various market policies and	3	1	3	3	3	-	-	-	-	-	2	2	2	2

Course Code	Competency and COs			POs and PSOs									
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	standards associated with EVs.														

XVII NAMES OF RESOURCE PERSONS

a) Mahammadsoaib Saiyad
b) ………………………………………………….

B. Tech.

ELECTRICAL ENGINEERING

Syllabus

AY: 2023-24

M & V Patel Department of Electrical Engineering

Chandubhai S Patel Institute of Technology

CHARUSAT

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Authored By: Charusat

ACADEMIC REGULATIONS & SYLLABUS

2023-24

(Choice Based Credit System)

Faculty of Technology & Engineering

M & V Patel Department of Electrical Engineering

Bachelor of Technology Programme

(Electrical Engineering)

Chandubhai S Patel Institute of Technology

Vision

To become a leading institute for creation and dissemination of knowledge in the frontiers of technology

Mission

To prepare world-class technocrats and researchers and facilitate enhanced deployment of technology for betterment of lives

M & V Patel Department of Electrical Engineering

Vision

To become an internationally recognized, research intensive education entity, serving as a workforce developer for the nation by providing the best education in close collaboration with society.

Mission

To create world class technocrats with strong theoretical foundations and practical engineering skills who can contribute through the technological applications and research for the betterment of lives.

THE PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO 1	**Preparation:** To prepare the students with sound academically background, intensive development of knowledge and skills, and with an uncompromising attitude to achieve excellence in the electrical engineering.
PEO 2	**Core Competence:** To build a broad engineering core competence and to enable the students to solve the problems systematically by

	providing them exposure to different dimensions of specializations.
PEO 3	**Learning Environment:** To provide conducive learning environment instilling sound foundation in mathematics, computing techniques, science and humanities along with professional courses for futuristic thinking in an increasingly technology dependent society.
PEO 4	**Professionalism:** To provide a platform to the students to plan and shape their career in the chosen field and to imbibe ethical values with thorough professionalism and encouragement towards entrepreneurial thinking and traits.

PROGRAM ARTICULATION MATRIX

(-Not Matching, 1-Low, 2-Medium, 3-High)

	PEO 1	PEO 2	PEO 3	PEO 4
To create world class technocrats with strong theoretical foundations and practical engineering skills who can contribute through the technological applications and research for the betterment of lives.	3	3	3	3

PROGRAM OUTCOMES (POs)

At the end of the program, the student will be able to:

PO 1	**Engineering knowledge:** Apply knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO 2	**Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO 3	**Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health and safety, and the cultural, societal, and environmental considerations.

PO 4	**Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO 5	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO 6	**The engineer and society:** Apply reasoning obtained by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO 7	**Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO 8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO 9	**Individual and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO 10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO 11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one’s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO 12	**Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

At the end of the program, the student will be able to:

PSO 1	Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.
PSO 2	Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

ACADEMIC REGULATIONS

Bachelor of Technology (Electrical Engineering) Programme

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
Phone: 02697-265011, Email: info@charusat.ac.in
www.charusat.ac.in

Year – 2023-2024

CHARUSAT

FACULTY OF TECHNOLOGY AND ENGINEERING
ACADEMIC REGULATIONS
Bachelor of Technology Programmes

Choice Based Credit System

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

Choice based Credit System with Semester pattern of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a course works in the chosen subject of specialization and also complete a project/dissertation if any. Apart from the Programme Core courses, provision for choosing University level electives and Programme/Institutional level electives are available under the Choice based credit system.

2. Duration of Programme

Undergraduate programme (B. Tech.)

Minimum	8 semesters (4 academic years)
Maximum	12 semesters (6 academic years)

3. Eligibility for admissions:

As enacted by Govt. of Gujarat from time to time.

4. Mode of admissions

As enacted by Govt. of Gujarat from time to time.

5. Programme structure and Credits

As per annexure – 1 attached

6. Attendance

6.1 All activities prescribed under these regulations and enlisted by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student regarding attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Principal.

6.2 Student's attendance in a course should be 80%.

7 Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

7.1.1. Internal evaluation by the course faculty member(s) based on continuous assessment, the continuous assessment will be conducted by the respective department /institute.

7.1.2 Final end-semester examination by the University through written paper or practical test or oral test or presentation by the student or a combination of these.

7.1.3 The weightages of continuous assessment and End-semester university examination in overall assessment shall depend on individual course as approved by Academic Council through Board of Studies.

7.1.4 The performance of candidate in continuous assessment and in end-semester examination together (if applicable) shall be considered for deciding the final grade in a course.

7.1.5 In order to earn the credit in a course a student has to obtain grade other than FF.

7.2 Performance in continuous assessment and end-semester University Examination

7.2.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course.
Details of minimum percentage of marks to be obtained in the examinations (internal/external) are as follows

Minimum marks in University Exam per course	Minimum marks Overall per course
40%	45%

7.2.2 If a candidate obtains minimum required marks in each course but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

8 Grade Point System

8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade **on** a ten-point scale as per the following scheme:

Table: Grading Scheme (UG)

Range of Marks (%)	≥80	**<80 ≥73**	**<73 ≥66**	**<66 ≥60**	**<60 ≥55**	**<55 ≥50**	**<50 ≥45**	**<45**
Corresponding Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his/her performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

(i) SGPA = $\Sigma C_i G_i / \Sigma C_i$ where C_i is the number of credits of course i
G_i is the Grade Point for the course i
and i = 1 to n, n = number of courses in the semester

(ii) CGPA = $\Sigma C_i G_i / \Sigma C_i$ where C_i is the number of credits of course i
G_i is the Grade Point for the course i
and i = 1 to n, n = number of courses of all semesters up to which CGPA is computed.

9 Award of Class

☞ The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Award of Class	CGPA Range
Distinction	CGPA ≥ 7.5 & ≤ 10.0
First Class	CGPA ≥ 6.0 & < 7.5
Second Class	CGPA ≥ 5.0 & < 6.0
Pass Class	CGPA < 5.0

Grade sheets of only the final semester shall indicate the class. In case of all the other semesters, it will simply indicate as Pass / Fail.

9.1 Maximum duration allowed for Completion of a programme

- Maximum duration to allow for completion of a particular programme shall not be more than twice the normal duration of the respective programme. For example, a 6-Semester programme should be completed within not more than 12 semesters.

10 Detention Criteria

- No student will be allowed to move further in next semester if CGPA is less than 3 at the end of an academic year.

- A student will not be allowed to move to third year if he/she has not cleared all the courses of first year.

- A student will not be allowed to move to fourth year if he/she has not cleared all the courses of second year.

11 Transcript:

- The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

Annexure – 1

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY											
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN ELECTRICAL ENGINEERING											
Sem	**Course Code**	**Course title**	**Teaching Scheme**				**Examination Scheme**				**Total**
			Contact Hrs.			**Credits**	**Theory**		**Practical**		
			Theory	**Practical**	**Total**		**Internal**	**External**	**Internal**	**External**	
VII	EE441	Power System Operations and Control	4	2	6	5	30	70	25	25	150
	EE442	Power System Protection	4	2	6	5	30	70	25	25	150
	EE443	Electrical Machine Design	4	2	6	5	30	70	25	25	150
	EE444	Simulation Lab		4	4	2			50	50	100
	EE450	Summer Internship II		3	3	3			75	75	150
		Elective - I	3	2	5	4	30	70	25	25	150
	Assignment Practice/ Student Counselling		0	0	4	0	0	0	0	0	0
			15	**15**	**34**	**24**	**120**	**280**	**225**	**225**	**850**
VIII	EE458	Major Project	0	40	40	20	0	0	400	400	800
	Student Counselling		0	0	2	0	0	0	0	0	0
			00	**00**	**40**	**20**	**0**	**0**	**400**	**400**	**800**

Code	**Elective - I**	**Code**	**Elective - I**
EE446.01	Commissioning and Testing of Electrical Equipment	EE476.01	Advances in Power System
EE447.01	Modeling and Control of Renewable Energy Sources	EE481	Industry 4.0
EE472.01	Communication Engineering	EE482	Smart Grid Technologies and Applications

B. Tech. (Electrical Engineering) Programme

SYLLABI

(Semester – VII)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Charotar University of Science and Technology, Gujarat

Course Code: **EE441**
Course Title: **Power System Operations and Control**

UG Engineering Programme	Semester
Electrical Engineering	Sevan

I RATIONALE

Electrical engineers have to deal with the power system operations and controls. Calculations and simulations are performed to verify power system stability, power flow analysis, short circuit studies, and Economic operation of generators. This curriculum course structure is developed such that the student is able to evaluate the power flow analysis, power system stability studies, short circuit studies, economic dispatch, automatic generation control, and power system security for the interconnected power system.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Evaluate the performance of power system stability analysis and power flow analysis for the interconnected electric power system**.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) ***Evaluate*** *planning and designing of the new interconnected power system through power flow analysis with practical simulation and with software tools.*
2) ***Evaluate*** *short circuit studies of an inter connected power system and design the circuit breaker ratings.*
3) ***Evaluate*** *the small signal stability, transient stability, and voltage stability for the electric power system with practical simulation and with software tools.*
4) ***Interpret*** *the transients condition in power system.*
5) ***Choose*** *the automatic generation control, power system security, and economic dispatch for ensuring the smooth and efficient operation of an electric power system.*

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)	Total Credits (L+T+P)	Examination Scheme (Anna 2021)		
		Theory Marks	Practical Marks	Total Marks

L	T	P	C	ESE	PA	ESE	PA	150
4	0	2	5	70	30*	25	25	

***Legends: L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

***(*):** Under the **Theory PA**, out of 30 marks, 15 **marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 15 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Formulate the bus admittance (Ybus) matrix using singularity and non-singular transformation method for the given power system network.	02
2	1	Find power flow analysis using NR, GS method for the given power system network.	02
3	2	Formulate the bus impedance(Zbus) matrix using building algorithm method.	02
4	4	Find the Transients in power system.	02
5	3	Evaluate the Small signal stability for the given power system network.	02
6	3	Evaluate the Transient stability for the given power system network.	02
7	5	Evaluate the economic operating schedule for the given generator units.	02
8	1	Simulate the power flow analysis in MATLAB and Power World Simulator for the given power system network.	02
9	3	Simulate the Small signal stability in MATLAB for the given power system network.	02
10	3	Simulate the Transient stability in MATLAB for the given power system network.	02
11	3	Simulate Voltage stability in MATLAB for the given power system network.	02
12	5	Simulate economic dispatch using Power World Simulator for the given generator units.	02
		Total	**24**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB Simulation software (version 2021a)	8,9,10,11
2	Power World Simulator (Version 22)	8,12

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy balances.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Load Flow Analysis	**CO 1. *Evaluate planning and designing of the new interconnected power system through power flow analysis with practical simulation and with software tools.*** *PrO 1.* Formulate the bus admittance (Ybus) matrix using singularity and non-singular transformation method for the given power system network. *PrO 2.* Find power flow analysis using NR, GS method for the given power system network. *PrO 8.* Simulate the power flow analysis in MATLAB and Power World Simulator for the given power	1.1 Introduction to Load Flow Analysis 1.2 Formulation of Bus Admittance Matrix and Examples using Singular transformation method 1.3 Gauss – Siedel Load Flow Method and examples, Newton – Raphson Load Flow Method 1.4 Fast Decoupled Load Flow Method 1.5 Computer aided simulation of load flow analysis, Examples

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	system network. 1a. Justify the need of Load Flow Analysis for the given power system network. 1b. Evaluate Bus Admittance Matrix for the given power system network. 1c. Determine power flow analysis for the given network using GS and NR method. 1d. Determine power flow analysis for the given network using FDLF method. 1e. Justify the method of power flow analysis for the given power system network.	
Unit– II Optimal Dispatch of Generation	**CO *5. Choose the automatic generation control, power system security, and economic dispatch for ensuring the smooth and efficient operation of an electric power system.*** *PrO 7. Evaluate the economic operating schedule of generator.* *PrO 12.* Simulate economic dispatch using Power World Simulator for the given power system. 2a. Describe the optimal dispatch generation 2b. Determine the equality and inequality constraints for the given type of generation unit 2c. Evaluate the Operating cost of thermal power plant. 2d. Discriminate the economic dispatch without losses and different generation limits for the given thermal power plant units. 2e. Evaluate the Economic dispatch including losses for the given thermal power plant units.	2.1 Introduction to optimal dispatch generation 2.2 equality and inequality constraints. 2.3 Operating cost of thermal power plant 2.4 economic dispatch neglecting losses and no generation limits and with consideration of generation limits 2.5 Economic dispatch including losses and examples
Unit– III Automatic Generation Control (AGC)	**CO 5. *Choose the automatic generation control, power system security, and economic dispatch for ensuring the smooth and efficient operation of an electric power system*.** 3a. Describe the basic generation control loops and automatic generation control. 3b. Formulate generator model, prime mover model, load model, and governor model for the given system. 3c. Propose AGC in a single area and in	3.1 basic generation control loops and automatic generation control 3.2 generator model, prime mover model, load model, and governor model 3.3 AGC in a single area and in multi-area system 3.4 Reactive power and voltage control

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	multi-area power system network. 3d. Demonstrate Reactive power and voltage control for the given system. 3e. Prepare the amplifier model, exciter model, generator model, sensor model, and excitation system stabilizer for the power system network.	3.5 amplifier model, exciter model, generator model, sensor model, and excitation system stabilizer
Unit-IV Transients in Power System	**CO 4. *Interpret the transients condition in power system.*** *PrO 4. Find the Transients in power system.* 4a. Analyse Traveling waves on transmission lines. 4b. Examine surge impedance and wave velocity. 4c. Evaluate the Reflection and refraction of wave for the given transmission line with-the specified condition. 4d. Recognize the Reflection and refraction of the given transmission line with the use of Bewley lattice diagram. 4e. Define the Lightning phenomena and insulation coordination.	4.1 Traveling waves on transmission lines, wave equations, and specifications of traveling waves 4.2 surge impedance and wave velocity. 4.3 Reflection and refraction of traveling waves, and typical cases of line terminations 4.4 Successive reflections and Bewley lattice diagram 4.5 Lightning phenomena, protection of power system against lightning surges, and insulation coordination
Unit – V Short Circuit Studies using Bus Impedance Matrix	**CO 2. *Evaluate short circuit studies of an inter connected power system and design the circuit breaker ratings.*** *PrO 3. Formulate the bus impedance(Zbus) matrix using building algorithm method.* 5a. Compare the Bus admittance and incidence matrices for the given power system network. 5b. Review Thevenin's theorem and Z_{BUS} for the given power system network. 5c. Calculate Modification of existing Z_{BUS} for the given power system network. 5d. Unbalanced fault analysis using Z_{BUS}.	5.1 Bus admittance and incidence matrices. 5.2 Thevenin's theorem and Z_{BUS} 5.3 Modification of existing Z_{BUS} after addition or removal of branches and Direct determination of Z_{BUS} 5.4 Unbalanced fault analysis using Z_{BUS}, Examples
Unit – VI Power System Stability	**CO 3. *Evaluate the small signal stability, transient stability, and voltage stability for the electric power system with practical simulation and with software tools.*** *PrO 5. Evaluate the Small signal stability for the given power system network.* *PrO 6. Evaluate the Transient stability for the given power system network.*	6.1 Derivation of swing equation, M and H constants, equivalent H constant. 6.2 Classification of power system stability. 6.3 Steady state stability, derive the expression for

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	PrO 9. Simulate the Small signal stability in MATLAB for the given power system network. *PrO 10. Simulate the Transient stability in MATLAB for the given power system network.* *PrO 11. Simulate Voltage stability in MATLAB for the given power system network.* 6a. Illustrate the swing equation for rotor dynamics. 6b. Classification of power system stability. 6c. Evaluate the Steady state stability analysis for given power system network. 6d. Evaluate the Transient stability analysis for given power system network. 6e. Justify Voltage stability analysis for power system stability. 6f. Voltage stability analysis for power system stability	natural frequency of oscillations, damped frequency of oscillations, and Derivation of synchronizing power coefficient, examples. 6.4 Transient stability and factors affecting it. Equal area criteria, Cases of sudden application of mechanical power input, application of three phase fault at the middle of the line or at the end of the line, Example based on equal area criteria. 6.5 Voltage stability, Classification of Voltage Stability, Reasons for Voltage instability may arise. Transmission System Characteristic, P-V Curve, Q-V Curve, Generator Characteristic, Load Characteristic, and Characteristic of reactive power compensating device. Voltage collapse, Prevention of voltage collapse.
Unit – VII Power System Security	**CO 5. *Choose the automatic generation control, power system security, and economic dispatch for ensuring the smooth and efficient operation of an electric power system*.** 7a. Describe the Operating states of power system and Factors affecting power system security. 7b. Explain Overview of security analysis. 7c. Illustrate the AC power flow methods and Contingency relaxation for given power system network.	7.1 Operating states of power system, Factors affecting power system security, and Contingency analysis. 7.2 Overview of security analysis and Linear sensitivity factors. 7.3 AC power flow methods and Contingency relaxation.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Load Flow Analysis	10	0	2	2	4	4	0	12
II	Optimal Dispatch of Generation	08	0	2	2	2	2	0	08
III	Automatic Generation Control (AGC)	08	0	2	2	2	2	0	08
IV	Transients in Power System	04	0	2	0	0	2	0	04
V	Short Circuit Studies using Bus Impedance Matrix	06	0	2	2	2	2	0	08
VI	Power System Stability	20	2	8	6	4	6	0	26
VII	Power System Security	04	0	2	2	0	0	0	04
Total		**60**	**0**	**10**	**14**	**20**	**16**	**0**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **Formulation of bus admittance matrix:** Collect the Transmission network data and formulate the bus admittance matrix using singular and non-singular type method for power flow analysis.

2) **Perform power flow analysis:** Collect the real network data or the IEEE standard test system for the power flow analysis using different methods like NR, GS and FDLF.
3) **Formulation of bus impedance matrix:** Collect the Transmission network data and formulate the bus impedance matrix using building algorithm method for short circuit studies.
4) **Perform small signal stability analysis:** Collect the real network data or the IEEE standard test system for and perform small signal stability analysis.
5) **Perform transient stability analysis:** Collect the real network data or the IEEE standard test system for and perform transient stability analysis.
6) **Economic Dispatch of generator:** Collect the data from the different generator units and perform economic dispatch.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different types of power system stability.
b) Prepare 15 min. power point presentation for voltage stability analysis.
c) Prepare 15 min. power point presentation for automatic generation and control for power system.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Load Flow Analysis	1	*PrO 1.* Formulate the bus admittance (Ybus) matrix using singularity and non-singular transformation method for the given power system network. 1a. Justify the need of Load Flow Analysis for the given power system network. 1b. Evaluate Bus Admittance Matrix for the given power system network.	Lecture with media, Tutorial	PPTs, black board
	2	*PrO 2. Find power flow analysis using NR, GS method for the given power system network.* 1c. Determine power flow analysis for the given network using GS and NR method.	Lecture With media, Tutorial	PPTs, black board
	3	*PrO 8.* Simulate the power flow analysis in MATLAB and Power World Simulator for the given power system network. 1d. Determine power flow analysis for the given network using FDLF method. 1e. Justify the method of power flow analysis for the given power system network.	Lecture with media, Tutorial	PPTs, black board
Unit– II Optimal Dispatch of Generation	**4**	*PrO 7. Evaluate the economic operating schedule of generator.* 2a. Describe the optimal dispatch generation 2b. Determine the equality and inequality constraints for the given type of generation unit. 2c. Evaluate the Operating cost of thermal power plant.	Lecture with media, and Tutorial	PPTs, black board
	5	PrO 12. Simulate economic dispatch using Power World Simulator for the given power system. 2d. Discriminate the economic dispatch without losses and different generation limits for the given thermal power plant units. 2e. Evaluate the Economic dispatch including losses for the given thermal power plant units.	Lecture with media, Tutorial	PPTs, black board
Unit– III Automatic Generation Control (AGC)	**6**	3a. Describe the basic generation control loops and automatic generation control. 3b. Formulate generator model, prime mover model, load model, and governor model for the given system. 3c. Propose AGC in a single area and in multi-area power system network.	Lecture with media	PPTs
	7	3d. Demonstrate Reactive power and voltage control for the given system. 3e. Prepare the amplifier model, exciter model, generator model, sensor model, and excitation	Lecture with media	PPTs, Video

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		system stabilizer for the power system network.		
Unit-IV Transients in Power System	**8**	4a. Analyse Traveling waves on transmission lines. 4b. Examine surge impedance and wave velocity. 4c. Evaluate the Reflection and refraction of wave for the given transmission line with- the specified condition.	Lecture with media, and Tutorial	PPTs, black board
	9	*PrO 4. Find the Transients in power system.* 4d. Recognize the Reflection and refraction of the given transmission line with the use of Bewley lattice diagram. 4e. Define the Lightning phenomena and insulation coordination.(Self-study)	Lecture with media, Tutorial	PPTs, black board
Unit – V Short Circuit Studies using Bus Impedance Matrix	**10**	*PrO 3. Formulate the bus impedance(Zbus) matrix using building algorithm method.* 5a. Compare the Bus admittance and incidence matrices for the given power system network. 5b. Review Thevenin's theorem and Z_{BUS} for the given power system network. 5c. Calculate Modification of existing Z_{BUS} for the given power system network. 5d. Unbalanced fault analysis using Z_{BUS}.	Lecture with media, Tutorial	PPTs, black board
Unit – VI Power System Stability	**11**	*PrO 5. Evaluate the Small signal stability for the given power system network.* *PrO 9. Simulate the Small signal stability in MATLAB for the given power system network.* 6a. Illustrate the swing equation for rotor dynamics. 6b. Classification of power system stability. 6c. Evaluate the Steady state stability analysis for given power system network.	Lecture with media, Tutorial	PPTs, black board
	12	*PrO 6. Evaluate the Transient stability for the given power system network.* *PrO 10. Simulate the Transient stability in MATLAB for the given power system network.* 6d. Evaluate the Transient stability analysis for given power system network.	Lecture with media, Tutorial	PPT, Pictures , black board
	13	*PrO 6. Evaluate the Transient stability for the given power system network.* *PrO 10. Simulate the Transient stability in MATLAB for the given power system network.* 6d. Evaluate the Transient stability analysis for given power system network.	Lecture with media, Tutorial	PPTs, black board
	14	*PrO 11. Simulate Voltage stability in MATLAB for the given power system network.* 6e. Justify Voltage stability analysis for power system stability. 6f. Voltage stability analysis for power system	Lecture with media	PPTs, Video

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		stability		
Unit – VII Power System Security	**15**	7a. Describe the Operating states of power system and Factors affecting power system security. 7b. Explain Overview of security analysis. 7c. Illustrate the AC power flow methods and Contingency relaxation for given power system network.	Lecture with media	PPTs, Video

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Power System Analysis	Hadi Saadat	Tata McGraw Hill
2	Power System Stability and Control	P. Kundur	Tata McGraw Hill
3	Power System Analysis and Design	B.R. Gupta	S. Chand
4	Power System Analysis	Grainger & Stevenson	Tata McGraw Hill
5	Power Generation, Operation and Control	J. Wood and B.F. Wollenberg	John Wiley & Sons, New York, USA
6	Power System Voltage Stability	C.W. Taylor	Tata McGraw Hill

XV SOFTWARE/LEARNING WEBSITES

1. https://cusp.umn.edu/power-systems/power-generation-operation-control
2. http://nptel.ac.in/downloads/108101040/
3. https://www.montefiore.uliege.be/cms/c_4570395/en/electric-power-and-energy-systems
4. http://labs.ece.uw.edu/pstca/
5. https://www.ece.ucf.edu/~qu/ma-opendss/

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and teamwork	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE441	**Semester VII**	**Power System Operations and Control** Mark '**H'=3** for high, '**M**'=2 for medium or **'L'=1'** for the relevant correlation of each PO or PSO, Competency or CO													
441	**Competency** Evaluate the performance of power system stability analysis and power flow analysis for the interconnected electric power system	***3***	***3***	***3***	***2***	***2***	***1***	***1***	**-**	***2***	***1***	***2***	**-**	***1***	***1***
441.1	Evaluate planning and designing of the new interconnected power system through power flow analysis with practical simulation and with software tools.	3	2	2	2	2	-	-	-	-	-	-	-	2	-
441.2	Evaluate short circuit studies of an inter connected power system and design the circuit breaker ratings	2	1	3	1	2	-	-	-	-	-	-	-	2	-
441.3	Evaluate the small signal stability, transient stability, and voltage stability for the electric power system with practical simulation and with software tools.	1	3	2	1	2	-	-	-	-	-	-	-	2	-
441.4	Interpret the transients condition in power system.	1	1	2	-	-	-	-	-	-	-	-	-	2	-
441.5	Choose the automatic generation control, power	1	-	1		1	-	-	-	-	-	-	-	2	-

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and teamwork	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	system security, and economic dispatch for ensuring the smooth and efficient operation of an electric power system.														

XVII NAMES OF RESOURCE PERSONS

a) Mihir R. Patel
b) ……………………………………………………..

Course Code: **EE442**
Course Title: **Power System Protection**

UG Engineering Programme	**Semester**
Electrical Engineering	Seventh

I RATIONALE

Electrical engineers have to deal with various types of electrical machines and power system. Different types of protection systems are used to protect various types of machines and power system for safe operation and desired result. This curriculum course structure is developed such that the student is able to evaluate the performance of the protection schemes, relays, CTs & PTs, and suggest the setting of relay for different type of protective schemes.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Evaluate the performance of various electrical protection systems used in industry and power system**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) ***Use*** the protection schemes for apparatus and power system protection**.**
2) ***Evaluate*** *the performance of the relays and Instrument transformers.*
3) **Design** the suitable protection system for the apparatus and power system protection with relay setting.
4) ***Conduct*** the different types of tests on relay.
5) ***Interpret*** *the numerical relaying*.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			**Total Credits (L+T+P)**	**Examination Scheme**				
				Theory Marks		**Practical Marks**		**Total Marks**
L	**T**	**P**	**C**	**ESE**	PA	**ESE**	**PA**	150
4	0	2	5	70	30*	25	25	

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

(*): *Under the* ***Theory PA****, out of 30 marks,* ***10 marks*** *are for* ***micro-project assessment*** *to facilitate integration of COs and the remaining 20 marks is converted from 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Create the protection zone for the given power system network.	02
2	2	Set the operating time of the given electromagnetic relay for the given system.	02
3	2	Set the given numerical relay for the given system.	02
4	2	Set the given numerical relay for voltage-based operation.	02
5	3	Design the relay coordination for the given radial feeder.	02
6	3	Design the relay coordination for the given parallel feeder.	02
7	3	Design the protection system for the given transformer for different configuration.	02
8	3	Set the given numerical relay for *the given* generator.	02
9	3	Set the given numerical relay for *the given* induction motor.	02
10	3	Design the protection system for the given busbar configuration.	02
11			
12			
13			
14			
15			
		Total	**20**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB Simulation software (version 2021a) **(Optional)**	1
2	Test rig of electromagnetic relays	2
3	Test rig of Numeric relay consisting of (Siemens relay REJ525) • Microprocessors, • ADC • DAC	3
4	Test rig of Numeric relay consisting of (ABB SPAU 130C) • Microprocessors, • ADC • DAC	4
5	Test rig of radial feeder with IDMT relays.	5
6	Test rig of parallel feeder with IDMT and directional relays.	6
7	Test rig of transformer differential protection schemes	7
8	Test rig of Generator protection schemes	8
9	Test rig of Induction motor protection schemes	9

S. No.	Equipment Name with Broad Specifications	PrO No.
10	Test rig of Busbar protection schemes	10
11		

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Protective Relaying System	**CO 1. *Use*** the protection schemes for apparatus and power system protection. *PrO 1.* *reate the protection zone for the given power system network.* 1a. Justify the need of protection scheme for the given situation 1b. Evaluate the basic tripping circuit for the specified condition 1c. Justify the requirements of protection in the given system 1d. Choose protection scheme for the specified situation.	1.1 Principles and need for protective schemes 1.2 Basic tripping circuit, Characteristic of protective system 1.3 Basics to design protective system.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit– II **Instrument Transformer**	**CO 2.** ***Evaluate*** *the performance of the relays and Instrument transformers.* 2a. Determine the CT and PT rating for the specified application. 2b. Choose the correct CT for the specified protective system for the given type of protection 2c. Justify the selection for the VT or CCVT. For the specified conditions.	2.1 Saturation characteristic of current transformer 2.2 Difference between measurement CT and Protective CT. 2.3 Potential Transformer
Unit– III **Overcurrent Protection**	**CO 2.** ***Evaluate*** *the performance of the relays and Instrument transformers.* *PrO 2. Set the operating time of the given electromagnetic relay for the given system.* *PrO 3.* Set the given numerical relay for the given system. *PrO 4. Set the given numerical relay for voltage-based operation.* 3a. Choose the relevant electromagnetic or numerical relay for the given situation 3b. Determine the relay operating time for the specified fault current levels for the given type of electromagnetic relay 3c. Justify the need for directional features of electromagnetic relay for the specified conditions. 3d. Evaluate the performance of the given electromagnetic relay for the specified application.	3.1 Operating principles of relays. 3.2 Different operating characteristics of different relays. 3.3 Over current and directional relays
Unit-IV **Protection of Transmission Line.**	**CO 3. Design** the suitable protection system for the apparatus and power system protection with relay setting. *PrO 5.* Design the relay coordination for the given radial feeder. *PrO 6.* Design the relay coordination for the given parallel feeder. 4a. Select the relevant relays for the specified application. 4b. Evaluate the performance of the given type of relay configurations for the specified condition. 4c. Design the radial feeder protection system for the given set of radial transmission lines.	4.1 Different configuration for the relays for transmission line protection. 4.2 Distance protection. 4.3 Carrier current protection.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	4d. Design the parallel feeder protection system for the given set of parallel transmission lines. 4e. Prepare the drawing and specification of a carrier current protection for the given situation.	
Unit – V Transformer Protection	**CO 3. Design** the suitable protection system for the apparatus and power system protection with relay setting. PrO 7. Design the protection system for the given transformer for different configuration. 5a. Distinguish between fault and abnormalities from the given set of data. 5b. Select the proper connection of control circuit for the given configuration. 5c. Design the transformer protection scheme for the given transformer and relay.	5.1 Faults and abnormalities in the transformer. 5.2 Percentage biased differential protection of transformer.
Unit – VI Protection of Generator	**CO 3. Design** the suitable protection system for the apparatus and power system protection with relay setting. *PrO 8. Set the given numerical relay for the given generator.* 6a. Choose the relevant protection scheme for the given generator 6b. Develop control circuit for the protection of the given generator. 6c. Choose the correct relay for the given protection of generator. 6d. Differentiate between class A, B and C protection from the given data.	6.1 Generator stator and rotor protection. 6.2 Negative phase sequence protection. 6.3 Reverse power protection and field failure protection.
Unit– VII Induction Motor Protection	**CO 3. *Interpret* the suitable protection system for the apparatus protection.** *PrO 9. Set the given numerical relay for the given induction motor.* 7a. Develop control circuit for the protection of the given induction motor. 7b. Choose the correct relay setting the given induction motor.	7.1 Protection of induction motor
Unit-VIII Bus bar Protection	**CO 3. Design the suitable protection system for the apparatus and power**	

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	system protection with relay setting. *PrO 10. Design the protection system for the given busbar configuration.* 8a. Develop control circuit for the protection of the given busbar configuration.	8.1 Protection of busbar
Unit – IX Testing, commissioning and Maintenance of Relays	***CO 4. Conduct*** the different types of tests on relay. 9a. Perform the different types of tests on relay depending upon the specified conditions.	9.1 Different types of tests for relays.
Unit –X Numerical Relaying	***CO 5. Interpret*** *the numerical relaying*. 10a. Select the best algorithm for the given application. 10b. Develop the block diagram of numerical relay for the given application. 10c. Estimate the correct phasor by the given method.	10.1 Numerical relay hardware 10.2 DSP and data acquisition system. 10.3 Fourier algorithm

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Protective Relaying System	04	0	2	2	0	0	0	04
II	Instrument Transformer	05	0	2	0	2	0	0	04
III	Overcurrent Protection	08	0	2	2	2	0	0	06
IV	Protection of Transmission Line	16	0	2	0	4	0	6	12
V	Transformer Protection	07	0	0	3	0	6	0	09
VI	Protection of Generator	05	0	0	5	5	0	0	10
VII	Induction Motor Protection	02	2	0	0	5	0	0	07
VIII	Bus bar Protection	02	2	0	5	0	0	0	07

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
IX	Testing, commissioning and Maintenance of Relays	03	0	6	0	0	0	0	06
X	Numerical Relaying	08	2	3	0	0	0	0	05
Total		**60**	**06**	**17**	**17**	**18**	**06**	**06**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **Protection schemes:** Collect the specifications of different types of electromagnetic relays used in different protection situations in different types of industries.
2) **Relays:** Collect the specifications of different types of static relays used in different protection situations in different types of industries.
3) **Equipment protection:** Prepare case studies of protections failures in different types of industries.
4) Undertake the survey in reputed industry and submit a report on different type of protection schemes used in that industry.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different protection schemes used in industries.
b) Prepare 15 min. power point presentation on how to choose relay for the particular application.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Protective Relaying System	1	**PrO 1.** *Create the protection zone for the given power system network using MATLAB software.* 1a. Justify the need of protection scheme for the given situation 1b. Evaluate the basic tripping circuit for the specified condition 1c. Justify the requirements of protection in the given system *1d.* Choose protection scheme for the specified situation.	Lecture with media	PPTs, black board
Unit– II Instrument Transformer	2	2a. Determine the CT and PT rating for the specified application. 2b. Choose the correct CT for the specified protective system for the given type of protection 2c. Justify the selection for the VT or CCVT. For the specified conditions. **(Self-learn)**	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– III Overcurrent Protection	**3**	**PrO 2.** *Set the operating time of the given electromagnetic relay for the given system.* 3a. Choose the relevant electromagnetic or numerical relay for the given situation 3b. Determine the relay operating time for the specified fault current levels for the given type of electromagnetic relay	Lecture with media, Tutorial and Demonstration	PPTs, black board
	4	**PrO 3.** *Set the given numerical relay for the given system.* **PrO 4.** *Set the given numerical relay for voltage-based operation.* 3c. Justify the need for directional features of electromagnetic relay for the specified conditions. 3d. Evaluate the performance of the given electromagnetic relay for the specified application.	Lecture with media, Tutorial and Demonstration	PPTs, black board, Video
Unit-IV Protection of Transmission Line.	**5**	4a. Select the relevant relays for the specified application. 4b. Evaluate the performance of the given type of relay configurations for the specified condition.	Lecture with media, Tutorial	PPTs, black board
	6	**PrO 5.** *Design the relay coordination for the given radial feeder.* 4c. Design the radial feeder protection system for the given set of radial transmission lines.	Lecture with media	PPTs, black board
	7	**PrO 6.** *Design the relay coordination for the given parallel feeder.* 4d. Design the parallel feeder protection system for the given set of parallel transmission lines.	Lecture with media,	PPTs, black board
	8	4e. Prepare the drawing and specification of a carrier current protection for the given situation.	Lecture with media	PPTs, black board
Unit – V Transformer Protection	**9**	5a. Distinguish between fault and abnormalities from the given set of data. 5b. Select the proper connection of control circuit for the given configuration.	Lecture with media, + PPT	PPTs, black board
	10	**PrO 7.** *Design the protection system for the given transformer for different configuration.*		PPT, Demonstra

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		5c. Design the transformer protection scheme for the given transformer and relay.	Lecture with media, Tutorial	tion, black board
Unit – VI Protection of Generator	**11**	6a. Choose the relevant protection scheme for the given generator 6b. Develop control circuit for the protection of the given generator.	Lecture with media	PPTs, black board
	12	**PrO 8.** *Set the given numerical relay for the given generator.* 6c. Choose the correct relay for the given protection of generator. 6d. Differentiate between class A, B and C protection from the given data.	Lecture with media, case study	PPT, black board
Unit– VII Induction Motor Protection **Unit-VIII Bus bar Protection**	**13**	**PrO 9.** *Set the given numerical relay for the given induction motor.* **PrO 10.** *Design the protection system for the given busbar configuration.* 7a. Develop control circuit for the protection of the given induction motor. 7b. Choose the correct relay setting the given induction motor. 8a. Develop control circuit for the protection of the given busbar configuration.	Lecture with media	PPTs, black board
Unit – IX Testing, commissioning and Maintenance of Relays **Unit –X Numerical Relaying**	**14**	9a. Perform the different types of tests on relay depending upon the specified conditions. 10a. Select the best algorithm for the given application.	Lecture with media, Tutorial	PPTs, black board
	15	10b. Develop the block diagram of numerical relay for the given application. 10c. Estimate the correct phasor by the given method.	Lecture with media, Tutorial	PPTs, black board, Video

XVIII SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN

S. No.	Title of Book	Author	Publication including ISBN
1	Power System Protection and Switchgear	B.A. Oza, N.C. Nair, R.P. Mehta and V.H. Makwana,	Mcgraw Hill, 2010 ISBN: 9780070671188, 0070671184
2	Protection and Switchgear	Bhavesh Bhalja, R. P. Maheshwari and N. G. Chothani	Oxford University Press, 2018 ISBN:9780199470679, 0199470677
3	Fundamentals of Power System Protection	Paithankar, Y.G. and Bhide, S.R.	Prentice Hall of India Pvt. Ltd, New Delhi, 2010, ISBN: 9788120341234
4	Power System Protection and Switchgear	Ram, Badri and Vishwakarma, B. H.	Mc Graw Hill Education, New Delhi, 2019, ISBN: 9780071077743
5	Power System Protection Static Relays	T.S. Madhav Rao	Tata Mcgraw Hill, 1989 ISBN: 9780074603079, 0074603078
6	Switchgear and Protection	Rao, Sunil S.	Khanna Publishers, New Delhi, 2008, ISBN: 9788174092323
7	Power System Protection and Switchgear	Rabindranath, B. and Chander, N.	New Age International (P) Ltd., New Delhi, First Edition 2011, ISBN: 9780852267585
8	Switch Gear and Protection	Ingole, Arun	Pearson Education, Noida, 2017, ISBN:9789353063382

XIX SOFTWARE/LEARNING WEBSITES

6. https://onlinecourses.nptel.ac.in/noc21_ee110/preview
7. Protective Relaying https://www.youtube.com/watch?v=NEXWcOgqZOI
   https://www.youtube.com/watch?v=h8rpdVt9MUo

XX PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE442	Semester VII														
442	**Competency** *Evaluate the performance of various electrical*	***3***	***2***	***2***	***2***	***1***	***1***	-	-	-	-	-	***1***	***3***	-

Course Code	Competency and COs	**PO 1** Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	protection systems used in industry and power system														
442.1	Use the protection schemes for apparatus and power system protection	3	-	-	1	-	-	-	-	-	-	-	-	1	-
442.2	Evaluate the performance of the relays and instrument transformers	2	2	1	1	1	-	-	-	-	-	-	1	1	-
442.3	Design the suitable protection system for the apparatus and power system protection with relay setting.	3	3	3	2	2	2	-	-	1	-	-	1	3	-
442.4	Conduct the different types of tests on relay.	1	-	-	1	1	-	-	-	-	-	-	1	1	-
442.5	Interpret the numerical relaying.	3	1	-	-	1	-	-	-	-	1	-	-	1	-

XXI NAMES OF RESOURCE PERSONS

a) Dr. Nilaykumar A. Patel
b) ……………………………………………………

Course Code: **EE443**
Course Title: **Electrical Machine Design**

UG Engineering Programme	Semester
Electrical Engineering	7th

I RATIONALE

Electrical engineers have to deal with various types of electrical machines. Overall design of rotational and stationary machines is essential. This curriculum course structure is developed such that the student is able to design transformers, induction machines and synchronous machines.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Design various components of the electrical machines**.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) **To introduce the students with the fundamentals of designing an equipment system.**
2) **To learn how to design different part of transformer.**
3) **To learn how to design different part of induction machine.**
4) **To learn how to design different part of synchronous machine.**

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			**Total Credits (L+T+P)**	**Examination Scheme**				
				Theory Marks		**Practical Marks**		**Total Marks**
L	**T**	**P**	**C**	**ESE**	PA	**ESE**	**PA**	150
4	0	2	5	70	30*	25	25	

***Legends: L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

***(*):** Under the **Theory PA**, out of 40 marks, **20 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	To study various parts of AC machines	04
2	3	To solve the design problems related to the main dimensions of Induction machine.	04
3	1	Basic consideration of electrical machine design based on heating time constant.	04
4	2	Various basic design concepts of transformer	02
5	3	To solve the problems of stator and rotor design of an induction machine.	04
6	3	To solve the rotor design of a slip ring induction machine.	04
7	2	To solve and calculate the dimensions of various parts of a transformer	04
8	4	To solve and calculate various components for an alternator.	04
		Total	**30**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Matlab for coding	7
2	Excel for calculating parameters	2,5,6

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Follow designing norms.
3) Practice energy conservation.
4) Function as a team member.
5) Function as a team leader.
6) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain* *(to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I **Basic Considerations in Electrical Machine Design**	**CO 1. *To introduce the students with the fundamentals of designing an equipment system*.** *PrO 1. To study various parts of AC machines.* PrO 3. *Basic consideration of electrical machine design based on heating time constant.*	1.1 Design factors 1.2 Limitations in design 1.3 Modern trends in design of electric machines 1.4 Temperature rise
Unit– II **Design of Transformer**	**CO 2. *To learn how to design different part of transformer.*** *PrO 4. Various basic design concepts of transformer.* *PrO 7. To solve and calculate the dimensions of various parts of a transformer*	2.1 Output equation of a transformer. 2.2 Output equation – volt per turn 2.3 Stacking factor 2.4 Ratio of iron loss to copper loss 2.5 Relation between core area and weight of iron and copper 2.6 Optimum designs 2.7 variation of output and losses in transformer with linear dimensions 2.8 Design of core 2.9 Choice of flux density and current density 2.10 Choice of window space factor, window dimensions 2.11 Design of yoke 2.12 Overall dimensions 2.13 Design of high voltage and low voltage winding 2.14 Primary & Secondary resistance, Leakage reactance of windings, Regulation 2.15 Mechanical forces 2.16 No load current calculation 2.17Change of parameters with change of frequency 2.18Temperature rise of transformer 2.19 Design of tank 2.20 Design of tank with cooling tubes.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain* *(to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit– III Induction Machines Design	**CO 3. *To learn how to design different part of induction machine.*** *PrO 2.* To solve the design problems related to the main dimensions of Induction machine. *PrO 5. To solve the problems of stator and rotor design of an induction machine.* *PrO 6. To solve the rotor design of a slip ring induction machine.*	3.1 Choice of specific electrical loadings 3.2 Choice of specific magnetic loadings 3.3 Output equation 3.4 Separation of D & L 3.5 Stator Design: Turns per phase, Stator conductors, Shape of stator slots, Number of stator slots, Area of stator slots, Length of mean turn, Stator teeth, Stator core 3.6 Length of air gap, Relation for calculation of length of air gap. 3.7 Squirrel Cage Rotor Design: Number of rotor slots, Rules for selecting rotor slots, Reduction of harmonic torques, Design of rotor bars & slots, Design of end rings 3.8 Wound Rotor Design: Number of rotor slots, number of rotor turns, area of rotor conductors, Rotor windings, Rotor teeth, design of rotor core 3.9 Estimation of operating characteristics- No load current calculation, short circuit current calculation, Stator and rotor resistance and leakage reactance calculation 3.10 Dispersion coefficient – Effect on maximum power factor and overload capacity, Effect of change of air gap length, number of poles and frequency, Relation between D&L for best power factor
Unit-IV Synchronous Machines Design	**CO 4. *To learn how to design different parts of Synchronous machine.*** **PrO 8.** To solve the problems of stator design of the alternator	4.1 Choice of specific electrical loadings 4.2 Choice of specific magnetic loadings 4.3 Output equations 4.4 Main dimension 4.5 Short Circuit Ratio Effect of SCR on machine performance 4.6 Length of air gap and shape of pole face 4.7 Armature design: Number of armature slots, Coil span, Turns per phase, Conductor section, Slots dimension, Length of mean turn. 4.8 Stator Core design 4.9 Elimination of harmonics, Estimation of air gap length. 4.10 Design of rotor, Height of pole, Design of damper winding, Height of pole shoe, Pole profile drawing 4.11 Design of magnetic circuit,

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain* *(to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
		Determination of full load field MMF 4.12 Design of field winding, Determination of Direct and Quadrature axis synchronous reactance.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Basic Consideration in Electrical Machine Design	04	2	2	1	0	0	0	05
II	Design of Transformer	19	2	4	4	5	4	3	22
III	Induction Machines Design	19	2	4	4	5	4	3	22
IV	Synchronous Machines Design	18	2	3	5	5	3	3	21
Total		**60**	**8**	**13**	**14**	**15**	**11**	**9**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **Transformer Core Design:** Prepare a case study and design core of a transformer in excel and/or matlab.
2) **Induction Machine Design:** Prepare a case study and design the **stator** of a squirrel cage induction machine in matlab.
3) **Induction Machine Design:** Prepare a case study and design the **rotor** of a squirrel cage induction machine in matlab.
4) **Induction Machine Design:** Prepare a case study and design the **stator** of a wound rotor induction machine in matlab.
5) **Induction Machine Design:** Prepare a case study and design the **rotor** of a wound rotor induction machine in excel.
6) **Synchronous Machine Design:** Prepare a case study and design the synchronous machine in excel.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different design methods used in industries.
b) Prepare 15 min. PowerPoint presentation on how steps of designing a particular electrical machine.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Basic Considerations in Electrical Machine Design	1	**PrO 1.** *To study various parts of AC machines.* ***Pro 3**. Basic consideration of electrical machine design based on heating time constant* 1c. Design factors 1d. Limitations in design 1e. Modern trends in the design of electric machines	Lecture with media	PPTs, black board
Unit – II Design of Transformer	2	1f. Temperature rise, Expression for temperature rise, heating & cooling time constants. 1g. Examples **PrO 4.** Various basic design concepts of transformer. **PrO 7.** To solve and calculate the dimensions of various parts of a transformer Specification, Output equation of transformer 2a. Output equation-Volt per turn, Stacking factor	Lecture With media	PPTs, black board
	3	2b. Ratio of iron loss to copper loss 2c. Relation between core area and weight of iron and copper 2d. Optimum designs	Lecture with media Tutorial and Field visit	PPTs, black board
	4	2e. variation of output and losses in transformer with linear dimensions with examples 2f. Design of core, Choice of flux density and current density 2g. Choice of window space factor, window dimensions	Lecture with media, and Demonstration	Real samples of relays, PPTs, Videos
	5	2h. Design of yoke, Overall dimensions with examples 2i. Design of high voltage and low voltage winding 2j. No load current calculation	Lecture with media, Tutorial	PPTs
	6	2k. Change of parameters with the change of frequency 2l. Temperature rise of the transformer 2m. Design of tank and examples	Lecture with media	PPTs

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – III Induction Machines Design	**7**	**PrO 2.** To solve the design problems related to the main dimensions of Induction machine. **PrO 5.** To solve the problems of stator and rotor design of an induction machine. **PrO 6.** To solve the rotor design of a slip ring induction machine. 3a. Choice of specific electrical loadings, Choice of specific magnetic loadings 3b. Output equation, Separation of D & L with examples	Lecture with media, Field visit	PPTs, Video
	8	3c. Turns per phase, Stator conductors 3d. Shape of stator slots, Number of stator slots, Area of stator slots 3e. Length of mean turn, Stator teeth, Stator core 3f. Length of air gap, Relation for calculation of length of air gap	Lecture with media, and Tutorial	PPTs
	9	3g. Number of rotor slots, Rules for selecting rotor slots, Reduction of harmonic torques 3h. Design of rotor bars & slots, Design of end rings 3i. Estimation of operating characteristics 3j. No load current calculation	Lecture with media, + PPT	PPTs, Video
	10	3k. Short circuit current calculation 3l. Stator and rotor resistance and leakage reactance calculation 3m. Dispersion Co-efficient 3n. Effect of dispersion coefficient on maximum power factor and overload capacity.	Lecture with media, Tutorial	PPT, Video
	11	3o. Effect of change of air-gap length 3p. Effect of change of number of poles and frequency 3q. Relation between D & L for best power factors and examples.	Lecture with media	PPTs, black board
Unit– IV Synchronous Machine Design	**12**	**PrO 8.** To solve the problems of stator design of the alternator 4a. Choice of specific electrical loadings and Choice of specific magnetic	Lecture with media, case study	PPT, Pictures , black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		loadings 4b. Output equations and Main dimension 4c. Short Circuit Ratio, Effect of SCR on machine performance 4d. Length of air gap and shape of pole face		
	13	4e. Examples 4f. Number of armature slots and Coil span 4g. Turns per phase, Conductor section 4h. Slots dimension, Length of mean turn	Lecture with media, Tutorial	PPTs, black board
	14	4a. Stator Core, Elimination of harmonics 4b. Estimation of air gap length 4c. Height of pole, Design of damper winding, Height of pole shoe	Lecture with media, Tutorial	PPTs, Video
	15	4d. Design of magnetic circuit, Determination of full load field MMF 4e. Design of field winding 4f. Determination of Direct and Quadrature axis synchronous reactance 4g. Examples	Lecture with media, Tutorial	PPTs, Video

XXII SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	A course in Electrical machine design	A.K. Sawhney & A. Chakrabarti	Dhanpat Rai & Co. ISBN-10. 8177001019.
2	The performance and design of alternating current machines	M. G. Say	CBS Publishers & Distributors, ISBN: 9788123910277
3	Design of rotating electrical machines	Juha Pyrhonen, Tapani Jokinen, Valeria	Wiley, ISBN:9781118581575
4	Design of electrical machines	K.G. Upadhyay	New Age International, 2011. ISBN. 8122422829

S. No.	Title of Book	Author	Publication including ISBN
5	Design of electrical machines	V.N. Mittal & A. Mittal	Publisher, N.C. Jain, 2002. ISBN, 8180141268
6	Electric Machinery	A.E. Fitzerald, Charles Kingsley, Stephen D. Umans	Tata Mcgraw Hill, ISBN: 9780073660097
7	Elements of electrical design	J.G. Jamnani	Mahajan publishing house, ISBN: 9789381256046

XXIII SOFTWARE/LEARNING WEBSITES

1. https://www.youtube.com/playlist?list=PLux1Y9qIGZ3MAitb5PkczXLStSPHCwnia
2. Electrical Machine Design https://cusp.umn.edu/electric-machines-drives/electric-machines-design

XXIV PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE443	**Semester VII**	**Protection and Switchgear** **Mark 'H'=3** for high, '**M**'=2 for medium or **'L'=1'** for the relevant correlation of each PO or PSO, Competency or CO													
443.01	Course Outcome 1	3	1	3	2	2		2		1	1	2		2	
443.02	Course Outcome 2	3	2	2	2	1		2		2		1	1	2	
443.03	Course Outcome 3	3	2	2	2	1		2		2		1	1	2	
443.04	Course Outcome 4	3	2	2	2	1		2		2		1	1	2	

XXV NAMES OF RESOURCE PERSONS

a) Mihir Mehta

Course Code: **EE444**
Course Title: **Simulation Lab**

UG Engineering Programme	Semester
Electrical Engineering	Seventh

I RATIONALE

Simulations are recognized as an efficient and effective way of teaching complex and dynamic engineering systems. A simulation-based teaching environment enables students to acquire experience and consider their previous results. The use of Simulation labs has been gaining currency in the domains of engineering and technology programs

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Identify and define engineering problems, conduct experiments and investigate to analyze and interpret data to arrive at substantial conclusions**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1. ***Use*** a suitable package to solve the power flow problems for interconnected power systems.
2. ***Develop*** a program in a suitable package to assess the performance of power system parameters.
3. ***Use*** simulators as a learning aid and gain a profound insight into the working of various power electronic converters and drive systems.
4. ***Explore*** the behavior of different types of electrical machines.
5. **Validate** the dynamic models of existing as well as unknown circuits and systems.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			**Total Credits (L+T+P)**	**Examination Scheme**				
				Theory Marks		**Practical Marks**		**Total Marks**
L	**T**	**P**	**C**	**ESE**	PA	**ESE**	**PA**	
0	0	4	2	0	0	50	50*	100

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

(*): *Under the* ***Practical PA****, out of 50 marks,* ***25 marks*** *are for* ***Research Paper Publication*** *to facilitate integration of COs and the remaining 25 marks is converted from lab manual to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Analyze the different types of solvers used in MATLAB software	02
2	1	Analyze the use of symbolic Toolbox in MATLAB software.	02
3	2	Solve Linear equations and matrix solution using MATLAB software	02
4	1	Creation of bus Admittance and Impedance matrix Using MATLAB software.	02
5	2	Computation of transmission line parameter using MATLAB software.	02
6	3	Analyze the single phase half wave and full wave bridge uncontrolled rectifier with R, RL & RLE load.	02
7	3	Analyze the single phase half wave and full wave bridge controlled rectifier with R, RL & RLE load.	02
8	4	Simulate speed control of separately excited DC Machines	02
9	3	Simulate the three phase full wave circuit using MATLAB software	02
10	4	Implement Open Circuit, Short Circuit& Load tests on Single Phase transformer using MATLAB	04
11	5	Analyze the effect of fault using Power world simulator and using MATLAB software	02
12	4	Design of leakage inductance of transformer using FEM software	02
13	1	Simulate the Load flow analysis using Power World Simulator software	04
14	2	Implement relay Co- ordination for radial system using Mi Power software.	04
15	2	Computation of Long term load forecasting using Mi Power software.	04
16	3	Analyze the effect of Harmonic using MATLAB software	02
17	3	Introduction of PSIM simulation software	02
18	3	Design Boost convertor with PI controller	04
19	3	Design Multilevel inverter using PSIM software	04
20	3	Design Flyback converter using PSIM software	04
		Total	**54**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB Simulation software (version 2021a)	1,2,3,4,5,9,10,16
2	PSIM Simulation Software	6,7,8,17,18,19,20
3	Power World Simulator	11, 13
4	FEM Software	12
5	Mi Power Software	14,15

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Function as a team member.
3) Function as a team leader.
4) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1^{st} year
b) 'Valuing Level' in 2^{nd} year
c) 'Organising Level' in 3^{rd} year
d) 'Characterising Level' in 4^{th} year.

VIII SPECIAL INSTRUCTIONAL STRATEGIES (if any)

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'P' in item No. 4*** does not mean only the traditional practical method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the simulation*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.

IX SUGGESTED STUDENT ACTIVITIES

Other than the laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different software used in industries.
b) Prepare 15 min. power point presentation on how to choose software for the particular application.

XXVI SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Electric Machinery Fundamentals	Stephen. J. Chapman,	Mcgraw Hill, 2017 ISBN: 9780071070522

S. No.	Title of Book	Author	Publication including ISBN
2	Power System Analysis	Hadi Saadat	Mcgraw Hill, 1998 ISBN: 0075616343

XXVII SOFTWARE/LEARNING WEBSITES

1. http://www.prdcinfotech.com/business/software-engineering-group/software-products/mipower/
2. https://www.mathworks.com/products/matlab.html
3. https://www.powerworld.com/
4. http://www.femm.info

XXVIII PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE444	**Semester VII**														
444	**Competency** *Identify and define engineering problems, conduct experiments and investigate to analyze and interpret data to arrive at substantial conclusions*	***3***	***2***	***2***	***2***	***1***	**-**	***1***	**-**	**-**	**-**	**-**	***1***	***3***	**-**
444.1	Use a suitable package to solve the power flow problems for interconnected power systems	3	2	2	2	2	-	-	-	-	-	-	-	3	3
444.2	Develop a program in a suitable package to assess the performance of transmission line parameters	3	3	3	2	2	-	-	-	-	-	-	-	-	3
444.3	Use simulators as a learning aid and gain a	3	-	-	2	3	-	-	2	-	-	1	2	-	3

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	profound insight into the working of various power electronic converters and drive systems														
444.4	Explore the behavior of different types of electrical machines	3	2	3	1	-	-	-	-	-	-	-	-	3	2
444.5	Validate the dynamic models of existing as well as unknown circuits and systems	3	3	2	2	-	-	-	-	-	-	-	-	3	-

XXIX NAMES OF RESOURCE PERSONS

a) Dr. Jigar Sarda

Course Code: **EE450**
Course Title: **Summer Internship-II**

UG Engineering Programme	**Semester**
Electrical Engineering	Seventh

I RATIONALE

This course offers the opportunity for the young students to acquire on job the skills, knowledge, attitudes, and perceptions along with the experience needed to constitute a professional identity. This course includes planning of the tasks which are to be completed within the time allocated, and in turn, helps to develop ability to plan, use, monitor and control resources optimally and economically. By studying this course ability like creativity, imitativeness and performance qualities are also developed in students. Also give an insight into the working of the real organizations. Leadership development and supervision skills are also integrated objectives of learning this course. To appreciate the linkages among different functions and departments. This course helps the students in exploring career opportunities in their areas of interest. By attending this course student get deeper understanding in specific functional areas.

II COMPETENCY

The course should be taught and implemented with the aim to develop the required course outcomes (COs) so that students will acquire following competency needed by the industry:

- **Develop professional habit, attitude and ethics related to the industrial working environment**.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

Solve engineering problems by working in a team with across other disciplines.
Apply knowledge and skills learned in company/industry/organization to real-world problems.
Use experience related to professional and ethical issues in the work environment
Examine the impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context
Evaluate contemporary issues related with engineering in general by adopting new tools and technologies

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)	**Total Credits (L+T+P)**	**Examination Scheme**		
		Theory Marks	**Practical Marks**	**Total Marks**

L	T	P	C	ESE	PA	ESE	150
0	0	0	3	00	00	150	

***Legends: L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

V Course Details: Instructional Method and Pedagogy

Format of Summer Internship Report:

At the start of the semester student should prepare project report. Guideline of the Project report is as follow

The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- Chapter-1: Overview of company: Detail of the company including Location and spread of the company, Number of employees, engineers, technicians, administrators in the company, Divisions of the company, Administrative tree (if available), Customer profile and market share, details of process or services, various department and role of each department, detail of your department.
- Chapter-2: Overview of Internship Experience: In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out.
- Chapter 3: Observations: In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what

you write should be based on what you did and observed that truly belongs to the company/industry/organization. Present your observations.

- Chapter 4: Learning Outcomes: In the last section, summarize the summer internship activities, learning outcomes, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.
- References: List any source you have used in the document including books, articles and web sites in a consistent format.
- Appendices: If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it and waste tons of paper.

Note:

a) The report should contain as many diagrams and figures, charts etc. as relevant for the internship.
b) Originality of the report (written in own words) would be given more importance rather than quality of printing and use of glossy paper or multi-colour printing

VI ASSESSMENT OF Summer Internship

Assessment of summer internship is done at the beginning of term by the internal examiner and external examiner after successful completion of summer internship

A) Assessment of summer internship by Internal Examiner.

S. No.	Description	Marks
1	**Final Report including documentation**. (marks based on: clarity in presentation and organization; styles and language; quality of diagrams, drawings and graphs; accuracy of conclusion drawn; citing of cross references; suggestion for further research/project work)	**25**
2	**Presentation** (presentation skills including communication skills to be assessed by observing presentations and asking questions during presentation)	**25**
3	**Viva** (ability to give answers the methods/materials used and technical knowledge, and involvement of individual to be assessed by asking questions during presentation and viva/voce)	**25**
Total		**75**

B) Assessment of summer internship by External Examiner.

S. No.	Description	Marks
1	**Final Report including documentation**. (marks based on: clarity in presentation and organization; styles and language; quality of diagrams, drawings and graphs; accuracy of conclusion drawn; citing of cross references; suggestion for further research/project work)	**25**
2	**Presentation** (presentation skills including communication skills to be assessed by observing presentations and asking questions during presentation)	**25**
3	**Viva** (ability to give answers the methods/materials used and technical knowledge, and involvement of individual to be assessed by asking questions during presentation and viva/voce)	**25**
	Total	**75**

VIII PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	**PO 1** Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE450	**Semester VII**														
450	**Competency** The course should be taught and implemented with the aim to develop the required course outcomes (COs) so that students will acquire following competency needed by the industry:	***3***	***2***	***2***	***2***	***2***	***2***	***1***	***1***	***2***	***2***	***2***	***1***	***2***	***2***
450.1	Solve engineering problems by working in a team with across other disciplines.	3	-	1	-	-	1	-	1	-	1	1	1	-	1
450.2	Use experience related to professional and ethical issues in the work environment	3	3	2	2	1	2	-	-	3	2	1	2	2	-
450.	Use experience	3	2	1	-	-	-	-	3	1	-	2	-	-	1

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
3	related to professional and ethical issues in the work environment.														
450. 4	Examine the impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context.	3	-	-	3	3	3	-	-	2	1	3	2	3	3
450. 5	Evaluate contemporary issues related with engineering in general by adopting new tools and technologies	3	1	3	2	3	3	2	-	-	2	2	3	3	2

IX NAMES OF RESOURCE PERSONS

A) Mahammadsoaib Saiyad

Course Code: **EE446.01**
Course Title: **Commissioning and Testing of Electrical Equipment**

UG Engineering Programme	Semester
Electrical Engineering	Seventh

I RATIONALE

Power Systems and Industrial Plants consist of number of electrical drives, transformers, circuit breakers and other equipment which require installation, commissioning, testing and regular maintenance to prevent permanent breakdown. This course will enable the students to understand the concepts, principles and acquire basic skills of installation, commissioning, testing and maintenance of electrical equipment in power stations, substations and industry.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- Perform the different commissioning and high voltage tests on electrical equipment.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) ***Familiar*** *with specifications, electrical safety regulations and rules during installation and maintenance of electrical equipment.*
2) ***Perform*** *the high voltage tests on electrical equipment.*
3) ***Perform*** *the installation and commissioning tests on electrical equipment.*
4) ***Prepare*** *commissioning test report, troubleshooting chart and maintenance schedule of electrical equipment.*

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	
3	0	2	4	70	30*	25	25	150

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

(*): *Under the* ***Theory PA****, out of 30 marks,* ***15 marks*** *are for* ***micro-project assessment*** *to facilitate integration of COs and the remaining 15 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Understand the name plate and other specifications of transformer, dc machine, alternator and induction machine.	04
2	2	Determine the dielectric strength of the given transformer oil.	02
3	2	Perform the impulse voltage test on the given insulator.	04
4	2	Measure the dielectric loss factor and dc resistivity of the given transformer oil.	02
5	3	Perform the polarity test on single phase and three phase transformers.	02
6	3	Examine the reflection of load on secondary of three phase transformer.	02
7	3	Determine the number of pole of induction motor without energizing the motor.	02
8	4	Prepare a troubleshooting and maintenance chart for the given electrical equipment.	04
9	4	Prepare a commissioning test report for the given electrical equipment.	04
10	3	Perform the polarity test, ratio test and saturation characteristic on current transformer.	04
		Total	**30**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Induction Motor	2
2	Current Transformer	3
3	Single and Three phase transformers	4, 8
4	0-100 kV motorized oil BDV test kit	5
5	5-stage, 1.5kJ, 150kV Marx Impulse voltage generator	6
6	Oil dissipation factor and dc resistivity meter	7

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) <u>Follow</u> safe practices.
2) <u>Function</u> as a team member.
3) <u>Function</u> as a team leader.

4) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Extracts from Indian Electricity Rules – 1956 and Principles and Planning of Maintenance	**CO 1.** ***Familiar*** *with specifications, electrical safety regulations and rules during installation and maintenance of electrical equipment.* PrO 1. Understand the name plate and other specifications of transformer, dc machine, alternator and induction machine. 1a. Understand the specifications, electrical safety regulations and rules during installation and maintenance of electrical equipment.	1.1 Indian Electricity Act 1.2 Indian Electricity rules and regulations 1.3 Routine Maintenance, Periodical Maintenance & Maintenance on Fault
Unit– II Commissioning & Testing of Transformer	**CO 2.** ***Perform*** *the high voltage tests on electrical equipment.* ***CO 3. Perform*** *the installation and commissioning tests on electrical equipment.* ***CO 4. Prepare*** *commissioning test report, troubleshooting chart and maintenance schedule of electrical equipment.* *PrO 2.* Determine the dielectric strength of the given transformer oil. *PrO 3.* Perform the impulse voltage test on the given insulator. *PrO 4.* Measure the dielectric loss factor and dc resistivity of the given transformer oil. *PrO 5.* Perform the polarity test on single phase and three phase transformers. *PrO 6.* Examine the reflection of load on secondary of three phase transformer. *PrO 8.* Prepare a troubleshooting and	2.1 Measurement of winding resistance of transformer 2.2 Open circuit and short circuit test of transformer 2.3 Measurement of Insulation resistance 2.4 Power frequency voltage withstand test on transformer 2.5 Impulse voltage and partial discharge test 2.6 Parallel operation of transformers 2.7 Transformer oil purification and tests 2.8 Polarization index and

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	maintenance chart for the given electrical equipment. *PrO 9.* Prepare a commissioning test report for the given electrical equipment. 2a. Perform different high voltage tests on the given transformer oil and transformer. 2b. Perform commissioning and installation tests on the given transformer. 2c. Prepare the detailed report on commissioning, installation and testing of the given transformer.	dissipation factor measurement 2.9 Drying of transformers 2.10 Commissioning of transformers 2.11 Causes of faults, troubleshooting and maintenance of transformers
Unit– III Commissioning & Testing of Induction Machines	***CO 3. <u>Perform</u>*** *the installation and commissioning tests on electrical equipment.* ***CO 4. <u>Prepare</u>*** *commissioning test report, troubleshooting chart and maintenance schedule of electrical equipment.* *PrO 7.* Determine the number of pole of induction motor without energizing the motor. *PrO 8.* Prepare a troubleshooting and maintenance chart for the given electrical equipment. *PrO 9.* Prepare a commissioning test report for the given electrical equipment. 3a. Perform commissioning and installation tests on the given induction machine. 3b. Prepare the detailed report on commissioning, installation and testing of the given induction machine.	3.1 Insulation test and drying out of windings, Temperature rise test, 3.2 induction motor stator and rotor interaction, balance and vibration, ventilation and cooling, 3.3 Hammer test, Testing against variation of voltage/current/frequency 3.4 Testing of auxiliaries, Degree of protection (IP Grade) 3.5 AC Motor Troubles: Insulation failure due to transient voltages, low starting torque, pull out torque and stalling, low power factor, excessive slip, crawling, single phasing, reversal of one phase winding, starting transients in squirrel cage induction motors, overheating, pull over, vibration having electrical origins, slip ring wear, shaft currents, ball and roller bearing trouble. 3.6 Commissioning steps for Induction motor 3.7 Commissioning of Induction Generator.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit-IV Commissioning & Testing of Alternator	***CO 3. <u>Perform</u>*** *the installation and commissioning tests on electrical equipment.* ***CO 4. <u>Prepare</u>*** *commissioning test report, troubleshooting chart and maintenance schedule of electrical equipment.* *PrO 8.* Prepare a troubleshooting and maintenance chart for the given electrical equipment. *PrO 9.* Prepare a commissioning test report for the given electrical equipment. 4a. Perform commissioning and installation tests on the given alternator. 4b. Prepare the detailed report on commissioning, installation and testing of the given alternator.	4.1 Commissioning of alternator: Preparation and drying out of alternator windings before commissioning, insulation resistance measurements, high voltage tests, measurements of temperature of windings 4.2 extra high voltage alternators, alternators protective gear tests, trip circuit supplies and tripling tests, starting time and rate of picking up load, control of auxiliaries Alternator stator and rotor interaction, effect of excitation on stator current of an alternator in parallel with a large system 4.3 Alternator troubles: Instability of exciters, complete loss of field, failure of exciter voltage to build up, motor driven exciters, alternator instability, neutral inversion with grounded voltage transformer and un-grounded power system, heating of copper conductors, shaft currents and bearing troubles, slip ring wear on alternator rotor, Unbalanced stator currents, transient torques, heating of stator core end punching and clamps
Unit – V Commissioning & Testing of	***CO 3. <u>Perform</u>*** *the installation and commissioning tests on electrical equipment.* ***CO 4. <u>Prepare</u>*** *commissioning test report,*	5.1 Bus bar: Temperature rise test, rated short time current test, Power

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
Sub Station Equipment	*troubleshooting chart and maintenance schedule of electrical equipment.* *PrO 10.* Perform the polarity test, ratio test and saturation characteristic on current transformer. 5a. Perform commissioning and installation tests on the given substation equipment. 5e. Prepare the detailed report on commissioning, installation and testing of the given substation equipment.	frequency voltage withstand test, Impulse / surge testing, Vibration test. 5.2 Isolator Testing: Short circuit test, charging current, Making & Breaking test, Inductive current making & breaking test. 5.3 Circuit Breaker testing: Mechanical endurance test, Temperature rise test, Impulse & surge testing, Short circuit testing of circuit breakers, synthetic testing of circuit breakers, mechanical operation and adjustment, electrical auxiliaries, Troubleshooting & maintenance of circuit breakers 5.4 Protective relay tests: relay setting 5.5 C.T. & P.T. Testing, Station Batteries for D.C. Supply 5.6 Testing & Commissioning of Lightning Arrestor, and Substation Commissioning by Thermography

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Extracts from Indian Electricity Rules – 1956 and Principles and Planning of Maintenance	03	6	0	0	0	0	0	06
II	Commissioning & Testing of Transformer	12	2	2	4	6	6	2	22
III	Commissioning & Testing of Induction	09	0	0	4	2	6	0	12

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
	Machines								
IV	Commissioning & Testing of Alternator	09	2	2	0	2	2	0	08
V	Commissioning & Testing of Sub Station Equipment	12	0	2	4	4	4	8	22
Total		**45**	**10**	**06**	**12**	**14**	**18**	**10**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) Undertake the survey and submit a report on specifications and manufacturer of various electrical equipment.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.

c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.

d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding specifications and manufacturers of electrical equipment.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Breakdown mechanisms in gaseous, liquid and solid dielectrics	1	**PrO 1.** Understand the name plate and other specifications of transformer, dc machine, alternator and induction machine. 1a. Indian Electricity Act. 1b. Indian Electricity rules and regulations ***(self-learn)*** 1c. Routine Maintenance, Periodical Maintenance & Maintenance on Fault.	Lecture with media	PPTs, black board
Unit– II Commissioning & Testing of Transformer	2	**PrO 2.** Determine the dielectric strength of the given transformer oil. 2.1 Measurement of winding resistance of transformer 2.2 Open circuit and short circuit test of transformer 2.3 Measurement of Insulation resistance	Lecture With media	PPTs, black board
	3	**PrO 3.** Perform impulse voltage test on given insulator. 2.4 Power frequency voltage withstand test on transformer 2.5 Impulse voltage and partial discharge test	Lecture With media	PPTs, black board
	4	**PrO 4.** Measure the dielectric loss factor and dc resistivity of the given transformer oil. 2.6 Parallel operation of transformers 2.7 Transformer oil purification and tests 2.8 Polarization index and dissipation factor measurement	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
	5	**PrO 5.** Perform the polarity test on single phase and three phase transformers. **PrO 6.** Examine the reflection of load on secondary of three phase transformer. 2.9 Drying of transformers 2.10 Commissioning of transformers 2.11 Causes of faults, troubleshooting and maintenance of transformers	Lecture With media	PPTs, black board
Unit– III Commissioning & Testing of Induction Machines	6	**PrO 7.** Determine the number of pole of induction motor without energizing the motor. 3.1 Insulation test and drying out of windings, Temperature rise test, 3.2 induction motor stator and rotor interaction, balance and vibration, ventilation and cooling,	Lecture With media	PPTs, black board
	7	3.3 Hammer test, Testing against variation of voltage/current/frequency 3.4 Testing of auxiliaries, Degree of protection (IP Grade) 3.5 AC Motor Troubles: Insulation failure due to transient voltages, low starting torque, pull out torque and stalling, low power factor, excessive slip, crawling, single phasing, reversal of one phase winding	Lecture With media	PPTs, black board
	8	**PrO 8.** Prepare a troubleshooting and maintenance chart for the given electrical equipment. 3.6 AC Motor Troubles: starting transients in squirrel cage induction motors, overheating, pull over, vibration having electrical origins, slip ring wear, shaft currents, ball and roller bearing trouble. 3.7 Commissioning steps for Induction motor 3.8 Commissioning of Induction Generator.	Lecture With media	PPTs, black board
Unit-IV Commissioning & Testing of Alternator	9	4.1 Commissioning of alternator: Preparation and drying out of alternator windings before commissioning, insulation resistance measurements, high voltage tests, measurements of temperature of windings	Lecture With media	PPTs, black board
	10	**PrO 9.** Prepare a troubleshooting and maintenance chart for the given electrical equipment. 4.2 extra high voltage alternators, alternators protective gear tests, trip circuit supplies and tripling tests, starting time and rate of picking up load, control of auxiliaries Alternator stator and rotor interaction, effect of excitation on stator current of an alternator in parallel with a large system	Lecture With media	PPTs, black board
	11	**PrO 9.** Prepare a commissioning test report for the given electrical equipment.	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		4.3 Alternator troubles: Instability of exciters, complete loss of field, failure of exciter voltage to build up, motor driven exciters, alternator instability, neutral inversion with grounded voltage transformer and un-grounded power system, heating of copper conductors, shaft currents and bearing troubles, slip ring wear on alternator rotor, Unbalanced stator currents, transient torques, heating of stator core end punching and clamps		
Unit – V Commissioning & Testing of Sub Station Equipment	**12**	5.1 Bus bar: Temperature rise test, rated short time current test, Power frequency voltage withstand test, Impulse / surge testing, Vibration test. 5.2 Isolator Testing: Short circuit test, charging current, Making & Breaking test, Inductive current making & breaking test.	Lecture With media	PPTs, black board
	13	**PrO 9.** Prepare a commissioning test report for the given electrical equipment. 5.3 Circuit Breaker testing: Mechanical endurance test, Temperature rise test, Impulse & surge testing, Short circuit testing of circuit breakers, synthetic testing of circuit breakers, mechanical operation and adjustment, electrical auxiliaries, Troubleshooting & maintenance of circuit breakers	Lecture with media	PPTs, black board
	14	**PrO 10.** Perform the polarity test, ratio test and saturation characteristic on current transformer. 5.4 Protective relay tests: relay setting 5.5 C.T. & P.T. Testing, Station Batteries for D.C. Supply	Lecture with media	PPTs, black board
	15	5.6 Testing & Commissioning of Lightning Arrestor, and Substation Commissioning by Thermography	Lecture with media	PPTs, black board

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Testing, Commissioning and Maintenance of Electrical Equipments	S. S. Rao	Khanna Publisher, India, ISBN 978-8174091857

XV SOFTWARE/LEARNING WEBSITES

- http://nptel.iitm.ac.in/courses/IITMADRAS/Electrical_Machines_II/Testing/index.php
- http://www.ece.ualberta.ca/~knight/ee332/synchronous/s_main.html
- http://www.ece.ualberta.ca/~knight/ee332/induction/i_main.html

XVI COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE446.01	**Semester VI**	**Commissioning and Testing of Electrical Equipment** **Mark 'H'=3** for high, '**M**'=2 for medium or **'L'=1'** for the relevant correlation of each PO or PSO, Competency or CO													
446.01	**Competency** Perform the different commissioning and high voltage tests on electrical equipment.	3	2	-	2	-	1	-	-	2	1	-	2	3	-
CO1	Familiar with specifications, electrical safety regulations and rules during installation and maintenance of electrical equipment.	2	-	-	-	-	-	-	-	-	-	-	3	1	-
CO2	Perform the high voltage tests on - electrical equipment.	3	3	2	1	2	-	2	-	2	-	-	3	2	-
CO3	Perform the installation and commissioning tests on electrical equipment.	3	3	-	-	2	-	-	-	2	3	-	3	2	-
CO4	Prepare commissioning test report, troubleshooting chart and maintenance schedule of electrical equipment.	3	3	-	-	2	-	-	-	2	3	-	3	2	-

XVII NAMES OF RESOURCE PERSONS

a) Prof. Mihir A. Bhatt

Course Code: **EE447.01**

Course Title: **Modeling & Control of Renewable Energy Sources**

UG Engineering Programme	Semester
Electrical Engineering	Seventh

I RATIONALE

The renewable energy sources like solar photovoltaic, wind energy and fuel cell are emerging power generation sources. The course is developed such that student would get the knowledge from basics to control schemes of renewable energy sources.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Evaluate the performance of various control methods used in field**.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) ***Use*** the control technique for photovoltaic panel**.**
2) ***Evaluate*** *the performance of the photovoltaic and wind system*
3) **Interpret** the suitable control technique for photovoltaic system and wind system.
4) ***Evaluate*** the performance of the mppt converter used for power maximization.
5) ***Choose*** *the type of converter circuit for power control***.**

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			**Total Credits (L+T+P)**	**Examination Scheme**				
				Theory Marks		**Practical Marks**		**Total Marks**
L	**T**	**P**	**C**	**ESE**	PA	**ESE**	**PA**	150
3	0	2	4	70	30*	25	25	

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

(*): *Under the* ***Theory PA****, out of 30 marks,* ***15 marks*** *are for* ***micro-project assessment*** *to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the

students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Modeling of PV cell to determine I-V & P-V curve in MATLAB.	04
2	2	Programing of PV cell output parameters to determine I-V & P-V curve in MATLAB.	04
3	3	Simulation of Grid Connected PV system in PVSyst.	04
4	3	Simulation of stand-alone PV system in PVSyst.	04
5	3	To observe the output parameters of PV panel.	02
6	4	Simulate of Nernst equation to determine fuel cell output.	04
7	4	To study Wind Farm DFIG detailed model.	04
8	5	To Study Solid-oxide fuel cell connected to three phase-Electrical power system.	04
		Total	**30**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB Simulation software (version 2021a)	1,2,5,6,7,8
2	PVsyst Simulation software (Version 6.7)	3,4

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Introduction of Renewable Energy Sources	**CO 1. *Recognize the electric circuits connected to various Renewable Energy Sources.*** *PrO 1.* *Modeling of PV cell to determine I-V & P-V curve in MATLAB.* 1a. Evaluate the performance of solar cell circuits. 1b. Justify the characteristics of solar cell for various connections.	1.1 Global and Indian renewable energy scenario 1.2 various renewable energy sources 1.3 Energy units and conversions 1.4 Energy Storage 1.5 Introduction to IEEE 1547 standard.
Unit– II Solar Photovoltaic System	**CO 3. *Understand the construction, working and control of solar and wind energy system*** ***CO 5. Design & Analyze the performance of PV system.*** *PrO 2. Programing of PV cell output parameters to determine I-V & P-V curve in MATLAB.* *PrO3. Simulation of Grid Connected PV system in PVSyst.* *PrO 4. Simulation of stand-alone PV system in PVSyst.* *PrO 5. To observe the output parameters of PV panel.* 2a. Evaluate the performance of grid connected PV system 2b. Evaluate the performance of standalone PV system 2c. Justify the output characteristics of PV cell. 2d. Setup the circuit to determine output characteristics of PV cell.	2.1 History of Solar Energy Usage 2.2 Equivalent circuit of photovoltaic cell 2.3 From cell to module and from module to array 2.4 I-V curve under standard test condition, Fill Factor 2.5 impact of temperature and insolation on I-V curve 2.6 Shading impacts on I-V curve, I-V curve for various types of loads (resistor, DC motor, battery) 2.7 MPPT Concept, DC-DC converters (boost, buck, buck-boost) for MPPT 2.8 MPPT methods: reference cell methods, Perturb & Observation method, power slope method 2.9 standalone PV system design 2.10 Grid connected PV system principle 2.11 PV-Grid connection topologies: unfold circuit, double stage,

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
		single stage 2.12 classification of grid connected inverter: centralized inverter, string inverter & micro inverter 2.13 PV-Grid connected system architecture 2.14 PV-Grid connected system design 2.15 Economics of PV System, PV system in micro-grid 2.16 Emerging PV cell technologies.
Unit– III Wind Energy System	**CO 3.** ***Understand the construction, working and control of solar and wind energy system.*** *PrO 7. To study Wind Farm DFIG detailed model.* 3a. Determine the output from wind energy system. 3b. Choose the control methods for maximizing output power.	3.1 Introduction, Wind Characteristics 3.2 Wind turbines: fixed speed & variable speed wind turbine 3.3 components of wind energy conversion system 3.4 Types of generators: Induction generator, synchronous generator, Permanent Magnet Synchronous Generator, 3.5 Doubly Fed Induction Generator 3.6 Pitch angle control, Grid Integration of wind energy system 3.7 Economics of wind energy conversion system
Unit-IV Fuel Cell	**CO 4.** ***Aware with the performance characteristics of Renewable Energy Sources.*** *PrO 6. Simulate of Nernst equation to determine fuel cell output.* *PrO 8. To Study Solid-oxide fuel cell connected to three phase-Electrical power system.* 4a. Choose the relevant operating parameters of fuel cell for the given condition. 4b. Select the relevant control method for the	4.1 Construction & Operation of fuel cell 4.2 types of fuel cells, comparison of different component and characteristic of fuel cell 4.3 fuel cell equivalent circuit, 4.4 need for dynamic

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	specified application. 4c. Evaluate the performance of fuel cell for the specified condition.	modeling of fuel cell, 4.5 Polymer Electrolyte Membrane Fuel Cell (PEMFC) model structure, 4.6 Solid Oxide Fuel Cell (SOFC) model structure
Unit – V Renewable Energy Sources in Smart Grid & Electric Vehicle	**CO 5. *Acquire the knowledge regarding the fundamentals of Smart Grid and Electric Vehicle.*** 5a. Interpret the smart grid data with respect to the given power system. 5b. Determine vehicle charging control strategy. 5c. Prepare a design of solar water pumping system. 5d. Select the method for vehicle to grid connection.	5.1 Introduction of Smart Grid 5.2 History of Electric Vehicle 5.3 Architecture of PV based electric vehicle charging: Conventional method, Centralized architecture, Distributed architecture, single stage architecture 5.4 Control Strategy for EV charging 5.5 Impact of EV charging on Electrical Distribution System, V2G Concept 5.6 PV powered water pumping 5.7 Smart Meters

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction of Renewable Energy Sources	05	2	2	2	4	0	0	10
II	Solar Photovoltaic System	20	0	2	4	4	2	0	12
III	Wind Energy System	15	0	2	4	4	4	0	14
IV	Fuel Cell	05	0	4	4	4	6	0	18
V	Renewable Energy Sources in Smart Grid & Electric Vehicle	15	0	2	4	4	6	0	16
Total		**60**	**2**	**12**	**18**	**20**	**18**	**0**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **Solar PV MPPT:** Make and test the advanced version of solar MPPT methods.
2) **Design grid connected PV system and make economic evaluation.**
3) **Design stand alone PV system and make economic evaluation.**
4) **Create a solar water pumping system**.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different designs of grid connected renewables.
b) Prepare 15 min. power point presentation on control schemes for renewables.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Introduction of Renewable Energy Sources Unit– II	1	*PrO1. Modeling of PV cell to determine I-V & P-V curve in MATLAB.* 1a. Evaluate the performance of solar cell circuits.	Lecture with media	PPTs, black board
	2	1b. Justify the characteristics of solar cell for various connections.	Lecture With media	PPTs, black board
Solar Photovoltaic System Unit– III Wind Energy System	3	*PrO 2. Programing of PV cell output parameters to determine I-V & P-V curve in MATLAB.* 2c. Justify the output characteristics of PV cell.	Lecture With media	PPTs, black board
	4	2d. Setup the circuit to determine output characteristics of PV cell. (self learn)	Lecture With media	PPTs, black board
	5	*PrO3. Simulation of Grid Connected PV system in* *2a.* Evaluate the performance of grid connected PV system	Lecture With media	PPTs, black board
Unit-IV Fuel Cell Unit – V	6	*2c.* Justify the output characteristics of PV cell.	Lecture With media	PPTs, black board
	7	*PrO 4. Simulation of stand-alone PV system in PVSyst.* 2b. Evaluate the performance of stand alone PV system	Lecture With media	PPTs, black board
	8	2c. Justify the output characteristics of PV cell. (self learn)	Lecture With media	PPTs, black board
Renewable Energy Sources in Smart Grid & Electric Vehicle Unit – I Introduction of Renewab	9	*PrO 5. To observe the output parameters of PV panel.* 2d. Setup the circuit to determine output characteristics of PV cell.	Lecture With media	PPTs, black board
	11	*PrO 7. To study Wind Farm DFIG detailed model.* 3a. Determine the output from wind energy system.	Lecture With media	PPTs, black board
	12	3b. Choose the control methods for maximizing output power. (self learn)	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
le Energy Sources				
Unit– II	13	*PrO 6. Simulate of Nernst equation to determine fuel cell output.* 4a. Choose the relevant operating parameters of fuel cell for the given condition.	Lecture With media	PPTs, black board
	14	4b. Select the relevant control method for the specified application(self learn)	Lecture With media	PPTs, black board
	15	*PrO 8. To Study Solid-oxide fuel cell connected to three phase-Electrical power system* *4c.* Evaluate the performance of fuel cell for the specified condition. (self learn)	Lecture With media	PPTs, black board

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Renewable & Efficient Electric Power System	Gilbert M. Masters	Wiley, ISBN: 9780471280606
2	Design of Smart Power Grid Renewable Energy Systems	Ali Keyhani	Wiley, ISBN:9781119573265
3	Modeling and Control of Fuel Cells	M. Hashem Nehrir, Caisheng Wang	Wiley, ISBN: 9780470233283

XV SOFTWARE/LEARNING WEBSITES

1. https://onlinecourses.nptel.ac.in/noc18_ee35/course (Design of Photovoltaic System)

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	**PO 1** Used of mathematical method	**PO 2** Analysis mechanical system	**PO 3** Used engineering soft.	**PO 4** Solve problem in machine	**PO 5 Modern tool usage**	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and team work:	**PO 10** Communication	**PO11** Project management	**PO12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem
EE447.01	**Semester VII**														
CO1	Use the control technique for photovoltaic panel.	*3*		*1*		*1*				*1*	*1*		*1*	*2*	
CO2	Evaluate the performance of the photovoltaic and wind system	*3*	*2*	*2*	*1*	*1*		*2*		*1*		*1*		*2*	
CO3	Interpret the suitable control technique for photovoltaic system and wind system.	*3*	*2*	*2*	*1*	*1*	*1*	*2*		*1*		*1*	*1*	*2*	*1*
CO4	Evaluate the performance of the mppt converter used for power maximization.	*3*	*1*		*1*	*2*		*2*	*1*			*2*		*2*	*1*
CO5	Choose the type of converter circuit for power control.	*3*	*1*	*3*	*3*	*3*	*1*	*3*		*1*		*2*	*3*	*3*	*2*

XVII NAMES OF RESOURCE PERSONS

a) Prof. Pratik Mochi

b) ……………………………………………………….

Course Code: **EE472.01**
Course Title: **COMMUNICATION ENGINEERING**

UG Engineering Programme	Semester
Electrical Engineering	Seventh

I RATIONALE

The course has been designed to introduce the Various Analog Communication Fundamentals, describe Amplitude Modulation and Demodulation, understand Angle Modulation and Demodulation, Noise and Performance of Various Receivers, Probability Theory & Random Variables & Error Correction Codes, Digital Modulation Techniques along with communication Systems in Presence of Noise, Digital Multiplexing, Line Coding, and PSD of different signals.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Evaluate the analog and digital communication parameters**.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) To get Knowledge of Various Amplitude Modulation and Demodulation Systems.
2) Describe Different Angle Modulation and Demodulation Systems.
3) Able to get Depth Analysis in Noise Performance of Various Receivers.
4) Design a digital communication system.
5) Analyze Channel characteristics.
6) To simulate on MATLAB software.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	
3	0	2	4	70	30*	25	25	150

***Legends: L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

***(*):** Under the **Theory PA**, out of 30 marks to facilitate integration of COs and the average of 2 tests,attendance and assignments to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Illustrate Double Sideband AM Generator.	01
2	1	calculate modulation index of DSB wave by trapezoidal pattern	01
3	1	Illustrate Double Sideband AM Reception.	01
4	1	Demonstrates Diode Detector.	01
5	2	Frequency Modulation using Varactor modulator.	01
6	2	Demonstrates Operation of Detuned Resonant Circuit.	01
7	2	Demonstrates Operation of Quadrature Detector	01
8	2	Demonstrates Operation of Phase-Locked Loop Detector	01
9	3	Illustrate selectivity curve for radio receiver	01
10	4	Describes Basic Block Diagram of Digital Communication System.	01
11	4	Illustrate Pulse Amplitude Modulation and observe its waveform	01
12	4	Illustrate Pulse Position Modulation and observe its waveform.	01
13	4	Describes concept of Pulse Code Modulation and to observe the performance of PCM system.	01
14	4	Illustrates concept of Pulse Width Modulation (PWM) using MATLAB, observe its waveforms and/or implement same using available trainer kit.	01
15	5,6	Demonstrates sampling theorem and reconstruction of signal using MATLAB, observe its waveforms and/or implement same using available trainer kit.	01
16	5,6	Reviews concept of Pulse Code Modulation (PCM) using MATLAB, observe its waveforms and/or implement same using available training kit.	01
17	5,6	Performs concept of Delta Modulation (DM) technique using MATLAB, observe its waveforms and/or implement same using available trainer kit.	01
18	5,6	Forms a digital carrier system signals like Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK) and Binary Phase Shift Keying (BPSK) using MATLAB, observe its waveforms and/or implement same using available training kit.	01
19	5,6	Analyze Huffman code using MATLAB.	02
		Total	**20**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Sintech trainer board for communication	1 to 14
2	MATLAB	15 to 19

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy conservation.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Amplitude Modulation	**CO 1.** To get Knowledge of Various Amplitude Modulation and Demodulation Systems. Pro 1. Illustrate Double Sideband AM Generator. Pro 2. calculate modulation index of DSB wave by trapezoidal pattern. Pro 3. Illustrate Double Sideband AM Reception. 1a. Define Amplitude Modulation 1b. Describe Amplitude Modulation Index 1c. Derive Modulation Index for Sinusoidal AM	1.1 Amplitude Modulation 1.2 Amplitude Modulation Index, Modulation Index for Sinusoidal AM 1.3 Frequency Spectrum for Sinusoidal AM 1.4 Average Power for Sinusoidal AM, Effective Voltage and Current for Sinusoidal AM 1.5 Non sinusoidal

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	1d. Derive Average Power for Sinusoidal AM, Effective Voltage and Current for Sinusoidal AM 1e. Explain Non sinusoidal Modulation, Double-Sideband Suppressed Carrier (DSBSC) Modulation	Modulation, Double-Sideband Suppressed Carrier (DSBSC) Modulation
Unit– II **Angle Modulation**	**CO 2.** Describe Different Angle Modulation and Demodulation Systems. PrO 5. Frequency Modulation using Varactor modulator. Pro 7. Demonstrates Operation of Quadrature Detector 2a. Define frequency modulation. 2b. Describe Frequency Spectrum for Sinusoidal FM. 2c. Derive Average Power in Sinusoidal FM. 2d. 2d. Define Deviation Ratio, Measurement of Modulation Index for Sinusoidal FM, Phase Modulation. 2e. 2e. Differentiate Equivalence Between PM and FM 2f. 2f. Describe Sinusoidal Phase Modulation, Digital Phase Modulation 2g. 2g. Describe FM Transmitter, Angle Modulation Detectors 2h. 2h. Explain Automatic Frequency Control 2i. 2i. Describe Amplitude Limiters 2j. 2j. Differentiate Pre-Emphasis and De-Emphasis 2k. 2k. Describe FM Broadcast Receivers.	a. Frequency Modulation b. Frequency Spectrum for Sinusoidal FM, Average Power in Sinusoidal FM c. Non-Sinusoidal Modulation: Deviation Ratio, Measurement of Modulation Index for Sinusoidal FM, Phase Modulation d. Equivalence Between PM and FM e. Sinusoidal Phase Modulation, Digital Phase Modulation f. Angle Modulator Circuits g. FM Transmitter, Angle Modulation Detectors h. Automatic Frequency Control i. Amplitude Limiters, Pre-Emphasis and De-Emphasis, FM Broadcast Receivers.
Unit– III **Receivers**	**CO 3.** Able to get Depth Analysis in Noise Performance of Various Receivers. Pro 6. Demonstrates Operation of Detuned Resonant Circuit. Pro 9. Illustrate selectivity curve for radio receiver. 3a. Select the different types of receivers. 3b. Define Tuning range, Tracking, sensitivity and gain 3c. Explain Image rejection, Spurious	3.1 Homodyne, Heterodyne and Super Heterodyne Receivers 3.2 Tuning range, Tracking, sensitivity and gain 3.3 Image Rejection, Spurious Responses, Adjacent Channel Selectivity 3.4 AGC, Double Conversion

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	responses, Adjacent channel selectivity 3d. Describe different types of receivers.	3.5 Electronically Tuned Receivers (ETRS), Integrated-Circuit Receivers
Unit-IV Probability Theory & Random Processes	CO-5) Analyze Channel characteristics. CO-6) To simulate on MATLAB software. PrO10: Describes basic block diagram of communication system 4a. Calculates example of probability. 4b. Calculates example of random process. 4c. Analyzes communication system response in context with CDF, PDF and Gaussian PDF. 4d. Solved problems of binary schematic channel for BER issue.	4.1 Introduction to digital communication, Probability, Conditional Probability Of Independent Events 4.2 Relation between probability and probability density 4.3 CDF, PDF, Gaussian PDF 4.4 Threshold Detection, Random Variable, Variance of a random variable 4.5 Statistical Average, Chebyshev In Equality 4.6 Raleigh Probability Density, The Central Limit Theorem 4.7 correlation, random processes, PSD of random processes(along with examples) 4.8 Multiple random processes, band pass random processes
Unit – V Information Theory	*CO-4) Design a digital communication system.* *PrO19: Analyze Huffman code using MATLAB.* PrO 18. Forms a digital carrier system signals like Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK) and Binary Phase Shift Keying (BPSK) using MATLAB, observe its waveforms and/or implement same using available training kit. 5a. Computes capacity of channel. 5b. Evaluates Huffman code for systems.	5.1 Concept & Measure Of Information 5.2 entropy, Error Free Communication Over A Noisy Channel 5.3 The Channel Capacity Of A Discrete Memory Less Channel 5.4 Channel Capacity Of A Continuous Channel 5.5 Practical Communication Systems In Light Of Shannon's Equation

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – VI Principle of digital data Transmission	CO-6) To simulate on MATLAB software. PrO10: Describes Basic Block Diagram of Digital Communication System. **Pro 14:** Illustrates concept of Pulse Width Modulation (PWM) using MATLAB, observe its waveforms and/or implement same using available trainer kit. **Pro 15:** Demonstrates sampling theorem and reconstruction of signal using MATLAB, observe its waveforms and/or implement same using available trainer kit. **Pro 16.** Reviews concept of Pulse Code Modulation (PCM) using MATLAB, observe its waveforms and/or implement same using available training kit. **Pro 17.** Performs concept of Delta Modulation (DM) technique using MATLAB, observe its waveforms and/or implement same using available trainer kit. 6a. Evaluates various line codes efficiency. 6b. Design simplified systems using latest and effective components.	a. Line Coding, PSD of On/Off Signal, Bipolar Signal b. Duo Binary Signal, Pulse Shaping, Nyquist First and Second Criterion for Zero ISI c. Regenerative Repeaters, Detection Error Probability d. M-ary System, Scrambling e. Digital Carrier System, Digital Multiplexing

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Amplitude Modulation	10	0	2	4	5	4	0	15
II	Angle Modulation	10	0	2	4	5	4	0	15
III	Receivers	05	0	2	4	2	1	0	09
IV	Probability Theory & Random Processes	10	0	1	4	4	6	0	15
V	Information Theory	05	0	2	2	2	2	0	08
VI	Principle of digital data Transmission	05	0	0	2	3	3	0	08
Total		**45**	**0**	**09**	**20**	**21**	**20**	**0**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical

evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different communication technology.
b) Prepare 15 min. power point presentation on how to choose line code for the particular communication system.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Amplitude Modulation	1	Pro 1. Illustrate Double Sideband AM Generator. Pro 2. calculate modulation index of DSB wave by trapezoidal pattern. 1a. Define Amplitude Modulation 1b. Describe Amplitude Modulation Index 1c. Derive Modulation Index for Sinusoidal AM	Lecture With media	PPTs, black board
	2	Pro 3. Illustrate Double Sideband AM Reception. 1d. Derive Average Power for Sinusoidal AM, Effective Voltage and Current for Sinusoidal AM 1e. Explain Non sinusoidal Modulation, Double-Sideband Suppressed Carrier (DSBSC) Modulation	Lecture With media	PPTs, black board
Unit– II Angle Modulation	**3**	PrO 4. Frequency Modulation using Varactor modulator. 2a. Define frequency modulation. 2b. Describe Frequency Spectrum for Sinusoidal FM. 2c. Derive Average Power in Sinusoidal FM.	Lecture With media	PPTs, black board
	4	Pro 6. Demonstrates Operation of Detuned Resonant Circuit. 2d. Define Deviation Ratio, Measurement of Modulation Index for Sinusoidal FM, Phase Modulation. 2e. Differentiate Equivalence Between PM and FM 2f. Describe Sinusoidal Phase Modulation, Digital Phase Modulation	Lecture With media	PPTs, black board
	5	Pro 9. Illustrate selectivity curve for radio receiver. 2g. Describe FM Transmitter, Angle Modulation Detectors 2h. Explain Automatic Frequency Control 2i. Describe Amplitude Limiters	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
	6	Pro 7. Demonstrates the Operation of Quadrature Detector 2j. Differentiate Pre-Emphasis and De-Emphasis 2k. Describe FM Broadcast Receivers.	Lecture With media	PPTs, black board
Unit– III Receivers	7	**PrO10. Describes basic block diagram of communication system** 3a. Select the different types of receivers. 3b. Define Tuning range, Tracking, sensitivity and gain	Lecture With media	PPTs, black board
	8	**PrO10. Describes basic block diagram of communication system** 3c. Explain Image rejection, Spurious responses, Adjacent channel selectivity 3d. Describe different types of receivers.	Lecture With media	PPTs, black board
Unit-IV Probability Theory & Random Processes	9	**Pro 14:** Illustrates concept of Pulse Width Modulation (PWM) using MATLAB, observe its waveforms and/or implement same using available trainer kit. 4a. Calculates example of probability. 4b. Calculates example of random process.	Lecture With media	PPTs, black board
	10	**Pro 15:** Demonstrates sampling theorem and reconstruction of signal using MATLAB, observe its waveforms and/or implement same using available trainer kit. 4c. Analyzes communication system response in context with CDF, PDF and Gaussian PDF. 4d. Solved problems of binary schematic channel for BER issue.	Lecture With media	PPTs, black board
Unit – V Information Theory	11	PrO 18. Forms a digital carrier system signals like Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK) and Binary Phase Shift Keying (BPSK) using MATLAB, observe its waveforms and/or implement same using available training kit. 5a. Computes capacity of channel.	Lecture with media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
	12	PrO 18. Forms a digital carrier system signals like Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK) and Binary Phase Shift Keying (BPSK) using MATLAB, observe its waveforms and/or implement same using available training kit. 5a. Computes capacity of channel. 5b. Evaluates Huffman code for systems.	Lecture With media	PPTs, black board
Unit – VI Principle of digital data Transmission	13	**Pro 16.** Reviews concept of Pulse Code Modulation (PCM) using MATLAB, observe its waveforms and/or implement same using available training kit. 6a. Evaluates various line codes efficiency.	Lecture With media	PPTs, black board
	14	**Pro 17.** Performs concept of Delta Modulation (DM) technique using MATLAB, observe its waveforms and/or implement same using available trainer kit. 6b. Design simplified systems using latest and effective components.	Lecture With media	PPTs, black board
	15	*PrO19: Analyze Huffman code using MATLAB.* 5b. Evaluates Huffman code for systems.	Lecture With media	PPTs, black board

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Electronic Communications	Dennis Roddy & John Coolen	Prentice Hall of India Pvt. Ltd, New Delhi ISBN: 9780133120837
2	Modern Digital and Analog Communication System	Rabindranath, B. and Chander, N.	New Age International (P) Ltd., New Delhi, First Edition 2011, ISBN: 978-0-19-538493-2
3	Digital and Analog communication system	B.P.Lathi	LCBS Publication ISBN:9789353063382
4	Communication system	B. Carlson	TMH Publication ISBN: 9780071321174
5	Digital and analog communication system	Simon Haykin	Willey Publication ISBN: 978-8126536535

S. No.	Title of Book	Author	Publication including ISBN
6	Modern Digital and Analog Communication System	B.P.Lathi	Oxford Press 978-0-19-538493-2
7	Principle of communication system	Taub & Schilling,	TMH Publication ISBN: 978-1259029851
8	Embedded Systems Architecture	Tammy Noergaard	Elsevier ISBN: ***9780123821966***

XV SOFTWARE/LEARNING WEBSITES

1. http://adp.mmu.edu.my/enotes/asma/dtc5038/notes/lecture%20notes%203%2020chapter%203.pdf
2. http://web.cecs.pdx.edu/~ece2xx/ece223/slides/communicationsx4.pdf
3. http://stattrek.com/Lesson2/Binomial.aspx?Tutorial=sat http://www.youtube.com/watch?v=IYdiKeQ9xEI
4. http://www.complextoreal.com/tutorial.htm

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs						POs and PSOs									
		PO 1 Used of mathematical method	**PO 2** Analysis mechanical system	**PO 3** Used engineering soft.	**PO 4** Solve problem in machine	**PO 5** The engineer and society	**PO 6** Environment and sustainability	**PO 7** Ethics	**PO 8** Individual and team work:	**PO 9** Communication	**PO 10** Life-long learning			**PSO1** Evaluate the performance of Electrical Machines and Power systems Power Systems	**PSO2** Evaluate the performance of Electrical and Electronic Instrumentaton Systems	
EE472.01	**Semester VI**						**COMMUNICATION ENGINEERING** **Mark ‘H’=3** for high, ‘**M**’=2 for medium or **‘L’=1’** for the relevant correlation of each PO or PSO, Competency or CO									
472.01	**Competency** *Evaluate the performance communication system*	*3*	*1*	*3*	*3*	*1*	*0*	*0*	-	-	*1*	-	-	*1*	-	-
472.01.1	To get Knowledge of Various Amplitude Modulation and Demodulation Systems.	3	1	2	2	1	-	-	-	-	-	-	-	1	3	-
472.01.2	Describe Different Angle Modulation and Demodulation Systems.	3	2	2	1	1	-	-	-	-	-	-	-	2	2	-
472.01.3	Able to get Depth Analysis in Noise Performance of Various Receivers.	3	2	1	2	1	-	-	-	-	-	-	-	2	2	-
472.01.4	Design a digital communication system.	3	1	2	2	1	1	1	-	-	-	-	-	1	2	-

472.01.5	Analyze Channel characteristics	3	2	1	1	1	-	-	-	-	1	-	-	1	1	-
472.01.6	To simulate on MATLAB software	3	2	3	2	-	-	-	-	-	-	-	-	2	2	-

XVII NAMES OF RESOURCE PERSONS

Mr. Ankur Patel

Course Code: **EE476.02**
Course Title: **Advances in Power System**

UG Engineering Programme	Semester
Electrical Engineering	Seven

I RATIONALE

Electrical engineers have to deal with the power system operations and controls. Power systems are living in an era of major changes, pushed forward by the emergence of new technical issues and the availability of innovative technologies, enabling new functions for monitoring and control of transmission network operation and stability. This curriculum course structure is developed such that the student is able to recognize the Restructuring of Power System, State Estimation in Power System, Substation Automation, Smart Grid Technology, virtual power plant, Load Forecasting for the interconnected power system.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Advances in power system through the understanding of restructuring of power system, smart grid technology and virtual power plant.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) ***Describe*** *past, present and future of Power System with Restructuring of an Indian Power System, and* ***analysis*** *of power system state estimation with methodologies used for state estimation.*
2) ***Illustrate*** *the fundamentals and detail knowledge regarding smart grid technologies with operations and control.*
3) ***Describe*** *the fundamentals and detail knowledge regarding virtual power plant with operations and control.*
4) ***Apply and validate*** *various algorithm to predict future energy demand for smart grid and learn data analysis.*
5) ***Express*** *the* ***formulation*** *for substation automation and distribution, system monitoring, control and design.*

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)	Total Credits (L+T+P)	Examination Scheme (Anna 2021)		
		Theory Marks	Practical Marks	Total Marks

L	T	P	C	ESE	PA	ESE	PA	150
3	0	2	4	70	30*	25	25	

***Legends: L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

***(*):** Under the **Theory PA**, out of 30 marks, 15 **marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 15 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Illustrate National Power Portal (NPP)	02
2	1	Describe Restricting of Power System	02
3	1	Describe and Evaluate power system state estimation	02
4	2	Describe Smart Grid Technology	02
5	3	Describe Virtual Power Plant	02
6	4	Describe Load Forecasting	02
7	5	Describe Distribution Automation	02
8	2	Simulate Battery energy management system for micro grid	02
9	2	Simulate load flow analysis for hybrid AC/DC micro grid	02
10	4	Energy Demand Forecasting for EAI data in Python: • Using Stats Model like AR, ARMA, ARIMA and SARIMA • Using ML model like RNN and LSTM	02
11	1	Simulate Power System State Estimation in Torrit using WLSE method the given power system network.	02
12	2	Design and simulate Hybrid AC/DC Micro-grid for Residential Application Using Homer PRO.	02
		Total	**24**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB Simulation software (version 2021a)	8,9
2	Python	10
3	Torrit	11
4	Homer PRO	12

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) Follow safe practices.
2) Practice energy balances.
3) Function as a team member.
4) Function as a team leader.
5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I Restructuring of Power System	**CO 1.** ***Describe past, present and future of Power System with Restructuring of an Indian Power System, and analysis of power system state estimation with methodologies used for state estimation.*** *PrO 1. Illustrate National Power Portal (NPP).* *PrO 2. Describe Restricting of Power System.* 1a. Describe history of electric power and Traditional power industry. 1b. Justify Restructuring power system network and discuss Electricity market entities and model. 1c. Illustrate Regional Load Dispatch Centre and discuss the Benefits of Deregulation. 1d. Explain Congestion management and Available Transfer capabilities for Power market.	1.1 Brief history of electric power, Indian History of Power System, Traditional power industry 1.2 Motivation for Restructuring, Players in Restructuring Process, Electricity market entities and model 1.3 Regional Load Dispatch Centre, Benefits of Deregulation, Basic terminologies in deregulation 1.4 Congestion, Available Transfer capabilities 1.5 Ancillary Services, Growth of power sector in India

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	1e. Explain Ancillary Services, and recognise the Growth of power sector in India.	
Unit– II State Estimation in Power System	**CO 1. *Describe past, present and future of Power System with Restructuring of an Indian Power System, and analysis of power system state estimation with methodologies used for state estimation.*** *PrO 3. Describe and Evaluate power system state estimation.* *PrO 11. Simulate Power System State Estimation in Torrit using WLSE method the given power system network.* 2l. Describe Power system state estimation. 2m. Determine Weighted least square estimation and evaluate State estimation of AC network. 2n. Detect and identify bad data in measurement set. 2o. Network observability and demonstrate state estimation.	2.1 Power system state estimation: Past, Present and Future 2.2 Weighted least square estimation, Hessian matrix formulation, State estimation of AC network. 2.3 Detection and identification of bad data measurement, estimation of quantities not being measured, 2.4 Network observability and Pseudo measurement, Application of state estimation.
Unit– III Introduction to Smart Grid Technology	**CO 2. *Illustrate the fundamentals and detail knowledge regarding smart grid technologies with operations and control*.** *PrO 4. Describe Smart Grid Technology.* *PrO 12. Design and simulate Hybrid AC/DC Micro-grid for Residential Application Using Homer PRO.* 3a. Discriminate the conventional and smart grid technology, and discuss the promoters of smart grid. 3b. Identify the functions, components, communication methods, and technology used in smart grid. 3c. Discuss benefits, disadvantages, application, and challenges for smart grid. 3d. Explain Microgrid architecture, components, and discuss its benefits. 3e. Illustrate application, challenges, importance for Microgrid, also, outline Smart Grid Projects in India.	3.1 Structure of conventional grid, Conventional grid to Smart grid, Comparison, Smart Grid Enablers/Promoters, what is smart grid? 3.2 Functions of Smart Grid, Components of Smart Grid, Communication Methods in Smart Grids, Technologies used in Smart Grid 3.3 Benefits of Smart Grid, Disadvantages of Smart Grid, Applications of Smart Grid, Base elements and challenges 3.4 Microgrid: A Review, Microgrid Architecture, Components of Microgrid, Benefits of Microgrids 3.5 Application of Microgrid, Challenges in Microgrid,

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
		R&Ds in Microgrid, Microgrid Importance, Smart Grid Projects in India
Unit-IV Introduction to Virtual Power Plant	**CO 3. *<u>Describe</u> the fundamentals and detail knowledge regarding virtual power plant with operations and control.*** *PrO 5. <u>Describe</u> Virtual Power Plant.* 4a. <u>Identifying current scenarios of renewable energy resources and discuss its challenges in Indian power system.</u> 4b. <u>Describe</u> virtual power plant with its control architectures. 4c. <u>Discuss</u> VPP benefit to the various stakeholder of power system and <u>outline</u> the business opportunity in the worldwide as well as in India.	4.1 Introducing the current scenarios of renewable energy resources and its challenges in Indian power system 4.2 Virtual Power Plant, Classification of Virtual Power Plant (VPP), Control Architectures of VPP, VPP differences due to grid location 4.3 VPP benefit to the various stakeholder of power system, the business opportunity in the worldwide as well as in India.
Unit – V Load Forecasting	**CO 4. *<u>Apply</u> and <u>validate</u> various algorithm to predict future energy demand for smart grid and learn data analysis.*** *PrO 6. <u>Describe</u> Load Forecasting.* *PrO 10. Energy Demand Forecasting for EAI data in Python:* *a) Using Stats Model like AR, ARMA, ARIMA and SARIMA* *b) Using ML model like RNN and LSTM* 5a. <u>Explain Load Forecasting and its different methods.</u> 5b. <u>Discriminate</u> data, information, knowledge, and wisdom. Also, <u>explain</u> time series for data <u>analysis</u>. 5c. <u>Describe</u> and <u>formulate</u> Statistical Models (AR, ARMA, ARIMA, SARIMA) Used for Load Forecasting. 5d. <u>Describe</u> and <u>formulate</u> Machine Learning Methods (RNN, LSTM) Used for Load Forecasting. 5e. <u>Validation</u> of load forecasting and <u>compare</u> different cases studied for load forecasting.	5.1 Introduction to Load Forecasting, Method Used for Load Forecasting. 5.2 what is Data, Information and Knowledge? Time Series 5.3 Statistical Models Used for Load Forecasting, AR, ARMA, ARIMA, SARIMA 5.4 Machine Learning Methods Used for Load Forecasting, RNN, LSTM 5.5 Error in forecasting, Different Cases Studied for Load Forecasting

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – VI Distribution Automation	***CO 5. Express the formulation for substation automation and distribution, system monitoring, control and design.*** *PrO 7. Describe Substation Automation.* 6a. Explain distribution automation(DA) and its Functionality. 6b. Discuss typical integrated distribution utility and key design parameters of SCADA/DMS system. 6c. Explain levels, application and benefits of DAS. Also, illustrate Advanced Distribution Management System 6d. Describe Fiber Optic Technology, power line carrier communication, and explain typical components of PLCC.	6.1 Introduction to distribution automation(DA), Automation Systems and Functionality, 6.2 Typical Integrated Distribution Utility, Key Design Parameters of SCADA/DMS system, 6.3 Levels of Automation, Applications and benefits of DAS. Advanced Distribution Management System (ADMS), 6.4 Fiber Optic Technology(FOT) in dedicated communication system, power line carrier communication(PLCC) Explain typical components of PLCC.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Restructuring of Power System	08	2	3	3	3	0	1	12
II	State Estimation in Power System	06	2	2	2	2	2	0	10
III	Introduction to Smart Grid Technology	12	2	6	4	4	1	0	17
IV	Introduction to Virtual Power Plant	06	2	4	2	2	0	0	10
V	Load Forecasting	08	2	3	2	2	2	1	12
VI	Distribution Automation	05	2	2	2	3	0	0	09
Total		**45**	**12**	**20**	**15**	**16**	**05**	**02**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **Prepare case study report on Indian power system:** Collect the information and prepare the case study report on Indian power system from the national power portal and ministry of power website.
2) **Design Hybrid AC/DC Micro-grid for Residential Application:** Collect the real network data or the IEEE standard test system and design the micro-grid for residential application using Homer PRO simulation platform.
3) **Energy Demand Forecasting:** Collect the past energy consumption data and using statistical/ machine learning algorithm predict next day/week/month energy demand. Any data analytics simulation tool/platform is use.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) With respect to ***section No.12***, teachers need to ensure by creating opportunities and provisions for ***co-curricular activities.***

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding restructure and deregulation of power system.
b) Prepare 15 min. power point presentation for smart grid technology.
c) Prepare 15 min. power point presentation for Micro-grid.
d) Prepare 15 min. power point presentation for virtual power plant

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Restructuring of Power System	**1**	*PrO 2. Illustrate National Power Portal (NPP).* 1d. Describe history of electric power and Traditional power industry. 1e. Justify Restructuring power system network and discuss Electricity market entities and model. 1f. Illustrate Regional Load Dispatch Centre and discuss the Benefits of Deregulation.	Lecture with media, Tutorial	PPTs
	2	*PrO 3. Describe Restricting of Power System.* 1g. Explain Congestion management and Available Transfer capabilities for Power market. 1h. Explain Ancillary Services, and recognise the Growth of power sector in India.	Lecture With media, Tutorial	PPTs
Unit– II State Estimation in Power System	**3**	*PrO 3. Describe and Evaluate power system state estimation.* 2f. Describe Power system state estimation. 2g. Determine Weighted least square estimation and evaluate State estimation of AC network.	Lecture with media, and Tutorial	PPTs, black board
	4	*PrO 11. Simulate Power System State Estimation in Torrit using WLSE method the given power system network.* 2h. Detect and identify bad data in measurement set. 2i. Network observability and demonstrate state estimation.	Lecture with media, Tutorial	PPTs
Unit– III Introduction to Smart Grid Technology	**5**	*PrO 4. Describe Smart Grid Technology.* 3f. Discriminate the conventional and smart grid technology, and discuss the promoters of smart grid.	Lecture with media	PPTs, Video
	6	3g. Identify the functions, components, communication methods, and technology used in smart grid. 3h. Discuss benefits, disadvantages, application, and challenges for smart grid.	Lecture with media	PPTs
	7	*PrO 12. Design and simulate Hybrid AC/DC*	Lecture	PPTs,

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		Micro-grid for Residential Application Using Homer PRO. 3i. Explain Microgrid architecture, components, and discuss its benefits.	with media, and Tutorial	Video
	8	3j. Illustrate application, challenges, importance for Microgrid, also, outline Smart Grid Projects in India.	Lecture with media, Tutorial	PPTs
Unit – IV Introduction to Virtual Power Plant	9	*PrO 5. Describe Virtual Power Plant.* 4a. Identifying current scenarios of renewable energy resources and discuss its challenges in Indian power system.	Lecture with media, Tutorial	PPTs, Video
	10	4b. Describe virtual power plant with its control architectures. 4c. Discuss VPP benefit to the various stakeholder of power system and outline the business opportunity in the worldwide as well as in India.	Lecture with media, Tutorial	PPTs
Unit – V Load Forecasting	11	*PrO 6. Describe Load Forecasting.* 5a. Explain Load Forecasting and its different methods. 5b. Discriminate data, information, knowledge, and wisdom. Also, explain time series for data analysis.	Lecture with media, Tutorial	PPTs, black board
	12	5c. Describe and formulate Statistical Models (AR, ARMA, ARIMA, SARIMA) Used for Load Forecasting. 5d. Describe and formulate Machine Learning Methods (RNN, LSTM) Used for Load Forecasting.	Lecture with media, Tutorial	PPT, Pictures , black board
	13	*PrO 10. Energy Demand Forecasting for EAI data in Python:* *c) Using Stats Model like AR, ARMA, ARIMA and SARIMA* *d) Using ML model like RNN and LSTM* 5e. Validation of load forecasting and compare different cases studied for load forecasting.	Lecture with media, Tutorial	PPTs, black board
Unit – VII Distribution Automation	14	*PrO 7. Describe Substation Automation.* 6a. Explain distribution automation(DA) and its Functionality. 6b. Discuss typical integrated distribution utility and key design parameters of SCADA/DMS system.	Lecture with media, Tutorial	PPTs, Video
	15	6c. Explain levels, application and benefits of DAS. Also, illustrate Advanced Distribution Management System. 6d. Describe Fiber Optic Technology, power line carrier communication, and explain typical	Lecture with media, Tutorial	PPTs, Video

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		components of PLCC.		

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Power System Analysis	Hadi Saadat	Tata McGraw Hill
2	Restructured Power System	S.A. Khaparde and A.R. Abhyanka	Narosa
3	Electric Power Distribution Systems	A.S. Pabla	6th Edition, Tata McGraw Hill
4	Smart Grids and Microgrids: Technology Evolution	Prajof Prabhakaran, S. Mohan Krishna, J. L. Febin Daya, Umashankar Subramaniam, P. V. Brijesh	ISBN: 978-1-119-76059-7 Willy
5	Smart Grid Fundamentals	Radian Belu	eBook ISBN 9780429174803 Taylor & Francis Group
6	Scheduling and Operation of Virtual Power Plants	Ali Zangeneh, Moein Moeini-Aghtaie	eBook ISBN: 9780323852685 Paperback ISBN: 9780323852678 ELSEVIER

XV SOFTWARE/LEARNING WEBSITES

1. https://www.ci2lab.com/
2. https://sourceforge.net/directory/windows/?q=smart+grid
3. https://onlinecourses.swayam2.ac.in/arp19_ap60/announcements?force=true
4. https://www.statsmodels.org/stable/tsa.html
5. https://enode.com/blog/virtual-power-plant-software
6. https://www.global.toshiba/ww/products-solutions/renewable-energy/products-technical-services/vpp.html
7. https://transparency.entsoe.eu/dashboard/show?loggedUserIsPrivileged=false

XVI PO-COMPETENCY-CO MAPPING

		POs and PSOs

Course Code	Competency and COs	**PO 1** Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and teamwork	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE476.02	**Semester VII**	**Power System Operations and Control** Mark **'H'=3** for high, '**M**'=2 for medium or **'L'=1'** for the relevant correlation of each PO or PSO, Competency or CO													
476.02	**Competency** Advances in power system through the understanding of restructuring of power system, smart grid technology and virtual power plant.	***3***	***3***	***3***	***2***	***3***	***1***	***1***	-	***2***	***1***	***2***	-	***1***	***1***
476.02.1	Describe past, present and future of Power System with Restructuring of an Indian Power System, and analysis of power system state estimation with methodologies used for state estimation.	3	3	1	1	1	-	-	-	-	-	1	-	2	1
476.02.2	Illustrate the fundamentals and detail knowledge regarding smart grid technologies with operations and control.	3	2	3	1	2	-	1	-	-	-	-	-	3	-
476.02.3	Describe the fundamentals and detail knowledge regarding virtual power plant with operations and control.	3	2	2	1	2	-	-	-	-	-	-	-	2	1
476.02.4	Apply and validate various algorithm to predict future energy demand for smart grid and learn data analysis.	3	3	2	2	3	-	-	-	-	-	-	-	2	2
476.02.5	Express the formulation for substation automation and	3	2	3	1	2	-	-	-	-	-	-	-	2	1

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and teamwork	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	distribution, system monitoring, control and design.														

XVII NAMES OF RESOURCE PERSONS

a) Mihir R. Patel

b) ……………………………………………………

Course Code: **EE481**
Course Title: **Industry 4.0**

UG Engineering Programme	Semester
Electrical Engineering	Seventh

I RATIONALE

The industries are now shifting to have more power of automation and industrial revolutions is making the industrial process efficient and easy to control without special skilled man power. The old day's industries are slowly adopting the deployment of new industry revolutions and hence engineers are required for suggesting technologies for these revolutions, deploying it, train the operators and manpower for this new edge technology and efficiently operate the industrial process through it. The syllabus focuses on backbone technologies in industry 4.0, such as dynamic simulation, digital twin and cyber physical system, augmentation of AI and ML algorithm and industrial robot programming. With these all, the student will be able to help in its procurement, installation as well as to use these technologies in an industry and train the manpower to use it as well.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- Lead the industry 4.0 deployment and train the man power to effectively use it.
- Use(/work with) the smart manufacturing technologies, 3D printers, smart sensors, IIoT solutions, Industrial robot etc. by networking them and programming them.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) Find opportunity of applying industry 4.0 Solutions for various industrial process, advise deployment on basis of the cost involved in deployment and possible return on investment.
2) Design the 3D model of an object to be 3D printed.
3) Write Python and C Code for smart sensor, Industrial and IIoT communications.
4) Develop dynamic Simulation for a physical system and test it. Develop Digital twin and Cyber Physical System(CPS).
5) Write code for preventive/predictive maintenance.
6) Program Industrial robot for required action and motion.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme (CHARUSAT 2021)				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
3	0	2	4	70	30*	25	25	

***Legends: L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

***(*):** Under the **Theory PA**, out of 30 marks, **20 marks** are for **attendance**, **micro-project assessment** and other **assignment** if given any, to facilitate integration of COs and the remaining 10 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	2,4	*Make 3 D model in various freely available software supporting 3 D printing.*	02
2	4	*Simulate Mechanical and Electrical Systems in industries using software like ANSYS and MATLAB.*	03
3	3	*Develop IoT solution for Industries.* *a) Integrate sensor and Write code to collect the data using M2M communication.* *b) Write code to interconnect and interact among smart devices/PLC/Robots with IOT systems.*	04
4	4,5	*Write code to apply AI ML algorithm and verify with various data set.*	Planned to be covered during class room
5	4,5	*Write code to use statistics for feature extraction. Apply to various industrial process.*	Planned to be covered during class room
6	4,5	*Apply Fourier, Wavelet and similar transform to Mechanical Vibration data set, generate features and Write code for Predictive maintenance of machines, using AI ML applications.*	Planned to be covered during class room
7	6	*Design Mechanical System for various types of industrial Robots and make 3 D model, simulate the response of robot.*	06
8	6	*Program Industrial Robot for various types of kinematics and movements.*	06
9	4	Develop *Digital Twin(DT) and Cyber* Physical Systems (CPS) for various	06

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
		industrial task and process. a) Develop Digital Twin for industrial processes with ANSYS, MATLAB and Amazon AWS/PTC Thingswork/or suitable cloud offering. b) Develop CPS and use Library support available in Amazon AWS for implementing CPS.	
10	4	Write MATLAB code for parameter tuning of digital twin using Least Square Estimation (LSQ), Weighted Least Square Estimation (WLS) and Maximum Like-hood(ML) methods.	03
		Total	**30**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

Sr. No.	Equipment Name with Broad Specifications	PrO No.
1	IIoT Gateways, Vibration sensor, ESP series kits	3,9
2	PLC	3
3	*Industrial Robot*	6
4	*3 D Printer*	1,7

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) "One cannot always expect to goal in first attempt". Physical processes are difficult to model. Test – verify and remodify the simulation for best responses.
2) "Think out of box but have empathy and follow ethics". The deployment of new industry revolution needs thinking out of box. The success got because of technology must not cut the job. It is moral duty of the industry to providing proper training to already available man power and sustain them by assigning suitable task. The prospective engineers, needs to be trained for these new technologies and make them employable. The student enrolled in this course will try to look to their internship at various industries with this approach and is expected to try their best.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year
c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit 1: Background and Framework for Industry 4.0	*Co1. Find opportunity of applying industry 4.0 Solutions for various industrial process, Advise deployment on basis of the cost involved in deployment and possible return on investment.* *Co 2: Design the 3D model of an object to be 3D printed.* *Co4. Develop dynamic Simulation for a physical system and test it. Develop Digital twin and Cyber Physical System(CPS).* *Pro1. Make 3 D model in various freely available software. (Simple model as well as several products like Ultra capacitor, BLDC motor, Fuel Cell etc)* *Pro 2: Simulate Mechanical and Electrical Systems in industries, using software like ANSYS and MATLAB.* *1a. Understand the origin and key benefits of deploying Industry 4.0* *1b. Understand design principles and Pillars for Industry 4.0.* *1c. Transform a small and medium enterprises (SMEs) into Industry 4.0. Do cost benefit and ROI analysis for deployment of industry 4.0.* *1d. Develop mathematical model/ simulation/Digital twin for the given physical process or object.* *1e. Select appropriate additive manufacturing process/3D printer for given application.* *1f. Understand Environmental Manage-ment and Technology potential for Environmental Management 4.0. and hence, Apply design principles looking to it.* *1g. Understand Technical and Human Resource view of Industry 4.0 and incorporate it in deployment and management strategies.*	*1. 1 Basics (Covering topics like, Industry 4.0: origin and key understandings, Industry 4.0 design principles, Industry 4.0 Pillars and Role) and related other topics helping in, technology selection, cost analysis and ROI for deployment of Industry 4.0.* *1. 2 Additive Manufacturing techniques, Various types of 3D printers and Example applications.* *1. 3 Example implementation of Cyber Physical Systems (CPS) and digital Twin for Industries.*

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
Unit 2: **Industrial IoT and Industrial Communication**	*Co3. Write Python and C Code for smart sensor, Industrial and IIoT communications.* *Pro 3: Develop IoT solution for Industries.* *a) Integrate sensor and Write code to collect the data using M2M communication.* *b) Write code to interconnect and interact among smart devices/PLC/Robots with IOT systems.* *2a. Understand Various Industrial Sensor and smart sensors.* *2b. Understand various industrial communication protocols.* *2c. Understand difference between IoT and M2M communication. Understand basics of M2M communication.* *2d. Understand structural of IIoT and key features. Accordingly, select appropriate cloud and software offering for IIoT.*	*2. 1 Comparison for various Industrial IoT solution offerings* *2. 2 Industrial Sensors* *2. 3 Various Industrial Communications.* *2. 4 M2M communication.* *2. 5 Interconnect Industrial Sensor, PLC, Robot etc with IIoT systems.*
Unit 3: AI-ML application in Industry 4.0	*Co4. Develop dynamic Simulation for a physical system and test it. Develop Digital twin and Cyber Physical System(CPS).* *Co5. Write code for preventive/predictive maintenance.* *Pro 4: Write code to apply AI ML algorithm and verify with various data set.* *Pro 5: Write code to use statistics for feature extraction. Apply to various industrial process.* *Pro 6: Apply Fourier, Wavelet and similar transform to Mechanical Vibration data set, generate features and Write code for Predictive maintenance of machines, using AI ML applications.* *3a. Understand role and various application of of AI and ML in Industry 4.0.* *3b. Understand various statistical feature extraction criteria and code it.* *3c. Write code to Apply Fourier and Wavelet on given data set.* *3d. Understand LSTM, Random forests, Decision Trees, and logistic regression and Write code for it using ready libraries.* *3e. Understand classification algorithms like SVM, and code it. Write code using ready*	*3.1 Application of AI ML in to industrial process.* *3.2 Various AI ML algorithm (LSTM, Random forests, Decision Trees, and logistic regression, classification algorithms like SVM etc) and Their coding* *3.3 Generating features from real time data for predictive maintenance, Application of various Algorithm for Predictive Maintenance.*

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	libraries, for Predicting machines health using classification algorithm.	
Unit 4: Industrial Robotics and Programming	*Co6. Program Industrial robot for required action and motion.* *Pro 7: Design Mechanical System for various types of industrial Robot and make 3 D model, simulate the response of robot.* *Pro 8: Program Industrial Robot for various types of kinematics and movements.* *Pro 9: Develop Digital Twin(DT) and Cyber Physical Systems (CPS) for various industrial task and process.* *a) Develop Digital Twin for industrial processes with ANSYS, MATLAB and Amazon AWS/PTC Thingswork/or suitable cloud offering.* *b) Develop CPS and use Library support available in Amazon AWS for implementing CPS.* *Pro 10. Write MATLAB code for parameter tuning of digital twin using Least Square Estimation (LSQ), Weighted Least Square Estimation (WLS) and Maximum Like hood(ML) methods.* *4a. Understand various types of Industrial Robot and their expected role and Functions in various industrial processes.* *4b. Understand various parts of robot: Robot Manipulator and Robot Controller, Sensors and actuators in Robot, Other mechanical arrangement.* *4c. Understand various Specification of Industrial Robot. Learn key features of Various Robot (Model) from leading Industrial Robot Manufacturer.* *4d. Understand Mathematical back-ground behind various types of motion and develop the simulation for evaluating the response.* *4e. Program Industrial robot for various types of motion using Rapid, ROS and PLC ladder*	*4.1 Types of Industrial Robot and their expected role and Functions,* *4.2 Various Robot (Model) from leading Industrial Robot Manufacturer* *4.3 Robot Manipulator and Robot Controller, Sensors and actuators in Robot, Other mechanical arrangement,* *4.4 Specification of Indus trial Robot and Procure ment (Payload, Reach, Max. TCP Velocity, Max TCP Acceleration, Accel eration Time, Repeat ability, Services available and Programmable),* *4.5 Mathematical back-ground and Progra mming using Rapid, ROS and PLC ladder logic.*

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	logic.	

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Background and Framework for Industry 4.0	10	--	15	--	--	--	--	15
II	Industrial IoT and Industrial Communication	10	4	4	7	--	--	--	15
III	AI-ML application in Industry 4.0	15	10	8	7	--	--	--	25
IV	Industrial Robotics and Programming	10	4	4	4	3	--	--	15
Total		**45**	**18**	**31**	**18**	**3**	**00**	**00**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. It should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

Just to give idea, A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here:

7) **Augmented Reality for Training the Industrial work force:** the AR can help to teach the newly recruited and old days (but not much experience to work with digital equipment) man power and can make them ready to efficiently use the industry 4.0 for industrial system.

8) **3D Printing of complex products:** Analyze the internal of several real products like Ultra capacitor, BLDC motor, Fuel Cell, Semiconductor switch etc. Develop the 3D model after understanding the internal and manufacture it with help of industries having specialized 3D printers for it.
9) **Modeling and verification of industrial processes**: The MATLAB and ANSYS can help a lot to develop the simulation for a physical system. It is not much covering the ready single blocks to simulate the industrial system presently (2023). The libraries can be built and verification can be done by comparing the simulated results with real system data logging.
10) **Optimization routines and its deployment in industry 4.0**: Various optimization techniques are researched till date and they can be used to optimize the industrial process. Such routines can be called as API during some layer of Industry 4.0.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

The teacher can use following strategies, to accelerate the attainment of the various types of learning outcomes in this course:

a) Tutorials that need hand calculation and testing in Software.
b) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance may to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.
d) The teacher can suggest several research paper to read and refer. This can be assign to specially to evaluate the capacity of individual, to understand topics related to signal processing but not in syllabus. The capacity can be judge by the way or quality of efforts put, the simulation developed to verify the algorithm and brain storming done and presented to evaluator in form of presentation.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Make a report by literature review on new Embedded boards
b) Identify the possible reduction in mathematical burden required in particular AI/ML algorithm, modify the code and test verify the speed for new algorithm.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit 1: Background and Framework for Industry 4.0	1	*1a. Understand the origin and key benefits of deploying Industry 4.0 1b. Understand design principles and Pillars for Industry 4.0. 1c. Transform a small and medium enterprises (SMEs) into Industry 4.0. Do cost benefit and ROI analysis for deployment of industry 4.0.*	*Lecture Only*	*Mixture of black board and ICT, Videos from Internet*
	2	*Pro 1: Make 3 D model in various freely available software. Pro 2: Simulate Mechanical and Electrical Systems in industries, using software like ANSYS and MATLAB. 1d. Develop mathematical model/ simulation/Digital twin for the given physical process or object. 1e. Select appropriate additive manufacturing process/3D printer for given application.*	*Lab only*	*Mixture of black board and ICT*
	3	*1f. Understand Environmental Manage-ment and Technology potential for Environmental Management 4.0. and hence, Apply design principles looking to it. 1g. Understand Technical and Human Resource view of Industry 4.0 and incorporate it in deployment and management strategies.*	*Lecture Only*	*Mixture of black board and ICT*
Unit 2: Industrial IoT and Industrial Communication	4	*2a. Understand Various Industrial Sensor and smart sensors. 2b. Understand various industrial communication protocols.*	*Lecture only.*	*Mixture of black board and ICT*
	5	*Pro 3: Develop IoT solution for Industries. a) Integrate sensor and Write code to collect the data using M2M communication. 2b. Understand various industrial communication protocols. (continued from last week).*	*Lab and Lecture both.*	*Mixture of black board and ICT*
	6	Pro 3: *Develop IoT solution for Industries. b) Write code to interconnect and interact among smart devices/PLC/Robots with IOT systems.*	*Lab and Lecture both.*	*Mixture of black board and ICT*

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		2c. Understand difference between IoT and M2M communication. Understand basics of M2M communication. *2d. Understand structural of IIoT and key features. Accordingly, select appropriate cloud and software offering for IIoT.*		
Unit 3: AI-ML application in Industry 4.0	7	*3a. Understand role and various application of of AI and ML in Industry 4.0.* *3b. Understand various statistical feature extraction criteria and code it.*	*Lecture Only*	*Mixture of black board and ICT*
	8	*Pro 4: Apply Fourier, Wavelet and similar transform to Mechanical Vibration data set, generate features and Write code for Predictive maintenance of machines, using AI ML applications.* *3c. Write code to Apply Fourier and Wavelet on given data set.*	*Lab and Lecture both.*	*Mixture of black board and ICT*
	9	*3d. Understand LSTM, Random forests, Decision Trees, and logistic regression and Write code for it using ready libraries.*	*Lecture Only*	*Mixture of black board and ICT*
	10	*3e. Understand classification algorithms like SVM, and code it. Write code using ready libraries, for Predicting machines health using classification algorithm.*	*Lecture Only*	*Mixture of black board and ICT*
Unit 4: Industrial Robotics and Programming	11	*4a. Understand various types of Industrial Robot and their expected role and Functions in various industrial processes.* *4b. Understand various parts of robot: Robot Manipulator and Robot Controller, Sensors and actuators in Robot, Other mechanical arrangement.* *4c. Understand various Specification of Industrial Robot. Learn key features of Various Robot (Model) from leading Industrial Robot Manufacturer*	*Lecture Only*	*Mixture of black board and ICT*
	12	*4d. Understand Mathematical back-ground behind various types of motion and develop the simulation for evaluating the response.*	*Lecture Only*	*Mixture of black board and ICT*
	13			

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		Pro7. Design Mechanical System for various types of industrial Robots and make 3 D model, simulate the response of robot. *Pro8. Program Industrial Robot for various types of kinematics and movements.* *4e. Program Industrial robot for various types of motion using Rapid, ROS and PLC ladder logic.*	*Laboratory Only*	*Mixture of black board and ICT*
	14	*Pro9. Develop Digital Twin(DT) and Cyber Physical Systems (CPS) for various industrial task and process.* *a) Develop Digital Twin for industrial processes with ANSYS, MATLAB and Amazon AWS/PTC Thingswork/or suitable cloud offering.*	*Laboratory Only*	*Mixture of black board and ICT*
	15	*Pro9. Develop Digital Twin(DT) and Cyber Physical Systems (CPS) for various industrial task and process.* *b) Develop CPS and use Library support available in Amazon AWS for implementing CPS.* *Pro 10. Write MATLAB code for parameter tuning of digital twin using Least Square Estimation (LSQ), Weighted Least Square Estimation (WLS) and Maximum Like hood(ML) methods.*	*Laboratory Only*	*Mixture of black board and ICT*

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	*Industry 4.0: Current Status and Future Trends,*	*Edited by Jesús Hamilton Ortiz, William Gutierrez Marroquin, Leonardo Zambrano Cifuentes*	*An IntechOpen Publication, Published in London, United Kingdom* *DOI: 10.5772/intechopen.90396*

S. No.	Title of Book	Author	Publication including ISBN
2	*Industry 4.0 The Industrial Internet of Things*	*Alasdair Gilchrist,*	*Apress Publication ISBN: 978-1-4842-2047-4*
3	*Industrial Robots Programming: Building Applications for the Factories of the Future.*	*J. Norberto Pires, Mechanical Engineering Department, University of Coimbra, Portugal*	*SpringerLink, ISBN 0-387-23325-3 ISBN 9780387233253*
4	*Additive Manufacturing Technologies: 3D Printing, Rapid Prototyping, and Direct Digital Manufacturing,*	*Ian Gibson, David Rosen and Brent Stucker*	*SpringerLink, Second Edition ISBN: 978-1-4939-2113-3*
5	*Machine Learning using Python*	*Manaranjan Pradhan, U Dinesh Kumar*	*Wiley, ISBN: 978-81-265-7990-7 ISBN: 978-265-8855-8(ebk)*

XV SOFTWARE/LEARNING WEBSITES

1. *NPTEL Courses by Prof. Sudip Misra, IIT Kharagpur* *https://onlinecourses.nptel.ac.in/noc20_cs69/preview*
2. *OpenSim/OpenSimulator, FreeScad, Tinkercad*
3. *MATLAB, ANSYS, RobotStudio 2.2*

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE 481	Semester VI	**Protection and Switchgear** Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1' for the relevant correlation of each PO or PSO, Competency or CO													
481.1	*Find opportunity of applying industry 4.0*	*2*	*2*	*2*	*2*	*2*	*1*	*1*	*1*	*1*	*1*	*2*	--	--	--

Course Code	Competency and COs	**PO 1** Engineering knowledge	**PO 2** Problem analysis	**PO 3** Design/development of solutions.	**PO 4** Conduct investigations of complex problems	**PO 5** Modern tool usage	**PO 6** The engineer and society	**PO 7** Environment and sustainability	**PO 8** Ethics	**PO 9** Individual and teamwork	**PO 10** Communication	**PO 11** Project management and finance	**PO 12** Life-long learning	**PSO1** Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	**PSO2** Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	Solutions for various industrial process, advise deployment on basis of the cost involved in deployment and possible return on investment.														
481.2	*Design the 3D model of an object to be 3D printed.*	**3**	**2**	**2**	**2**	**3**	--	--	--	**1**	--	--	--	--	--
481.3	*Write Python and C Code for smart sensor, Industrial and IIoT communications.*	**2**	**1**	**2**	**1**	**2**	--	--	--	**1**	--	--	--	---	--
481.4	*Develop dynamic Simulation for a physical system and test it. Develop Digital twin and Cyber Physical System(CPS).*	**3**	**3**	**3**	**2**	**2**	--	--	--	--	**1**	**1**	**2**	--	**2**
481.5	*Write code for preventive/predictive maintenance.*	**3**	**3**	**2**	**3**	**2**	--	--	--	--	--	--	--	---	**2**
481.6	*Program Industrial robot for required action and motion.*	**3**	**3**	**3**	**2**	**3**	--	--	**1**	**1**	**1**	--	--	--	**2**

XXX NAME OF RESOURCE PERSONS

a) Jivanadhar A. Joshi

Course Code: **EE482**
Course Title: **Smart Grid Technologies and Applications**

UG Engineering Programme	**Semester**
Electrical Engineering	Seventh

I RATIONALE

The smart grid can help measure and reduce energy consumption and costs. One of the visions for the smart grid is that consumers will be able to monitor their energy consumption in real time, or at least near real time, via the Web. This curriculum course structure is developed such that the student is able to understand the basic concepts, components, architecture, various measurement technologies of smart grid and the importance of renewable energy in smart grid.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a ***macro-level industry-oriented task*** for this course**:**

- **Evaluate the performance of various renewable energy sources and energy storage equipment which are used in industry and power system.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1) ***Understand*** *the basic concepts, components and architecture of smart grid*.
2) ***Compare*** *the various measurement technologies in smart grid.*
3) **Design** renewable energy sources which are used in smart grid.
4) ***Illustrate*** about battery technology and energy.
5) ***Interpret*** *about role of Electric Vehicles in smart grid*.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			**Total Credits (L+T+P)**	**Examination Scheme**				
				Theory Marks		**Practical Marks**		**Total Marks**
L	**T**	**P**	**C**	**ESE**	PA	**ESE**	**PA**	
3	0	2	4	70	30*	25	25	150

Legends: L*-Lecture;* ***T*** *– Tutorial/Teacher Guided Theory Practice;* ***P*** *- Practical;* ***C*** *– Credit,* ***ESE*** *- End Semester Examination;* ***PA*** *- Progressive Assessment*

(*): *Under the* ***Theory PA****, out of 30 marks,* ***10 marks*** *are for* ***micro-project assessment*** *to facilitate integration of COs and the remaining 20 marks is converted from 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	**Practical Outcomes (Practicum)** (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	<u>Create</u> the hybrid AC/DC microgrid test system in grid-connected mode using MATLAB software.	04
2	2	<u>Computes</u> the PMU benchmark model.	02
3	3	<u>Test</u> the MPPT analysis of PV and wind generator using P & O method.	02
4	3	<u>Analyze</u> the effect of partial shading on a PV generation.	02
5	3	<u>Design</u> the energy management system of microgrid using MATLAB software.	04
6	3	<u>Design</u> the deep neural network for MPPT analysis using MATLAB software.	04
7	4	<u>Create</u> test scenarios for battery management system using Simulink Test	02
8	5	<u>Design</u> vehicle-to-grid system to control microgrid system parameters using MATLAB software.	04
		Total	**24**

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB Simulation software (version 2021a)	1,5,6,7,8
2	Test rig of doubly fed induction generator based wine energy conversion system	2
3	Test rig of MPPT boost converter for solar PV	3,4

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

1) <u>Follow</u> safe practices.
2) <u>Practice</u> energy conservation.
3) <u>Function</u> as a team member.
4) <u>Function</u> as a team leader.
5) <u>Follow</u> ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

a) 'Responding Level' in 1st year
b) 'Valuing Level' in 2nd year

c) 'Organising Level' in 3rd year
d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the ***'Apply'*** and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
Unit – I **Introduction**	**CO 1.** ***Understand** the basic concepts, components and architecture of smart grid.* *PrO 3. Create the hybrid AC/DC microgrid test system in grid-connected mode using MATLAB software.* 1f. *Prepare* the drawing of smart grid for the given situation *1g. Design* the smart grid system for the given area 1h. *Determine* the components used for the specified smart grid application	1.6 Today's Gird Versus Smart Grid, Rationale for Smart Grid, 1.7 Computational Intelligence, Power System Enhancement, 1.8 Communication and Standards. 1.9 Environment and Economics, Shareholders Roles and Function, 1.10 Architecture of Smart Grid 1.11 Smart Grid components functions.
Unit– II **Sensors and Measurement**	**CO 2.** ***Compare** the various measurement technologies in smart grid.* *Pro 2. Computes the PMU benchmark model.* 2p. Determine the sensors for the specified smart grid application. 2q. Choose the correct sensor for the specified application. 2r. Design the PMU for the given rating of smart grid.	2.6 Sensors for Smart Grid 2.7 Monitoring and Measurement Technologies 2.8 PMU, Smart meters, Smart Appliances 2.9 Multi Agent Systems (MAS) Technology 2.10 Micro grid and Smart grid comparison 2.11 Wide Area Measurement
Unit– III **Distributed Generation**	**CO 3. Design** renewable energy sources which are used in smart grid. *Pro 3. Test the MPPT analysis of PV and wind generator using P & O method.* *Pro 4. Analyze the effect of partial shading on a PV generation.* *Pro 5. Design the energy management system of microgrid using MATLAB*	3.6 Solar Energy, PV Systems 3.7 Wind turbine Systems 3.8 Biomass, Small and Micro Hydro Power 3.9 Fuel Cell 3.10 Geothermal heat pumps

Unit No. & Title	*Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)*	*Topics and Sub-topics*
	software. *Pro 6.* Design the deep neural network for MPPT analysis using MATLAB software. 3e. Determine the maximum power generation point from the given type of PV array. **3f.** Evaluate the effect of partial shading on the PV array. **3g.** Design the energy management system for the given smart grid/ microgrid system. **3h.** Design the deep neural network for the given application.	
Unit-IV **Energy Storage**	**CO 4. *Illustrate*** about battery technology and energy. *Pro 6. Create test scenarios for battery management system using Simulink Test* 4f. Select the relevant battery for the specified application. 4g. Evaluate the performance of the given type of battery for the specified condition. 4h. Design the battery management system for the specified application.	4.9 Batteries, Flow Batteries 4.10 Fuel Cell and hydrogen electrolytes 4.11 Flywheel 4.12 Super conduction magnetic energy storage systems, super capacitors, Simulation and case studies
Unit – V **Electric Vehicles**	**CO 5. *Interpret*** *about role of Electric Vehicles in smart grid.* *Pro 7. Design vehicle-to-grid system to control microgrid system parameters using MATLAB software.* 5c. Evaluate the performance of the plugin electric vehicle for the specified condition. 5d. Evaluate the performance of the hybrid electric vehicle for the specified condition. 5e. Design the vehicle to grid system to control the parameters of microgrid.	5.6 Plugin Electric Vehicles and hybrid, 5.7 Vehicle classes, 5.8 Vehicle Architecture, 5.9 Gird to Vehicle (G2V) Charging, Grid Impacts, 5.10 Vehicle to Grid (V2G).

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit	Unit Title	Teaching	Distribution of Theory Marks

No.		Hours	R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction	09	4	4	4	2	0	0	14
II	Sensors and Measurement	09	4	4	4	2	0	0	14
III	Distributed Generation	09	0	4	2	2	2	4	14
IV	Energy Storage	09	2	2	4	2	0	4	14
V	Electric Vehicles	09	0	4	2	2	4	2	14
Total		**45**	**10**	**18**	**16**	**10**	**6**	**10**	**70**

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be ***individually*** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should ***not exceed three.***

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***15 (fifteen) student engagement hours*** during the course.

A suggestive list of micro-projects ***for a minimum Learning effort of 15 hours per semester*** are given here. Similar micro-projects could be added by the concerned faculty of the course:

1) **Electric Vehicle:** Collect the specifications of different types of electric vehicles manufacturing by different types of industries.
2) **Battery:** Collect the specifications of different types of battery used in different Electric vehicle by different types of industries.
3) **Renewable generator:** Prepare case studies on control issues in renewable energy integration and microgrid.
4) Undertake the survey in reputed industry and submit a report on different type of sensors and standard used to implement smart grid.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

a) Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b) ***'L' in item No. 4*** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
c) About ***15-20% of the topics/sub-topics*** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for ***self-directed learning***, but to be progressively assessed for their integration into the COs.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related ***co-curricular*** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

a) Library survey regarding different protection schemes used in industries.
b) Prepare 15 min. power point presentation on how to choose relay for the particular application.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I **Introduction**	1	**Pro 1.** *Create the hybrid AC/DC microgrid test system in grid-connected mode using MATLAB software.* 1a. Prepare the drawing of smart grid for the given situation	Lecture with media	PPTs, black board
	2	*1b. Design* the smart grid system for the given area	Lecture With media	PPTs, black board
	3	1c. *Determine* the components used for the specified smart grid application	Lecture with media,	PPTs, black board
Unit– II **Sensors and Measurement**	**4**	2a. Determine the sensors for the specified smart grid application.	Lecture with media, Tutorial	PPTs, black board
	5	2b. Choose the correct sensor for the specified application.	Lecture with media, Tutorial	PPTs, black board
	6	**Pro 2.** *Computes the PMU benchmark model.* 2c. Design the PMU for the given rating of smart grid.	Lecture with media	PPTs, black board
Unit– III **Distribut**	**7**	**Pro 3.** *Test the MPPT analysis of PV and wind generator using P & O method.*	Lecture with media,	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
ed Generation		**Pro 4.** *Analyze the effect of partial shading on a PV generation.* 3a. Determine the maximum power generation point from the given type of PV array. 3b. Evaluate the effect of partial shading on the PV array.		
	8	**Pro 5**. *Design the energy management system of microgrid using MATLAB software.* 3c. Design the energy management system for the given smart grid/ microgrid system.	Lecture with media	PPTs, black board
	9	**Pro 6**. Design the deep neural network for MPPT analysis using MATLAB software. 3d. Design the deep neural network for the given application.	Lecture with media, + PPT	PPTs, black board
Unit-IV Energy Storage	10	4a. Select the relevant battery for the specified application.	Lecture with media, Tutorial	PPT, black board
	11	4b. Evaluate the performance of the given type of battery for the specified condition.	Lecture with media	PPTs, black board
	12	**Pro 6.** *Create test scenarios for battery management system using Simulink Test* 4c. Design the battery management system for the specified application	Lecture with media, case study	PPT, black board
Unit – V Electric Vehicles	13	5a. Evaluate the performance of the plugin electric vehicle for the specified condition.	Lecture with media	PPTs, black board
	14	5b. Evaluate the performance of the hybrid electric vehicle for the specified condition.	Lecture with media, Tutorial	PPTs, black board
	15	**Pro 7**. *Design vehicle-to-grid system to control microgrid system parameters using MATLAB software.* 5c. Design the vehicle to grid system to control the parameters of microgrid	Lecture with media, Tutorial	PPTs, black board, Video

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Smart Grid: Fundamentals of design and analysis	James Momoh	John Wiley & sons, 2012. ISBN: 9780070671188, 0070671184
2	Smart Grid: Technology and	Janaka Ekanayake, Kithsiri Liyanage, Jianzhong Wu,	John Wiley. ISBN: 9788126557356

S. No.	Title of Book	Author	Publication including ISBN
	Applications	Akihiko Yokoyama, Nick Jenkins	
3	Smart Grid Applications, Communications, and Security	Lars T. Berger, Krzysztof Iniewski	Wiley, 2012 ISBN: 978-1-118-00439-5
4	Smart Grid: Integrating Renewable, Distributed and Efficient Energy	Fereidoon Sioshansi	Academic Press, 2012, ISBN: 9780071077743
5	Innovative Testing and Measurement Solutions for Smart Grid	Qi Huang, Shi Jing, Jianbo Yi, Wei Zhen	Wiley, 2016 ISBN: 978-1-118-88992-3

XV SOFTWARE/LEARNING WEBSITES

1. Introduction to Smart Grid: - https://nptel.ac.in/courses/108107113
2. Fundamentals of Electric Vehicles: Technology & Economics: - https://nptel.ac.in/courses/108106170

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE482	Semester VII														
482	**Competency** Evaluate the performance of various renewable energy sources and energy storage equipment which are used in industry and power system	***3***	***2***	***2***	***2***	***1***	**-**	***1***	**-**	**-**	**-**	**-**	***1***	***3***	**-**
482.1	Understand the basic concepts, components and architecture of smart grid.	3	3	3	2	1	-	2	-	-	-	1	1	3	2
482.2	Compare the various measurement technologies in	3	3	1	1	3	-	1	-	-	-	2	-	3	1

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	smart grid														
482.3	Design renewable energy sources which are used in smart grid	3	3	1	2	-	-	-	-	-	-	2	-	3	2
482.4	Illustrate about battery technology and energy	3	2	2	-	1	-	-	-	-	-	-	-	2	-
482.5	Interpret about role of Electric Vehicles in smart grid	3	2	2	2	-	1	2	-	1	-	1	-	2	1

XVII NAMES OF RESOURCE PERSONS

a) Dr. Jigar Sarda

B. Tech. (Electrical Engineering) Programme

SYLLABI

(Semester – VIII)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Charotar University of Science and Technology, Gujarat

(use as template)

Re-designed Course Structure of UG in Electrical Engineering

Course Code: **EE458**

Course Title: **Major Project**

UG Engineering Programme	Semester
Electrical Engineering	Eight

I RATIONALE

This course enables the students to apply some of the knowledge and/or skills developed during the programme to new problem for which there are number of engineering solutions. This course include planning of the tasks which are to be completed within the time allocated, and in turn, helps to develop ability to plan, , use, monitor and control resources optimally and economically. By studying this course abilities like creativity, imitativeness and performance qualities are also developed in students. Leadership development and supervision skills are also integrated objectives of learning this course.

II COMPETENCY

The course should be taught and implemented with the aim to develop the required course outcomes (COs) so that students will acquire following competency needed by the industry:

- **Find the solution of a real-life problem**.

III COURSE OUTCOMES (COs)

Depending upon the nature of the projects undertaken, the following could be some of the major course outcomes that could be attained, although, in case of some projects few of the following course outcomes may not be applicable.

1) Apply the engineering knowledge to analyze, provide solution and carried out investigation of the problems by individual or team.
2) Identify the engineering problem; solve, report and present it by teamwork and work with other discipline.
3) Use experience related to professional and ethical principles and responsibility during their project.
4) Evaluate the problem in commercial, societal, health, safety, legal, cultural and environment aspects before reach to final solution.
5) Apply the engineering skills to contemporary issues related to engineering in general with advanced tools and technologies

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)	Total Credits (L+T+P)	Examination Scheme		
		Theory Marks	Practical Marks	Total Marks

L	T	P	C	ESE	PA	ESE	PA	800
0	0	36	20	00	00	400	400	

***Legends: L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment*

***(*):** Under the **Theory PA**, out of 40 marks, **20 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.*

V COURSE DETAILS

i. Major Project showcases that, a student during the final semester will able to apply the engineering knowledge to a practical problem.

ii. Students can carry out their project work in-house at institute/ industry/ research organization/govt. organization.

iii. Project should be related to the Product/prototype design and development or Experimentation, Simulations and evaluation of any system etc.

iv. A student is required to carry out the project work related to Electrical Engineering or some interdisciplinary work under the guidance of a faculty member and the supervisor of the concerned industry/institute/organization.

v. The major project work provides student an opportunity to develop something on their own or group of maximum 3 students. Final decision related to the number of students in one group is based on internal supervisor recommendation.

vi. Project will be evaluated at least twice during the semester by internal faculty as a part of continuous evaluation and final evaluation will be carried out by external examiner at the end of semester.

vii. Student should submit the project report at the end of the semester duly signed by project supervisor and head of department. Report comprising literature review, objective, methodology, scope of the project work undertaken, outcome of project and conclusions.

viii. The students are required to identify their problem and they are required to follow all the rules and instruction issued by department/industry.

ix. Students should finalize the project at the end of seven semester (before end of November)

VI PROJECT REPORT

At the end of the semester student should prepare project report. Guideline of the Project report is as follow

a) Certificate
b) Acknowledgements
c) Abstract (in one paragraph not more than 150 words)
d) Content Page
e) Chapter-1 Introduction and background of the Industry or User based Problem/Task
f) Chapter-2 Literature Survey for Problem/Task Identification and Definition,

g) Chapter-3 Proposed Detailed Methodology of solving the problem/completing the task
h) Chapter-4 Methodology actually followed to solve the problem/complete the task
i) Chapter-5 The solution, its development and its implementation/execution (with details including drawings, software, used for or developed for the solution etc. in detail). In case of survey/experiment type of project this chapter will include data analysis, information presentation and analysis.
j) Chapter-6 Final conclusions and recommendations if any for future projects.

Note:

a) The report should contain as many diagrams and figures, charts etc. as relevant for the project.
b) Originality of the report (written in own words) would be given more importance rather than quality of printing and use of glossy paper or multi-colour printing

VII ASSESSMENT OF PROJECT WORK

Assessment of Project work also has two components, first is Progressive Assessment, while another is End of the Term Assessment.

i. Progressive Assessment Criteria

a) Project guide or the group of internal faculty are supposed to carry out this assessment. It is a continuous process, during which for developing desired qualities in the students, faculty should orally give informal feedback to students about their performance and interpersonal behaviour while guiding them on their project work every week.

S. No.	Criteria	Marks
	First Progressive Assessment at the end of 4th week	
1	**Project Proposal** for work to be completed at the end of semester: Including problem definition and methodology adapted, literature review adopted (in case of working models prepared then quality of workman ship, appropriateness of raw materials/components/ technology being used, functioning of the prototype etc. has to be also assessed)	**40**
	General Behaviour (initiative, resourcefulness, reasoning ability, imagination/creativity, self-reliance)	**Only written feed-back/suggestions**
	Second Progressive Assessment at the end of 12th week	
2	**Execution of Plan during semester** (marks to be also given based on ability to collect relevant information, ability to follow correct procedure, manipulative skills, ability to observe, record & interpret, ingenuity in the use of material and equipment, target achievement) In case of working models, quality of workman ship (including accuracy in dimensions, shape, tolerance limits), appropriateness of raw materials/components/ technology being used, functioning of the prototype, cost effectiveness, marketability, modernity etc. has to be also	**120**

S. No.	Criteria	Marks
	assessed	
3	**Log book** (for work done during semester- detailed and regular entry would be basis of marks)	**24**
4	**General Behaviour** (persistence, interest, confidence, problem solving ability, decision making ability, initiative to act, team spirit, sharing of material etc., participation in discussions, completion of individual responsibilities, leadership)	**Only written feed-back./ suggestions**
	Third Progressive Assessment at the end of 14th week	
5	**Portfolio for Self-learning and reflection** (for work done in semester- marks based on amount of reflection and completion of the portfolio)	**48**
6	**Project Report including documentation -** for work done in semester (marks based on: clarity in presentation and organization; styles and language; quality of diagrams, drawings and graphs; accuracy of conclusion drawn; citing of cross references; suggestion for further research/project work)	
7	**Presentation** (presentation skills including communication skills to be assessed by observing presentations and asking questions during presentation for work done during semester)	**48**
8	**Defence** (ability to defend the methods/materials used and technical knowledge, and involvement of individual to be assessed by asking questions during presentation and viva/voce for work done during semester)	**56**
	Total	**400**

Note: Oral feedback on general behaviour may also be given whenever relevant/ required during day to day guidance and supervision.

ii. END SEMESTER EXAMINATION

a) Criteria of Marks for ESE for Capstone Project Execution.

S. No.	Description	Marks
1	**Problem Identification/Project Title** (Innovation /Utility of the Project for industry/ User/Academia) marks to be also given based on (i) Accuracy or specificity of the scope and (ii) Appropriateness of the work with reference to desired course outcomes.	**16**
2	**Industrial Survey and Literature Review** (marks to be given based on extent/volume and quality of the survey of Industry / Society / Institutes/Literature/Internet for Problem Identification and possible solutions)	**24**
3	**Project Proposal** (In case of working models, detailed design and planning of fabrication/assembly of the prototype has to be also assessed). Marks to be given also based on appropriateness, flexibility, detail and clarity in methods/planning.	**24**
4	**Execution of Plan** (marks to be also given based on ability to collect	**120**

S. No.	Description	Marks
	relevant information, ability to follow correct procedure, manipulative skills, ability to observe, record & interpret, ingenuity in the use of material and equipment, target achievement) In case of working models, quality of workman ship (including accuracy in dimensions, shape, tolerance limits), appropriateness of raw materials/components/ technology being used, functioning of the prototype, cost effectiveness, marketability, modernity etc. has to be also assessed.	
5	**Log book** (detailed and regular entry would be basis of marks)	**16**
6	**Portfolio for Self learning and reflection** (marks based on amount of reflection and completion of the portfolio)	**40**
7	**Final Report including documentation**. (marks based on: clarity in presentation and organization; styles and language; quality of diagrams, drawings and graphs; accuracy of conclusion drawn; citing of cross references; suggestion for further research/project work)	**80**
8	**Presentation** (presentation skills including communication skills to be assessed by observing presentations and asking questions during presentation)	**40**
9	**Defence** (ability to defend the methods/materials used and technical knowledge, and involvement of individual to be assessed by asking questions during presentation and viva/voce)	**40**
Total		**400**

X Guidelines for Portfolio Preparation and assessment

The main purpose of portfolio preparation is learning based on self-assessment and portfolio is not to be used for assessment in traditional sense.

a) Each student has to prepare his/her portfolio separately. However, he/she can discuss with the group members about certain issues on which he/she wants to write in the portfolio.

b) Whatever is written inside the portfolio is never to be used for assessment, because if teachers start giving marks based on whatever is written in the portfolio, then students would hesitate in true self-assessment and would not openly describe their own mistakes or shortcomings.

c) Some marks are allocated for portfolio, these marks are to be given based on how sincerely portfolio has been prepared and not based on what strengths and weaknesses of the students are mentioned in the portfolio.

d) Portfolio has to be returned back to the students after assessing it (assessment is only to see that whether portfolio is completed properly or not) by teachers. Because student is the real owner of the portfolio.

e) Students mainly learn during portfolio preparation, but they can further learn if they read it after a gap. And hence they are supposed to keep the portfolios with them even after completion of the diploma because it is record of their own experiences (it is like diary some people write about their personal experiences), because they can read it again after some time and can revise their learning (about their own qualities)

XI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE458	**Semester VIII**														
458	**Competency** *Find the solution of a real-life problem.*	***3***	***2***	***2***	***2***	***2***	***2***	***1***	***1***	***2***	***2***	***2***	***1***	***2***	***2***
458.1	Apply the engineering knowledge to analyze, provide solution and carried out investigation of the problems by individual or team.	3	-	1	-	-	1	-	1	-	1	1	1	-	1
458.2	Identify the engineering problem; solve, report and present it by teamwork and work with other discipline.	3	3	2	2	1	2	-	-	3	2	1	2	2	-
458.3	Use experience related to professional and ethical principles and responsibility during their project.	3	2	1	-	-	-	-	3	1	-	2	-	-	1
458.4	Evaluate the problem in commercial, societal, health, safety, legal, cultural and environment aspects before reach to final solution.	3	-	-	3	3	3	1	-	2	1	3	2	3	3
458.5	Apply the engineering skills to contemporary issues related to engineering in general with advanced tools and technologies	3	1	3	2	3	3	-	-	-	2	2	3	3	2

XII NAMES OF RESOURCE PERSONS

a) Dr. Nilay Patel
b) Prof. Soaib Saiyad